

The hypercube Ramsey number has linear order

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Abstract

We prove that the two-color Ramsey number of the n -dimensional binary cube is at most $C2^n$, where C is an absolute constant. This resolves positively the hypercube Ramsey conjecture of Burr and Erdős.

1 Introduction

Let Q_n be the graph with vertex set $\{0, 1\}^n$ in which two vertices are adjacent when they differ in exactly one coordinate. For a finite graph H , let $R(H)$ be the least positive integer M such that every red-blue coloring of the edges of K_M contains a monochromatic subgraph isomorphic to H . The copy is not required to be induced.

Theorem 1.1. *There is an absolute constant $C > 0$ such that*

$$R(Q_n) \leq C2^n \quad \text{for every integer } n \geq 0.$$

1.1 The problem and its historical development

In their 1975 paper, Burr and Erdős called a family of graphs an L -set if one constant bounds the Ramsey number of every member by that constant times its order [3, Section 1]. The arboricity of a graph is the minimum number of forests into which its edges can be partitioned. They conjectured that every family of uniformly bounded arboricity is an L -set. A graph is d -degenerate if every nonempty subgraph has a vertex of degree at most d ; uniformly bounded arboricity and uniformly bounded degeneracy are equivalent boundedness conditions for families [3, Section 3]. Burr and Erdős raised the cube question separately in Section 7, asking whether the cubes form an L -set. Theorem 1.1 resolves that question positively.

The cube question sits at a natural density boundary. The first-moment lower bound used by Burr and Erdős implies that a family with one linear Ramsey constant must satisfy $e(H)/|V(H)| = O(\log |V(H)|)$ as its graph orders grow, where $e(H)$ is the number of edges [3, Lemma 4.3 and Section 7]. For the cube, $e(Q_n)/|V(Q_n)| = n/2 = \frac{1}{2} \log_2 |V(Q_n)|$. Thus cubes reach the logarithmic order of edges per vertex that this necessary condition still permits. They test linear Ramsey growth beyond families with bounded degeneracy, at the largest possible order of this density parameter up to a constant factor.

For $n \geq 1$, parity of the coordinate sum gives the connected cube two bipartition classes of size 2^{n-1} . The standard lower-bound coloring with two blocks [6, Introduction] gives $R(Q_n) \geq 3 \cdot 2^{n-1} - 1$. Indeed, for $n \geq 2$, take blocks of sizes $2^n - 1$ and $2^{n-1} - 1$, color edges inside each block red, and color crossing edges blue. Each red component is too small for Q_n , while a blue copy would need 2^{n-1} vertices in each block. The case $n = 1$ is immediate. Theorem 1.1 therefore determines the

order of $R(Q_n)$. It gives no numerical value for C and does not determine whether $R(Q_n)/2^n$ converges or what its limiting value would be.

One line of progress established linear Ramsey growth when a sparsity parameter is fixed. Chvátal, Rödl, Szemerédi and Trotter proved in 1983 that graphs of any fixed maximum degree have Ramsey number linear in their order, with a constant depending on the degree [4]. Lee later proved the corresponding assertion for every fixed degeneracy, resolving the bounded-degeneracy form of the Burr–Erdős conjecture [26, Theorem 1.1]. These results did not settle the cube question because both the maximum degree and the degeneracy of Q_n equal n : their constants may grow with the dimension.

A second line sought quantitative estimates when the degree grows with the graph. Early work on cubes includes Beck’s 1983 paper [2]. Graham, Rödl and Ruciński proved that a bipartite graph H of maximum degree at most $\Delta \geq 1$ satisfies $R(H) < 8(8\Delta)^\Delta |V(H)|$ [18, Theorem 1]. For Q_n with $n \geq 1$, whose order is $m = 2^n$, this gives $R(Q_n) < 8(16n)^n$, or $m^{O(\log \log m)}$. Shi then obtained a bound polynomial in m , using a refinement of Kostochka and Rödl’s common-neighborhood lemma [25, 31], and later refined the estimate [32]. This moved the cube problem from a growing exponent in m to a fixed exponent.

The next estimates approached and then improved on the quadratic scale. For positive integers n , Fox and Sudakov obtained $R(Q_n) \leq n2^{2n+5}$ [14, Corollary 1.3], and Conlon, Fox and Sudakov obtained $R(Q_n) \leq 2^{2n+6}$ [7, Corollary 4.2]. Lee’s Theorem 1.3 and the paragraph following it give $R(Q_n) \leq 2^{2n} + n^2 2^n$ for all sufficiently large n [26]. Tikhomirov’s exponential improvement is $R(Q_n) \leq 2^{2n-cn+1} + 2$ for an absolute $c > 0$ and all sufficiently large n [34, Corollary 1.2]. In terms of $m = 2^n$, these estimates pass from an extra logarithmic factor above m^2 , to a constant multiple of m^2 , to $(1 + o(1))m^2$, and finally to an exponent below 2. The present theorem reaches a constant multiple of m .

Consequences for related graph families. The bound transfers to targets occupying a controlled fraction of an ambient cube. If H is a not necessarily induced subgraph of Q_d and $2^d \leq A|V(H)|$ for a fixed $A \geq 1$, monotonicity gives $R(H) \leq R(Q_d) \leq C2^d \leq CA|V(H)|$. The bound on the ambient order controls the constant for the smaller target.

For example, let $d \geq 0$ and $k \geq 1$ be integers, and let F be a spanning subgraph of Q_d . Its Cartesian power $F^{\square k}$ has vertex set $V(F)^k$; two tuples are adjacent when they agree in all but one coordinate and the two entries there are adjacent in F . Concatenating the k blocks of cube coordinates makes this product a spanning subgraph of Q_{dk} . Consequently,

$$R(F^{\square k}) \leq R(Q_{dk}) \leq C2^{dk} = C|V(F)|^k,$$

with the same absolute C for all permitted F, d, k .

This includes Cartesian powers of paths on 2^d vertices for $d \geq 1$, and cycles on 2^d vertices for $d \geq 2$. Starting with the order 0, 1 for Q_1 , obtain an order of Q_d by prefixing 0 to the order of Q_{d-1} and then 1 to its reverse. Consecutive vertices are adjacent, and for $d \geq 2$ the first and last vertices are also adjacent, giving the cycle. Writing P_a for the path on a vertices, Mota, Sárközy, Schacht and Taraz proved $R(P_a \square P_b) = (3/2 + o(1))ab$ as $ab \rightarrow \infty$ for positive integers a, b [29, Corollary 1.4]. Their framework fixes the degree and does not give a constant uniform in a growing number of Cartesian factors; the deduction above does so when the factor spans a cube.

1.2 Background on embedding and sampling

The common-neighborhood method addresses a basic extension step in bipartite embedding. Once one target class has been placed, each vertex in the other class needs an unused common neighbor

of its already placed neighbors. Under their respective density and size assumptions, Kostochka and Rödl’s deterministic lemma [25, Lemma 1] and Sudakov’s probabilistic argument [33, Section 2] find large host sets in which every subset of a prescribed size has many common neighbors; Sudakov explicitly drew on Gowers’s earlier sampling idea [17, Section 4, Lemma 11]. Later proofs allow a controlled collection of exceptional subsets and arrange the first embedding so that target neighborhoods avoid them. Conlon uses a hypergraph packing argument [5]; independently, Fox and Sudakov use an auxiliary hypergraph embedding [14, Lemmas 2.1–2.2]. Conlon, Fox and Sudakov use local avoidance of exceptional subsets and collisions [7, Section 4]. These refinements give bounds of the form $2^{O(\Delta)}m$ for bipartite targets of order m and maximum degree Δ , but the exponential dependence on $\Delta = n$ still costs a power of the cube order. Lee’s embedding records a penalty for small common neighborhoods, controls its average over target neighborhoods, and embeds the most constrained vertices first [26, Section 2].

Tikhomirov [34, Sections 1, 5 and 6] identifies an obstruction to mapping a whole cube bipartition uniformly into one prepared set. His proof instead distinguishes dispersed common neighborhoods, a denser subgraph, and a block structure suited to a facet-wise cube embedding. These density and structural alternatives provide a useful methodological comparison for the reductions below. His Theorem 1.1 is a density theorem for one color: for all sufficiently large n , it embeds Q_n in every bipartite graph of density at least $1/2$ whose two parts each have at least $2^{(2-c)n}$ vertices, for an absolute $c > 0$. The present proof uses the absence of a cube in both colors of a counterexample to derive discrepancy.

A distinct line studies $R(Q_n, K_s)$, the least order of a complete graph whose every red-blue edge coloring contains a red Q_n or a blue K_s . Conlon, Fox, Lee and Sudakov proved $R(Q_n, K_s) \leq c_s 2^n$ for every fixed $s \geq 3$ [6, Theorem 1.1]. Fiz Pontiveros, Griffiths, Morris, Saxton and Skokan obtained the triangle asymptotic $R(Q_n, K_3) = (1 + o(1))2^{n+1}$ [13, Theorem 1.1], and their theorem for a fixed clique gives

$$R(Q_n, K_s) = (s - 1)(2^n - 1) + 1$$

for each fixed $s \geq 3$ and all sufficiently large n [12, Theorem 1.1]. The cube tilings in the Conlon–Fox–Lee–Sudakov and triangle papers use subcubes obtained by fixing initial coordinates, then control the edges between adjacent pieces. The present proof uses that geometry with piece sizes chosen from retained patch masses and with different control of crossing edges. These conclusions for a fixed clique do not imply the diagonal cube theorem, where the target in both colors grows with n .

The task in the present diagonal problem is to retain a constant multiplicative loss when the host order is only a slowly diverging multiple of 2^n . We work with a hypothetical counterexample sequence having two disjoint host sides of size N each and $N/2^n \rightarrow \infty$, with no prescribed rate of divergence.

1.3 The four phases of the proof

A *role* is a vertex of the cube being embedded, and a *label* is a vertex of the host graph. We map the even and odd roles to disjoint host sides. The construction must preserve every cube edge in one color and assign distinct labels to distinct roles. After an injective assignment f of the odd roles, the possible labels for an even role v form the common-neighbor set

$$L_v = \bigcap_{u \sim v} N_G(f(u)),$$

where G is the chosen color. Most embedding arguments below construct probability laws p_v supported on L_v with $\sum_v p_v(x) \leq 1$ for every host label x . Summing over any set of roles then

gives Hall's condition and hence distinct representatives. In the last part of the proof, we instead construct random two-element subsets of L_v and bound directly the probability that any collection of these sets violates Hall's condition.

To describe the reductions, a *patch* will mean a pair of host sets, one on each side, with specified sampling laws μ, ν ; the required density properties vary between the constructions. Write

$$d_G(\mu, \nu) = \Pr\{xy \in G\}, \quad x \sim \mu, \quad y \sim \nu \quad \text{independently.}$$

The *width* of μ is $\log(N \max_x \mu(x))$: a width bound w allows no atom larger than e^w/N . In particular, the uniform law on M labels has width $\log(N/M)$. A discrepancy bound controls $|d_G(\mu, \nu) - 1/2|$ for every pair of laws within specified width budgets. An orientation specifies which host side receives the first law; reversing it interchanges the sides and their width budgets.

A patch property is *available* if, for some fixed $\kappa > 0$ and all sufficiently large n , a patch satisfying it exists after every removal of at most κN labels from each side. The patch may change with the removed sets.

1. **Reduce to a sequence and construct the probability tools.** If the theorem fails, Section 2 gives counterexamples with two disjoint host sides of size N , where $n \rightarrow \infty$, $N/2^n \rightarrow \infty$, and $N \leq n2^n$. The construction must work without any prescribed rate for the middle limit. Section 3 provides mixtures of patch laws with small average label masses, locally determined choices of random data, and ways to sample labels injectively while controlling their joint probabilities.
2. **Use density gains or deduce discrepancy.** By Lemma 3.2, we may pass to a subsequence and remove $o(N)$ labels from each side so that each tested patch property is either absent or available. At prescribed power widths, Section 4 then shows that available bias of at least a suitable negative power of n forces an available nearly monochromatic patch at doubled widths, unless a cube can already be embedded. The broad-side and small-grid constructions in Sections 5–7 then embed a cube under their respective concentration and high-degree hypotheses. These implications, applied in both colors of a counterexample, exclude bias throughout an initial range of widths: every permitted law pair has density close to one half. Sections 8–11 extend this range, allowing much greater concentration on one side than on the other. They also exclude the full-dimensional cluster witnesses of Section 10, in which many labels have unusually large pairwise common neighborhoods. Section 12 records the resulting discrepancy bounds with a small linear width budget on either side and a power width on the other. The cluster exclusion also bounds the scales of the smaller cluster patches used next.
3. **Allocate the cube to patches and use the larger gains.** Section 13 extracts disjoint patches and retains a family with one color, one orientation, and one of the finitely many constructions described there. It allocates the cube to these patches by subcubes obtained by fixing initial coordinates, with sizes proportional to the patch sizes. This is the geometry used in earlier cube-versus-clique tilings [6, 13]. The fixed coordinates identify the cube edges crossing between patches.

Smaller host sets and crossing edges reduce the available common neighborhoods, so the construction needs a compensating gain. For comparison, suppose that each of n independent restrictions retains each label with probability $1/2 + \delta$. A set of size M would then leave an expected $M(1/2 + \delta)^n$ labels. Relative to $N2^{-n}$, the factor is $(M/N)(1 + 2\delta)^n$: the accumulated density surplus can compensate for the smaller support. The proof obtains the needed gain either from edge density or from common-neighbor density inside cluster patches, and bounds the additional losses from crossing edges and dependent choices. Section 14 constructs the laws

used inside cluster patches. When the gain is large enough, Section 15 assigns the odd roles injectively and bounds the column sums of the even laws supported on L_v . Hall’s theorem then completes the embedding.

4. **Complete the cases with smaller gains.** Sections 16 and 17 first assign the odd roles and preserve lists for the even roles. A fixed linear map on cube words partitions the odd roles into syndrome classes; a chosen $O(\log n)$ of these classes receive provisional labels. Their labels are replaced later, one class at a time, from reserved host sets. Each even role has at most one neighbor in each late class. Section 18 controls how these replacements shorten its list and leaves two distinct candidates adjacent to all its assigned odd neighbors. A joint bound on the candidate pairs rules out Hall-deficient collections, giving distinct even labels.

Appendix A collects the quantitative branches, including their width, density, and loss budgets. It is a reference for the parameter choices after the corresponding constructions have been introduced.

The probability comparisons. A recurring construction samples a hidden k -tuple on the first host side and requires neighboring odd labels to be adjacent to every entry. Given these odd labels, the tuple’s conditional distribution is supported on the k th Cartesian power of their common neighborhood. A bound on its largest atom therefore gives a lower bound on the size of that neighborhood. The tuple can also influence which observations are selected. Its likelihood must include that selection event, with the selection rule recomputed when the tuple is varied. Lemma 3.7 formulates this comparison; Section 4 gives its first application.

Injectivity introduces another dependence. The calibrated sampler in Lemma 3.9 preserves individual marginals exactly and bounds probabilities of several specified outputs by their product probabilities times a controlled factor. Exact marginals are needed in later cluster constructions whose scales may remain fixed. The clock sampler in Lemma 3.10 also avoids specified local failures and provides a joint upper bound. Summing such bounds against nonnegative functions permits the common-neighbor and load estimates to be used after injective sampling. The later sequential assignment requires further conditional comparisons, developed where the successive observations are defined in Section 18.

Figure 1 records the dependencies between the reductions. In particular, the cluster exclusion is used both before the deep discrepancy conclusion and during the later patch extraction.

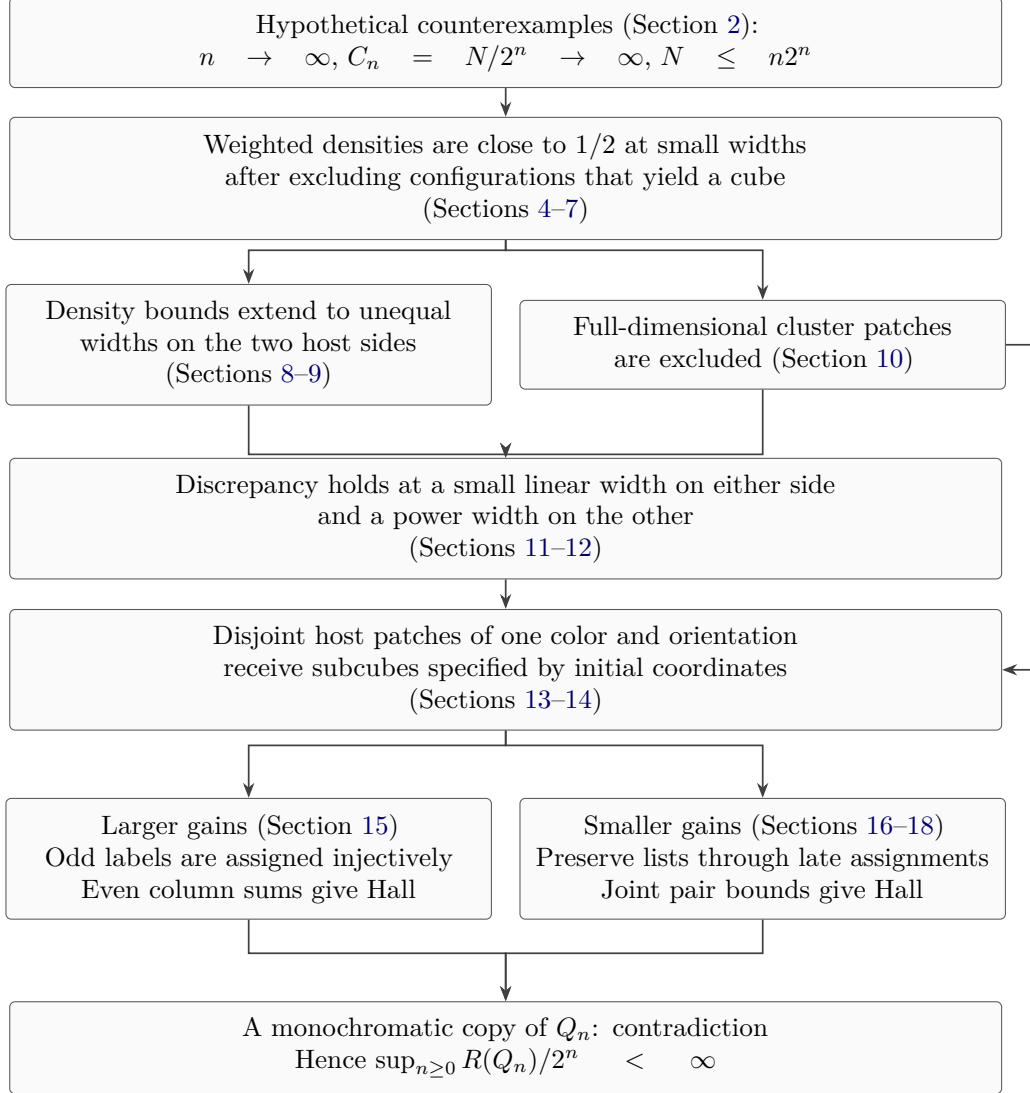


Figure 1: Logical dependencies of the proof. Each reduction is carried out along the hypothetical counterexample sequence, passing to subsequences and removing a vanishing fraction of labels where needed. The separate arrow from cluster exclusion to patch extraction records its second use: it bounds the measured scales of the remaining cluster patches. One common color and orientation are fixed before cube allocation. The final estimates apply to every sequence $C_n \rightarrow \infty$, with no prescribed rate of divergence.

1.4 Terminology

All constants and exponents are fixed before $n \rightarrow \infty$. A fixed threshold for a small patch scale need not grow with n .

Term or symbol	Meaning
Position or role	A vertex of the cube being embedded.
Label	A vertex of the host graph.
X, Y, N	Disjoint host sides, each initially of size N ; later width normalizations continue to use this original N .
C_n	The ratio $N/2^n$, tending to infinity at an unspecified rate.
Width of μ	$\log(N \max_x \mu(x))$, with natural logarithms.
Patch and M_i	In the final tiling, a pair of retained host sets of common size M_i , used for a portion of the cube. Earlier patches are specified by their laws.
Raw law	The specified law before the avoidance constraints of the stage being discussed.
Gate	An event indicator included in a probability or likelihood to restrict it to specified local conditions; it does not normalize the law.
Profile	A probability distribution on a finite set of permitted local choices, chosen to balance expected label masses.

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2 Reduction and notation

We prove Theorem 1.1 by contradiction along sequences. All constants in an asymptotic argument, including exponents unless explicitly indicated otherwise, are independent of n .

Lemma 2.1 (Counterexample sequence). *If $R(Q_n)/2^n$ is unbounded, there is a sequence of dimensions and red-blue colorings between disjoint sets X, Y , each of size N , with no monochromatic Q_n across the two sides, such that*

$$n \longrightarrow \infty, \quad C_n = N/2^n \longrightarrow \infty, \quad N \leq n2^n.$$

Proof. Each fixed dimension has finite Ramsey number by the finite Ramsey theorem [30]. Thus an unbounded ratio has a subsequence along which both the dimensions and the ratios tend to infinity. For each such n , take a coloring of $K_{R(Q_n)-1}$ containing no monochromatic Q_n , and set

$$N = \min \left\{ \left\lfloor \frac{R(Q_n) - 1}{2} \right\rfloor, n2^n \right\}.$$

Choose $2N$ of its vertices and split them into equal sides. Restriction to the cross edges cannot create a monochromatic cube. Moreover, $N/2^n \rightarrow \infty$, since both terms in the defining minimum have ratios tending to infinity. $\square$

Fix such a sequence for the remainder of the proof. We will contradict its existence without imposing any rate of growth on C_n .

Write A and B for the even and odd vertices of Q_n . We call these vertices *roles* or *positions*, and call host vertices *labels*. Usually the embedding sends A to X and B to Y ; we may interchange the host sides. Distances between cube positions are Hamming distances.

After labels have been discarded, all size and law normalizations still use the original side size N , unless another normalization is stated. For a probability law μ on either host side, define its *width* by

$$\text{width}(\mu) = \log(N \max_x \mu(x)).$$

Thus the uniform law on a set of M labels has width $\log(N/M)$. All logarithms are natural.

For a color G and laws μ on X , ν on Y , put

$$d_G(\mu, \nu) = \sum_{x \in X} \sum_{y \in Y} \mu(x) \nu(y) \mathbf{1}_{\{xy \in G\}}.$$

This is the edge density seen by independent draws from the two laws. A label in an argument of d_G denotes its point mass. A label *hits* another label in G when the two are adjacent in that color. We leave integer roundoffs in scales and layer counts implicit when the accompanying asymptotic inequalities have slack.

The tools in Section 3 are used throughout. Sections 4–7 establish the initial discrepancy bound (7.2); Sections 8–11 exclude asymmetric purity, intermediate bias, full-dimensional clusters, and the linear-budget jump. Section 12 records the resulting deep discrepancy and interaction estimates. Sections 13–15 extract patches, allocate the cube, and exclude the high modes. Sections 16–18 handle the bounded and low modes, ending with the two-endpoint Hall estimate. Figure 1 shows the dependencies between these reductions.

3 General conventions and probabilistic tools

This section supplies the finite tools used throughout the proof. We first retain witness families after small removals and balance their expected label loads. We then prove conditional probability comparisons, a local construction for choosing centers, and two samplers that assign distinct labels. Each later application specifies the laws and local tests to which these tools are applied.

Definition 3.1 (Availability). A witness has specified supports on the current host sides. After a pair of label removals, it remains a witness if both supports are wholly retained. A family is **available** along a sequence if some constant $\kappa > 0$ has the following property: eventually, after every removal of at most κN labels from each side, at least one witness remains. The witness may be chosen anew for each pair of removals.

Lemma 3.2 (Stabilization and finite menus). *For a countable list of existence properties defined by witnesses intrinsic to their supports, we can pass to a subsequence and discard $o(N)$ labels so that each property is either eventually absent or available. Every fixed earlier availability survives with a smaller fixed removal tolerance. An available finite union has an available member after passing to a subsequence. An available menu can be replaced by a finite menu at each dimension.*

Proof. Treat the properties in order. If the current property can be eliminated by $o(N)$ additional removals along a subsequence, make those removals and pass to that subsequence. Otherwise it is available: failure of every fixed tolerance would let us choose successive dimensions at which removals of size at most N/j eliminate the property, producing just such a subsequence. Diagonalize these nested subsequences, choosing each term sufficiently far out that the accumulated removals have size $o(N)$. For each fixed earlier stage, the later removals eventually use less than half of its tolerance. Its availability therefore persists with half that tolerance, while an absence persists under every further removal.

For an available union of k families with tolerance κ , use κ/k as the removal tolerance for each member. If no member were available on a subsequence, we could combine their eliminating sets and eliminate the union within its tolerance. Finally, there are only finitely many pairs of label subsets at any fixed dimension. For each allowed removal pair choose one surviving witness. These representatives form a finite menu with the same availability. $\square$

Lemma 3.3 (Balanced mixtures and simultaneous profiles). *Let a finite patch menu be available with tolerance $\kappa > 0$, and let patch i carry laws (μ_i, ν_i) on its two supports. There is a probability distribution on the menu such that, coordinatewise,*

$$\mathbb{E}_i \mu_i, \mathbb{E}_i \nu_i \leq K/N$$

for a constant K .

More generally, let finitely many players have finite action sets and choose their actions independently according to probability distributions called profiles. Conditional on the realized actions, fix each player's output-vector rule independently of the profiles. Suppose that, for every choice of the other players' profiles, a player can meet its fixed expected-vector upper bounds by a mixture of its actions. Then there is a profile for every player under which all these bounds hold simultaneously.

Proof. For the first assertion, assign nonnegative prices to labels on the disjoint union of the host sides, and let P be their total. If $P = 0$ there is nothing to prove. Otherwise, fewer than κN labels on either side have price greater than $2P/(\kappa N)$. Availability gives a patch avoiding all these labels.

Each of its two laws then has expected price at most $2P/(\kappa N)$, so its pair-vector has price at most $4P/(\kappa N)$. If the convex hull of the pair-vectors contained no vector bounded coordinatewise by $4/(\kappa N)$, separation from that downward-closed region would give nonnegative prices contradicting this inequality. Thus $K = 4/\kappa$ suffices.

For the second assertion, write X_j for player j 's output vector and b_j for its prescribed upper bound. With the other profiles p_{-j} fixed, its feasible responses form

$$\mathcal{R}_j(p_{-j}) = \{q_j : \mathbb{E}_{q_j, p_{-j}} X_j \leq b_j\}.$$

These sets are nonempty by hypothesis, and are closed and convex because expectation is linear in q_j . Since the action sets are finite and the output rules are fixed, the expectations are continuous in all profiles; hence the response correspondence has closed graph. The product of these correspondences has a fixed point by Kakutani's theorem [24]. At that point every player's expected-vector bound holds. $\square$

A kernel below may be a *subprobability*, a nonnegative measure of mass at most one. For example, a row p_v , indexed by a cube role v , may be set to zero when its local construction is invalid. Auxiliary experiments may instead use a specified fallback draw to continue sampling. The final rows on a successful outcome must be probability measures. If they are supported on the allowed labels and $\sum_v p_v(y) \leq 1$ for each label y , then for every set S of roles,

$$|S| = \sum_y \sum_{v \in S} p_v(y) \leq \left| \bigcup_{v \in S} \text{supp } p_v \right|.$$

Hall's theorem [20] therefore gives distinct allowed representatives.

Definition 3.4 (Raw experiments, gates, and stage histories). A *raw experiment* is the specified sampling law before the avoidance constraints currently under discussion are imposed. A *local gate* is an event determined by the declared local inputs; retaining the gate means multiplying by its indicator, rather than conditioning the reference law on it. A *successful prehistory* is an entering stage history on which that stage's stated hypotheses hold. Comparisons are made separately at each such history.

The next comparisons bound expectations of arbitrary nonnegative tests. An absolute estimate for the probability of failure would not suffice when the test itself has very small probability.

Lemma 3.5 (Conditional avoidance comparison). *Let E_i be finitely many bad events with a symmetric neighbor relation $\sim$. Suppose that, whenever defined, the probability of E_i conditioned on avoidance of any subset of its nonneighbors other than itself is at most p_i . Independent underlying variables, with events joined when their variable scopes overlap, satisfy this hypothesis with $p_i = \Pr(E_i)$. Assume*

$$p_i \leq x_i \prod_{j \sim i} (1 - x_j), \quad 0 \leq x_i < 1.$$

Then all the events can be avoided with positive probability. For any set S of indices not containing i , $\Pr(E_i \mid \bigcap_{j \in S} E_j^c) \leq x_i$.

Write $A_S = \bigcap_{j \in S} E_j^c$. For disjoint index sets S, T and any nonnegative random variable W ,

$$\mathbb{E}[W \mid A_{S \cup T}] \leq \prod_{j \in T} (1 - x_j)^{-1} \mathbb{E}[W \mid A_S].$$

If an additional event F has conditional probability at most p_F under avoidance of any subset of its nonneighbors, then

$$\Pr(F \mid A_S) \leq p_F \prod_{j \in S: j \sim F} (1 - x_j)^{-1}.$$

Proof. The local lemma and its conditional induction go back to Erdős and Lovász [9, Section 2]. Erdős and Spencer introduced the lopsided nonneighbor condition [10, Section 1]; see Lu and Székely [27, Lemma 3] for the asymmetric formulation used here. Here is the induction for the stated hypotheses. Split a conditioning set into nonneighbors and neighbors of E_i . The numerator is bounded by p_i after discarding neighbor-avoidance conditions. By induction, each neighbor-avoidance condition has conditional probability at least $1 - x_j$, so the denominator is at least their product. This proves $\Pr(E_i \mid A_S) \leq x_i$, and simultaneously proves positivity of all avoidance probabilities. In particular,

$$\Pr(A_T \mid A_S) \geq \prod_{j \in T} (1 - x_j).$$

Inserting $\mathbf{1}_{A_T} \leq 1$ in the conditional expectation of W , then dividing by this lower bound, proves the first comparison. The same argument for F removes only its neighbors and uses its own conditional nonneighbor bound for the remaining numerator. $\square$

For independent underlying variables, removing all avoidance constraints that touch a specified set of variables makes those variables independent of the remaining avoidance event. They can then be integrated under their raw laws. In applications we impose avoidance separately at each entering history; we do not reweight earlier draws by the probability of later avoidance.

Lemma 3.6 (Scattered moments). *Let $Z_v \geq 0$, indexed by a finite role set U , be bounded by L on a success event. Each role has a near set, including itself, of size at most $f|U|$. Suppose that for any sequence of at most n indices, each outside the near sets of its predecessors, the joint product expectation with the success indicator is at most K^n times the product of deterministic comparison means d_v , where $K \geq 1$. Then*

$$\mathbb{E} \left[\mathbf{1}_{\text{success}} (|U|^{-1} \sum_v Z_v)^n \right] \leq K^n (|U|^{-1} \sum_v d_v + n f L)^n.$$

Proof. Expand the power by drawing n indices independently and uniformly from U . Mark a position as removed if its index lies in the near set of any preceding index. On success each removed factor costs at most L ; the unremoved indices satisfy the joint-comparison hypothesis. For each removal pattern, apply that comparison, then discard the separation conditions at the unremoved positions. Sum over the indices in reverse order. At an unremoved position the factor is $|U|^{-1} \sum_v d_v$. At a removed position the necessary proximity condition permits at most an $n f$ fraction of indices, so its factor is at most $n f L$. Summing over all removal patterns gives the claimed binomial expansion. $\square$

We may first apply the joint comparison conditional on a successful prehistory, retaining its indicator, and only afterwards discard nonlocal restrictions for an upper bound. Typically $n f L = o(1)$, while K and the average of the d_v 's are bounded. Markov's inequality at a sufficiently large constant threshold then permits a union over $\exp(O(n))$ columns. For $Z_v = N p_v(y)$, the resulting bound on the average gives

$$\sum_{v \in U} p_v(y) = \frac{|U|}{N} \left(|U|^{-1} \sum_{v \in U} Z_v \right) = o(1)$$

when $|U| = O(2^n)$, since $N = C_n 2^n$ and $C_n \rightarrow \infty$. The exceptional probability here includes the success indicator. We also use this argument with patch-specific sizes.

Lemma 3.7 (Gated posterior comparison). *Fix any earlier information in a reference experiment. Let a candidate z have prior law π , and let $F_z(t)$ be the likelihood of data t together with a required event. Thus F_z includes the event indicator and may have total mass less than one. Set*

$$m(t) = \int F_z(t) d\pi(z).$$

Use one common reference measure for all data densities; in a finite space these may simply be masses. If Q is any reference probability law on the data fixed by the earlier information and $\epsilon > 0$, then the joint probability of the required event and $\{m(t) < \epsilon Q(t)\} \cup \{m(t) = 0\}$ is at most ϵ . If also $F_z(t) \leq e^s Q(t)$, the posterior $F_z(t) d\pi(z)/m(t)$ outside this exception is dominated by $e^s \epsilon^{-1} \pi$. Integrating the posterior against the required-event data measure recovers the prior experiment with its gate retained.

Proof. The data measure with the required event has density m . Its mass on $\{m < \epsilon Q\}$ is at most $\epsilon \int Q = \epsilon$, and its mass where $m = 0$ is zero. On the complement, $F_z/m \leq e^s/\epsilon$. Finally, for any nonnegative test h ,

$$\int m(t) \int h(z, t) \frac{F_z(t)}{m(t)} d\pi(z) dt = \int \int h(z, t) F_z(t) dt d\pi(z),$$

with the left integrand set to zero where $m = 0$. This is the asserted cancellation. $\square$

The gate and every selection rule in F_z must be recomputed as z varies. To use this estimate after an assignment, we first compare the joint law of the required outputs with the reference experiment. Only then may we remove nonlocal prehistory restrictions, keeping the local gate inside F_z .

3.1 Geometric height device

We next choose centers in Hamming balls while keeping the choices at nearby sites geometrically compatible. A center has an ID consisting of its location and level. The construction gives neighboring sites levels that differ by at most one, and bounds the number of active IDs they can consult. Its path-maximum rule is related to the admissible-path construction of Lipschitz percolation [8, Sections 4–5]. Here eligibility may depend on all the sampled center positions, so the independence needed in the proof comes only after separating those positions from the later activations.

Let the queried sites form a subset S of Q_d , where $d = \Theta(n)$. A horizontal step may join queried sites at Hamming distance at most a fixed positive integer D , including a step that stays at the same site. Prospective center locations range over all of Q_d . Write V for the volume of a radius- r ball in Q_d . For levels $0, \dots, H$, the random experiment has three stages:

- (i) At each vertex of Q_d and each level, independently, put a prospective center with probability λ/V , where $\lambda = n^{J_0}$ and the fixed constant $J_0 > 2$. Write P for all these prospective positions.
- (ii) Using P and any auxiliary information, specify for every site-level (v, j) an eligible set $E(v, j)$ of prospective centers in its same-level r -ball. The required size is $|E(v, j)| \geq \lambda/3$.

- (iii) After those choices, activate each prospective center independently with probability n^{b_0}/λ . Write A for these activation variables, where $0 < b_0 < b < 1$.

A site-level is *bad* if its eligible set contains no active center, or if its same-level $(r + D)$ -ball contains more than n^b active centers. We allow either of the following radii:

- $r = \Theta(d)$ with $r \leq (1/4 + o(1))d$;
- $r = \lfloor n^{1-\rho} \rfloor$, for fixed $0 < \rho < 1$, with $b > b_0 + (D + 1)\rho$.

Choose fixed $0 < \sigma < \zeta < 1$, $0 < \theta < 1$, and $a > 0$ such that

$$b_0 > a > \zeta + \sigma + (1 - \theta), \quad b > a + 4\sigma.$$

A space-level path stays in levels $[0, H]$. Its permitted steps are an upward vertical step from a bad site-level, or a downward step of one level together with a horizontal step. For a queried site v , restrict the whole path to spatial distance $2DH$ from v . Define the *long height* $h(v)$ to be the largest j for which (v, j) is reachable by such a path from any level-zero start. A rule with a shorter consultation radius restricts the same paths to the smaller ball. If the resulting height equals H or its site-level is bad, the rule makes no selection.

Lemma 3.8 (Local height selection). *In this experiment, take*

$$R_0 = \lceil \log^2 n \rceil, \quad R_i = MR_{i-1}, \quad M = \lceil n^\sigma \rceil,$$

and let $H = R_h$ be the first scale at least $n^{1-\zeta}$. Thus $H = O(n^{1-\zeta+\sigma})$.

- (i) *Except on an event of probability at most $\exp(-n^{1+c})$, for some fixed $c > 0$, whenever all eligible sets have the required sizes, the long heights lie in $\{0, \dots, H - 1\}$, their site-levels are good, and their values differ by at most one on horizontal steps. Computing a long height uses only bad statuses within spatial distance $2DH$.*
- (ii) *At a fixed site, on correct eligible-set sizes throughout its consultation domain, the probability of positive long height is at most $\exp(-n^c)$ for some fixed $c > 0$. This remains true if one specified prospective center is forced present.*
- (iii) *If $\theta > .9$, $m = n^{\alpha+o(1)}$ for fixed $\alpha > 0$, and $\sqrt{m} \leq H$, replacing the consultation radius by $D\lfloor \sqrt{m} \rfloor$ changes the path maximum with probability $o(\exp(-2m^{1/5}))$, on the same correct sizes. This estimate also allows one forced prospective center.*

All bounds on correct sizes concern intersection with that event, not conditioning on it. The local bounds in (ii) and (iii) also hold when, for each fixed P , one takes the supremum of the conditional activation probability over fixed legal eligibilities, and only then averages over P .

At a good chosen height, independent pre-randomized site-level ties can select a uniform eligible active ID. For fixed P, E of the required sizes and a specified present ID c at level zero,

$$\Pr_{A, \text{ties}}(h(v) = 0 \text{ and } c \text{ is selected} \mid P, E) \leq 3/\lambda.$$

The long and short rules may use the same ties and invalid-choice conventions.

Proof. Step 1: the path event to be estimated. Use the metric

$$\text{dist}((v, j), (v', j')) = \max(|j - j'|, \lceil d_H(v, v')/D \rceil).$$

There are a bounded number of scales. Choose $1/4 \leq \eta_h < \dots < \eta_0 \leq 1/2$ with fixed positive gaps. A scale- i failure from a specified point is a path stopped when it first reaches metric distance R_i ,

with net rise at least $-\eta_i R_i$. We will bound its probability by $\exp(-n^a R_i^\theta)$, in the uniform form specified below. The proof first finds many separated short failures inside a long one. Their center balls may overlap. Removing the shared regions, while allowing a small loss in the size and crowd thresholds, makes the short failures depend on disjoint random inputs.

Fix a deterministic domain $\mathcal{D}$ of allowed site-level paths and a deterministic set $\mathcal{C}$ of location-levels that may provide centers. Both eligibility and crowd counts use only $\mathcal{C}$. At scale i , require eligibility size at least $s_i \lambda$ and call a crowd bad above $t_i n^b$, where s_i increase strictly within $[1/8, 1/4]$ and t_i increase strictly within $[1/3, 3/4]$, with fixed positive gaps. Let $F_i(P, E, A)$ be the scale failure from the specified start in these domains with these thresholds. With $\mathcal{D}, \mathcal{C}$ fixed, define

$$B_i(P) = \sup_E \Pr_A(F_i(P, E, A) \mid P, E),$$

where eligible sets must have the stated sizes within metric distance R_i of the start; the supremum is zero if no such sets exist. This defines a random variable depending only on the prospective positions. The induction claim, uniform in the two deterministic domains, is

$$\mathbb{E}_P B_i(P) \leq \exp(-n^a R_i^\theta). \quad (3.1)$$

The actual thresholds are stronger: actual legal sets have size at least $\lambda/3$, and every actual bad site-level remains bad when the crowd threshold is lowered to $t_i n^b$. Hence the same upper bound holds for failures in the original experiment on correct sizes.

Step 2: the smallest scale. A scale-zero failure must contain an up-step. Without one, reaching distance R_0 requires at least R_0 downward steps and gives net rise at most $-R_0$. An up-step identifies a bad site-level in the metric ball of the start. There are at most $\exp(O(R_0 \log n))$ possible such points.

For any fixed legal eligible set, a hole has conditional probability at most $\exp(-s_0 n^{b_0})$. The crowd test does not depend on eligibility. Unconditionally its count is binomial, with mean at most $n^{b_0} O(1)$ in the linear-radius case and $n^{b_0} O(n^{D\rho})$ in the sublinear case, by successive Hamming-layer ratios. The assumed gap below b gives a binomial upper tail with polynomial slack. The same union bound is valid after taking the eligibility supremum: the hole bound is uniform in the eligible set, and the crowd bound is independent of it. Since $b_0 > a$, these tails and the count of possible points prove the induction at scale zero.

Step 3: separated child paths and ball intersections. Put $R' = R_{i-1}$. A scale- i failure contains at least $q = \lfloor c' M \rfloor$ starts of scale- $(i-1)$ failures separated by KR' , with all child paths still inside the parent's metric ball. Here the child paths temporarily use the parent's bad statuses; K is a large fixed constant and $c' > 0$ is sufficiently small.

To see this, take a maximal separated set of child-failure starts. If it has fewer than q members, their KR' -balls cover every such start. Follow the parent path. Whenever it enters the cover, skip to its last visit to the currently hit ball, accounting separately for one outgoing step if necessary. No ball is used twice. These skips are bookkeeping: each has endpoint displacement, and hence possible upward displacement, at most $2KR'$. Outside the cover, partition the path into pieces stopped on first reaching distance R' , with one unfinished piece per intervening stretch. Every full piece has rise less than $-\eta_{i-1} R'$, since its start is not a child-failure start. The parent displacement forces at least $M - O((q+1)K)$ full pieces. Its total rise is consequently at most

$$-\eta_{i-1} M R' + O((q+1)K R'),$$

which is less than $-\eta_i M R'$ when c' is small relative to the gap $\eta_{i-1} - \eta_i$. This contradicts the parent failure.

There are at most $\exp(O(qR_i \log n))$ choices of these starts. A child's center domain has levels within R' of its start and spatial radius at most $r + DR' + D$. For two separated children, either their level intervals are disjoint or their spatial centers have distance s at least a large multiple of R' . At each common level, the intersection of their enlarged balls has volume $V \exp(-\Omega(R'))$.

For this last estimate, a uniform point of one enlarged ball lies within $s/10$ of its outer layer except with probability $\exp(-\Omega(s))$, by the binomial-layer ratios. If it also lies in the other ball, it must then flip at least .44s of the s coordinates on which the centers differ. The enlarged radius is at most $(1/4 + o(1))d$, so this is a hypergeometric upper tail of probability $\exp(-\Omega(s))$. For sublinear radius, the enlarged ball has radius at most $n^{1-c''}$ for some fixed $c'' > 0$; layer ratios and a subset union bound for the differing coordinates improve both exponents to $\Omega(s \log n)$. Enlarging the original ball costs a factor $\exp(O(R'))$, or $\exp(O(R' \log n))$ in the sublinear case. Taking K large makes the intersection bound valid relative to V in either case.

Step 4: removing shared centers. For a fixed configuration of child starts, delete every pairwise overlap from each child center domain. The remaining regions are deterministic and disjoint. Choose a small fixed $\varepsilon > 0$ below all the gaps $s_i - s_{i-1}$ and $t_i - t_{i-1}$. Except on an event of probability $\exp(-\Omega(n^b R'/q))$ per pair and level, an overlap contains at most $\varepsilon \lambda/q$ prospective centers and $\varepsilon n^b/q$ active centers. Indeed their means are at most $\lambda e^{-\Omega(R')}$ and $n^{b_0} e^{-\Omega(R')}$, respectively, so the claimed estimates follow from binomial upper tails.

If none of these count exceptions occurs, each child loses at most $\varepsilon \lambda$ eligible centers and εn^b active centers. Its eligible sets still have size at least $s_{i-1} \lambda$. A hole stays a hole, while a crowd above $t_i n^b$ remains above $t_{i-1} n^b$. Thus all the child failures transfer to the private regions with the child thresholds.

The probability calculation must not condition on successful overlap counts. Let C be the event that the chosen child configuration occurs, let $B_{\text{pos}}, B_{\text{act}}$ be the prospective and active overlap exceptions, and let P_ℓ be the positions in the private region of child ℓ . For every fixed legal eligibility assignment E ,

$$\Pr_A(C \mid P, E) \leq \mathbf{1}_{B_{\text{pos}}} + \Pr_A(B_{\text{act}} \mid P) + \prod_\ell b_\ell(P_\ell),$$

$$b_\ell(P_\ell) = \sup_{E_\ell} \Pr_A(\text{private child failure} \mid P_\ell, E_\ell).$$

The transferred events depend on disjoint activations once P, E are fixed. Their probabilities are bounded by the displayed private suprema, even if E used positions outside those regions. After paying for the overlap exceptions we have discarded their complements, so no conditioning remains to destroy independence. The inequality is uniform in E ; taking its supremum and averaging P factors the last product, because the P_ℓ 's are independent. Induction bounds the product of their expectations by $\exp(-qn^a(R')^\theta)$.

The exponent exceeds both the logarithm of the configuration count and the desired parent exponent. In fact, using $R_i = MR'$, $R' \leq n$, and $q \asymp M$,

$$\frac{n^a(R')^\theta}{R_i \log n} \geq \frac{n^{a-\sigma-(1-\theta)+o(1)}}{\log n} \longrightarrow \infty, \quad \frac{q}{M^\theta} \longrightarrow \infty.$$

The ratio of an overlap-error exponent to the child exponent is at least $\Omega(n^{b-a-2\sigma}(R')^{1-\theta})$, which also tends to infinity because $b > a + 4\sigma$. The number of pair-level exceptions is polynomial. The union over configurations therefore proves (3.1) and the position-averaged induction claim.

If one prospective center is forced present, erase its location-level from the deterministic center domain. The remaining positions have their original independent laws. Each eligible set and crowd

loses at most one center, and $\lambda/3 - 1 \geq s_i \lambda$, $n^b - 1 \geq t_i n^b$ for large n . Holes persist. Every original bad-path witness therefore remains a failure at the relaxed thresholds, uniformly in the erased center's activation. The same induction applies.

Step 5: heights and local approximations. At the top scale, $n^a H^\theta \geq n^{a+(1-\zeta)\theta}$, whose exponent is greater than one. Thus (3.1) permits a union over all sites with error $\exp(-n^{1+c})$. A path from level zero that first displaces by H would be a scale- h failure, since its endpoint still has nonnegative level. Outside the exceptional event no such path exists. The spatially unrestricted path maximum is therefore below H , and all paths relevant to a queried maximum lie in its $2DH$ consultation tube. A maximizing site-level is good, since otherwise the path could extend upward. If neighboring maxima differed by more than one, a downward horizontal step from the higher one would contradict maximality at the lower site. This proves (i).

For a fixed query, classify paths reaching it by their maximum metric displacement from their level-zero start. Below R_0 , positive height requires an up-step at one of $\exp(O(R_0 \log n))$ possible points, so Step 2 applies. For displacements in $[R_i, R_{i+1})$, the first prefix reaching distance R_i is a scale- i failure. There are at most $\exp(O(R_{i+1} \log n))$ possible starts near the query. Hence these paths have total probability at most

$$\sum_{i < h} \exp(O(R_{i+1} \log n) - n^a R_i^\theta).$$

Displacements at least H contribute at most $\exp(O(n) - n^a H^\theta)$. The exponent comparisons in Step 4 show that the sum, together with the base-scale contribution, is at most $\exp(-n^c)$. All arguments hold in deterministic restricted path domains and for position-averaged eligibility suprema. This proves (ii) with the stated local and forced-center uniformity.

A path reaching the query but leaving its $D\lfloor\sqrt{m}\rfloor$ -tube has displacement from its start at least $\sqrt{m}/3$. Only scales with $R_i \geq \sqrt{m}/(3M)$, or a top-scale failure, need be counted. The previous sum now has exponent much larger than $m^{1/5}$, since

$$\frac{n^a (\sqrt{m}/M)^\theta}{m^{1/5}} = n^{a-\sigma\theta+\alpha(\theta/2-1/5)+o(1)} \longrightarrow \infty.$$

Here $a > \sigma + (1 - \theta) > \sigma\theta$ and $\theta > .9$. This proves (iii).

Finally, ignore the height and crowd requirements when bounding the probability of selecting a specified level-zero ID. For fixed P, E , independent activation and a uniform tie give each member of $E(v, 0)$ the same probability of selection, at most $1/|E(v, 0)| \leq 3/\lambda$. Reinstating the requirements only decreases the joint probability. Couple the long and short constructions with the same positions, activations, eligibilities and ties. Whenever their path maxima agree, the selected ID and its validity agree as well, as required when a short rule is used inside a hypothetical-candidate calculation. $\square$

3.2 Near-product injections

The two samplers below use finite row sets and finite output spaces. Row i has a full output $o \in \Omega_i$, with a specified label coordinate in a common finite label set. Write $\hat{p}_i(o)$ for its full-output law and $p_i(y)$ for the label marginal. When there is no side data these are the same law. An assignment is injective when its label coordinates are distinct. A pointwise upper bound for prescribed outputs can be summed against any nonnegative function of those outputs; this is the form of comparison used later.

Preserving individual marginals while choosing distinct representatives is an established feature of bipartite dependent rounding; see Gandhi, Khuller, Parthasarathy and Srinivasan [16,

Properties (P1)–(P3)]. Their negative-correlation guarantee concerns edges incident to a common vertex. The next lemma additionally controls prescribed outputs on arbitrary distinct rows, with a relative error even when their probabilities are very small.

Lemma 3.9 (Calibrated near-product injections). *Let a finite set of rows have laws on d labels, with $p_i(y) \leq d^{-.95}$ and $\sum_i p_i(y) \leq .4$ for every label y . For sufficiently large d , there is an injective random assignment $(O_i)_i$ with exact individual full-output marginals $\hat{p}_i$ and, for every set J of $l \leq d^{.025}$ queried rows and prescribed outputs $(o_i)_{i \in J}$,*

$$\Pr(O_i = o_i \text{ for all } i \in J) \leq \exp(d^{-.04}l) \prod_{i \in J} \hat{p}_i(o_i). \quad (3.2)$$

Targets with repeated labels have probability zero. After sampling the labels, side data can be drawn independently from their conditional laws given those labels. We use this sampler for assignments within small bins.

Proof. It suffices to construct the labels: the independent conditional draws then give both assertions for full outputs.

Step 1: complete the columns and order the rows. First allow label atoms $O(d^{-.95})$ and column sums at most $1/2$. If there are m real rows, then $m \leq d/2$. Put $t = \lceil 2d/3 \rceil$ and write $c_y = \sum_{i=1}^m p_i(y)$. Add $t - m$ identical dummy rows with law

$$p_{\text{dum}}(y) = \frac{t/d - c_y}{t - m}.$$

The numerator is nonnegative and sums to $t - m$, so this is a probability law. Its atoms are $O(1/d)$, and every completed column sums to t/d .

Choose an order of the t rows, then number them in that order, such that every column prefix satisfies

$$\sum_{k < j} p_k(y) = s_j + O(d^{-1/8}), \quad s_j = (j - 1)/d.$$

A uniform random order has this property with positive probability. Indeed the expectation is s_j , and exposure without replacement has martingale increments bounded by the column atom range. Bounded-difference concentration [23, 1], followed by a union over j, y , gives the asserted error. Fix one such order.

Step 2: track the mass already used. At step j , sample row j 's law conditional on its label still being free. For each row a , let

$$D_j^a = \sum_{k < j} p_a(Y_k), \quad e_j = \max_{a, k \leq j} |D_k^a - s_k|.$$

Thus D_j^a is the mass that row a assigns to labels used before step j . Stop at the first $e_j > 1/20$; any undefined continuation after a stop may be declared a failure. We claim that, with probability $1 - \exp(-d^{\Omega(1)})$, there is no stop and $D_j^a = s_j \pm d^{-1}$ for all a, j .

Before stopping, all sampling denominators are bounded away from zero. Conditional on the first $j - 1$ draws, the drift of D_j^a is

$$\mathbb{E}[D_{j+1}^a - D_j^a \mid Y_1, \dots, Y_{j-1}] = \sum_{y \text{ free}} \frac{p_a(y)p_j(y)}{1 - D_j^j}.$$

Its martingale increments are $O(d^{-.95})$. Bounded-increment concentration gives cumulative errors $O(d^{-1/8})$, simultaneously for all rows and times, outside an event of probability $\exp(-d^{\Omega(1)})$.

When summing the drifts through step $b - 1$, replacing $1 - D_j^j$ by $1 - s_j$ costs at most

$$O\left(\sum_y p_a(y) \sum_{j < b} e_j p_j(y)\right) = O\left(d^{-1/8} + d^{-1} \sum_{j < b} e_j\right).$$

For the equality, use summation by parts on the column-prefix error against the bounded nondecreasing sequence e_j . A second summation by parts replaces $p_j(y)$ by $1/d$ in the remaining drift sum: the sequence $\mathbf{1}_{\{y \text{ free before } j\}}/(1 - s_j)$ has bounded variation. The resulting sum is

$$d^{-1} \sum_{j < b} \frac{1 - D_j^a}{1 - s_j} + O(d^{-1/8}).$$

Subtracting s_b now gives, even at a potential stopping index,

$$e_b \leq K d^{-1/8} + \frac{K}{d} \sum_{j < b} e_j.$$

Iteration bounds e_b by $O(d^{-1/8})$. This is smaller than both $1/20$ and d^{-1} for large d , proving the claim. Call this successful tracking event G , and condition the sampler on G .

Step 3: compare specified targets multiplicatively. Fix l distinct positive-probability target labels y_i on queried rows. Define a second sampler that forces $Y_i = y_i$ at each queried step and, at every ordinary step, excludes all target labels whose queried steps are still pending. On a path realizing the targets and satisfying G , the likelihood of the original unconditioned sampler divided by that of the forcing sampler is the product of

$$\frac{p_i(y_i)}{1 - D_i^i} \quad (i \text{ queried}), \quad 1 - \frac{\sum_{i > j, i \text{ queried}} p_j(y_i)}{1 - D_j^j} \quad (j \text{ ordinary}).$$

After extracting $\prod_i p_i(y_i)$, the target-step denominators and the ordinary-step factors cancel to small relative error. For each target, the prefix estimate and summation by parts give

$$\sum_{j < i} \frac{p_j(y_i)}{1 - s_j} = -\log(1 - s_i) + O(d^{-1/8}).$$

Tracking permits replacement of s_j by D_j^j at error $O(d^{-1})$; omitting queried steps costs $O(ld^{-.95})$ per target. Thus the linear terms in the ordinary-step logarithms cancel $-\log(1 - D_i^i)$. The quadratic terms have total size $O(l^2 d^{-.95})$, because each subtracted fraction is $O(ld^{-.95})$ and their sum is $O(l)$. The total log-error per target is $O(d^{-1} + ld^{-.95})$.

Integrating this ratio under the forcing law, and dividing by $\Pr(G) = 1 - \exp(-d^{\Omega(1)})$, gives the joint upper comparison $\exp(O(d^{-.09}l)) \prod_i p_i(y_i)$. For a single target the same ratio also gives a lower comparison $(1 - O(d^{-.09}))p_i(y_i)$: the forcing sampler itself tracks with exponentially small failure probability. Indeed, before a constant-error stop, excluding the one pending label changes a step's distribution in total variation by $O(p_j(y_i))$. Since the tracked increment is $O(d^{-.95})$, the cumulative drift change is $O(d^{-.95} \sum_j p_j(y_i)) = O(d^{-.95})$; the forced step contributes another $O(d^{-.95})$. The proof of Step 2 therefore still applies. The target mass was extracted before this tracking exception was bounded; no absolute error is divided by a tiny atom.

Step 4: calibrate the individual marginals exactly. Return to the original .4 column bound. For arbitrary real prices $c_i(y)$ on the real row-label pairs, put $\bar{c}_i = \sum_y p_i(y)c_i(y)$ and $\delta = d^{-.05}$. Perturb row i by multiplying its probabilities by $1 + \delta \text{sign}(c_i(y) - \bar{c}_i)$, then normalize. Writing p'_i for the result, its gain in centered expected price is

$$\begin{aligned} \sum_y p'_i(y)(c_i(y) - \bar{c}_i) &= \frac{\delta \sum_y p_i(y)|c_i(y) - \bar{c}_i|}{1 + \delta \sum_y p_i(y) \text{sign}(c_i(y) - \bar{c}_i)} \\ &\geq \frac{1}{2} \delta \sum_y p_i(y)|c_i(y) - \bar{c}_i|. \end{aligned}$$

The perturbed rows satisfy the relaxed hypotheses of Step 1. The tracking sampler's two-sided singleton error is $O(d^{-.09})$ relative to p'_i , so its error in this centered price is at most $O(d^{-.09}) \sum_y p_i(y)|c_i(y) - \bar{c}_i|$, smaller than the gain. Its joint upper bound, measured against the *original* inputs, costs $\exp(O(d^{-.05}l + d^{-.09}l))$, which fits (3.2) for large d .

Let $\mathcal{C}$ be the set of all laws on injections of the real rows satisfying the joint upper inequalities relative to those original inputs. These are linear inequalities on a finite probability simplex, so $\mathcal{C}$ is compact and convex. Every price-dependent construction just made belongs to this same set, and its marginal vector has price at least that of the original marginal vector. If the original vector were outside the marginal image of $\mathcal{C}$, a separating linear functional would contradict that property. It therefore belongs to the image, giving exact marginals while preserving all joint bounds. $\square$

3.3 Clock lemma

The next sampler also enforces local restrictions on full outputs. It gives a joint upper comparison with the product law, without the exact marginals of Lemma 3.9. We retain the finite row and full-output spaces, and the notation $\hat{p}_a, p_a$, from the preceding subsection. A predicate's *scope* is the set of rows whose full outputs it reads.

Lemma 3.10 (Clock sampling). *Let there be g labels, with $\log g = O(n)$, and finitely many probability rows $\hat{p}_a$. For each fixed $B \geq 1$, there are sufficiently large constants A_*, P_* and a small absolute constant $\theta_0 > 0$ with the following property. Suppose*

$$\sum_a p_a(y) \leq \theta_0, \quad p_a(y) \leq n^{-A_*}$$

for every label and row. Let a finite list of failure predicates have probability at most n^{-P_} under independent row outputs. Assume each predicate reads at most n^B rows and each row belongs to at most n^B predicate scopes. Then there is a random assignment with distinct labels avoiding all predicates, such that for every set J of at most n^B queried rows,*

$$\Pr(O_a = o_a \text{ for all } a \in J) \leq (1 + o(1)) \prod_{a \in J} \hat{p}_a(o_a).$$

The error is uniform over the rows and targets for fixed B , the chosen constants, and a fixed bound in $\log g = O(n)$. Noninjective targets have probability zero. Unneeded predicates and an empty assignment may be ignored.

Proof. Choose K_0 sufficiently large in terms of B , then choose A_*, P_* sufficiently large in terms of K_0, B . We first construct a greedy matching from independent edge clocks and use finite backward explorations to detect failures. The proof gives two estimates: a joint product bound for prescribed

outputs, and a small total probability for bad exploration outcomes involving any fixed row or label. The first controls the raw matching law. The second lets Lemma 3.5 enforce a match at every real row and avoidance of every predicate, at a multiplicative cost $1 + o(1)$ on the queried outputs.

Step 1: trim rare outputs and complete the columns. For each row a and incident failure predicate F , discard outputs o for which

$$\Pr(F \mid O_a = o) > n^{-P_*/2}$$

under the product law. Markov's inequality bounds the discarded mass for each predicate by $n^{-P_*/2}$, so the total loss per row is at most $\varepsilon = n^{B-P_*/2}$. Renormalize the retained laws. A test or query involves at most n^B rows, so this changes its probability by at most $(1 - \varepsilon)^{-n^B} = 1 + o(1)$. Under the new laws, every predicate has probability at most $n^{-P_*/3}$, both unconditionally and after any one row in its scope is pinned to a retained output. We continue to denote these laws by $\hat{p}_a, p_a$.

Put $\theta = \lceil 10^{-6}g \rceil / g$, and take $\theta_0 = 10^{-8}$, for example. If there are m real rows and $c_y = \sum_a p_a(y)$, add $\theta g - m$ identical dummy rows with law $(\theta - c_y)/(\theta g - m)$. This is a probability law because its numerator sums to the denominator. The denominator is a positive integer of order g , even after the small trimming loss. Every completed column sums to θ , and the dummy atoms are $O(1/g)$. The nonempty case has $g \geq n^{A_*}$, since an original real row is a probability law with atoms at most n^{-A_*} . Thus the largest edge rate in the completed row-label graph is $\lambda_* = O(n^{-A_*})$.

Step 2: matching and its backward explorations. Run an independent Poisson process of rate $p_a(y)$ on every positive row-label edge (a, y) . Each arrival has an independent mark drawn from the conditional law of the full output given label y . Through time $T_0 = K_0 \log n$, greedily match an arriving edge if both endpoints are free. Only its first arrival can matter: if that arrival is rejected, an endpoint is already matched and never becomes free again.

There are two types of tests. A predicate test fails if all rows in its scope are matched and their outputs violate the predicate. A real singleton-row test fails if that row is unmatched. For each test we also forbid an excessively large exploration of the clocks needed to determine its outcome.

Start that exploration with the scope rows as roots, each requesting its matching history through time T_0 . An endpoint's requested horizon is the latest time up to which its status is needed. Process the pending endpoint with largest horizon. In a fixed order, inspect its incident edges for first arrivals by that horizon, reusing information already revealed. A hit at time t requests the other endpoint through a time strictly before t ; retain the largest request if it is requested again. Horizons are processed in decreasing order, so a processed endpoint can never receive a larger request. These inspections determine the matching decisions at the roots by backward closure through all earlier events they require.

The active endpoints are the roots and every endpoint reached by a hit, including pending and processed endpoints. Stop with a *giant* failure as soon as their number reaches $L = \lceil n^{1K_0} \rceil$. Register the endpoint of the last hit before stopping. Consequently every inspected hit always has both endpoints active, even on a giant leaf.

Step 3: exploration leaves and nonneighbor conditioning. Eventually we discretize first-arrival times with a sufficiently fine finite mesh and a fixed tie order. A leaf of an exploration then specifies a product rectangle in the independent edge data. A hit pins an arrival bin and mark on an edge whose two endpoints are active. A negative inspection requires absence up to a specified cutoff on an edge with at least one active endpoint. Every realization in this rectangle follows the same inspections and reaches the same leaf.

Use the individual bad leaves as forbidden events, joining two leaves when their active endpoint sets intersect. Leaves with disjoint active sets can share an inspected edge only if both require absence on it. To force a positive-probability leaf, condition its edges separately. On absence edges couple the conditional first time to be no earlier than the old time; on hit edges fix the required data; leave other edges unchanged. Every already occurring nonneighbor leaf still occurs. This holds simultaneously for any collection of nonneighbors, even if they overlap with one another. If A is avoidance of such a collection and E is the forced leaf, the coupling gives $\Pr(A \mid E) \leq \Pr(A)$, hence $\Pr(E \mid A) \leq \Pr(E)$ whenever defined.

It remains to show, uniformly in an endpoint u , that

$$\sum_{E \text{ bad leaf: } u \text{ active in } E} \Pr(E) \leq n^{-.6K_0}.$$

Equivalently, sum over tests the probability that the test has a bad outcome and its exploration reaches u . We prove this in continuous time before choosing the mesh.

Step 4: inserted paths bound the incidence sum. Reaching u from a test's roots requires a path of time-decreasing arrivals from one of its root rows to u . Such a certificate uses distinct edges, since the exploration inspects first arrivals; the empty path is allowed. For a fixed edge path, count its possible distinguished arrival tuples. Poisson insertion expresses their expected count, with a specified bad outcome, as the integral of the product of edge rates over ordered times in $[0, T_0]$, multiplied by the bad-outcome probability in an ordinary realization with those arrivals inserted. This identity follows directly by summing over the Poisson counts and distinguishing one of their uniform-time points on each edge. Inserted marks may be fixed anywhere in their support for uniform estimates.

To sum certificates, reverse them from u and allow all walks. Total outgoing rates alternate between 1 at a row and θ at a label. The integrated rate sum for a start at a row is

$$\sum_{j \geq 0} \theta^{\lfloor j/2 \rfloor} \frac{T_0^j}{j!} = \cosh(\sqrt{\theta}T_0) + \theta^{-1/2} \sinh(\sqrt{\theta}T_0) = O(e^{\sqrt{\theta}T_0}/\sqrt{\theta}).$$

Starting at a label only decreases this upper bound. Each root row belongs to at most $n^B + 1$ tests. Lengths above $K_0^2 \log n$ contribute less than $n^{-K_0^2}$ before this incidence factor, by the factorial tail of the series.

For shorter paths, first dispose of those meeting two distinct rows of the same test, bounding the bad-outcome probability simply by one. Choose their two positions on the path and a test incident to the earlier row. In the normalized reversed walk, a subsequent label-to-row transition enters that test's scope with probability at most $n^B \lambda_*/\theta$. Their total contribution is therefore at most

$$O(e^{\sqrt{\theta}T_0}/\sqrt{\theta})(1 + \log n)^{O(1)}(n^B + 1)n^B \lambda_*/\theta.$$

The remaining short insertions meet at most one row of the test. We now bound bad outcomes uniformly under each such insertion.

Step 5: the exploration rarely becomes giant. Include the endpoints of all inserted arrivals as extra full-horizon roots. Every reached endpoint is connected to one of these roots by ordinary arrivals: take the part of its reaching path after the last inserted hit. There are at most $n^B + O(\log n)$ roots.

During largest-horizon processing, an edge from the current endpoint to an unprocessed endpoint is fresh: neither of its endpoints could have inspected it earlier. Superpose the fresh arrivals and, when necessary, independent extra arrivals to reach total rate 1 at row endpoints and

θ at label endpoints. They are dominated by a two-type branching forest. A node with horizon t has Poisson children of the opposite type at times in $[0, t]$, each child inheriting its arrival time as horizon. To handle repeated requests for a graph endpoint, attach its sole processing to the node making its highest request; its offspring have not yet been generated. Unused tree nodes can be filled with independent subtrees. This deferred construction dominates the active endpoint count by the forest's total progeny.

For $\delta = n^{-.01K_0}$, let $H_{\text{row}}(t)$ and $H_{\text{label}}(t)$ be the exponential moments of single-root progeny, including the root. They satisfy

$$\begin{aligned} H_{\text{row}}(t) &= \exp \left(\delta + \int_0^t (H_{\text{label}}(s) - 1) ds \right), \\ H_{\text{label}}(t) &= \exp \left(\delta + \theta \int_0^t (H_{\text{row}}(s) - 1) ds \right). \end{aligned}$$

Induction on truncated generations bounds their excesses above one by $4\delta e^{2\sqrt{\theta}t}/\sqrt{\theta}$ and $4\delta e^{2\sqrt{\theta}t}$, respectively. Substitution proves these bounds using $e^x - 1 \leq 2x$ for the resulting small positive exponents. With $R \leq n^B + O(\log n)$ roots, exponential Markov gives

$$\Pr(\text{giant}) \leq \exp \left(-\delta L + O(R\delta e^{2\sqrt{\theta}T_0}/\sqrt{\theta}) \right).$$

Here $\delta L \geq n^{.09K_0}$, while the second term is $O(n^{B-(.01-2\sqrt{\theta})K_0})$. Choosing $(.1-2\sqrt{\theta})K_0 > B$ makes this tail superpolynomially small, uniformly in the fixed short insertion.

Step 6: track the free mass in a background process. Fix a prescription removing at most $2n^B$ endpoints and inserting $O(\log n)$ arrivals between the remaining endpoints. In the resulting background matching define, even for original endpoints that were removed,

$$X_a(t) = \sum_{y \text{ free in background}} p_a(y), \quad R_y(t) = \sum_{a \text{ free in background}} p_a(y).$$

These are the available label mass of a row and the available row mass at a label. The deterministic comparison functions solve

$$z' = -qz, \quad q' = -\theta zq, \quad z(0) = q(0) = 1.$$

In particular $q = 1 - \theta + \theta z$ and $0 < q, z \leq 1$. We claim, with error probability at most $\exp(-n^{A_*/3})$, that throughout $[0, T_0]$, uniformly over original endpoints,

$$X_a(t) = q(t) + o(n^{-3B}), \quad R_y(t) = \theta z(t) + o(n^{-3B}).$$

The estimate is uniform for each fixed removal and insertion prescription; we do not take a union over all such prescriptions.

A free label y is matched at ordinary rate R_y , and a free row a at rate X_a . Hence the ordinary drifts of these masses are $-\sum_{y \text{ free}} p_a(y)R_y$ and $-\sum_{a \text{ free}} p_a(y)X_a$. If $E(t)$ is the maximum of the claimed approximation errors, each drift differs from its ODE drift by at most $2E(t)$. Removals cause initial error $O(n^B\lambda_*)$, and inserted matches cause total jumps $O(\log n\lambda_*)$.

The drift-compensated ordinary-jump martingales have jumps at most λ_* and predictable variance at most $T_0\lambda_*$. Their running absolute maxima exceed $n^{-A_*/4}$ with probability at most $\exp(-n^{A_*/2-o(1)})$ per endpoint. For completeness, compensating e^{vM_t} costs log-rate at most v^2 times the variance rate when $|v|\lambda_*$ is small; this follows from the exponential series for each jump.

Use both signs of $v = n^{-A_*/4}/(2T_0\lambda_*)$ and stop at the first crossing. This is the bounded-jump exponential estimate used here; compare [15]. A union over $O(g)$ endpoints is permitted by $\log g = O(n)$. Finally drift iteration yields

$$E(t) \leq \left[O((n^B + \log n)\lambda_*) + n^{-A_*/4} \right] e^{2T_0} \leq n^{-A_*/4+2K_0+o(1)}.$$

Taking $A_*/4 > 2K_0 + 3B$, with slack, proves the claim and its stated exception probability.

Step 7: ordinary target matches have a product upper bound. Fix $k \leq n^B$ full-output targets with distinct labels, also allowing any fixed short insertion from Step 4. Count tuples of ordinary arrivals that realize all the target matches. Insert their distinguished arrivals at candidate times $t_1, \dots, t_k$. Their intensities factor out the entire product $\prod_a \hat{p}_a(o_a)$, including all prescribed marks, before any probability estimate is made.

If these are the target matches, none of their endpoints ever matches to an outside endpoint. The outside process therefore agrees exactly with the background obtained by removing the queried rows and target labels, keeping the fixed inserted arrivals between remaining endpoints. Given this background, an ordinary arrival from a target endpoint to a free outside endpoint is forbidden before its candidate matching time. These are disjoint clocks independent of the background. Ignoring other necessary conditions gives the following upper bound for the remaining time integrand:

$$\mathbb{E} \exp \left(- \sum_{(a, y_a)} \int_0^{t_a} (X_a(s) + R_{y_a}(s)) ds \right).$$

The ODE identity $(\log(zq))' = -(q + \theta z)$ and Step 6 bound this, on the tracking event, by $(1 + o(1)) \prod_a z(t_a)q(t_a)$: its logarithmic error is at most $kT_0 o(n^{-3B}) = o(1)$. After integrating the times, the ordinary-target probability is at most

$$\left(\prod_a \hat{p}_a(o_a) \right) \left[(1 + o(1)) \left(\int_0^{T_0} z(t)q(t) dt \right)^k + T_0^k e^{-n^{A_*/3}} \right] \leq (1 + o(1)) \prod_a \hat{p}_a(o_a).$$

Indeed $\int_0^{T_0} zq = 1 - z(T_0) \leq 1$, and the second term is $o(1)$ for our choice of A_* . In particular the tracking exception still carries the full target product; the bound is relative even for arbitrarily small positive atoms.

For an unmatched singleton, remove just its row. The same absence calculation gives $\mathbb{E} \exp(-\int_0^{T_0} X_a(s) ds)$ as an upper bound. Since $q \geq 1 - \theta$, its probability under the short insertion is at most $n^{-(1-\theta)K_0+o(1)}$.

Step 8: bound bad leaves and impose avoidance. For the remaining short path and predicate pairs, at most one row of the predicate lies on the path. If all its matches are ordinary, sum Step 7 over its failing output tuples. Otherwise one match uses an inserted arrival at that row. Pin its output, discard its match requirement, and sum the ordinary-target bounds for the other rows using the one-pinned failure estimate from Step 1. There are $O(\log n)$ inserted arrivals to consider. In either case the matched-failure probability is at most $O(1 + \log n)n^{-P_*/3}$.

Multiply these estimates and the giant bound by the path and test factor $O((n^B + 1)n^{\sqrt{\theta}K_0}/\sqrt{\theta})$ from Step 4. The unmatched contribution is at most $n^{B-(1-\theta-\sqrt{\theta})K_0+o(1)}$. The two-row exception is at most $n^{2B+\sqrt{\theta}K_0-A_*+o(1)}$, and long paths cost $O(n^{B-K_0^2})$. Choose K_0 so that $B < (.4 - \theta - \sqrt{\theta})K_0$, in addition to Step 5's condition; then take A_*, P_* large enough for the other terms and tracking. Every contribution has a fixed power of slack below $n^{-.6K_0}$. This proves the active-endpoint bound from Step 3.

Choose the time mesh sufficiently fine for this bound to persist. Continuous arrival times are distinct almost surely; matching decisions and truncated explorations therefore converge almost surely as the mesh is refined. There are finitely many tests, endpoints and full-output target products. We may also preserve the raw joint target bound, with no path insertions, with negligible relative loss at every positive product. The discretized exploration now has finitely many leaves.

Assign each bad leaf charge twice its raw probability. A leaf has at most L active endpoints, so its total neighboring charge is at most $2Ln^{-.6K_0} = O(n^{-.5K_0})$. Thus its probability is at most its charge times the product of its neighbors' avoidance factors, and Lemma 3.5 supplies simultaneous avoidance. All real rows are then matched, all listed predicates are avoided, and all the required explorations are nongiant.

Finally partition any joint target event by the non-giant singleton leaves of its queried rows. Their intersections are product rectangles with at most $n^B L$ active endpoints. They have the same forcing property against bad leaves with disjoint active sets. The total neighboring charge is at most $2n^B L n^{-.6K_0}$, so the avoidance comparison costs $\exp(O(n^{B-.5K_0})) = 1 + o(1)$. Summing the rectangles gives the raw joint target bound from Step 7. The initial trimming normalization costs another $1 + o(1)$, proving the assertion for the original full-output laws. $\square$

4 Bias versus purity at power widths

At prescribed power widths, the next lemma combines an available density surplus with the absence of nearly monochromatic patches at doubled width budgets to construct a cube. Its contrapositive will be the first reduction toward discrepancy.

Lemma 4.1 (Bias versus purity). *Fix constants $0 < \beta \leq \gamma < 1$. For $h = h(\beta, \gamma) > 0$ chosen below, the following two hypotheses cannot both hold along a bad sequence:*

- (i) *Pairs of laws at widths $\leq n^\beta, n^\gamma$ on X, Y respectively with absolute bias $\geq n^{-h}$ from half are available;*
- (ii) *On the current sets there is **no** pair at respective widths $\leq 2n^\beta, 2n^\gamma$ having defect $\leq \exp(-n^h)$ in either color. Defect for a color means density of its complement.*

Equivalently, their simultaneous validity yields a monochromatic cube.

Proof. We will first prepare pairs of laws (μ_i, ν_i) with a density surplus in one color G , and distribute these pairs over the cube. An even role will share a hidden tuple of X -labels with nearby even roles. The odd roles receive laws on Y restricted to common neighbors of the tuples they encounter. After assigning distinct odd labels, the posterior law of an even role's tuple will show that those labels have a large common neighborhood in X . Masks on the patch laws will balance the resulting label loads, allowing Hall's theorem to assign the even roles distinctly as well.

Step 1: a density plateau. Set

$$\omega = \min(\beta, 1 - \gamma)/1000, \quad h = \omega/10^6, \quad k = \lceil n^{\omega/3} \rceil, \quad L = n^h, \quad a_* = n^{-h}.$$

Here k will be the tuple length, e^{-L} the minimum acceptable fraction retained by a single common-neighbor restriction, and a_* the density surplus we must preserve.

Take residual label sets containing a witness in hypothesis (i). For each integer $0 \leq j \leq \lfloor n^{\omega/2}/2 \rfloor$, let P_j be the largest density in either color among laws on those sets with width budgets

$$(s_X(j), s_Y(j)) = (n^\beta, n^\gamma) + j(n^{\beta-\omega/2}, n^{\gamma-\omega/2}).$$

These maxima exist by compactness and form a nondecreasing sequence in $[0, 1]$, starting at least at $1/2 + a_*$. If every consecutive increase were at least $n^{-\omega/3}$, their sum would exceed one for large n . Choose a step with $P_{j+1} - P_j < n^{-\omega/3}$, and a maximizing pair at its smaller budgets. Write these budgets as s_X, s_Y , its color as G , its first law as μ_i , and its density as $p = 1/2 + a_i$, where $a_i \geq a_*$.

We can trim the second law to a law ν_i such that

$$d_G(\mu_i, y) \geq p - n^{-\omega/5} \quad (y \in \text{supp } \nu_i), \quad \text{width}(\nu_i) \leq s_Y + 1.$$

To see this, first consider the columns with degree greater than $p + n^{-\omega/3}$. If their mass exceeded $\eta = \exp(-n^{\gamma-\omega/2}/2)$, conditioning on them would increase width by less than $n^{\gamma-\omega/2}/2$ and produce a density larger than P_{j+1} . Their mass is therefore at most η . If t is the mass of columns below $p - n^{-\omega/5}$, the original mean p gives

$$t(n^{-\omega/5} + n^{-\omega/3}) \leq n^{-\omega/3} + \eta.$$

Thus $t = o(1)$, so retaining the other columns costs less than one unit of width. We also retain the following upper density bound: any laws supported in this patch with respective widths at most

$$s_X + \frac{1}{2}n^{\beta-\omega/2}, \quad s_Y + \frac{1}{2}n^{\gamma-\omega/2}$$

have G -density at most $p + n^{-\omega/3}$. Both laws fit the next budget pair, which proves this bound independently of the trimming. The index i records the laws, budgets, and density just obtained.

This construction works after every removal allowed by hypothesis (i), so the resulting patches are available. Lemmas 3.2 and 3.3 allow us to pass to a subsequence, fix one available color G , and use a finite balanced mixture of its patches.

Step 2: a coordinate key distributing the patches. We need a function g on cube roles whose values are spread over many keys, while only a small number of keys occur among the neighbors of any one role. It will use $\lfloor n^{\gamma-\omega} \rfloor$ disjoint coordinate gadgets. In each gadget take s' chunks of $(s')^6$ bits, where $S = s' + 1$ is a power of two and $s' \asymp n^{2\omega}$. The number of bits used by all the gadgets is

$$m = \Theta(n^{\gamma+13\omega}) = o(n).$$

For one gadget, clip each chunk's count of ones to a central interval consisting of S^2 integer unit steps, and translate that interval to $[0, S^2]$. Denote the clipped values by $t_1, \dots, t_{s'}$, with chunk labels retained. Each endpoint has probability at least $1/3$: the counts are symmetric, and their maximum binomial atom is $O((s')^{-3})$, so the unclipped central interval has probability $O(S^2(s')^{-3}) = o(1)$.

Sort these values by rank and add sentinel ranks $0, S$ with values $0, S^2$. A binary search starts from the rank interval $[0, S]$. At every step its rank endpoints a, b have values enclosing $[aS, bS]$. At the middle rank, compare its count with the fixed value $(a + b)S/2$. If the count is at least this value, continue in the left half; otherwise continue in the right half. The invariant is preserved. At a leaf, two consecutive ranks enclose one elementary interval of length S . The gadget outputs the index of that interval and the labeled subset of chunks above its midpoint.

There are two useful properties of this output. First, all one-bit changes to a fixed input produce only $O(\log n)$ possible outputs. If the search path is unchanged, membership above the leaf midpoint is unchanged. Otherwise consider the first comparison that changes on the original path. One clipped count must move between a multiple of S and the integer immediately below it. Once that comparison and the direction of the move are specified, the new multiset of counts is

fixed. The identity of a tied chunk making the move does not affect the output: its count stays on the same side of every elementary midpoint. There are only $O(\log n)$ comparisons on the original path and two directions. Second, for any fixed output, each chunk is required to lie on a specified side of a fixed midpoint. Either requirement has probability at most $2/3$, because the opposite clipped endpoint has probability at least $1/3$. Independence of the chunks bounds the probability of that output by $(2/3)^{s'}$.

Let g be the joint output of all gadgets, and for odd u let $Z_u = \{g(v) : v \sim u\}$. Flipping one bit changes at most one gadget, so

$$|Z_u| \leq n^{\gamma-\omega+o(1)}, \quad \max_g \Pr(g(v) = g) \leq \exp(-\Omega(n^{\gamma+\omega})).$$

In the second estimate, v is a uniform role on either fixed cube parity. The unused coordinates allow the special bits to remain uniform after parity is fixed.

Assign a patch index i_g to each key so that the averages of $\mu_{i_g(v)}$ over $v \in A$, and of $\nu_{i_g(u)}$ over $u \in B$, are pointwise at most K/N for a fixed K . Such an assignment follows by drawing the tags independently from the balanced mixture. For example, if α_g is the fraction of even roles with key g , each summand in the normalized mean at a fixed label is bounded by

$$\alpha_g N \mu_{i_g}(x) \leq \exp(-\Omega(n^{\gamma+\omega}) + 2n^\gamma).$$

The same bound holds on the odd side. The summand bound is at most $\exp(-cn^{\gamma+\omega})$ for some fixed $c > 0$, and the total mean is bounded. Bounded-summand concentration therefore bounds a fixed positive deviation from that mean by $\exp(-\Omega(\exp(cn^{\gamma+\omega})))$. Since there are at most $2N = \exp(O(n))$ label coordinates, one assignment meets all the bounds. Fix it henceforth.

4.1 Hidden tuples and height choices

Step 3: tuples and the masses retained by their filters. The height device will let nearby even roles choose from a common set of tuple indices. Use Lemma 3.8 on the even sites of Q_n , with

$$D = 2, \quad b = \omega/30, \quad b_0 = b/4, \quad \rho = b_0/10, \quad r = \lfloor n^{1-\rho} \rfloor, \quad \lambda = n^{10}, \quad \zeta = b_0/20, \quad \sigma = \zeta/10.$$

Choose θ, a there to satisfy its inequalities, and use its height bound H . The logarithm of the volume of an $O(r + H)$ -neighborhood is $o(a_* n)$. Set $T = \lceil 3n^b \rceil$; it bounds the number of indices that an odd role's neighbors will collectively select.

A mask for a law is a subset of its support of mass at least $1/2$; the superscript M denotes conditioning on that subset. For now give each mask an arbitrary distribution, independently of all other masks. For every possible center ID c (a location and a level) and key g , choose a mask for μ_{i_g} and draw a tuple $W_{c,g}$ of k independent labels from $\mu_{i_g}^M$. Tuples and masks for different pairs (c, g) are independent. Each odd role u also has an independent mask for $\nu_{i_g(u)}$. We generate these data even for IDs whose centers will be absent. They are independent of prospective center positions, activations, and the ties used to choose active centers.

The same tuple $W_{c,g}$ will be used in every odd filter that mentions (c, g) . If an even role v selects c , and its odd neighbors eventually receive labels $\mathbf{y} = (y_u : u \sim v)$, its available label set is

$$S_{v,c}(\mathbf{y}) = \{x \in \text{supp } \mu_{i_g(v)}^M : xy_u \in G \text{ for every } u \sim v\},$$

where the mask is the one belonging to $(c, g(v))$. We will bound the posterior density of $W_{c,g(v)}$, supported on this set's k th power, to prove that the set is large. The next estimates quantify how much an odd filter changes when this entire shared tuple is removed.

For an odd role u and an ID set $\mathcal{D}$ with $1 \leq |\mathcal{D}| \leq T$, define the retained mass

$$R_u(\mathcal{D}) = \sum_y \nu_{i_{g(u)}}^M(y) \prod_{c \in \mathcal{D}} \prod_{g' \in Z_u} \prod_{x \text{ an entry of } W_{c,g'}} 1_{\{xy \in G\}}.$$

Let $R_u^{-(c,g')}(\mathcal{D})$ be the same expression with all requirements from the single indexed tuple $W_{c,g'}$ omitted. Other tuples with ID c remain in this expression. We call the set $\mathcal{D}$ valid at u if

$$R_u(\mathcal{D}) \geq \exp(-Lk|\mathcal{D}||Z_u|)$$

and, for every $c \in \mathcal{D}$ and $g' \in Z_u$,

$$\frac{R_u(\mathcal{D})}{R_u^{-(c,g')}(\mathcal{D})} \geq \begin{cases} \exp(k(-\log 2 + c_1 a_*)), & g' = g(u), \\ \exp(-Lk), & g' \neq g(u), \end{cases}$$

where $c_1 > 0$ is a fixed small absolute constant. The first condition ensures that every denominator in these ratios is positive. These filters use the common-neighborhood sampling operation of dependent random choice [14, Lemma 2.1]. We also need their mass ratios because deleting one tuple will later remove dependence on a hypothetical even candidate.

For fixed masks, positions, and $\mathcal{D}$, the probability that $\mathcal{D}$ is invalid is at most $\exp(-n^{\omega/5})$. We prove this by exposing the tuple entries successively, with any specified tuple placed last. Call an exposed prefix successful if every hit fraction in it is at least e^{-L} ; no condition is imposed on later entries. After a successful prefix, let η be the law on Y remaining after those hit requirements. Their total width cost is at most

$$O(kT|Z_u|L) = o(n^{\gamma-\omega/2}).$$

For the next entry x , drawn from its masked first law, put $q(x) = d_G(x, \eta)$. This is the fraction of current mass retained by that entry. Its probability of being below e^{-L} is at most $\exp(-n^{\beta-\omega/2}/4)$. Otherwise conditioning the incoming first law on these values of x would cost at most $n^{\beta-\omega/2}/4$ additional width. Together with η , it would give an opposite-color patch of defect below e^{-L} , within the doubled budgets, contrary to hypothesis (ii).

When this entry has key $g(u)$, both laws lie in the same prepared patch $i = i_{g(u)}$. The intrinsic upper density bound from Step 1 also gives

$$\Pr\{q(x) > p + n^{-\omega/3}\} \leq \exp(-n^{\beta-\omega/2}/4).$$

Indeed conditioning on a larger exceptional set would fit the first width margin, and η fits the second margin. For the lower mean bound, note that $2\mu_i - \mu_i^M$ is a probability law of density at most two relative to μ_i . Since η is supported on the trimmed second support, the column lower bound and then the intrinsic upper bound give

$$\begin{aligned} \mathbb{E}_{x \sim \mu_i^M} q(x) &= 2d_G(\mu_i, \eta) - d_G(2\mu_i - \mu_i^M, \eta) \\ &\geq 2(p - n^{-\omega/5}) - (p + n^{-\omega/3}) \geq p - 3n^{-\omega/5}. \end{aligned}$$

These estimates hold conditional on each fixed successful prefix. Comparing the mean with the upper bound for q , and allowing the exponentially small exceptional set, shows that

$$\Pr\{q < p - a_i/4 \mid \text{previous successful entries}\} = O(n^{-\omega/5}/a_i) = o(a_*/L).$$

Call such an entry a dip. Its indicator is set to zero once an earlier entry has had $q < e^{-L}$. This stopped sequence is adapted to the successive exposures, and its conditional expectation bound remains valid throughout; we do not condition on eventual validity. The bounded-increment inequality [1] shows that an own-key last tuple has at most $a_*k/(20L)$ dips, except with probability $\exp(-\Omega(ka_*^2/L^2))$. On a path with no step below e^{-L} , each dip costs at most L in the negative logarithm of the retained fraction; at every other own-key entry, $\log q \geq -\log 2 + .7a_i$. Multiplying the k fractions gives the own-key ratio above, for sufficiently small c_1 . For a cross-key tuple the lower bound $q \geq e^{-L}$ already gives its ratio. The same bound on all entries gives the lower bound for $R_u(\mathcal{D})$. There are only polynomially many entries and choices of the last tuple, and $ka_*^2/L^2 = n^{\omega/3-4h+o(1)}$. A union bound therefore gives the claimed $\exp(-n^{\omega/5})$ failure estimate.

Step 4: excluding failed sets and selecting heights. We now define eligibility before activating centers. For each odd role u and consecutive pair of levels, consider all ID sets of size at most T from the prospective centers in the balls of its incident even sites at these levels. Among the sets invalid at u , take a maximal disjoint family, using a fixed local rule, and mark every ID in its union forbidden for those even sites, at that ID's own level. Every invalid set intersects this union; otherwise it could be added to the family. Consequently every set consisting entirely of the remaining eligible IDs is valid.

Binomial tails show that, with probability tending to one, every site-level ball contains between $\lambda/2$ and 2λ prospective centers. Fix such positions and all masks. For a given odd role and level pair, a prospective list of at most T IDs can be chosen in $\exp(O(T \log n))$ ways. Invalidity events for disjoint ID sets depend on disjoint tuples, hence are independent under this conditioning. The probability of a disjoint family of n invalid sets is therefore at most

$$\exp(n[O(T \log n) - n^{\omega/5}]).$$

Since $T = O(n^{\omega/30})$, this bound still tends to zero after the union over all odd roles and level pairs.

Thus each such family has fewer than n members, each with at most T IDs. An even site has n neighboring odd roles, so at most $O(n^2T)$ IDs are forbidden in any one of its level balls. This is $o(\lambda)$, leaving at least $\lambda/3$ eligible centers. Activate the centers and apply the long local height rule of Lemma 3.8, with uniform eligible active choices. With probability $1 - o(1)$, the stated counts hold and the rule gives correct heights everywhere. Around an odd role, these heights occupy at most two consecutive levels because its even neighbors are at distance two from one another. At each occupied level the crowd bound in a representative site's $(r+2)$ -ball allows at most n^b selected IDs. The actual set therefore has size at most $2n^b < T$ and, by the marking argument, is valid. Call this global count, eligibility, and height outcome *geometric success*.

At an odd role u with valid incident choices, let p_u be $\nu_{i_{g(u)}}^M$ restricted to the corresponding tuple hits and normalized. If the choices fail, or their ID set is too large or invalid, define $p_u = 0$. Thus it is always a subprobability. Auxiliary product experiments may use fixed fallback samples for these zero rows; the fallbacks do not replace the zero output in a load estimate. Every rule uses only data within distance $O(r+H)$: height choices consult their long tubes, and marks consult the nearby balls, tuples, and odd masks. No rule consults geometric success itself. On that success all odd rows are probabilities, and the retained-mass bound always gives

$$Np_u(y) \leq \exp(O(n^\gamma)).$$

Step 5: the posterior tuple and even common neighbors. Fix an even role v and a possible reference ID c in its ball at some level. Let E be the following local event: c is present and is legitimately selected at v by the long path rule and its tie, at a good height below H ;

eligibility sizes are correct throughout v 's height tube; all neighboring odd kernels are valid; and the prospective counts in every neighboring-filter ball at all levels are at most 2λ . These conditions hold for the selected ID on geometric success.

Hold all preparatory data except $z = W_{c,g(v)}$ fixed, including its mask. Its prior π remains the original product of k masked first laws. For the vector $\mathbf{y} = (y_u : u \sim v)$ of odd-neighbor labels, define

$$F_z(\mathbf{y}) = 1_E \prod_{u \sim v} p_u(y_u), \quad M_c(\mathbf{y}) = \int F_z(\mathbf{y}) d\pi(z).$$

Selection, validity, and all other local rules in this formula are recomputed for each replacement of z . Thus F_z is the likelihood including the event of selecting this tuple, not the likelihood conditional on its actual selection.

To remove dependence on z from the reference law, form a mixture at each neighbor u . Its terms range uniformly over ID sets of size at most T containing c , drawn from the present IDs in the incident balls at all levels. If the prospective count condition fails or this family is empty (in particular, if c is absent), E is impossible and we use a fixed reference probability instead of the whole mixture. Otherwise the count condition makes the logarithm of the number of terms $O(T \log n)$. In each term omit *every* requirement from the indexed tuple $W_{c,g(v)}$, then normalize the remaining filtered law; use a fixed default if undefined or empty. These terms do not depend on z . In particular, they do not require the proposed set to pass its tests or to be selected. Let Q be the product of these mixtures over the neighbors of v .

On E , each actual odd kernel compares to its corresponding mixture term by the mass ratio from Step 3. All but at most m neighbors have $g(u) = g(v)$, since only flips in special coordinates can change the key. The comparison cost for the product is consequently at most

$$\exp((n-m)k(\log 2 - c_1 a_*) + mkL + O(nT \log n)).$$

Here $T \log n = o(ka_*)$ and $mL = o(a_* n)$. For a fixed $0 < c_2 < c_1$, these two error terms fit within $(c_1 - c_2)a_* kn$, giving

$$F_z(\mathbf{y}) \leq \exp((\log 2 - c_2 a_*)kn) Q(\mathbf{y}).$$

The entire tuple has been deleted in Q , while true selection and validity remain in F_z . This is precisely the distinction needed for Lemma 3.7.

Require, for the selected reference, that

$$M_c > 0, \quad M_c \geq e^{-c_2 a_* kn/4} Q.$$

The posterior $F_z d\pi(z)/M_c$ then has density relative to the uniform product law on the original N -label side X , with current laws extended by zero to discarded labels, at most

$$\exp((\log 2 - c_2 a_*/2)kn).$$

Indeed division by the displayed threshold uses $c_2 a_* kn/4$ of the gain, and the masked first width is $o(a_* n)$, leaving the stated bound after multiplying the k prior density factors. Writing $S = S_{v,c}(\mathbf{y})$, the posterior is supported on S^k , so

$$1 \leq \exp((\log 2 - c_2 a_*/2)kn)(|S|/N)^k, \quad |S| \geq N \exp(-(\log 2 - c_2 a_*/2)n).$$

Define p_v to be uniform on S if the selected reference meets E and the predictive threshold, and zero otherwise. This even subprobability uses only the local preparatory data and neighboring odd labels. Whenever it is nonzero, its support is a common neighborhood in the same color G .

4.2 Balance and load transfer

Step 6: choosing mask distributions to balance the rows. For this step use the raw experiment with conditionally independent odd draws given the preparatory data, including fallbacks where necessary, and without conditioning on geometric success. We will choose the mask distributions so that

$$\mathbb{E}p_u \leq 2\nu_{i_g(u)} \quad (u \in B), \quad \sum_{v \in A} \mathbb{E}p_v \leq K' \sum_{v \in A} \mu_{i_g(v)}$$

coordinatewise, for a fixed K' .

The player choosing the odd mask at u controls the output vector $\mathbb{E}p_u$. Given nonnegative label prices on Y , retain the labels with price at most twice the $\nu_{i_g(u)}$ -mean price. They have mass at least one half. Every nonzero p_u is supported there and has mass at most one, so its expected price is at most twice that mean, regardless of the other mask profiles. Separation therefore gives a mixture of masks meeting this player's vector bound.

The player for (c, g) instead controls the sum of outputs from even rows with key g selecting ID c . Let $n_{c,g}$ be the number of even roles with key g whose radius- r balls contain the location of c , regardless of its presence or selection. This count is deterministic. Put

$$w_c = (\lambda/V) \begin{cases} 3/\lambda, & \text{level zero,} \\ \exp(-n^{c_3}), & \text{otherwise,} \end{cases}$$

where V is the radius- r ball volume and $c_3 > 0$ is small and fixed. The probability that one such row selects c with event E is at most w_c . Presence of c costs λ/V ; given presence, Lemma 3.8 bounds its level-zero choice probability by $3/\lambda$, and a positive height by $\exp(-n^{c_3})$. The local size condition in E is retained in this application. The lemma's position-averaged bound is uniform over the legal eligibility sets, so the bound remains valid for the masks and tuple-dependent eligibility used here.

Given prices on X , this player can likewise choose a mask whose prices are at most twice their μ_{i_g} -mean. Every even output attributed to this player is supported in that mask. Multiplying its price bound by the probability w_c of selecting c shows that the player's expected output vector can be bounded by $2n_{c,g}w_c\mu_{i_g}$, again by separation. The output rules for fixed masks do not depend on their profile probabilities: in particular, the posterior computations condition on the realized masks. Thus Lemma 3.3 supplies simultaneous profiles for all players.

For any fixed even role v , summing w_c over IDs in its range costs at most 3 at level zero and at most $H\lambda\exp(-n^{c_3}) = o(1)$ over the other levels. Summing the players' vector bounds proves the even inequality. The fixed patch-tag assignment from Step 2 now shows that the parity averages of $N\mathbb{E}p_u(y)$ and $N\mathbb{E}p_v(x)$ are bounded by constants at every label.

Step 7: assigning odd labels and controlling even loads. First bound odd column sums in the preparatory experiment. The weights $Np_u(y)$ have cap $\exp(O(n^\gamma))$, bounded average means by Step 6, and independent local inputs for roles separated by more than $O(r + H)$. A near set has parity fraction at most $\exp(-n \log 2 + o(a_*n))$. Hence its contribution times n and the cap tends to zero. Lemma 3.6, followed by Markov's inequality and the union over labels, bounds every normalized average odd load by a constant with probability tending to one. Since $|B|/N = 1/(2C_n) \rightarrow 0$, the actual column sums are then at most $1/10$.

Write $\mathcal{H}$ for the preparatory history and S_{pre} for geometric success together with the odd column bound. On S_{pre} , all odd rows are probabilities and meet the hypotheses of Lemma 3.9 with $d = N$: their maximum atom is $N^{-1}\exp(O(n^\gamma)) \leq N^{-.95}$ eventually, the column bound is less

than .4, and $n^2 \leq N^{.025}$. Let $J_{\mathcal{H}}$ be the resulting odd injection law, and let $P_{\mathcal{H}}$ be the auxiliary product law of the odd draws. For every nonnegative function Φ of at most n^2 odd outputs, the joint target bound gives

$$\int \Phi(\mathcal{H}, \mathbf{y}) dJ_{\mathcal{H}} \leq \exp(N^{-.04} n^2) \int \Phi(\mathcal{H}, \mathbf{y}) dP_{\mathcal{H}} \quad (\mathcal{H} \in S_{\text{pre}}).$$

The comparison factor is $1 + o(1)$, uniformly in the entering history.

We next show that all selected even references meet the predictive threshold of Step 5 with probability tending to one. For a fixed even role v and possible reference c , let $\epsilon = e^{-c_2 a_* kn/4}$ and let D_c be the set of neighboring data with $M_c = 0$ or $M_c < \epsilon Q$. Apply the comparison with $\Phi = 1_E 1_{D_c}$, then average the preparatory histories with $1_{S_{\text{pre}}}$ retained. Removing this last indicator increases the resulting product-draw integral. Its remaining indicator 1_E makes the neighboring likelihood exactly F_z , because the odd kernels are valid on E . With the other preparatory data fixed, integration over the original masked tuple prior therefore gives

$$\sum_{\mathbf{y} \in D_c} \int F_z(\mathbf{y}) d\pi(z) = \sum_{\mathbf{y} \in D_c} M_c(\mathbf{y}) \leq \epsilon \sum_{\mathbf{y} \in D_c} Q(\mathbf{y}) \leq \epsilon.$$

There are at most $|A|(H+1)V = \exp(O(n))$ role-ID pairs, whereas $a_* kn = n^{1+\omega/3-h+o(1)}$. Thus their total predictive-failure probability is $o(1)$. The probability here is under the experiment that first draws the preparatory history and then, on S_{pre} , draws the odd injection. Failure to reach S_{pre} also has probability $o(1)$; earlier histories have not been reweighted by the probability of later success.

Let S_{good} be S_{pre} intersected with the event that every selected even reference passes its predictive requirement. It remains to control the even column sums after this odd injection. On S_{good} , the normalized even rows have cap $\exp((\log 2 - c_2 a_*/2)n)$. With the same near fraction as above,

$$n \exp(-n \log 2 + o(a_* n)) \exp((\log 2 - c_2 a_*/2)n) = o(1).$$

For the separated factors in an n th moment, their even kernels consult at most n^2 odd rows in total. The event S_{good} also tests references at other roles, so first use

$$1_{S_{\text{good}}} \prod_{j=1}^{\ell} Np_{v_j}(x) \leq 1_{S_{\text{pre}}} \prod_{j=1}^{\ell} Np_{v_j}(x), \quad \ell \leq n.$$

Each factor on the right still contains its own local validity and predictive requirements. Conditional on the entering history, the product of these factors is therefore a function of at most n^2 odd outputs, to which the displayed injection comparison applies. Only after that comparison remove $1_{S_{\text{pre}}}$, which restricts the preparatory history. In the resulting raw experiment the separated kernels depend on disjoint center data, masks, and odd draw seeds. The patch tags are already fixed. Their expectations thus factor, and their individual means are those bounded in Step 6. Explicitly, for separated $v_1, \dots, v_{\ell}$, $\ell \leq n$, and a fixed label x ,

$$\mathbb{E} \left[1_{S_{\text{good}}} \prod_{j=1}^{\ell} Np_{v_j}(x) \right] \leq (1 + o(1)) \prod_{j=1}^{\ell} \mathbb{E}_{\text{raw}}[Np_{v_j}(x)].$$

This is the separated-product hypothesis of Lemma 3.6; the displayed cap estimate handles nearby factors.

It follows that, except for probability $o(1)$ on S_{good} , every even column sum is at most one. Intersect this event with the preceding success events. All even rows are now probabilities supported on common G -neighbors of the assigned odd labels, so Hall's theorem [20] assigns them distinct representatives. The even images lie in X and the odd images in the disjoint side Y ; every cube edge has color G . This gives the required contradiction and proves the assertion, including $\beta = \gamma$. $\square$

5 A broad side of constant width

To turn Lemma 4.1 into initial discrepancy, we need to exclude available nearly pure patches with two small power widths. We approach this through two broad-side constructions. The present constant-width lemma handles an opposite-color alternative in the proof of Lemma 6.1. The latter allows the broad-side law to have a small polynomial density cap, or equivalently logarithmic width. After stabilization, it supplies the lower bound on both color densities in (7.1), which the grid construction of Section 7 uses to exclude these available nearly pure patches.

Lemma 5.1 (A broad side of constant width). *Work in the counterexample sequence fixed above. Suppose for one color G there is a finite random tag I with laws μ_I on X of width $\leq n^\gamma$ ($0 < \gamma < 1$ fixed), $\mathbb{E}\mu_I \leq K'_0/N$ pointwise. For each I , at least χN columns $y \in Y$ have $d_G(\mu_I, y) \geq .95$, with $K'_0, \chi > 0$ fixed. This yields a G -cube for large n .*

Proof. We construct probability rows on Y for the odd roles, assign them injectively, and then construct rows on their common neighborhoods in X . The individual laws μ_I may be too concentrated to serve a constant fraction of the cube. Their balanced average will instead control the mean loads of short random tuples in X .

Each such tuple will be sampled to see several random columns. These column requirements must vary across odd roles to spread their loads, while only a small number of distinct requirements may occur around one even role. We arrange this using counts and majority signs of disjoint chunks of cube coordinates. Posterior distributions then relate three objects: columns given streams of first-side labels, stream blocks given those columns, and odd labels given the resulting blocks. All these posteriors refer to specified raw experiments. We impose their numerical bounds later, one sampling stage at a time.

5.1 Parents, column indices and even types

Call y good for I if $d_G(\mu_I, y) \geq .95$, and define

$$(y, y') \in \mathcal{H} \iff \Pr_I(y \text{ and } y' \text{ are good for } I) \geq \chi^2/2.$$

The average of this probability over uniform $y, y' \in Y$ is at least χ^2 , by squaring the size of each tag's good-column set. Hence $\mathcal{H}$ contains a positive constant fraction of all pairs. Consequently a positive constant fraction of labels have a positive constant fraction of partners in $\mathcal{H}$. Draw V_0 uniformly from these labels. For each bin w , to be defined below, independently draw A_w uniformly from the partners of V_0 , conditional on V_0 . Both the law of V_0 and each conditional law of A_w have atoms at most $O(1)/N$. All draws use the retained labels under consideration.

Fix constants

$$a_0 = 1/3 < a_1 < \dots < a_8 < \tau_0 < \tau_1 < \log 2,$$

and take $\delta > 0$ sufficiently small relative to their gaps. Given the parents, draw a stream W_w at each bin, independently across bins and in independent segments of a fixed large length q_0 . To draw one segment, first choose a fresh tag conditioned on both parents being good, and then draw q_0 independent labels from μ_I conditioned on adjacency to both parents. The labels in a segment are independent conditional on its tag. Each has simultaneous-hit probability at least .90 before this last conditioning. Thus the segment law has density at most

$$(2/\chi^2)(.90)^{-q_0} \leq e^{a_0 q_0}$$

relative to $R'_0 = \mathbb{E}_I \mu_I^{\otimes q_0}$, upon increasing the fixed q_0 . For a length u divisible by q_0 , write $R[u]$ for the product of u/q_0 such reference segments. Each of its coordinate marginals is at most K'_0/N , and its joint density relative to product uniform measure is at most e^{un^γ} . Only finite streams are needed. All prefix and block lengths below are rounded upward to multiples of q_0 .

Counts, signs, and column indices. Partition some cube coordinates into $d_c = 300$ coarse chunks, each of length $\lfloor n^{1/5} \rfloor$. In each chunk, partition the possible counts of ones into consecutive intervals, or bins, of probability at most $2n^{-.04}$, with $O(n^{.04})$ endpoints. The binomial atom bound $O(n^{-.1})$ permits this choice. Let P be the vector of counts and $w(P)$ its vector of bins. The coarse key $i(P)$ records $w(P)$ and a flag: it is interior if no single bit flip changes a bin, and boundary otherwise. An interior key uses parent A_w and the list $C_i = \{w\}$; a boundary key uses parent V_0 and the list consisting of w and its grid neighbors. In particular, $w(P) \in C_{i(P')}$ whenever $P' = P$ or P' results from one coarse bit flip.

Use $m = \lceil n^\alpha \rceil$ further disjoint chunks, each of the same odd length asymptotic to n^3 , where $\alpha > 0$ is a small fixed constant. Their majority signs give $t \in Q_m$. Let F be the set of chunks whose count is at distance $1/2$ from mid-weight; these are exactly the signs that one bit flip can change. Let j count the chunks whose count is at distance at most $R_f = 5.5$ from mid-weight. Thus j counts chunks, not coordinates within them, and its cutoff is independent of the coarse boundary flag. A role is low if $j \leq J = \lfloor m^{1/20} \rfloor$, and high otherwise. All remaining cube coordinates are residual. On either actual cube parity the signs are uniform. Binomial atom bounds and a union over subsets of chunks give

$$\Pr(j \geq h) \leq n^{-.13h} \quad (1 \leq h \leq m), \quad \Pr(i(P) \text{ is boundary}) \leq n^{-.05}.$$

The hidden columns are indexed by $\ell = (i, t, j)$, $0 \leq j \leq J$, and by $\ell = (i, *)$ in the high case. An odd role uses the index determined by its counts and signs. Given the base data (V_0, A, W) , independently draw U_ℓ for all these indices. A low U_ℓ is one column; a high U_ℓ is a tuple of $s = \lceil K_s J \log m \rceil$ independent columns. Write $s_\ell = 1$ or s for its length. Its coordinate law π_ℓ is the following posterior in the raw parent-and-stream experiment:

- at an interior key, the law of A_w given $V_0, W_w[1 : b_j]$;
- at a boundary key, the law of V_0 given $(A_{w'}, W_{w'}[1 : b_j])_{w' \in C_i}$.

Here

$$u_j = \lceil K_1(j+4) \log m \rceil, \quad b_j = u_{j+1},$$

with segment rounding, and high keys use $j = J$. We draw these variables also at indices that occur only in the padded lists defined next.

Even types and the lengths of their arrays. For a low even role, with coarse data P, w , let S consist of the distinct keys

- $(i(P'), t, j)$, where $P' = P$ or is obtainable by one coarse flip;
- $(i(P), t^h, j)$, $h \in F$, and $(i(P), t, j \pm 1)$ when in range, replacing severity $J + 1$ by the high key. Here t^h is t with sign h reversed.

Its type is $K = (w, S, j)$. At a high even role take $K = (w, S, *)$, where S contains the high keys at the same coarse range of $i(P')$'s. If $j = J + 1$, there is also an optional low neighbor key $(i(P), t, J)$. This optional key is excluded from both S and the type K . These lists cover the keys at all actual neighbors, since a sign flip leaves j unchanged. Every key paired with w has $w \in C_i$.

We now fix the lengths and indices of the arrays; their block law P_K will be defined in Step 2. At each possible center ID, arrays of different types will be independent conditional on the base and hidden columns. For a low type, blocks have length $u = u_j$, and the array has total length

$$k_j = u_j \left\lceil K_2((j + 4) \log m + m^{.02})/u_j \right\rceil.$$

For a high type, put $u = u_* = \lceil \eta \log m \rceil$, with segment rounding, and $k_* = u_* \lceil m^{.005}/u_* \rceil$. Its full pool has $\lceil e^{K_h u_*} k_*/u_* \rceil$ blocks, of total length M_* . If an optional low key is designated, the used tuple is formed from the first k_*/u_* blocks all of whose entries hit that column. Without an optional key it consists of the first k_*/u_* blocks. At fixed base, the full pool law will use only the keys in the type K , so it is independent of the optional column's value before this extraction.

A tuple reference c specifies its ID and type, and in the high case also the block index subset. Its length k_c is k_j or k_* . The constants K_h, K_1, K_2, K_s are fixed and large; after K_h choose η so small that $M_* = o(m^{.03})$. We give the remaining order of choices below, taking α sufficiently small last. Undefined numerical formulas on unused or failure branches use fixed fallback values depending only on their declared local inputs.

The raw sampling order is

$$V_0 \longrightarrow (A_w, W_w)_w \longrightarrow (U_\ell)_\ell \longrightarrow \text{center arrays}.$$

Prospective positions, activations and choice ties are sampled separately; an array is generated even at an ID whose center is absent. Each posterior uses this raw experiment and only the observations explicitly listed in its definition. When a candidate value changes, every later raw kernel depending on it is evaluated at that new value. In particular, unobserved arrays are integrated under their new laws. Successful-history conditions are included only when explicitly retained as local gates.

5.2 Predictive estimates at the first two levels

Step 1: deleting a stream prefix from a parent posterior. For a key ℓ , remove the observation $W_w[1 : u]$ from the specified definition of π_ℓ , retaining its other observations, and call the resulting posterior $\pi_{\ell, -w, u}$. The shift $b_j = u_{j+1}$ ensures that each prefix needed for a listed or optional key lies within the observed stream. Given the retained observations, the incoming prefix has likelihood at most $e^{a_0 u}$ relative to $R[u]$, for every parent value in support: it comprises complete segments, and the partner is among the retained data.

The marginal density of this prefix is smaller than $e^{-\delta u}$ on a set of raw probability at most $e^{-\delta u}$. Off that set, Bayes' formula and the gap $a_1 - a_0$ give the test

$$\pi_\ell(y) \leq e^{a_1 u} \pi_{\ell, -w, u}(y) \quad \text{for every } y.$$

We also require $N \max \pi_\ell \leq \exp(K' b_j)$, for a sufficiently large fixed K' . Its raw exception is at most $e^{-\delta b_j}$. At an interior key this follows from the full stream likelihood conditional on V_0 . At a boundary key use product uniform measure for the bounded list $(A_{w'})_{w' \in C_i}$ and the segment references for their streams. The likelihood for each candidate V_0 is at most $(O(1)e^{a_0 b_j})^{|C_i|}$. The prior atom bound and the same small-denominator estimate give the asserted cap.

Step 2: reconstructing a stream block from hidden columns. Fix a type K and all base data except $W_w[1 : u]$. Its conditional raw law P_0 is the product of the true-parent segment laws. Let $\mathcal{G}_K$ be the set of alternative blocks z for which the Step 1 comparisons at (w, u) hold for every key in S . At a high type, also require them for all potentially used optional keys. This involves all coarse parts represented in S , but not the signs: the low $j = J$ and high priors have the same coordinate law. In particular, $\mathcal{G}_K$ is defined without observing any U .

Set $\rho_\ell = \pi_{\ell, -w, u}^{\otimes s_\ell}$. Relative to ρ_ℓ , let $L_\ell(z; \theta_\ell)$ be the likelihood of $U_\ell = \theta_\ell$ when the base block is replaced by z . Off the support of ρ_ℓ the gated likelihood can be set to zero. Relative to the product reference $\prod_{\ell \in S} \rho_\ell$, the observations U_S , with the event $W_w[1 : u] \in \mathcal{G}_K$ retained, have subdensity

$$m_S(\theta) = \int 1_{\mathcal{G}_K}(z) \prod_{\ell \in S} L_\ell(z; \theta_\ell) dP_0(z).$$

Define P_K by normalizing this integrand in z at the observed U_S . Dropping the factor for ℓ , but retaining the same gate, defines $m_{S-\ell}$ and $P_{K,-\ell}$.

We require positivity and the denominator bounds

$$m_S \geq e^{-\delta u \sum_S s_\ell}, \quad m_S / m_{S-\ell} \geq e^{-\delta u s_\ell}.$$

On the true-block gate, each exception has raw probability at most its threshold. For the ratio test, for example,

$$\int_{\{m_S < \varepsilon m_{S-\ell}\}} m_S d\left(\prod_{\ell' \in S} \rho_{\ell'}\right) \leq \varepsilon \int m_{S-\ell} d\left(\prod_{\ell' \in S} \rho_{\ell'}\right) \leq \varepsilon.$$

The last integral is at most one, and all unlisted keys have simply been integrated out. On the true-block gate $m_S > 0$ almost surely, which also implies $m_{S-\ell} > 0$. For a high type, adding one optional low key gives the analogous test $m_{S+\ell}/m_S \geq e^{-\delta u_*}$, with the same probability bound.

Combining these lower bounds with Step 1 gives

$$\frac{dP_K}{dP_{K,-\ell}} \leq e^{a_2 u s_\ell}, \quad \frac{dP_K}{dR[u]} \leq A_K^u, \quad A_K = \exp(K''(1 + \sum_{\ell \in S} s_\ell)),$$

where K'' is a sufficiently large fixed constant. The first comparison holds also at any replacement value of θ_ℓ passing its ratio test. We can now generate each center array as independent P_K -blocks of the lengths already specified.

The support of this posterior enforces the required adjacencies. Every z in the support of P_0 leaves the base data with positive raw probability. If a listed column has positive posterior likelihood at that base, the stream-generation rule says that it sees every entry of z . For an optional column the added likelihood is at most $e^{a_1 u_*}$, whereas its expectation under P_K is $m_{S+\ell}/m_S \geq e^{-\delta u_*}$. Its positive-likelihood set therefore has P_K -mass at least $e^{-(a_1 + \delta) u_*}$, and every block in that set hits the optional column. Choosing K_h large makes the probability of too few such blocks in a high pool $o(1)$, uniformly at histories passing these tests.

5.3 State neighborhoods and odd predictive estimates

We group actual cube roles into states that retain their residual bits, exact coarse counts, and the count in every majority-sign chunk, with one exception: counts at distances 5.5 and 6.5 on the same side of mid-weight are merged. A state also records the exact value of j . Encode each count field and the j -field by a one-hot vector. Together with the residual bits this embeds the states in Q_d , $d = (1 + o(1))n$.

A state determines parity despite the merged counts. Give each merged symbol its outer representative, at distance 6.5, as a baseline. Replacing it by its inner representative changes the count by one and reverses parity. The number of replacements is j minus the number of unmerged counts within the cutoff. This determines the fine-coordinate parity; the coarse counts and residual bits are exact. It also determines the key, type and optional designation. The exact coarse counts here are more information than will be retained in Lemma 6.1.

Join an odd state to an even state when some actual cube edge realizes that pair. Each state has degree $O(n)$, and any two even states adjacent to one odd state have ambient distance at most $D = 8$. Indeed, one bit flip changes only boundedly many one-hot entries, including the possible change of j . A flip changing j passes between distances 5.5 and 6.5. Thus all low neighbors of a high odd state, when present, form one state; the same holds for the high neighbors of a low state. The padded lists above cover all this incidence.

Use even states as height sites and choose one center ID at each, shared by all actual roles represented there. An ID is a location-level pair in the ambient cube. We aim to use at most $T = \lceil m^{.001} \rceil$ distinct IDs around each odd state. For the height device take $r = \lfloor \rho n \rfloor$, $\lambda = n^{10}$, and parameters with $2n^b \leq T$ and $\theta > .9$. Its exponents are chosen after α , with $b > b_0 > 0$ much smaller than $.001\alpha$, and $\sigma, \zeta, 1 - \theta$ small enough for its inequalities. These height exponents are unrelated to the stream lengths b_j . Choose $\rho > 0$ so small that the binary ball entropy at 2ρ , in natural logs, is less than $\log 2 - \tau_1$. We will also use the short height computation of radius $D \lfloor \sqrt{m} \rfloor$.

Observation records and their counts. Fix an odd state b and a mapping of its neighboring even states to at most T IDs. Its observation record lists distinct ID/type arrays, so repeated uses of the same array are recorded only once. At a low state we observe every used full low tuple and, if a high neighbor is present, its full pool and the block subset selected to hit U_ℓ . Call that subset the mask. At a high state we observe all used full high pools and the possible one used low tuple. We also record which high type/optional-key combinations are needed at each ID, including the absence of an optional key. Their block subsets are computed from the observed pools and optional columns. If there are too few hits, the test uses the first blocks as a fallback; actual usable tuples will be ensured separately. A tuple reference c is therefore a specified set of primitive blocks in one of the observed arrays.

There are $O(T + j)$ low tuples around a low state. To see this, low types with unchanged t, F have boundedly many generic variants, determined by coarse incidences and severity; each variant uses a subset of the ID list. Every other variant changes a sign or F , and hence comes from a chunk at distance at most 1.5 from mid-weight. There are at most j such chunks and constantly many relevant flips per chunk. Each of their $O(j)$ neighboring states uses one ID. Ambiguous merged chunks do not change signs or F , so they produce only generic variants regardless of how many there are.

At a high state, there are $O(T)$ full pools of boundedly many types. Optional choices vary only near the interface, where $j \leq J + 2$. Their generic variants retain the sign, while the other

variants arise from $O(J)$ relevant neighboring states. Thus there are only $O(T + J)$ needed high tuple references, even though several may use subsets of the same pool.

The counts of actual records below assume that the r -ball of each even state adjacent to b has at most 2λ prospective centers at every level, as will be required in the local validity event. At fixed center presence, this leaves polynomially many possible IDs. Choosing their lists and assigning the exceptional states gives logarithmic record counts at most

$$\begin{aligned} O(T \log n + (j + 1) \log T + k_* 1_{j=J}), & \quad b \text{ low,} \\ O(T \log n + (J + 1) \log T), & \quad b \text{ high.} \end{aligned}$$

The low mask contributes at most $\log(M_*/u_*) = O(k_*)$. At high states we count optional-key assignments, then compute their subsets; we do not enumerate all subset combinations. If only the full observation list is recorded, its logarithmic count is $O(T \log n)$.

For history tests, before positions have been sampled, use abstract records with the IDs renamed in $[T]$. Group them by coarse bin, central sign and severity. Their logarithmic counts are

$$\begin{aligned} O(T \log T + (j + 1) \log m + k_* 1_{j=J}), & \quad b \text{ low,} \\ O(T \log T + (J + 1) \log m), & \quad b \text{ high.} \end{aligned}$$

Besides generic variants, a low record specifies $O(j + 1)$ chunk indices and their finitely many near-center statuses. A high record needs such information only at the interface. Coarse data enter through a bounded neighborhood of bins. In particular, regular Step 2 low-type patterns number $m^{O(j+1)}$; the small-prefix u_* tests have boundedly many patterns per bin modulo the sign of their one optional key. These counts distinguish the full arrays observed, the tuple references extracted from them, and the records used to enumerate tests.

Step 3: reconstructing an odd label from center arrays. Fix the base and every hidden column except the target $\vartheta = U_\ell$ at b . Its raw prior is $\pi_\ell^{\otimes s_\ell}$. At a low state, condition also on the possible high pool, whose law does not use this optional low key. The remaining observations are the low-array blocks. At a high state, the observations instead comprise every full high pool and the possible one low tuple. In either case these observed blocks are independent given ϑ , with laws $P_K(\cdot | \vartheta)$; their types have $\ell \in S$.

For each observed block use $P_{K,-\ell}$ as a reference law. Let $g_e(\vartheta)$ be the likelihood of block e relative to this law. The candidate gate $\Theta(\vartheta)$ requires positive $m_{S-\ell}$ and the Step 2 lower ratio $m_S/m_{S-\ell}$ for every observed type. These conditions use no array entries. At a low state with a high pool, also include the indicator that ϑ selects the specified successful mask from that fixed pool. There is no mask indicator at a high state. Thus, with c a same-mode tuple reference,

$$M = \int \Theta(\vartheta) \prod_e g_e(\vartheta) d\pi_\ell^{\otimes s_\ell}(\vartheta), \quad M_{-c} = \int \Theta(\vartheta) \prod_{e \notin c} g_e(\vartheta) d\pi_\ell^{\otimes s_\ell}(\vartheta).$$

At low states the products use only low blocks; at high states they use all the observed blocks, and deletion may remove a strict subset of one pool. On the candidate gate each block density is at most $e^{a_2|e|s_\ell}$, where $|e|$ is its length.

Let k'_j be the minimum k_l with $|l - j| \leq 1$. Require positivity and

$$M/M_{-c} \geq e^{-\delta k_c s_\ell}, \quad M \geq \begin{cases} e^{-\delta k'_j}, & b \text{ low,} \\ e^{-\delta s}, & b \text{ high.} \end{cases}$$

For a fixed observation record and fixed subsets, each exception intersected with the true-target gate costs at most its threshold in the raw joint experiment. This is the subdensity calculation of Step 2, now using $\int M_{-c} \leq 1$. At low states it includes occurrence of the specified successful mask. At high states an individual ratio test can be unioned over all subsets of one pool at cost $\exp(O(k_*))$, which is smaller than its $\delta k_* s$ exponent. No union over all joint choices of subsets is needed.

Let $\mathcal{P}$ and $\mathcal{Q}_c$ be the normalized full and deleted target integrands. Their gates do not include the denominator tests just imposed. At a low state, $\mathcal{P}$ is supported on labels adjacent to all needed tuples, including the high tuple specified by the mask. The block likelihood and denominator bounds give $\mathcal{P} \leq e^{a_3 k_c} \mathcal{Q}_c$. We reserve a further multiplier $e^{\delta k'_j}$ for selection, still below $e^{a_4 k_c}$. Even after that multiplier the normalized atom cap is e^{D_L} , where $D_L = m^{15}$: its logarithm costs only $O(b_j + (T + j) \max_{|l-j| \leq 1} k_l)$. The independent high pool is not part of this likelihood cost.

At a high state the bounds are

$$\log(\mathcal{P}/\mathcal{Q}_c) \leq a_3 k_c s, \quad \log(N^s \max \mathcal{P}) \leq K_0 s J \log m.$$

There are $O(TM_*) = o(J)$ full high-pool entries and at most one low tuple of length $O(J \log m)$; together with the prior and denominator bounds these give the second inequality. The fixed K_0 may depend on K_1, K_2 , but not on K_s .

High rows: a common law for all deletion costs. For a high row we need one law on index–label pairs that has small expected likelihood cost against every needed high deletion reference. Fix the mapping and observed data in the true-target-gated raw model, with the preceding denominator tests satisfied. The remaining target ϑ has law $\mathcal{P}$. The needed block subsets are now fixed by the full pools and the other, optional low keys; they do not depend on the remaining target draw. Let $r_{\text{ref}} = O(T + J)$ be their number. If it is zero, the cost constraints below are vacuous: restrict the index–label law to the density-good set defined below and normalize, without using a grid.

For a supported true prefix $\vartheta_{<h}$, let $\mathcal{P}_h$ and $\mathcal{Q}_{c,h}$ be the next-coordinate conditionals. They are functions of this prefix. Joint domination ensures that the deletion conditional is defined there. Smooth it with a small fixed uniform component to obtain $\tilde{\mathcal{Q}}_{c,h}$, and put

$$E_h^c(y) = k_c^{-1} [\log(\mathcal{P}_h(y)/\tilde{\mathcal{Q}}_{c,h}(y))]_+.$$

If Y has law p , then $\Pr_p(\log(q(Y)/p(Y)) > z) \leq e^{-z}$: on that set $p(Y) < e^{-z} q(Y)$. The positive part of this negative log has exponential moment at parameter 1/2 at most 2. Applying this conditionally at successive true prefixes shows that the sum of negative parts is $O(s)$, except with probability $e^{-\Omega(s)}$ per reference. By the chain rule, the sum of the full unsmoothed logs is at most $a_3 k_c s$. Smoothing increases a positive log by at most a fixed constant. Consequently

$$\frac{1}{s} \sum_{h=1}^s E_h^c(\vartheta_h) \leq a_3 + o(1).$$

The same argument against uniform measure gives

$$\frac{1}{s} \sum_{h=1}^s \frac{[\log(N \mathcal{P}_h(\vartheta_h))]_+}{D_H} \leq K_0/K_D + o(1), \quad D_H = K_D J \log m.$$

We choose K_D large enough to make this last bound a small fixed constant.

These are averages at the labels actually drawn. To control the averages of the conditional laws, fix a convex weight vector ξ on the references and a large fixed L_0 . For the filtration $\mathcal{F}_h = \sigma(\vartheta_1, \dots, \vartheta_h)$, conditional on the fixed observations, set

$$C_h^\xi(y) = \min\{L_0, \sum_c \xi_c E_h^c(y)\}, \quad D_h^\xi = C_h^\xi(\vartheta_h) - \sum_y \mathcal{P}_h(y) C_h^\xi(y).$$

The D_h^ξ are martingale differences bounded in absolute value by L_0 . On the preceding actual-path bounds, failure of

$$\frac{1}{s} \sum_h \sum_y \mathcal{P}_h(y) C_h^\xi(y) \leq a_3 + \delta'$$

forces $\sum_h D_h^\xi < -\delta's/2$, for large n . Its probability is at most $\exp(-\Omega_{\delta', L_0}(s))$. This concentration is applied before conditioning on any successful path. Applying it also to $\min\{1, [\log(N\mathcal{P}_h(y))]_+/D_H\}$ bounds the average conditional mass of labels with $N\mathcal{P}_h(y) > e^{D_H}$ by a small fixed constant η_D .

Here is a finite set of directions sufficient for all the cost constraints. Fix a small $\varepsilon_g > 0$, and for ξ in the reference simplex set

$$a_c = \lceil r_{\text{ref}} \xi_c / \varepsilon_g \rceil, \quad A = \sum_c a_c, \quad \widehat{\xi}_c = a_c / A.$$

Then $A \leq (1 + \varepsilon_g^{-1})r_{\text{ref}}$ and $\xi_c \leq (1 + \varepsilon_g)\widehat{\xi}_c$ for every c . The number of nonnegative integer vectors with this bound on their sum is $\exp(O_{\varepsilon_g}(r_{\text{ref}}))$, by the composition count $\binom{\lfloor (1 + \varepsilon_g^{-1})r_{\text{ref}} \rfloor + r_{\text{ref}}}{r_{\text{ref}}}$. Thus all the predictable bounds hold on this grid, with total failure $e^{-c's}$ for a fixed $c' > 0$, since $T + J = o(s)$. This is a conditional bound at fixed data; integrating against the gated data subdensity and adding the earlier denominator exceptions gives the same form of bound for each mapping. The later union over abstract mappings has a different count and will determine how large K_s must be.

Now fix a path passing these tests, and define the probability law $\nu(h, y) = \mathcal{P}_h(y)/s$ and set $G_D = \{(h, y) : N\mathcal{P}_h(y) \leq e^{D_H}\}$. For each grid direction restrict ν to G_D and to $\sum_c \xi_c E_h^c(y) \leq L_0$. The removed mass is at most

$$\beta = \eta_D + (a_3 + \delta')/L_0.$$

After normalization, the unclipped expected cost in that direction is at most $(a_3 + \delta')/(1 - \beta)$. Choose the small constants and L_0 so that $(1 - \beta)^{-1} \leq 1 + \varepsilon'$, with $\varepsilon' > 0$ small. All these restricted laws belong to the same compact convex set

$$\mathfrak{F} = \left\{ R \geq 0 : \sum_{h,y} R(h, y) = 1, \quad R(h, y) \leq (1 + \varepsilon')\nu(h, y)1_{G_D}(h, y) \right\}.$$

Nonnegative costs and the coordinatewise grid bound show that for every convex direction ξ , some member of $\mathfrak{F}$ has weighted expected cost at most a_3 plus a chosen small error. If no member satisfied all the individual bounds, separation of the compact convex set of cost vectors from the region below these bounds would give a nonnegative separating vector. After normalizing it, this would contradict the directional conclusion. Hence one law $\mathcal{R} \in \mathfrak{F}$ satisfies all the cost bounds.

Its support hits every required tuple: a positive $\mathcal{P}_h(y)$ has a positive posterior continuation, all of whose coordinates hit the observed blocks. Moreover,

$$\left[\log \frac{\mathcal{R}(h, y)}{\widehat{\mathcal{Q}}_{c,h}(y)/s} \right]_+ \leq \log(1 + \varepsilon') + k_c E_h^c(y).$$

The gap from a_3 to a_4 therefore yields

$$\mathcal{R}(h, y) \leq 2e^{D_H}/(sN), \quad \mathbb{E}_{\mathcal{R}}\left(\left[\log(\mathcal{R}(h, y)/(\tilde{Q}_{c,h}(y)/s))\right]_+\right) \leq a_4 k_c.$$

Choose one such law deterministically from the local data. A deleted reference uses only the full observation list with those primitive blocks removed, together with the fixed history and true prefixes. It does not use other optional subset assignments. True prefixes are legitimate fixed data here because the hidden columns were sampled before the arrays. The clip parameters can be chosen with a rate $c' > 0$ independent of K_s . The path tests include positivity and true support; these hold almost surely in the true gated model.

5.4 Making history tests simultaneous

We next sample the base and hidden columns so that Steps 1 and 2 hold throughout. At the resulting history, every abstract mapping must also have Step 3 failure probability over independent center arrays at most

$$e^{-c_L k'_j} \quad \text{at low states,} \quad e^{-c_H s} \quad \text{at high states,}$$

for fixed small $c_L, c_H > 0$. We achieve this in five stages, restricting only the newly drawn variables at each entering history. Later avoidance probabilities never reweight earlier histories.

For Step 2 a failure always means the failed numerical test intersected with its true-block gate $W_w[1 : u] \in \mathcal{G}_K$. For Step 3 it means failure at the specified record intersected with its true-target gate, at a base passing Step 1. In a low record with a specified high mask, the event includes occurrence of that successful mask. The earlier stages will enforce the block and key-ratio gates; usable hits for selected centers are enforced below. These definitions keep raw subdensity bounds as intersection bounds, rather than conditioning them on future success.

A regular Step 1 or 2 test has scale $L = u_j$, including the prior cap at severity j , with the high prior cap assigned $j = J$. A small-prefix test has scale $L = u_*$. Each raw failure probability is at most $e^{-\delta L}$. The regular pattern count per bin and central sign is $m^{O(j+1)}$, so a bound such as $e^{-\delta u_j/16}$ sums to $o(1)$ over all regular patterns and severities if K_1 is large. Small-prefix tests have boundedly many patterns modulo sign translation, and $e^{-\delta u_*/16} \rightarrow 0$.

An alarm at an entering history H is an event

$$\left\{ \Pr_{\text{raw future}} (B \cap \text{local gate} \mid H) > q \right\}.$$

If the expectation of this conditional probability is at most p , Markov bounds the alarm probability by p/q . Its scope comprises the current-stage variables occurring in the canonical record: base bins in the relevant C_i -lists, hidden keys in the relevant type lists, and any specified optional keys. Future array variables are integrated out. This definition determines the scopes used in each local lemma below.

For these alarm families, the gated likelihoods are evaluated only where their defining ratios have positive denominators. Auxiliary continuations may still reach undefined branches. Fix their fallback rules invariant under the bin, sign and ID renamings used in the abstract records: for example, fix an order on retained host labels, use the first label and repeated copies when a probability law is undefined, and fixed numerical values for undefined scalar formulas. These choices depend on no additional keys or arrays. Thus the symmetry assertions below hold also for the completed experiments.

Conditioning stage 1: the global parent. For each abstract Step 1 or 2 pattern, the conditional failure probability given V_0 is a function of this one draw. Its mean is at most $e^{-\delta L}$. Restrict V_0 so that every such function is at most $e^{-\delta L/2}$; the alarm probability per pattern is at most $e^{-\delta L/2}$.

The pattern unions just given apply globally here. Conditional on V_0 , the bin variables are iid, every formula reads a bounded-radius bin neighborhood, and only boundedly many coarse incidence shapes occur. Likewise translating all signs does not change a raw failure probability: the low keys being integrated have independent laws that do not depend on their signs. No factor 2^m , or factor for all bin names, is needed. The total excluded parent mass is $o(1)$. The retained draw does not change any of the posterior definitions.

Conditioning stage 2: the coarse base. Fix the selected V_0 and start from the independent variables (A_w, W_w) . Exclude Step 1 failures outright. Also exclude the alarms that a Step 2 gated failure, averaged over its future hidden keys, has probability greater than $e^{-\delta L/4}$. Stage 1 bounds their expectations by $e^{-\delta L/2}$, so the Markov cost is at most $e^{-\delta L/4}$ per alarm.

Group the events at each bin. Their scopes are the base variables in the bounded union of the record's C_i -lists. The dependency degree is bounded, and the pattern sums are $o(1)$. The conditional avoidance lemma therefore applies with charges $o(1)$. It gives a base passing Step 1 and the stated conditional Step 2 bounds, separately at each selected V_0 .

Conditioning stage 3: the high hidden keys. Given a successful base, the high U 's are initially independent. The one-target calculations of Step 3 give, for any fixing of the other keys, raw gated exceptions at most $e^{-c_{L0}k'_j}$ or $e^{-c_{H0}s}$, with fixed small positive constants c_{L0}, c_{H0} . These estimates integrate the target and arrays. A high record specifies type/optional-key appearances and computes their subsets from the pools; it does not condition on their extraction in advance.

Restrict the high keys to pass high-only Step 2 tests. For tests involving low keys, require their conditional raw failure probabilities to be at most $e^{-\delta L/8}$. Stage 2 and Markov give alarm probabilities at most $e^{-\delta L/8}$. Also require each high Step 3 failure, still averaged over independent low keys and arrays, to have probability at most $e^{-c_{H0}s/2}$. Its alarm costs at most $e^{-c_{H0}s/2}$.

These events read only high keys in nearby coarse bins, hence have bounded dependency degree after grouping by bin. The high Step 3 abstract count is $\exp(O(T \log T + J \log m))$. Taking K_s large makes its product with $e^{-c_{H0}s/2}$ tend to zero. All grouped probabilities are thus $o(1)$, and another bounded-degree local lemma provides the high-key history. The abstract alarms retain the sign and ID symmetries; no later spatial selection is used in them.

Conditioning stage 4: separate optional pretrims. Given this base and high history, separately restrict each low U_ℓ to satisfy its optional small-prefix tests $m_{S+\ell}/m_S \geq e^{-\delta u_*}$. Only that low variable remains random in such a test, and each is involved in boundedly many coarse patterns. Their probabilities from Stage 3 are $o(1)$, so the low variables remain independent with a density multiplier $1 + o(1)$ per variable.

A regular Step 2 pattern uses $O(j+1)$ low keys, so its probability multiplies by $(1 + o(1))^{O(j+1)}$ and remains at most $e^{-\delta u_j/16}$. A high Step 3 pattern reads only $O(J)$ low keys: the possible low tuple's S -list and the optional keys named in the record. Its bound remains $e^{-c_{H0}s/3}$. For low Step 3 the one-target bound holds at every fixing of the other low keys; only the target's own distortion matters. Its bound remains $e^{-c_{L0}k'_j/2}$.

Conditioning stage 5: the low hidden keys. Start from the product of these trimmed low laws. Exclude all remaining Step 2 failures. For Step 3 impose the conditional array-failure bounds $e^{-c_L k'_j}$ and $e^{-c_H s}$, taking, for example, $c_L = c_{L0}/4$ and $c_H = c_{H0}/6$. Markov applied to the Stage 4 bounds gives alarms of these same exponential orders.

Group by central sign, coarse bin and severity. The current variables in a low group's scope are exactly the low keys in its listed types and optional entries. They have bounded sign and severity distance and lie in nearby bins. High groups involving low variables occur only near the interface and read the optional keys and the possible low tuple's S -list; high requirements with no low variables already hold. Both the dependency degree and the number of groups touching one low variable are polynomial in m , with fixed exponents.

After the pattern unions, each group can have probability at most $m^{-K_{\text{deg}}}$ for any prescribed fixed K_{deg} : choose K_1 large for the regular tests, then K_2 large for the low Step 3 logarithmic count $O(T \log T + (j+1) \log m + k_* 1_{j=J})$, and K_s large for the high count. Polynomially small charges with total $o(1)$ at each variable give the final local-lemma restriction. This completes the construction of the key history and its uniform conditional bounds over independent center arrays.

Order of constants. First fix q_0 , the gaps between the a_i 's, δ , and the path clip parameters other than K_D . Choose K_h for optional block hits and then η so small that $K_h \eta + .005 < .03$. The coefficients in the history pattern counts are fixed before choosing K_1, K_2, K_s . Next take K_1 large for regular history tests, then K_2 large for low Step 3, then K_D large relative to K_0 , and finally K_s large for the high pattern unions. Increasing K_2 or K_s need not decrease the respective Step 3 exponential rates before counting patterns.

The scales give

$$TM_* = o(J), \quad T \log n = o(k_*), \quad T \log n = o(m^{.02}), \quad T + J = o(s).$$

Allow enough slack in the constants that each actual low configuration count is smaller than its required fixed margin times k'_j , and that each high testing count is smaller than its required margin times s . These relations hold for every fixed $\alpha > 0$; choose α sufficiently small after all the likelihood constants, including the following truncation constant.

Prior-heavy labels. For a block law P_K , define its average coordinate marginal by $\bar{P}_K(x) = u^{-1} \sum_{i=1}^u P_K(z_i = x)$. Call a label prior-heavy for K if $N \bar{P}_K(x) > B_K$, where $B_K = A_K^{K_B}$. Fix a small $v_0 > 0$ with $v_0 + \tau_0/\tau_1 < 1$. At fixed successful key history and a fixed tuple reference c , the probability that more than $v_0 k_c$ entries are prior-heavy is at most

$$2^{k_c} A_K^{k_c} (q_0 K'_0 / B_K)^{v_0 k_c / q_0}.$$

Indeed there are at most N/B_K heavy labels. A reference segment hits this set with probability at most $q_0 K'_0 / B_K$, and that many heavy entries require at least $v_0 k_c / q_0$ segments to hit it. The segments of $R[k_c]$ are independent; the block domination contributes $A_K^{k_c}$, and the subset union contributes at most 2^{k_c} . Choose K_B large enough that this is $o(1)$, even after the union over high subsets.

The following are instead deterministic averages over actual cube roles: for sufficiently small α ,

$$\mathbb{E}_{v \in A} B_{K(v)} \exp(O(k_* 1_{j(v) > J})) = O(1), \quad \Pr_{v \in B}(j(v) > J) e^{D_H} = o(1).$$

At low severity $\log B_K = O(j+1)$, except for an extra $O(s)$ at $j = J$; at high severity it is $O(s)$. The rarity bound $\Pr(j \geq h) \leq n^{-.13h}$ outweighs these costs once α is small. The constants

multiplying k_* are already fixed. Also choose α so that the Step 1 low-prior caps have logarithms at most $.02(j+1)\log n$.

5.5 Height choices and the odd kernels

Fix a successful history through the hidden columns. Generate prospective center presence and independent arrays according to their raw laws. At an even site-level, first remove any center in its r -ball whose tuple for that state fails the prior-heavy fraction bound or has too few optional hits. Each candidate fails with probability $o(1)$, and these tests use independent arrays across centers.

For each odd state and each consecutive pair of levels, consider all mappings of its neighbors into their prospective centers using at most T IDs. Test Step 3 for each mapping, including the specified successful-mask event when required at low states. Choose a maximal family of failed ID sets that are pairwise disjoint, by a deterministic local rule. Mark their union forbidden at all adjacent sites on those levels. A site's eligible set is what remains of its prospective centers after these marks and its singleton removals.

With conditional probability $1 - \exp(-\Omega(n))$, every site-level ball has between $\lambda/2$ and 2λ prospective centers, singleton losses are small, and every such maximal family has fewer than n members. The first two claims follow from binomial tails. For the third, fix presence within the count bounds. Disjoint ID sets read disjoint arrays, which are independent given the key history. The history's conditional failure bounds and the actual configuration counts therefore bound the probability of n disjoint failures by $\exp(-\Omega(nk'_j))$ at a low star and $\exp(-\Omega(ns))$ at a high star, including the union over their records. These bounds also sum over all stars and level pairs. Since a site is incident to $O(n)$ stars, the marks remove at most $O(n^2T)$ centers per site-level. The eligible sets thus have size at least $\lambda/3$ everywhere, with the stated exception.

Activations have not been used in defining eligibility. Apply the long height rule and uniform eligible-active tie choices. With probability $1 - o(1)$, neighboring even states around any odd state use at most two consecutive levels. At either level all their choices lie in one $(r+D)$ -ball, so the crowd bound limits their total number of IDs to $2n^b \leq T$. Every actual mapping passes Step 3: otherwise its failed ID set would be disjoint from the marked maximal family, contradicting maximality. The singleton removals ensure its required hits.

Local validity under replacement. Use the same deterministic marking and singleton rules also at hypothetical key and array values. Markings use only the displayed likelihood tests and their records, not the history alarms or the odd selection adjustment below. Long and short height computations use the same pre-activation eligible sets and ties. Either returns no choice if its maximum is H or if its resulting site-level is bad.

For either computation, call an odd state locally valid when all the following hold at its star:

- every adjacent even state has a legitimate choice, and the r -ball centered at each such state has at most 2λ prospective centers at every level $0, \dots, H$;
- the mapping uses at most T IDs, and all required singleton and hit tests pass;
- the true-target gate, support conditions, and Step 3 numerical and path tests for this mapping pass.

This is the same function of the local choices and star data for both truncations. It does not require global path correctness or eligibility sizes in a larger region. Thus equality of the adjacent path maxima, with the common ties, gives equality of choices and validity. The long calculation is

valid everywhere on the global geometry event above. Set odd kernels to zero on local invalidity, with the fixed local fallbacks used only in auxiliary computations.

In particular, hypothetical computations do not retest global Step 1 or 2 success. The optional-prior comparisons in a high $\mathcal{G}_K$ do not read realized low-key values. Eligibility reads optional values only in types or mappings at the consulted sites and incident stars. Marks at a site-level do not test eligibility at other sites. These facts ensure that the two height computations really use the same eligible sets and that unconsulted keys remain unobserved.

Low rows: correcting for the selection event. The following rows are attached to actual odd roles, even when several roles use one state computation. We continue to write b for the role when taking loads, with its state understood. A high role uses the law $\mathcal{R}$ for its actual observation record. A low role requires an adjustment because its observed arrays were selected using the hidden column.

Fix the key history except for its target U_ℓ , and fix prospective center presence. Use the original prior π_ℓ for the target, and simulate the raw center, activation and tie experiment with the short height rule. Let $\mathfrak{d}$ be a canonical observation record, o its observed array data, and $Q_\mathfrak{d}$ the product of its Step 3 block reference laws. When a high pool is present, include its target-independent law in this reference. For target $U_\ell = y$, its gated observation likelihood is $F_y(o) = \Theta(y) \prod_e g_e(y)$.

Let $\mathbb{P}_y$ denote this raw experiment with the target replaced by y . Conditional on the recorded observations, the probability that short selection validly presents exactly this record is a number $a_y(o) \in [0, 1]$. More precisely,

$$\mathbb{P}_y(\text{valid proxy presentation of } \mathfrak{d}, O_\mathfrak{d} \in do) = F_y(o) a_y(o) Q_\mathfrak{d}(do).$$

The presentation event implies the required ratio gates and any successful mask. Given these gates, the recorded low arrays have their product block laws, independently of a recorded high pool. Integrating the remaining selection event defines a_y . In doing so, unrecorded arrays whose laws depend on y are generated at that candidate value; they are not fixed at their realized values. Mappings with the same record are combined into one presentation event, and the events for different records are disjoint. Values on zero-mass conditioning data can be fixed arbitrarily. The resulting table $(a_y)_y$ is a function of the fixed inputs and observed data, not a new observation of the actual target.

Set $\varepsilon = e^{-\delta k'_j}$. At valid data define

$$\bar{a} = \frac{\sum_y \pi_\ell(y) F_y a_y}{\sum_y \pi_\ell(y) F_y}.$$

If $\bar{a} \geq \varepsilon$, use the adjusted posterior proportional to $\pi_\ell F a$; otherwise retain $\mathcal{P}$, proportional to $\pi_\ell F$. For observations presented by the long rule use this same short-rule table. In either case the row is at most $\varepsilon^{-1} \mathcal{P}$, preserving the deletion multiplier $e^{a_4 k_c}$ and cap e^{D_L}/N .

Let p_b^{pr} be the proxy row, extended by zero on invalidity. For each fixed output label x , integrate the original target prior and the raw center experiment. Cancellation of the adjusted normalizer against the actual presentation density gives

$$\mathbb{E}[p_b^{\text{pr}}(x) 1_{\{\text{adjusted}\}}] \leq \pi_\ell(x) \sum_{\mathfrak{d}} \int F_x(o) a_x(o) dQ_\mathfrak{d}(o) \leq \pi_\ell(x).$$

The last step uses the disjoint presentation events. On fallback data, its presentation density is at most $\varepsilon \sum_y \pi_\ell(y) F_y$. Hence

$$\mathbb{E}[p_b^{\text{pr}}(x) 1_{\{\text{fallback}\}}] \leq \varepsilon \pi_\ell(x) \sum_{\mathfrak{d}} \int F_x(o) dQ_\mathfrak{d}(o) \leq \varepsilon (\#\text{records}) \pi_\ell(x).$$

Each gated observation density integrates to at most one; their gates need not be disjoint. The record count on the position gate is smaller than ε^{-1} , so the total mean is at most $2\pi_\ell(x)$.

For a fixed key history H , define the normalized means

$$\bar{p}_b^{\text{pr}}(y; H) = \mathbb{E}_{\text{raw centers}|\mathbf{H}}[Np_b^{\text{pr}}(y)], \quad \bar{p}_b^{\text{long}}(y; H) = \mathbb{E}_{\text{raw centers}|\mathbf{H}}[Np_b^{\text{long}}(y)].$$

Here “centers” includes presence, arrays, activations and ties. Thus presence is held fixed in the lookup calculation above and then averaged in these means. The proxy mean uses low variables only at sign distance $O(\sqrt{m})$ from the target, with the base and high history fixed. Indeed the short rule reads statuses in its short tube, and the incident-star marking rules add only bounded state distance. The one-hot encoding bounds the corresponding sign distance. Although a consulted center may be farther away, its array has one of these local types; all arrays not read by the presentation event integrate out. The validity and lookup calculations have this same locality.

At a successful key history, compare long and short calculations with the same randomness. On correct eligibility sizes, the height lemma bounds mismatch at any one of the $O(n)$ neighboring states by $o(e^{-2m^{1/5}})$. Incorrect sizes have probability $e^{-\Omega(n)}$. Since $D_L = m^{15}$,

$$\bar{p}_b^{\text{long}}(y; H) \leq \bar{p}_b^{\text{pr}}(y; H) + O(e^{D_L})[O(n)o(e^{-2m^{1/5}}) + e^{-\Omega(n)}] = \bar{p}_b^{\text{pr}}(y; H) + o(1).$$

The height estimate bounds an intersection with correct sizes, not a conditional law given size success. With either rule, the unaveraged row and its lookup table use center randomness only within ambient radius $r + o(n)$ once the key history is fixed.

Odd column sums through the three histories. We now apply scattered moments three times, always averaging over actual roles in B and setting weights to zero outside the specified class. This distinction matters because many roles can use the same state.

First consider $Z_b(y) = N\pi_{\ell(b)}(y)$ for low roles, under the coarse-base law given the selected V_0 . The boundary roles and roles with $j \geq 1$ contribute deterministically at most

$$O\left(n^{-.05+.02} + \sum_{h \geq 1} n^{-.13h+.02(h+1)}\right)$$

by their rarity and prior caps. For interior $j = 0$, the cap is $n^{.02}$. Before coarse avoidance, the mean posterior at a bin is the bounded candidate prior, by integration of the raw conditional law. Same-bin pairs have fraction at most $(2n^{-.04})^{d_c}$, so the repeat cost $n(2n^{-.04})^{d_c}n^{.02}$ tends to zero. For distinct bins, drop the coarse constraints touching them at constant cost per bin and integrate independent raw bin variables. The scattered-moment bound, followed by Markov and the union over labels, shows that the average of these weights is bounded at all labels with high probability under the conditioned base law.

Next fix such a base and the high-key history. The weights are now $Z_b(y) = \bar{p}_b^{\text{pr}}(y; H)$, viewed as functions of the low keys. Their cap is e^{D_L} . Their near relation is sign distance $O(\sqrt{m})$, of fraction $2^{-m+o(m)}$; hence the repeat term is $n2^{-m+o(m)}e^{D_L} = o(1)$. For separated rows, remove the final low-key avoidance constraints touching their target variables. The charge sum gives cost $O(1)^n$. Conditional on all other low keys, the targets are independent and no other retained row reads them, by short locality. The one-target proxy bound gives comparison means $O(N\pi_{\ell(b)}(y))$; the optional pretrim adds only $1 + o(1)$ per target. Their averages are bounded by the first calculation. Scattered moments therefore bound all proxy-mean averages with high probability

under the low-key law. The additive comparison above gives the same conclusion for the long means.

Finally fix one of these key histories and use weights $Z_b(y) = Np_b^{\text{long}}(y)$, now random over raw centers. Low rows again have cap e^{D_L} . Rows separated in residual coordinates by more than $2r + o(n)$ have disjoint center scopes and comparison means $\bar{p}_b^{\text{long}}(y; H)$. Residual-ball rarity is exponential in $-n$, so its product with ne^{D_L} tends to zero. High rows contribute deterministically at most $2\Pr_B(j > J)e^{D_H} = o(1)$. Applying the moment and label-union bound at this last history gives bounded normalized average loads, while geometry success supplies valid probability rows. Since

$$\sum_{b \in B} p_b(y) = \frac{|B|}{N} \left(\frac{1}{|B|} \sum_{b \in B} Np_b(y) \right) \quad \text{and} \quad |B|/N = 1/(2C_n) \rightarrow 0,$$

all odd column sums are at most the clock threshold θ_0 , except on an event of probability $o(1)$ intersected with the preceding successes.

5.6 Actual outputs and even placement

Fix a history and geometry with valid odd rows and the column bounds. For each high even role v , let c be its selected tuple reference. Under the product of the odd row laws, the high neighbors' outputs (h_b, y_b) are independent. Their costs

$$\left[\log \frac{p_b(h_b, y_b)}{\tilde{Q}_{c, h_b}(y_b)/s} \right]_+$$

have total mean at most $a_4 k_* n$, by the defining property of $\mathcal{R}$, and each is at most $D_H + O(1)$ by smoothing and the cap. Require their sum to be at most $a_5 k_* n$. Bounded-summand concentration bounds each failure by an exponential with exponent of order

$$\frac{nk_*^2}{(D_H + O(1))^2} = n^{1-.09\alpha+o(1)}.$$

Each predicate reads at most n rows, and each row belongs to at most n predicates. All label atoms are $e^{o(n)}/N$, hence smaller than any fixed inverse power of n . Apply the clock lemma with scope exponent $B = 2$. It yields an injective odd assignment, retaining the high output indices, which satisfies all these budgets and has joint upper bound $(1 + o(1)) \prod p_b$ on at most n^2 queried rows.

Even rows: deletion of primitive block values. Fix an actual even role v , a possible center ID there, and a tuple reference c , including its block index subset at high type. Fix the successful key history and all primitive center data except the block values z belonging to c . Primitive data here are presence, ties, activations and the sampled blocks themselves. In particular, do not fix other extracted tuples that may be functions of these blocks. The raw conditional law P'_0 of z is a product of blocks of total length $k = k_c$, with density at most $(A_K e^{n^\gamma})^k$ relative to product uniform measure.

For neighboring odd outputs $\mathbf{y}$, including their indices at high rows, define the sublikelihood

$$F_z(\mathbf{y}) = 1_E \prod_{b \sim v} p_b(y_b).$$

Every choice, extracted subset and kernel is recomputed as z varies. The local gate E requires:

- presence of the specified center and legitimate long-rule selection of this c at v , with its specified block subset;
- valid neighboring rows with their count and mapping bounds, and eligible sets of size at least $\lambda/3$ throughout the long height consultation domain of v ;
- the prior-heavy fraction bound for c , and the high log budget at v when needed.

These conditions are local to ambient radius $r + o(n)$, also at hypothetical inputs. They hold for the actual selected reference on the global success event. No success elsewhere is part of E .

We construct a product reference on $\mathbf{y}$ independent of z . For each same-mode neighbor b , let $\mathcal{D}_b(c)$ be the finite list of possible observation configurations in its adjacent balls under the fixed presence-count gate. Enumerate ID/type lists without requiring that their selection is feasible for the present array values. At a low neighbor include the possible mask; at a high neighbor record only full pools and the possible low tuple. For each record $\mathfrak{d}$, let $q_{b,\mathfrak{d},-c}$ be the corresponding deleted reference, namely Q_c at low or $\tilde{Q}_{c,h}/s$ at high. Use its locally defined fallback if its formula is undefined. Set

$$Q_b = \frac{1}{|\mathcal{D}_b(c)|} \sum_{\mathfrak{d} \in \mathcal{D}_b(c)} q_{b,\mathfrak{d},-c}, \quad Q = \prod_{b \sim v} Q_b,$$

using uniform Q_b at opposite-mode neighbors. If the count gate fails, set E empty and take any fixed local reference instead.

Each deleted component uses only retained primitive data. At a low neighbor, deletion of the full low tuple leaves any mask indicator on a different, fixed high pool. At a high neighbor, every likelihood factor involving a block in the fixed subset c is removed from the one full-pool observation. The remaining target gate has no successful-mask indicator. A pool used for several optional tuples is still observed only once; those extracted tuples are recomputed with z , rather than held fixed. True hidden-column prefixes, sampled before all arrays, do remain fixed. These facts prove that Q is independent of z , and that its mixture contains the actual deletion component whenever E holds.

The likelihood costs now separate into three cases. At a low same-mode neighbor, the adjusted row is at most $e^{a_4 k}$ times its actual deleted component. The low record count, including a mask if needed, fits the reserved margin because $k'_j \leq k$. At a high same-mode neighbor use the imposed sum of positive-log costs, at most $a_5 kn$ over these neighbors. Each full-observation-list mixture costs only $O(T \log n) = o(k_*)$ in the logarithm: the subset c is fixed, and other optional subset decisions are not recorded in the reference. Finally there are at most $O(mn^3) = o(n^4)$ opposite-mode neighbors, each costing $O(D_L + D_H)$ against uniform measure, including the high index when present. Their total is $o(kn)$. The reserved gaps therefore give

$$F_z(\mathbf{y}) \leq e^{a_6 kn} Q(\mathbf{y}).$$

Let $m_c(\mathbf{y}) = \int F_z(\mathbf{y}) dP'_0(z)$. For each fixed (v, c) , the gated posterior lemma bounds the raw exception to

$$m_c(\mathbf{y}) > 0, \quad m_c(\mathbf{y}) \geq e^{-\delta kn} Q(\mathbf{y})$$

by $e^{-\delta kn}$. To apply it to the actual assignment, first retain its successful-prehistory indicator and compare the required odd outputs to product rows using the clock bound. Only after this comparison discard nonlocal center-success restrictions, retaining E , and integrate the raw centers at the successful key history. The bound is thus $(1 + o(1))e^{-\delta kn}$ per (v, c) on the entering success event. The union over roles, IDs and high subset indices costs $\exp(O(n) + O(k_*))$ and is $o(1)$, since $k \rightarrow \infty$.

On the complementary event use the posterior $F_z dP'_0/m_c$. Its density relative to product uniform is at most $e^{\tau_0 kn}$: the prior contributes $k(\log A_K + n^\gamma) = o(kn)$, and the other exponents fit the gap below τ_0 . Let $\bar{\mu}$ be its average coordinate marginal. The support of each neighboring kernel implies that $\bar{\mu}$ is supported on common neighbors of the star.

Set $H_1 = \{x : N\bar{\mu}(x) > e^{\tau_1 n}\}$, whose uniform mass is at most $e^{-\tau_1 n}$. Choose $v_1 > \tau_0/\tau_1$ with $v_1 + v_0 < 1$. The joint density bound and a subset union give

$$\Pr_{\text{post}}(\#\{i : z_i \in H_1\} > v_1 k) \leq 2^k e^{\tau_0 kn - v_1 k \tau_1 n} = e^{-\Omega(kn)}.$$

It follows that

$$\bar{\mu}(H_1) = \mathbb{E}_{\text{post}} \frac{1}{k} \sum_i 1_{\{z_i \in H_1\}} \leq v_1 + e^{-\Omega(kn)}.$$

The gate separately bounds the marginal mass on prior-heavy labels by v_0 . Removing both sets leaves a mass bounded away from zero. Normalize the restriction to obtain the even row, with normalized cap $O(e^{\tau_1 n})$. Use only the actually selected reference, and set the row to zero if its actual gate or its m_c test fails.

Even comparison means and the Hall assignment. We first estimate the row under product odd sampling and raw centers at a successful key history. For a fixed potential ID and reference c , cancellation of the posterior denominator against the gated data subdensity bounds its contribution at a prior-light label x by a constant times

$$\mathbb{E}_{\text{centers}} \left[\left(k^{-1} \sum_i 1_{\{z_i = x\}} \right) \sum_{\mathbf{y}} F_z(\mathbf{y}) \right].$$

The constant accounts for the truncation normalization. The sum over $\mathbf{y}$ is at most the indicator that this center is present and legitimately selected with the size gate. We now bound that selection probability before averaging the tuple marginal.

Force this center present and fix its tuple values, but continue to average all other prospective presence and activation variables. At level zero, conditional on presence, the eligible-active tie bound is $3/\lambda$. At positive levels the forced-center height estimate gives e^{-n^c} for some fixed $c > 0$. At each presence configuration this estimate first takes the supremum over fixed pre-activation eligible sets of the required sizes, then averages over presence. Fixing the tuple may change these sets, but does not change the position/activation product law, since arrays were generated independently. Thus this is neither a bound at every fixed presence configuration nor a posterior conditioned on selection.

The sum of presence probabilities over possible locations is λ at each level. There are polynomially many levels, so summing the selection bounds costs $O(1)$ per fixed subset. A tuple reference consists of whole iid blocks, and its unselected average marginal is $\bar{P}_K$, at most B_K/N on a prior-light label. The number of possible high subsets is $\exp(O(k_*))$. Consequently the normalized comparison mean for the even row at v is at most

$$d_v = O(B_{K(v)} \exp(O(k_* 1_{j(v) > J}))),$$

whose deterministic average over actual even roles is bounded by the cube-rarity calculation above.

For the final scattered-moment calculation set $Z_v(x) = Np_v(x)$ for the even rows and retain the actual success indicator. The cap is $O(e^{\tau_1 n})$. A row's near set consists of residual-coordinate distance at most $2r + o(n)$. With $H_{\text{bin}}(t) = -t \log t - (1-t) \log(1-t)$, its repeat cost is at most

$$n \exp\{-n(\log 2 - H_{\text{bin}}(2\rho)) + o(n)\} O(e^{\tau_1 n}) = o(1).$$

For up to n separated rows, at most n^2 odd outputs are needed. At a successful prehistory and fixed center data, apply the clock comparison to these outputs. Then remove nonlocal success restrictions but retain each local gate, and integrate the raw centers. The separated rows have disjoint long-rule scopes, so their integrals factor and give

$$\mathbb{E} \left[1_{\text{success}} \prod_{v \in U} Z_v(x) \mid \text{successful key history} \right] \leq (1 + o(1)) \prod_{v \in U} d_v$$

for such a separated set U , with harmless fixed factors included in d_v . Scattered moments and the label union give bounded normalized average even loads. Since $|A|/N \rightarrow 0$, every even column sum is at most one, except on an event of probability $o(1)$ intersected with the preceding successes.

The key-history, geometry, odd-load and predictive estimates together have success probability tending to one under their successive stage laws. Thus with positive probability the odd assignment is injective and every even row is a probability law on its common neighborhood, with all even column sums at most one. Hall's theorem gives distinct even representatives. The two assignments form the required G -cube. $\square$

6 A broad side with a small polynomial density cap

Lemma 6.1 (A broad side with a small polynomial density cap). *Work in the counterexample sequence fixed above. There is a universal $D_* > 0$ with the following property. Let $0 < \gamma < 1$, $p_0 > 0$ be fixed. Suppose a finite random tag of law Λ , with pairs (μ_i, ν_i) on X, Y , satisfies*

$$N\mathbb{E}_\Lambda \mu_i, N\mathbb{E}_\Lambda \nu_i \leq K, \quad \text{width}(\mu_i) \leq n^\gamma, \quad N \max \nu_i \leq n^{D_*},$$

where K is a constant and both average inequalities hold pointwise. In one color G assume on every support $y \in \text{supp } \nu_i$, $d_G(\mu_i, y) \geq 1 - \epsilon$, $\epsilon = n^{-p_0}$. Then one of the colors contains the cube for large n .

Proof. The preliminary reduction either produces a cube in the opposite color or supplies pairs of parent labels that both see a common first-side law. In the latter case we construct first-side tuples hitting those parents, then use their observations to assign the odd roles. The posterior of a whole tuple after its odd neighbors are assigned will supply the even row for Hall's theorem. This follows the parent and height construction of Lemma 5.1, with two changes: each tuple carries a tag that must be deleted along with its entries, and a high odd row redraws an actual parent rather than a hidden vector. The constant D_* is universal; only subsequent auxiliary choices and the sufficiently large dimension threshold may depend on the fixed input parameters.

6.1 Local base histories and hidden tags

The second-side mixture and its tag posterior are

$$\Pi(y) = \sum_i \Lambda(i) \nu_i(y), \quad \eta_y(i) = \frac{\Lambda(i) \nu_i(y)}{\Pi(y)}, \quad B_y = \sum_i \eta_y(i) \mu_i$$

when $\Pi(y) > 0$. Thus B_y is the first-side law obtained by observing y in the second-side mixture. Restrict Π and normalize to obtain Π' on $\{y : N\Pi(y) \geq n^{-D_*}\}$. The discarded mass is at most n^{-D_*} , and $\Pi' \leq O(K)/N$.

Suppose first that two independent Π' -draws satisfy $d_G(B_y, y') < c_0 = .01$ with probability at least .1. A set of first draws of positive Π' -mass then has a positive Π' -mass of such partners. Each partner set contains $\Omega(N)$ labels because $\Pi' \leq O(K)/N$. The laws B_y have width at most n^γ , and their average over these first draws is at most $O(1)/N$: indeed $\mathbb{E}_{y \sim \Pi} \eta_y = \Lambda$. Their degrees to the partners in the other color exceed .99. Lemma 5.1 therefore gives a cube in that color.

In the remaining case the directed bad-pair probability is less than .1. A union bound for the two directions shows that the relation

$$y \succ y' \iff d_G(B_y, y'), d_G(B_{y'}, y) \geq c_0$$

has $(\Pi' \otimes \Pi')$ -mass greater than .8. In particular a positive Π' -mass of labels has partner mass at least .1. Draw V_0 from Π' restricted to those labels. Given V_0 , draw the candidates A_w independently from $\Pi'|_{\{y: y \succ V_0\}}$, with bins w defined below. Both the initial prior and every conditional candidate prior are bounded by constant multiples of Π .

Use the coarse and fine chunks of Lemma 5.1: there are $d_c = 300$ coarse chunks of length $\lfloor n^{1/5} \rfloor$, and $m = \lceil n^\alpha \rceil$ fine chunks of odd length asymptotic to n^3 . Their majority signs, exactly flippable coordinates, and number of counts within $R_f = 5.5$ of mid-weight are denoted by t, F, j . Here use $J = \lfloor m^{.04} \rfloor$, and call $j \leq J$ low. The signs remain uniform on either cube parity, while $\Pr(j \geq q) \leq n^{-.13q}$ and the coarse boundary fraction is at most $n^{-.05}$.

The coarse key of a site x is $h_x = (w, \text{int}/\text{bdy})$. Define its named primary variable by

$$P_{(w, \text{int})} = A_w, \quad P_{(w, \text{bdy})} = V_0,$$

and let P_h^* be the other member of $\{A_w, V_0\}$. Two keys have matching primaries when these are the same random variable, not merely when their realized labels agree. Give the coarse keys a symmetric relation connecting each key to itself, the two keys at a bin, and boundary keys at adjacent bins; let $C(h)$ include all these neighbors and h itself. A coarse flip stays in this relation: if it changes the bin, both endpoints are boundary keys. Within $C(h)$, only the opposite key at the same bin can have a nonmatching primary.

Given the parents, draw independent base tags I_h from η_{P_h} restricted to $d_G(\mu_i, P_h^*) \geq c_1 = c_0/2$. The mean of this degree under η_{P_h} is at least c_0 , so the restriction has mass bounded below by a positive constant. On the parent support,

$$\eta_y(i) = \Lambda(i) \frac{\nu_i(y)}{\Pi(y)} \leq n^{2D^*} \Lambda(i).$$

Thus the conditional base-tag law is at most $O(n^{2D^*})\Lambda$.

Next, given the base (V_0, A, I) , draw independent hidden scalars Z_ℓ , indexed by $\ell = (h, t)$. Their laws $\pi_\ell = \pi_h$ are parent posteriors in this base model:

- at $h = (w, \text{int})$, redraw A_w given V_0 and the tags I_s , $s \in C(h)$;
- at $h = (w, \text{bdy})$, redraw V_0 given the candidates A_u at bins of $C(h)$, and the tags I_s , $s \in C(h)$.

Only low odd positions will use a hidden scalar Z_ℓ as their target label. A high position will instead use its actual named primary. Thus there are no high hidden vectors or stream pools in this construction.

An even position x of coarse key $h = h_x$ uses a type β carrying h , low/high and the following observation list S (also retain j for low types):

- low: all (s, t) , $s \in C(h)$, and (h, t^a) , $a \in F$ with t^a the one-sign flip;

- high: just (h, t) when $j = J + 1$, and an empty list otherwise.

Write $u_\beta = j + 1$ for low, 1 for high. All adjacent low odd keys are covered (the high-to-low transition can only be at the paired fringe with sign unchanged).

For every potential center ID we will generate one tuple per type β , independent across ID/type pairs given history through Z , consisting of one new tag i and $k = \lceil \kappa J \log n \rceil$ labels on X . The tag has a restricted posterior law T_β defined below. Given this tag, draw iid from μ_i conditioned individually to hit:

- P_h and all Z_S for low;
- P_h, P_h^* and all Z_S for high.

These requirements include the actual primaries needed by adjacent high odd roles. At a low-to-high transition the coarse key is unchanged. Step 2 will prove that the common-hit mass is at least $c_1/2$, so the sampler is defined on its success gates.

The raw generation order is

$$V_0 \longrightarrow (A_w, I_{(w, \text{int})}, I_{(w, \text{bdy})})_w \longrightarrow (Z_\ell)_\ell \longrightarrow \text{center tuples.}$$

Given V_0 , the bin variables in the second stage are independent; given the base, so are the hidden scalars; given both, so are the tuples at distinct ID/type pairs. Generate tuples also for IDs whose centers are not present, independently of center positions, activations and ties. On undefined branches use fixed local fallbacks respecting the bin, sign and ID symmetries. Whenever a defining sampling formula is valid, use it without checking numerical tests elsewhere. Later avoidance changes the stage law, but does not redefine any raw posterior formula.

A tuple observation is the entire primitive pair consisting of the new center tag and its k label entries. Deleting a tuple will remove that whole observation. When the pair is varied in the even reconstruction, all choices and observations derived from it are recomputed. The old base tag I_h remains in the history: the fresh tag sampled from T_β does not replace it. Likewise equal realized values of two primary or hidden variables do not identify their roles in a conditioning formula.

Parameters can be chosen as follows. Take the small fixed $\kappa > 0$ so small that $\kappa(1 + \log(4/c_1)) < .001$. Use very small successive constants $d_2, \delta_2, d_1, \delta_1, d_0, D_* > 0$, each sufficiently small in terms of previous ones and the absolute coarse dimension (in particular $d_2 \ll \kappa$). Choose $2D_* < d_0$, so the fixed multiplier in the base-tag bound is absorbed and that bound is $n^{d_0}\Lambda$ for large n . If C is an absolute constant with $|S| \leq Cu_\beta$, require $Cd_1 + \delta_2 + d_0 < d_2$, with slack; also take $d_0, \delta_1 \ll d_1$. Finally $\alpha > 0$ is sufficiently small in terms of all these constants and p_0 . The target fan is $T = \lceil m^{.001} \rceil$. Constants hidden in pattern counts below are absolute for the fixed coarse dimension; in particular choosing D_* does not require K, γ, p_0 . We detail the needed predictive tests.

Step 1: deleting one base tag from a hidden-key posterior. For $\ell = (h, t)$, let $\pi_{\ell, -s}$ be the defining parent posterior for π_ℓ with the observation I_s omitted and all its other listed observations retained. We require

$$N \max \pi_\ell \leq n^{d_1}, \quad \pi_\ell \leq n^{d_1} \pi_{\ell, -s} \quad (s \in C(h)).$$

To estimate a deletion failure, fix the retained observations of this parent posterior. Given any supported candidate parent value, the incoming tag has likelihood at most $n^{d_0}\Lambda(i)$, by conditional independence of the base tags and their uniform domination. Its predictive density relative to Λ is

smaller than $n^{-\delta_1}$ with probability at most $n^{-\delta_1}$. Outside that event Bayes' formula bounds the posterior multiplier by $n^{d_0+\delta_1} \leq n^{d_1}$, simultaneously at every parent label.

For the absolute cap, use all observations in the parent posterior. In the boundary case reference the bounded list of candidate labels to uniform laws; reference all incoming tags to Λ . The parent and conditional candidate priors have normalized caps $O(1)$, and only boundedly many tag factors occur. The same predictive-denominator calculation, with $d_0, \delta_1 \ll d_1$, proves the absolute test. Each Step 1 failure has probability $O(n^{-\delta_1})$ in its raw parent experiment.

Step 2: reconstructing the tag of a center tuple. For a type β at h , fix the base except I_h . Let T_0 be the original conditional law used to generate that base tag from its two parents. We first replace I_h by a hypothetical value i to define a posterior law on tags. Given the resulting Z_S , a center later samples a fresh tag from this law. The posterior formula itself does not consult the actual value of I_h again. Restrict alternative tags i to G_β requiring the step-1 pointwise ratio comparisons for deletion of I_h at the keys in S , computed with its value i (laws compared on all labels, so G_β does not read realized Z 's). Write

$$m_S = \int 1_{G_\beta}(i) \prod_{\ell \in S} \frac{\pi_\ell(Z_\ell; i)}{\pi_{\ell, -h}(Z_\ell)} dT_0(i).$$

Notation $m_{S-\ell}$ drops one factor with the same gate. Use their normalized integrands (against T_0) as $T_\beta, T_{\beta, -\ell}$. Enforce positivity where needed and $m_S, m_S/m_{S-\ell} \geq n^{-\delta_2 u_\beta}$. Conditional on the base with I_h omitted, the reference law of Z_S is $\prod_{\ell \in S} \pi_{\ell, -h}$. The true-event data subdensity relative to this reference is m_S . Both m_S and $m_{S-\ell}$ integrate to at most one, so for $a = n^{-\delta_2 u_\beta}$,

$$\int_{m_S < a} m_S \leq a, \quad \int_{m_S < a m_{S-\ell}} m_S \leq a \int m_{S-\ell} \leq a.$$

These are bounds on the joint event with true $I_h \in G_\beta$. Reference atoms of mass zero may be omitted on that gate. Division of the normalized integrands now gives

$$T_\beta \leq n^{d_2 u_\beta} \Lambda, \quad T_\beta \leq n^{d_2 u_\beta} T_{\beta, -\ell}.$$

Indeed the absolute ratio to Λ is at most $n^{d_0+d_1|S|+\delta_2 u_\beta}$, and the ratio to the deleted posterior is at most $n^{d_1+\delta_2 u_\beta}$. The stated order of constants places both below $n^{d_2 u_\beta}$. The same bounds hold at alternative observations satisfying the gates. Also every tag in T_β 's support sees P_h with degree $\geq 1 - \epsilon$, P_h^* with degree $\geq c_1$, and each Z_S with degree $\geq 1 - \epsilon$ if its primary matches, or $\geq c_1$ if not (degree here into μ_i). Indeed positive likelihood for $Z_{(h', t')}$ entails a positive case in the base model of $P_{h'}$ having that value with $I_h = i$, since $h \in C(h')$. In that case the base tag at h was drawn to see that value with degree $\geq 1 - \epsilon$ if its primary is the same named variable $P_h = P_{h'}$, or degree at least c_1 in the crossing case $P_{h'} = P_h^*$. For a low tuple, the only required column that can have degree merely c_1 is the single Z -key with nonmatching primary. For a high tuple it is P_h^* ; its optional Z -key has matching primary. Every other required column has defect at most ϵ . Consequently their common neighborhood has μ_i -mass at least $c_1 - O(m\epsilon) \geq c_1/2$. This proves that the tuple sampler is defined on all the stated gates.

6.2 Odd observation configurations and deletions

A state retains the residual bits and the fine count fields and total j of the preceding lemma, with counts at distances $R_f, R_f + 1$ merged on each side of mid-weight. On the coarse chunks

it retains only the bin vector, encoded by one-hot fields, and the interior/boundary flag. The ambient encoded cube still has dimension $(1 + o(1))n$. Unlike the earlier state, this one need not determine parity, because it omits exact coarse counts.

Let q_{st} be this quotient map, and set $\mathcal{S}_A = q_{\text{st}}(A)$, $\mathcal{S}_B = q_{\text{st}}(B)$. Use $\mathcal{S}_A$ as the height sites. For an odd state $b \in \mathcal{S}_B$, define its even neighborhood by actual-edge existence:

$$N_q(b) = \{a \in \mathcal{S}_A : \text{there are } u \in B, v \in A \text{ with} \\ q_{\text{st}}(u) = b, q_{\text{st}}(v) = a, u \sim v\}.$$

This definition remains valid when $\mathcal{S}_A \cap \mathcal{S}_B \neq \emptyset$. The random choice at each even image is shared by all its actual even representatives, and every actual odd role reads the entire neighborhood of its odd image. A coarse flip changes at most one bin field and the flag; a fine flip changes one count field and possibly the field for j ; a residual flip changes one bit. Thus the encoded distance of each edge is bounded, and two even neighbors of one odd image are within $D_0 = 10$. There are constantly many coarse possibilities and boundedly many possibilities per fine chunk, giving degree $O(n)$. The sets $\mathcal{S}_A, \mathcal{S}_B$ retain their even and odd roles even when they contain the same encoded point. All neighbors across a low/high transition form one state at the paired fine-count fringe. All nonmatching-primary neighbors also form one state, differing only in the coarse flag. These two facts limit the exceptional tuples below.

An observation descriptor records the distinct ID/type pairs used by the neighboring even states, together with the coarse and fine keys defining their laws. Repeated use of one ID/type pair observes the same tuple once; it does not create another tuple or another hidden scalar. The number of distinct observations is bounded as follows.

- At low center severity j , adjacent types observed with at most T distinct IDs altogether give $O(T + j + 1)$ tuples; as before generic variants are constantly many (same t, F , coarse choices and severity variants), while exceptional adjacent states changing F or t number $O(j + 1)$.
- At high, high observations have constant generic variants; exceptional sign-changing optional variants only matter at $j \leq J + 2$, with $O(J)$ exceptional states. There is at most one adjacent low tuple and at most one nonmatching-primary tuple.

Let $\mathcal{D}_b$ be the set of descriptors at a fixed odd state when the allowed IDs form a polynomial-size set. To specify one, list its at most T IDs, choose a subset of that list for each of the constantly many generic types, and give one ID for each exceptional neighboring state. Therefore

$$\log |\mathcal{D}_b| = O(T \log n + (J + 1) \log T).$$

For unions over histories, use abstract descriptors obtained by renaming these IDs in $[T]$ and translating the central sign. Their remaining fine data specify the $O(j + 1)$ coordinates at low states where a flip can change F , the sign, or the near-fringe status, with constantly many possible statuses per coordinate. At high states the same description uses $O(J)$ coordinates near the interface; away from it the types have no fine-key data. Consequently their log count per coarse bin and central sign is

$$O(T \log T + (J + 1) \log m).$$

Multiplicities of generic neighboring states do not change the observed tuple list or its law. The Step 2 type patterns alone have count $\exp(O(u_\beta \log m))$ per low severity and constant count at high.

Step 3: deleting a complete tagged tuple. For an odd state b fix such an observation configuration for its neighboring even states, and observe the full tuples of the indicated (ID,type) pairs. We form a posterior L_b of a target single label. We require for good configurations

$$L_b \leq e^{.16k} Q_{-c}$$

for each tuple c here in the same low/high mode and matching primary, where Q_{-c} is a reference posterior not using any entries or tag of that tuple. Absolute caps will be $N \max L_b \leq e^{m^{.15}/2}$ at low, and $N \max L_b \leq n^{.05J}$ at high. Here and below n is sufficiently large.

Low targets: a hidden scalar. Fix the base and all hidden scalars other than $\xi = Z_\ell$, where $\ell = (h_b, t)$. Its prior is π_ℓ . The observed data are the complete tagged tuples in the descriptor. For a candidate ξ , let $\Theta_b(\xi)$ require the positive support cases and the following Step 2 tests for each observed type, evaluated with this candidate: every defining denominator is positive, $m_S \geq n^{-\delta_2 u_\beta}$, and $m_S/m_{S-\ell'} \geq n^{-\delta_2 u_\beta}$ for every $\ell' \in S$. Include the Step 1 cap on the target prior. These conditions use the preceding base and hidden data and the type list, not any tuple tag or entry. The support and likelihood bounds proved in Steps 1–2 then hold on this gate over the whole support of each tuple law.

For a tuple of type β , use as reference its tag law $T_{\beta,-\ell}$, followed by its label sampler with the required hit to the named variable Z_ℓ omitted. The omitted normalizer and integrand do not read Z_ℓ , so this reference is independent of the candidate. Other required hits remain, even if their labels happen to equal the realized value of Z_ℓ . Undefined reference cases use fixed candidate-independent fallbacks. The tuple likelihood ratio is bounded by

$$n^{d_2 u_\beta} \begin{cases} (1 + 4\epsilon/c_1)^k, & \text{matching primary,} \\ (2/c_1)^k, & \text{nonmatching primary.} \end{cases}$$

The first factor is the tag comparison from Step 2. In the matching case, adding the omitted hit removes at most ϵ mass from a set of mass at least $c_1/2$; this gives the first label factor. The second case uses only the common-hit mass $c_1/2$.

Write $g_e(o_e | \xi)$ for these tagged-tuple likelihood ratios and Q^{data} for the product of their reference laws. Define

$$M(o) = \int \Theta_b(\xi) \prod_e g_e(o_e | \xi) d\pi_\ell(\xi), \quad M_{-c}(o) = \int \Theta_b(\xi) \prod_{e \neq c} g_e(o_e | \xi) d\pi_\ell(\xi).$$

Normalize the respective integrands to obtain L_b and Q_{-c} . The latter omits the complete tagged tuple c . Impose the data tests

$$M > 0, \quad M \geq e^{-.02k}, \quad M/M_{-c} \geq e^{-.02k}$$

for the needed deletions. These tests are not added to the candidate gate of either posterior. Because M is the true-gated subdensity and $\int M_{-c} dQ^{\text{data}} \leq 1$, each exception has joint mass at most $e^{-.02k}$, integrating the target and tuples in the raw model.

For a matching tuple in the same mode, normalization gives

$$\log \frac{L_b}{Q_{-c}} \leq d_2 u_\beta \log n + k \log(1 + 4\epsilon/c_1) + .02k < .16k.$$

Here $u_\beta \leq J + 1$ and $d_2 \ll \kappa$. For the absolute cap, there are $O(T + J)$ observed tuples, of which at most one has nonmatching primary. Adding all tag costs, that possible label cost, the prior cap and the denominator cost gives $O(J^2 \log n) = o(m^{.15})$. Thus $N \max L_b \leq e^{m^{.15}/2}$. Every supported candidate hits all observed tuple entries.

High targets: the actual primary in a local joint model. Now the target is the named primary $\xi = P_{h_b}$, so replacing it also changes some preceding base and hidden laws. Its prior π^0 is the original conditional candidate law given V_0 if it is A_w , and the original initial law if it is V_0 . Only in the first case is V_0 fixed from the outset.

Define the preceding observation list H_{loc} from the fixed tuple descriptor as follows. Include every hidden scalar in the union of the observed types' S -lists. Include the parent pair for each observed type, omitting the target itself. For each included $Z_{(h', t')}$, include the base tags indexed by $C(h')$ and all parents needed to generate those tags and to define its posterior, again omitting the target variable itself. Take the union, counting each primitive variable once. A deterministic bounded grid neighborhood may pad this list. There are $O(1)$ coarse bins and $O(J)$ hidden scalars: all high types have at most one hidden key, and only one neighboring low tuple can contribute its larger list.

Given ξ , generate this list by the original local base and hidden kernels. Unused exterior variables are marginalized. Let $g_*(H_{\text{loc}} | \xi)$ be its density relative to uniform laws for observed candidate labels and hidden scalars, and Λ for base tags. Let $\Theta(\xi, H_{\text{loc}})$ require positive support, all Step 1 cap and deletion tests for the included hidden keys, and the full Step 2 conjunction just specified for each observed type. This is a finite conjunction determined before tuple outcomes are observed. Replacing ξ recomputes every affected base-tag law, hidden prior, type law and gate from these inputs. The hidden data are fixed when the posterior is evaluated, but are integrated when its raw exception probability is estimated.

For a tuple use reference tag law Λ and the label sampler omitting the hit to the named variable ξ , with other required variables held at their observed values. Every tuple imposes that hit; undefined omission samplers use candidate-independent fallbacks. Hits to different variable names remain even if their realized values coincide with ξ . Write $g_e(o_e | \xi, H_{\text{loc}})$ for the conditional tagged-tuple likelihood relative to this reference. The absolute tag bound in Step 2 gives the same tuple comparisons as in the low calculation. In particular the possible low tuple has a matching primary for this hit.

The unnormalized target measure is

$$\Theta(\xi, H_{\text{loc}}) g_*(H_{\text{loc}} | \xi) \prod_e g_e(o_e | \xi, H_{\text{loc}}) d\pi^0(\xi).$$

Let M be its mass and L_b its normalization. Omitting the factor $e = c$ gives M_{-c} and Q_{-c} . The hidden likelihood and gate read no part of tuple c , so this deletion removes its tag and all its entries. After integrating the kept tuple factors against their conditional references, the deleted density has mass at most one: the remaining integral is a gated sublaw of H_{loc} . Hence the tests $M > 0$, $M \geq e^{-.02k}$, and $M/M_{-c} \geq e^{-.02k}$ have raw true-gated exception mass at most $e^{-.02k}$ each. This probability integrates H_{loc} as well as the target and tuples. As at low targets, the denominator tests are data tests, not additional candidate gates.

For a matching high tuple $u_\beta = 1$, so the previous deletion budget gives $L_b \leq e^{.16k} Q_{-c}$. The absolute-width budget has four contributions:

$$\begin{array}{ll} O(1 + d_0 \log n) & \text{from coarse observations,} \\ O(Jd_1 \log n) & \text{from hidden scalars,} \\ O((T + J)d_2 \log n) & \text{from tuple tags,} \\ o(k) + k \log(2/c_1) + .02k & \text{from labels and the denominator.} \end{array}$$

There is only one possible low tuple, so its $u_\beta = O(J)$ cost is included in the third line. Matching label costs total $O((T + J)k\epsilon/c_1) = o(k)$, and at most one tuple needs the weaker label factor.

The coefficient contributed by the last line to $J \log n$ is at most $\kappa(\log(2/c_1) + .02) + o(1) < .002$. Choosing d_2 and the smaller constants with the stated slack makes the total less than $.05J \log n$, proving $N \max L_b \leq n^{.05J}$. Its support again gives common adjacency.

6.3 Conditioning, center choices, and load estimates

We now sample histories on which the preceding local tests hold simultaneously. At each stage the new avoidance law is normalized at the fixed entering history; its success probability does not reweight any earlier draw. All posterior formulas remain those of the raw experiments. For a raw failure event E and an exposed history H , the alarm variable is $a_E(H) = \Pr_{\text{raw}}(E \mid H)$. Its expectation is the raw failure probability. Thus Markov's inequality changes a bound p into $\Pr(a_E > \sqrt{p}) \leq \sqrt{p}$. The events E below include their stated local true gates. After the union over deletions, each Step 3 configuration fails with raw joint mass at most $e^{-.018k}$ for large n . After its local factor union, Step 2 failure costs at most $n^{-\delta_2 u_\beta/2}$. We seek all Step 1 and Step 2 tests, and, at the final hidden history, a Step 3 failure probability over independent hypothetical center tuples at most $e^{-c_2 k}$ for each abstract descriptor. The deterministic posterior inequalities then follow from their gates.

Global parent. Restrict V_0 to values for which every conditional raw failure probability loses at most half its exponent. Conditional on V_0 , the bin variables are iid and the hidden laws are invariant under central-sign translation. Only finitely many coarse shapes and the abstract descriptors above need be tested. Markov's inequality and the unions over severities give costs bounded by

$$O(1)n^{-\delta_2/4} + \sum_{j \leq J} m^{O(j+1)} n^{-\delta_2(j+1)/4},$$

$$\exp(O(T \log T + (J+1) \log m) - .009k)$$

for Step 2 and Step 3 respectively. Both tend to zero for sufficiently small α ; Step 1 has only constantly many shapes. The tests that hypothetically redraw V_0 still use its original prior inside their formulas.

Coarse base. At a retained V_0 , start with the product of candidate-and-two-tag variables over bins. At each bin forbid failure of Step 1, or an alarm that the future Step 2 or Step 3 probability exceeds its next threshold. Conditional future probabilities have the same sign symmetry and use only a bounded bin neighborhood. The grouped bad-event probabilities are $o(1)$ and their dependency degree is bounded, so Lemma 3.5 supplies the desired law. The future bounds after this stage are $n^{-\delta_2 u_\beta/8}$ and $e^{-.0045k}$.

Hidden scalars. At a retained base, start with the product law of the Z 's. Forbid Step 2 failure and the alarm that a future Step 3 failure exceeds the next threshold, halving exponents by Markov once more. Group by bin and central sign, including all allowed severities in one group. Every group uses signs within bounded distance and nearby bins, so the dependency degree is $m^{O(1)}$. Even after the pattern unions these probabilities are at most n^{-c_3} for some $c_3 > 0$ fixed before α . High tests away from the interface have no hidden-key input and already satisfy the coarse-stage bound. We may take $c_2 = .001$, independently of the later choice of α .

For clarity, the hidden-stage pattern union has probability at most n^{-c_3} , and each group meets at most $m^C = n^{C\alpha+o(1)}$ other groups, for an absolute C . Choose α so that $C\alpha < c_3/4$. Charges $x = n^{-c_3/2}$ then satisfy

$$n^{-c_3} \leq x(1-x)^{m^C}, \quad m^C x = o(1).$$

The coarse stage has bounded degree and the same inequality with charges tending to zero. These checks give both avoidance and the later cost of deleting constraints touching a fixed local set.

On the admitted entering histories, the gated exceptions are exactly the failures that need to be excluded. In particular the high-target formula may redraw V_0 under its original prior, while its alarm after V_0 has been sampled is simply a function of the true local data.

Local height selection. The tuple sampler already imposes every required hit on valid inputs. There is no optional pool search in this construction, and the even mean will be controlled by the tag-mixture estimate below. Thus no singleton removals from Lemma 5.1 are needed. Use its center and height mechanism with linear radius $r = \lfloor \rho n \rfloor$, $\lambda = n^{10}$, and small fixed $\rho > 0$. Choose the other height parameters to give crowd at most $T/2$ and short-truncation error $o(e^{-2m^{1/5}})$. Given admitted hidden history, generate the independent tuple arrays. At each odd state and each pair of consecutive levels, take a maximal family of disjoint ID sets whose Step 3 descriptors fail. Mark their union forbidden at the adjacent even sites.

On the polynomial ball-count bounds let D_n bound the number of descriptors at that state and level pair. The preceding count gives $\log D_n = O(T \log n + (J+1) \log T) < c_2 k/2$, after taking α small. Disjoint ID sets use independent arrays, and hence

$$\Pr(\text{at least } n \text{ disjoint failed sets}) \leq D_n^n e^{-c_2 kn} \leq e^{-c_2 kn/2}.$$

This beats the $\exp(O(n))$ union over states and levels. After forcing one center present with a fixed tuple, at most one member of a disjoint family contains it; the other $n-1$ failures give the same negligible bound. Ball counts change by at most one.

There are therefore fewer than n marked sets per star and level pair. Their total loss at an even site-level is $O(n^2 T)$, leaving at least $\lambda/3$ eligible centers. Activate centers and make the uniform eligible-active choice at the long height. The height lemma implies global success with probability $1 - o(1)$; each odd state sees at most T IDs across at most two levels. A failed actual descriptor would be disjoint from the marked union, contradicting maximality.

One even state makes one shared choice. Each actual odd role uses the entire neighborhood of its odd state, not only the states represented by its own actual neighbors. This convention also defines the calculations under hypothetical replacements. A mark reads only the incident-star Step 3 tests, their base and hidden inputs, the listed center tuples and position counts. Eligibility does not consult future alarms, selection adjustments or global success. The long or short path rule then reads these eligibility sets and local activations and ties. Invalid choices produce zero rows, with fixed local fillers where a continued sampling experiment requires them.

Selected odd posteriors. From here on rows are indexed by actual odd roles b ; write L_b for the posterior at their state $q_{\text{st}}(b)$. At a high role use $p_b = L_b$ for the selected tuple descriptor. At a low role, selection itself carries information about the hidden scalar, so we use the short-height correction from Lemma 5.1. Fix the base, all hidden scalars except Z_ℓ , and the prospective center positions. In the raw experiment let $Z_\ell \sim \pi_\ell$, regenerate every consulted tuple whose law depends on it, and compute the short-height selections with their activation and tie randomness.

For a canonical descriptor $\mathfrak{d}$, let o be its complete tagged-tuple data and $Q_{\mathfrak{d}}^{\text{data}}$ their product reference from Step 3. If $F_\xi(o)$ is the gated tuple likelihood, then for the raw experiment $\mathbb{P}_\xi$ with target fixed at ξ ,

$$\mathbb{P}_\xi(\text{valid proxy presentation of } \mathfrak{d}, O_{\mathfrak{d}} \in do) = F_\xi(o) a_\xi(o) Q_{\mathfrak{d}}^{\text{data}}(do), \quad 0 \leq a_\xi \leq 1.$$

Here a_ξ integrates the remaining local selection event after these data are specified. Candidate-dependent unobserved tuples are regenerated under their changed laws; they are not fixed at

their actual values. Mappings with the same distinct ID/type observations are combined into one descriptor, so their presentation events are disjoint. Validity requires legitimate choices, the descriptor size bound, the true candidate gate, the Step 3 data tests, and position counts at most 2λ in the adjacent balls at all levels. These are the validity conditions of the previous lemma with its singleton and optional-pool conditions removed.

Write $M = \int F_\xi d\pi_\ell$ and $M^a = \int F_\xi a_\xi d\pi_\ell$. If $M^a \geq e^{-.02k}M$, use the posterior proportional to $F_\xi a_\xi d\pi_\ell$; otherwise use L_b . The adjustment table depends on the fixed inputs and presented data, not on a separate read of the actual Z_ℓ . Use this same table also for descriptors presented by the long selection. Invalid outcomes give zero rows. The possible multiplier $e^{.02k}$ leaves the bounds

$$p_b \leq e^{.2k} Q_{-c} \quad \text{for matching primary and mode,} \quad Np_b \leq \begin{cases} e^{m^{.15}}, & b \text{ low,} \\ n^{.05J}, & b \text{ high.} \end{cases}$$

All valid rows remain supported on common neighbors of the tuple entries.

In the raw proxy experiment, the expected mass assigned to a label y is at most $2\pi_\ell(y)$. To see the two contributions separately, the adjusted normalizer cancels against the actual presentation density, and disjointness gives

$$\mathbb{E}[p_b^{\text{pr}}(y) 1_{\text{adjusted}}] \leq \pi_\ell(y).$$

On fallback data, $M^a < e^{-.02k}M$, while each gated observation likelihood has integral at most one. Hence

$$\mathbb{E}[p_b^{\text{pr}}(y) 1_{\text{fallback}}] \leq e^{-.02k}(\#\text{descriptors})\pi_\ell(y) = o(\pi_\ell(y)).$$

The descriptor bound here uses the polynomial position-count gate and $O(T \log n + (J+1) \log T) < .01k$. This expectation uses the original prior π_ℓ , before avoidance of hidden-scalar alarms.

Let $\bar{p}_b^{\text{pr}}$ and $\bar{p}_b^{\text{long}}$ be the rows averaged over their raw center experiments at a fixed good hidden history. The proxy computation consults hidden keys at signs within $O(\sqrt{m})$ of the role's sign: eligibility is determined by incident stars, and the short tube contains only that many sign changes. Couple the two rules by the same positions, activations and ties. On equality of their adjacent path maxima, choices, validity and kernels agree. Use the short-truncation estimate in its intersection form, take a union over the $O(n)$ adjacent even sites, and multiply by the common cap. This gives

$$N\bar{p}_b^{\text{long}}(y) \leq N\bar{p}_b^{\text{pr}}(y) + O(n)e^{m^{.15}}o(e^{-2m^{1/5}}) + o(1) = N\bar{p}_b^{\text{pr}}(y) + o(1).$$

The last error also includes the exponentially small eligible-size exceptions. Both long and short rows, before center averaging, use center inputs only within spatial radius $r + o(n)$ at fixed history.

Odd loads. Three different averages enter the load estimate. We take all averages over actual odd roles, using zero outside the indicated low or high subset.

First expose the base. For a fixed label y , use the summand $1_{\{b \text{ low}\}} N\pi_{\ell(b)}(y)$, multiplied on raw inputs by its Step 1 validity indicator. Its cap is n^{d_1} . Boundary rows contribute at most $n^{-.05+d_1} = o(1)$. At an interior key, before coarse avoidance and conditional on V_0 , posterior cancellation bounds the mean by the candidate prior, hence by $O(1)$ after normalization. In an n -th moment, rows with bins within a fixed grid distance have fraction $O((2n^{-.04})^{d_c})$. Their cap cost is negligible. For separated bins, remove the coarse constraints touching their local base variables. Lemma 3.5 costs $O(1)^l$ for $l \leq n$ terms and then their raw means multiply. Lemma 3.6, Markov's inequality and a union over labels give a bounded base average simultaneously for all y , with probability $1 - o(1)$ in the admitted base law.

Next fix such a base and average the low proxy means over the avoided hidden-scalar law. Sign neighborhoods of radius $O(\sqrt{m})$ have fraction $2^{-m+o(m)}$, giving repeat cost $n2^{-m+o(m)}e^{m^{.15}} = o(1)$. For separated signs, remove the hidden-stage constraints touching each target scalar. With the other keys fixed, integrate those distinct targets under their original independent priors. The preceding proxy calculation bounds each mean by $2N\pi_{\ell(b)}(y)$, and no other retained function reads that target. Scattered moments give a bounded average of the proxy means, and the additive comparison gives the same for long means. High rows contribute deterministically at most $n^{-.13J}n^{.05J} = o(1)$.

Finally fix a typical admitted hidden history and take moments over centers, activations and ties. Long computations at residual distance more than $2r + o(n)$ use disjoint primitive randomness. Their comparison means are the long means just bounded; the absolute row caps handle close repeats. Thus the normalized average of $Np_b(y)$ is bounded for every y , with probability $1 - o(1)$ on center success. In actual column sums this is

$$\sum_{b \in B} p_b(y) = \frac{|B|}{N} \text{avg}_{b \in B} Np_b(y) = o(1),$$

so all odd columns are at most $1/10$ for large n .

The additional even history mean. For the even reconstruction we need a separate bound on the tag mixtures: with probability $1 - o(1)$ in the admitted hidden history,

$$\text{avg}_{x \in A} N \sum_i T_{\beta(x)}(i) \mu_i(a) = O(1) \quad \text{for every label } a.$$

The contribution outside interior low rows with $j = 0$ is bounded deterministically. Use $T_\beta \leq n^{d_2 u_\beta} \Lambda$, tag balance and the rarity bounds. For example the low rows with $j \geq 1$ contribute at most $O(\sum_{j \geq 1} n^{-.13j + d_2(j+1)})$; boundary and high rows are handled by their corresponding rarity bounds.

For an interior $j = 0$ row define

$$f_x(a; H_{\text{base}}, Z) = \mathbf{1}_{\text{local tests}} N \sum_i T_{\beta(x)}(i) \mu_i(a), \quad \phi_x(a; H_{\text{base}}) = \mathbb{E}_{Z, \text{raw}}[f_x \mid H_{\text{base}}],$$

where the indicator requires true $I_h \in G_\beta$, positivity, and the Step 2 tests, with value zero on undefined inputs. Then $f_x \leq Kn^{d_2}$, and ϕ_x depends only on a bounded bin neighborhood. Hold the base except I_h fixed. The true-gated subdensity of Z_S is the normalizer defining T_β , so its cancellation gives

$$\mathbb{E}_{I_h, Z_S \mid H_{\text{base}} \setminus I_h} [f_x(a; H_{\text{base}}, Z)] \leq N \sum_i T_0(i) \mu_i(a).$$

This uses the fact established in Step 2 that the posterior formula does not read the actual I_h a second time. Now $P_h = A_w$, and $T_0 \leq O(\eta_{A_w})$. Only after this cancellation average A_w under its prior dominated by Π . Since $\mathbb{E}_\Pi \eta_{A_w} = \Lambda$,

$$\mathbb{E}_{H_{\text{base}}, \text{raw}} [\phi_x \mid V_0] \leq O\left(N \sum_i \Lambda(i) \mu_i(a)\right) = O(1).$$

Apply scattered moments first to the ϕ_x 's under the coarse stage. The near-bin fraction is the one in the preceding odd base estimate, and the cap here is Kn^{d_2} ; at separated bins remove the touching coarse constraints before using the raw expectation. This bounds their cube average.

Given such a base, apply scattered moments to the f_x 's under the hidden stage. Each uses boundedly many keys at signs within bounded distance of the row's sign. The same near-sign count applies; removing the constraints touching separated key lists lets their means integrate to the ϕ_x 's. Thus the cube average of the tag mixtures is bounded, although an individual mixture need not have a constant pointwise cap.

Odd injection and even posterior reconstruction. On entering histories with odd columns at most $1/10$, apply Lemma 3.9 with $d = N$. The atom caps satisfy $N^{-1+o(1)} \leq N^{-.95}$, the column bound is below .4, and $n^2 \leq N^{.025}$. We obtain an odd injection whose outputs on at most n^2 queried roles are dominated by $1 + o(1)$ times their product law.

Fix an actual even role v and a possible center ID c , with the type required at its state. Hold the admitted history through Z and all other relevant primitive center randomness fixed. The remaining primitive variable is the complete tagged tuple $z = (i, x_1, \dots, x_k)$ at this ID/type, with raw law P_c . For odd-neighbor data $\mathbf{y}$ define

$$F_z(\mathbf{y}) = 1_E \prod_{b \sim v} p_b(y_b), \quad m_c(\mathbf{y}) = \int F_z(\mathbf{y}) dP_c(z).$$

All choices and kernel evaluations in this formula are recomputed at z . The gate E requires presence and legitimate long selection of c at v , valid kernel presentations on the entire odd-state stars of its actual neighbors, and correct eligible sizes throughout v 's height consultation domain. Thus it includes the descriptor size and position-count bounds. At a fixed hidden history these checks use center positions, tuples, activations and ties within radius $r + o(n)$, also after a hypothetical replacement of z .

For a neighbor b with matching mode and primary, let $\mathcal{D}_{b,c}$ be the possible descriptors containing this ID/type among the locally permitted IDs. The lists need not be selected validly; including all of them makes the index set independent of z . For each list use its posterior Q_{-c} , which deletes the tag and all entries of tuple c . Let q_b be the uniform average of these reference laws. On undefined cases choose fixed laws independent of z ; if the prerequisite position-count gate fails, E is empty. For nonmatching neighbors use the uniform law on Y . The product $Q = \prod_{b \sim v} q_b$ is independent of z . On E , for each matching neighbor,

$$p_b \leq e^{.2k} |\mathcal{D}_{b,c}| q_b, \quad \log |\mathcal{D}_{b,c}| = O(T \log n + (J+1) \log T) < .1k.$$

The deleted law uses only other tuple observations and the preceding hidden data. In particular its type-list gate never reads the removed tag or entries. Only coarse or fine chunk flips can produce the other neighbors, so their number is $o(n^4)$. Their absolute caps cost $o(kn)$ in total. We have proved

$$F_z \leq e^{.34kn} Q.$$

Retain the successful-prehistory indicator while applying the near-product inequality to the actual odd labels. After their joint law has been replaced by the product law, remove nonlocal center-success restrictions and retain E . Under this raw experiment the marginal data subdensity is m_c . Lemma 3.7 therefore bounds the joint mass of E and $m_c = 0$ or $m_c < e^{-.02kn} Q$ by $(1 + o(1))e^{-.02kn}$ for each fixed v, c . There are $\exp(O(n))$ such choices, and $\exp(O(n) - .02kn) = o(1)$. Thus the probability of a successful entering history together with any failed requirement for an actually selected tuple is $o(1)$.

The posterior proportional to F_z now has joint density on the k labels (marginalizing the tag) relative to uniform product at most $e^{.4kn}$, since the prior width per coordinate given a support tag

costs $\leq n^\gamma + \log(2/c_1) = o(n)$. Its average-coordinate marginal is supported on common neighbors: on every positive-mass input the same tuple reference is selected at v and each valid neighboring output hits **all** its label entries by the posterior supports at that star. Let $\mathcal{H}$ be the labels at which its average marginal exceeds $e^{.55n}/N$, so $|\mathcal{H}|/N \leq e^{-.55n}$. The joint cap gives

$$\Pr(\#\{j : x_j \in \mathcal{H}\} > .8k) \leq 2^k e^{.4kn} (|\mathcal{H}|/N)^{.8k} = o(1),$$

because $.8 \cdot .55 > .4$. The heavy marginal mass is the expected fraction of tuple entries in $\mathcal{H}$, hence at most $.8 + o(1)$. Restricting the average marginal to $\mathcal{H}^c$ and renormalizing therefore costs a constant factor. Use these row weights at v with the selected reference only, zero on local failures. Under auxiliary product odd sampling (with any fillers on invalid rows not used on E), their expected normalized mass at a , given good history and integrating centers as well, is bounded by

$$O\left(N \sum_i T_{\beta(v)}(i) \mu_i(a)\right).$$

Before truncation the cancellation, as a measure in z , is

$$\int m_c(\mathbf{y}) \frac{F_z(\mathbf{y}) dP_c(z)}{m_c(\mathbf{y})} d\mathbf{y} = dP_c(z) \int F_z(\mathbf{y}) d\mathbf{y}.$$

The last integral retains the selection indicator. After averaging the other raw center inputs it is bounded by the probability of selecting this fixed tuple on the stated gates. Coordinate averaging therefore gives the expected empirical mass of a in the selected tuple. With one forced present center in the ball and its tuple fixed, the probability with local size gates of its valid selection is $O(1/\lambda)$ on level 0 by uniform eligible active choice, and $\exp(-n^{\Omega(1)})$ on higher levels by the height bound. Here the tuple is fixed but the other prospective-center positions remain averaged. The positive-height bound is the position-averaged supremum over legal pre-activation eligibilities from Lemma 3.8, not a bound for each fixed position configuration. Summing presence probabilities thus costs a constant. The unconditional tuple marginal given history costs at most $2/c_1$ times the indicated tag mixture.

Even loads and completion. Let $w_v(a)$ be the final even row at label a , extended by zero on local failure. Fix an admitted history through Z for which the additional even mean is bounded. For at most n even roles separated in residual bits by more than $2r + o(n)$, keep the successful entering history while applying the joint odd-output comparison. Their actual odd-neighbor sets are disjoint, so this comparison replaces all queried outputs by their product law at cost $1 + o(1)$. Then remove nonlocal center-success restrictions, retaining each local gate E . The raw center integrals now factor, and the preceding cancellation gives comparison means $O(N \sum_i T_{\beta(v)}(i) \mu_i(a))$ for the normalized rows $Nw_v(a)$. Their cube average is bounded by the additional history estimate.

For close repeats choose ρ so that binary entropy in natural logs satisfies $H_{\text{bin}}(2\rho) < \log 2 - .55$. The near fraction then obeys

$$f \leq \exp(-(\log 2 - H_{\text{bin}}(2\rho))n + o(n)), \quad nf O(e^{.55n}) = o(1).$$

Lemma 3.6, Markov's inequality and the column union bound show that the probability of a retained success together with any normalized even load above a sufficiently large fixed constant is $o(1)$. Since $|A|/N \rightarrow 0$, a bounded normalized average gives actual even column sums at most one for large n . The total exceptional joint mass through the stagewise laws is $o(1)$, so these events have a common successful outcome. Hall's theorem supplies distinct even representatives adjacent to the already distinct odd labels, completing the embedding. $\square$

7 A small power key grid

We next exclude nearly pure patches whose two laws both have small power width. The construction uses a separate anchor for each cell in a grid. We first arrange that an anchor with a sufficiently spread-out law is unlikely to have very small degree into a narrow law.

Take rational constants $0 < D_0 < \min(.1, D_*/2)$, where D_* is from Lemma 6.1; here D_0 is an exponent, not a height step length. Fix $0 < d < D_0/1000$, $b = d/4$, $c = b/20$. We claim that, after stabilization and $o(N)$ discards, we may assume

$$\begin{aligned} \text{every } \sigma \text{ on } X, \tau \text{ on } Y \text{ with } N \max \sigma \leq n^{D_0}, \quad \text{width}(\tau) \leq n^{1/4} \\ \text{have both colors of density } > n^{-c}. \end{aligned} \quad (7.1)$$

To prove the claim, first suppose that a family violating this assertion were available. On an available subfamily one color G has density at least $1 - n^{-c}$. For each witness put

$$B_\sigma = \{x : d_G(x, \tau) < 1 - n^{-c/2}\}, \quad \sigma' = \sigma|_{X \setminus B_\sigma}.$$

Markov's inequality gives $\sigma(B_\sigma) \leq n^{-c/2}$, so this restriction increases the law by at most a factor of 2. In particular, $N \max \sigma' \leq 2n^{D_0} \leq n^{D_*}$ for large n , and every label in its support has degree at least $1 - n^{-c/2}$ into τ . The restricted witnesses are still available. Choose a balanced mixture by Lemma 3.3, and apply Lemma 6.1 with the sides interchanged: its narrow law is τ , its broad law is σ' , and its defect exponent is $p_0 = c/2$. This gives a monochromatic cube, a contradiction. Thus violations are not available, and Lemma 3.2 makes them absent after the claimed subsequence and discards.

Lemma 7.1 (Small-grid purity exclusion). *With the constants fixed above, on sets satisfying (7.1) there cannot be an available family of one-color patches (μ_i, ν_i) of widths $\leq n^{d/2}$ on X, Y , with every $y \in \text{supp } \nu_i$ having degree into μ_i in the given color at least $1 - e^{-n^p}$ for fixed $p > 0$.*

Proof. Step 1: the grid and its raw anchor laws. Suppose such a family exists. A finite menu retains availability by choosing a representative for each successful discard pair. Denote its given color by G .

Partition some cube coordinates into $s = \lceil n^d \rceil$ chunks of $\lfloor n^d \rfloor$ special bits each, with total length $m = \Theta(n^{2d})$, and set aside $q = \lceil n^{4d} \rceil$ further auxiliary bits. Bin each special chunk's count into consecutive intervals of probability at most $2n^{-b}$; this is possible since each binomial atom is $O(n^{-d/2})$. The vector of bin indices is a grid key g , and the auxiliary word is a second key $t \in Q_q$. Write $E(g)$ for the distinct grid keys obtained by moving one bin in one coordinate, so $|E(g)| \leq 2s$. Grid distance means the number of such one-bin moves. An actual cube position reads the cell indexed by its pair (g, t) ; many positions can read the same cell. A fixed g occurs with fraction at most $(2n^{-b})^s$. Since $m + q = o(n)$, unused residual bits remain, and t is uniform on either parity of the cube, also when g is fixed.

For each grid key independently sample a patch tag i_g . Conditional on these tags, sample independent anchors $W_{g,t} \sim \mu_{i_g}$, one for every cell. The distributions of the tags will be chosen below. For odd positions of key (g, t) , restrict ν_{i_g} to labels adjacent to all anchors in the two lists

$$W_{h,t} \quad (h \in E(g)), \quad W_{g,t'} \quad (d_H(t, t') \leq 1).$$

Call these the cross list and the own list, respectively. They cover all incident even anchors: a special-bit flip either leaves g unchanged or moves it to $E(g)$, an auxiliary-bit flip changes only t , and a residual-bit flip changes neither key.

We need to be able to omit any specified cross hit later. For this reason, process the cross list once with each possible element last, using a fixed order of the other elements independent of t . In every such order require each successive restriction to retain a fraction at least n^{-c} . The empty cross list is allowed. After the full cross restriction, require the own list, taken together, to retain a fraction at least $1 - n^{-2}$. Call the cell valid when all these requirements hold. On validity let $p_{g,t}$ be the normalized fully restricted law; otherwise let it be zero. Thus it is a subprobability supported on hits of all incident even anchors, and

$$Np_{g,t} \leq L := \exp(n^{d/2} + 2cs \log n + O(1)).$$

For an anchor variable w in the lists, let $p_{g,t}^{(-w)}$ be ν_{i_g} restricted to all listed hits except the hit of w , then normalized. If the restriction has zero mass, use a fixed fallback independent of w . Deletion is by the variable's name, even if two anchors happen to take the same label. On validity,

$$\begin{aligned} p_{g,t} &\leq (1 - n^{-2})^{-1} p_{g,t}^{(-w)} && \text{for an own hit,} \\ p_{g,t} &\leq n^c (1 - n^{-2})^{-1} p_{g,t}^{(-w)} \leq n^{2c} p_{g,t}^{(-w)} && \text{for a cross hit.} \end{aligned}$$

The second comparison uses the order in which that cross hit is last, followed by the restriction to the own list. Each deletion law is normalized without consulting the deleted variable's value.

Choose product profiles for the tags so that, for a constant K' ,

- (i) $\mathbb{E}N\mu_{i_g} \leq K'$ pointwise for every grid key g ;
- (ii) for each g , the expectation of $Np_{g,t}$, averaged over odd sites with this grid key, is pointwise $\leq K'$.

Both expectations refer to the raw independent tag and anchor model. To obtain these profiles, use one player per grid key in Lemma 3.3. Fix the other players' profiles and assign nonnegative prices to the two output vectors of this player. Availability permits a patch avoiding, on each side, all labels whose price exceeds a sufficiently large constant times total price divided by N . Its first output μ_{i_g} is supported on its first support; every possible second output $p_{g,t}$ is a subprobability on its second support. Both price contributions therefore have the required bound. Separation gives a mixture satisfying the two coordinatewise bounds, and the simultaneous-profile part of the lemma gives profiles for all players. Its hypothesis on the output rule holds because the filter and validity rules do not use the profile probabilities.

Step 2: filter tails and tag conditioning. Consider one step in one prescribed cross order, conditional on the starting tag and the preceding cross anchors, all of whose tests have passed. The current filtered law τ has width at most $\log L < n^{1/4}$. The next anchor comes from a distinct grid key, so its raw marginal law α is independent of this history and satisfies $N\alpha \leq K'$ pointwise. Put

$$H_\tau = \{x : d_G(x, \tau) < n^{-c}\}.$$

If $\alpha(H_\tau) \geq K'n^{-D_0}$, the normalized restriction $\sigma = \alpha|_{H_\tau}$ satisfies $N \max \sigma \leq n^{D_0}$, but $d_G(\sigma, \tau) < n^{-c}$, contrary to (7.1). Thus each cross step fails with probability $O(n^{-D_0})$. There are at most $O(s^2)$ order-step pairs, giving total cross-failure probability

$$O(s^2 n^{-D_0}) = n^{-D_0 + O(d) + o(1)}.$$

For the own test, condition on all tags and the cross anchors, with the cross tests passed. Each label in the support of the resulting law has degree at least $1 - e^{-n^p}$ into μ_{i_g} . The own anchors

are still independent draws from μ_{i_g} . The expected mass removed by their $q + 1$ constraints is at most $(q + 1)e^{-n^p}$, so Markov's inequality bounds own-test failure by $n^2(q + 1)e^{-n^p}$.

We first impose a condition on the tags alone. Let B_g be the event that, conditional on all tags, the cross-failure probability for cell (g, t) , including all orders and steps, exceeds $n^{-D_0/2}$. This probability is independent of t by auxiliary-key symmetry. The event B_g reads just the tags at g and $E(g)$, and

$$\Pr(B_g) \leq O(s^2 n^{-D_0}) n^{D_0/2} = n^{-D_0/2+2d+o(1)} \leq n^{-D_0/3}.$$

The overlap graph of these tag events has degree $O(s^2)$. Lemma 3.5 therefore gives a tag law avoiding all B_g , with charges $n^{-D_0/4}$, since $O(s^2)n^{-D_0/4} = o(1)$.

This tag law also has bounded average anchor loads with probability $1 - o(1)$. Here is the moment calculation. For a fixed $x \in X$, use even roles $v \in A$ and weights $Z_v = N\mu_{i_{g(v)}}(x)$, of cap $L_\mu = e^{n^{d/2}}$. Declare roles near when their grid keys are within a fixed sufficiently large distance. Each near set has fraction $f \leq n^{O(1)}(2n^{-b})^s$, and $nfL_\mu = o(1)$. For $l \leq n$ separated roles, remove the tag-avoidance constraints touching their tag variables. There are $O(ls)$ such constraints, so the comparison cost is

$$\exp(O(lsn^{-D_0/4})) = e^{o(l)}.$$

The remaining integral uses independent raw tags, each with mean weight at most K' by (i). Lemma 3.6, followed by Markov's inequality and a union over x , bounds $|A|^{-1} \sum_{v \in A} N\mu_{i_{g(v)}}(x)$ by a constant simultaneously for all x . Call such tags typical.

Step 3: predictive alarms before anchor conditioning. We next ensure that the eventual odd labels give a sufficiently large predictive denominator at every even role. Fix an even position v of key (g, t) , all tags, and all anchors except $z = W_{g,t}$. For a vector $\mathbf{y} = (y_u : u \sim v)$ of odd labels put

$$F_z(\mathbf{y}) = 1_{\mathcal{V}} \prod_{u \sim v} p_{g(u),t(u)}(y_u), \quad M_v(\mathbf{y}) = \int F_z(\mathbf{y}) d\mu_{i_g}(z),$$

where $\mathcal{V}$ requires validity of all the neighboring filters. Both validity and the kernels are recomputed as z varies. Let Q_v be the product of their deletion laws for the named variable $W_{g,t}$, with the fallbacks above when necessary. This is a probability law on $\mathbf{y}$ independent of z . At most m actual neighbors impose a cross hit of z , so the deletion bounds give

$$F_z \leq \exp(2cm \log n + O(n^{-1})) Q_v = e^{o(q)} Q_v; \quad m \log n = o(q).$$

Declare predictive failure when $M_v = 0$ or $M_v < e^{-.04q} Q_v$.

For fixed realized anchors $\mathbf{W}$, let $r_v(\mathbf{W})$ be the probability of this failure under independent neighbor draws, multiplied by $1_{\mathcal{V}}$; set it to zero if a neighboring filter is invalid. Invalid rows may use arbitrary local fallback probabilities in this auxiliary draw. Before any anchor conditioning, the data subdensity on $\mathcal{V}$ is exactly M_v . Hence, even with the tags and all other anchors fixed,

$$\mathbb{E}_{W_{g,t}} r_v \leq e^{-.04q}, \quad \Pr_{W_{g,t}}(r_v > e^{-.02q}) \leq e^{-.02q}.$$

These estimates retain $\mathcal{V}$ as an indicator; they do not condition the raw law on validity.

At a given key (g, t) we must impose the inequality $r_v \leq e^{-.02q}$ for all neighbor multiplicities that can occur at an even position with that key. Each auxiliary flip contributes once. The special-bit neighbors are described by at most $2s + 1$ counts: one for each possible adjacent grid bin and one for the unchanged key. The number of residual-bit neighbors is fixed. Thus there

are at most $(n+1)^{2s+1} = \exp(O(s \log n))$ distinct formulas, up to reordering the neighbor labels. A union over these formulas, rather than over cube positions, bounds the probability that a key violates an alarm by

$$\exp(-.02q + O(s \log n)) = \exp(-\Omega(q)),$$

uniformly in the tags.

Step 4: anchor conditioning and odd loads. Fix tags admitted by Step 2. Group, for each (g, t) , its filter failure and all its predictive alarms into one bad event on the raw independent anchors. Give the grid-times-auxiliary graph edges for one grid-bin move or one auxiliary-bit flip. Each grouped event reads anchors within distance two of its key: the filter reads distance one, and an alarm reads the filters at neighboring keys. The event graph therefore has degree $O((s+q)^4)$. Its failure probability is at most

$$n^{-D_0/2} + n^2(q+1)e^{-n^p} + \exp(-\Omega(q)).$$

Charges $n^{-D_0/4}$ again satisfy Lemma 3.5, since $16d < D_0/4$. Use the resulting anchor law at each fixed admitted tag history. Sampling in these two stages leaves the tag law unchanged; the anchor avoidance probability is not used to reweight it.

We show that the odd kernels have small column sums under this law. Fix $y \in Y$, and for an odd role u put

$$Z_u = N p_{g(u), t(u)}(y), \quad a_u(\mathbf{i}) = \mathbb{E}_{\text{raw anchors}|\mathbf{i}} Z_u, \quad d_u = \mathbb{E}_{\text{raw tags}} a_u(\mathbf{i}).$$

The cap is L . Take the near relation to mean that the grid keys are within a fixed distance large enough to contain the local scopes just described. Its fraction is again $f \leq n^{O(1)}(2n^{-b})^s$, and

$$nfL \leq \exp(-(b-2c+o(1))s \log n) = o(1),$$

because $b > 2c$.

For $l \leq n$ separated roles first integrate the anchors, keeping the tags fixed. To recover the independent raw anchor means, remove all avoidance constraints touching their anchor lists. A list has radius one and a constraint has radius two, so at most $O((s+q)^3)$ constraints touch each list. The lists are disjoint, and the resulting integral factors as $\prod_j a_{u_j}(\mathbf{i})$. Each factor uses only tags within grid distance one of its role. Next remove the tag-avoidance constraints touching these tag lists, at most $O(s^2)$ per role, and integrate the independent raw tags. Consequently the full two-stage law satisfies

$$\mathbb{E} \prod_{j=1}^l Z_{u_j} \leq \exp\left(O\left(l((s+q)^3 + s^2)n^{-D_0/4}\right)\right) \prod_{j=1}^l d_{u_j} \leq C^l \prod_{j=1}^l d_{u_j},$$

for a constant C independent of n . The exponent is $o(l)$, since $12d < D_0/4$. Profile constraint (ii) gives $|B|^{-1} \sum_{u \in B} d_u \leq K'$.

Lemma 3.6 now bounds the normalized average $|B|^{-1} \sum_u Z_u$ by a constant, simultaneously for all labels, with probability $1 - o(1)$. Here and in Step 2 the union is legitimate because $N \leq n2^n$: a sufficiently large fixed threshold in the n -th moment bound beats all N columns. The actual column sum is

$$\sum_{u \in B} p_{g(u), t(u)}(y) = \frac{|B|}{N} \left(\frac{1}{|B|} \sum_{u \in B} Z_u \right) = o(1),$$

since $C_n = N/2^n \rightarrow \infty$. It is therefore at most the clock-lemma constant θ_0 for large n . This property and typicality of the tags hold simultaneously with probability $1 - o(1)$.

Step 5: injective odd labels and posterior even rows. On a prehistory with the odd column bound, apply Lemma 3.10, for example with $B = 3$, to the actual odd-position kernels. Use predictive failure at each even position as a forbidden predicate. Each predicate reads its n odd neighbors, and each odd row belongs to n such scopes. Its product-law probability is at most $e^{-.02q}$ by the anchor alarms. The maximum label atom is at most L/N , with $\log L = o(q) = o(n)$. These bounds meet the lemma's polynomial requirements. We obtain an odd injection avoiding all predictive failures, with joint upper comparison $1 + o(1)$ on at most n^2 queried rows.

For each even role v , evaluated at its assigned neighbor data, use the probability row

$$p_v^X(x) = \frac{F_x(\mathbf{y})\mu_{i_g}(x)}{M_v(\mathbf{y})}, \quad g = g(v).$$

It is supported on common neighbors of those odd labels. Predictive success and the bound for F_x/Q_v give $Np_v^X \leq e^{.04q+o(q)}$, including the prior cap $N\mu_{i_g} \leq e^{n^{d/2}}$. In auxiliary experiments set the row to zero unless $\mathcal{V}$ and predictive success hold.

Step 6: posterior cancellation and Hall. Fix typical tags and a label $x \in X$. For the even load moments use $Z_v = Np_v^X(x)$, cap $L_{\text{even}} = e^{.04q+o(q)}$, and declare roles near when their auxiliary words are within Hamming distance five. Then

$$f \leq n^{O(1)}2^{-q}, \quad nfL_{\text{even}} \leq n^{O(1)}2^{-q}e^{.04q+o(q)} = o(1).$$

For $l \leq n$ separated roles their odd-neighbor sets are disjoint. Keep the successful-prehistory indicator while applying the clock lemma's joint comparison to these at most n^2 odd labels. This replaces their injection law by independent odd draws. We may now remove nonlocal success restrictions, retaining each local gate $\mathcal{V}$ in its star likelihood.

Next remove the anchor-avoidance constraints touching the l target variables $W_{g(v),t(v)}$. There are at most $O(l(s+q)^2)$ such constraints, and their total comparison cost is $\exp(O(l(s+q)^2n^{-D_0/4})) = e^{o(l)}$, in particular $O(1)^n$. The calculation at any other retained role does not consult a target being integrated: its anchor inputs have auxiliary words within distance two of its own word, whereas retained words are separated by more than five.

With all other anchors fixed, integrating one target and its product neighbor data gives the marginal data subdensity M_v . Thus, with zero terms where $M_v = 0$,

$$\sum_{\mathbf{y}} M_v(\mathbf{y}) \frac{F_x(\mathbf{y})\mu_{i_g}(x)}{M_v(\mathbf{y})} = \mu_{i_g}(x) \sum_{\mathbf{y}} F_x(\mathbf{y}) \leq \mu_{i_g}(x).$$

The last inequality uses the local validity gate; retaining predictive success can only decrease the integral. We can perform this cancellation successively at all separated roles. If $\mathcal{S}$ denotes the successful prehistory and assignment, the resulting comparison is

$$\mathbb{E} \left[1_{\mathcal{S}} \prod_{j=1}^l Z_{v_j} \mid \mathbf{i} \right] \leq C^l \prod_{j=1}^l N\mu_{i_{g(v_j)}}(x).$$

The comparison means have bounded average because the tags are typical. Lemma 3.6 and the same Markov and column union therefore bound the probability of $\mathcal{S}$ together with any normalized even load exceeding a sufficiently large fixed constant by $o(1)$, uniformly over typical tags. Average this estimate over the tag law. Step 4 gives typical tags and the odd column bound with probability $1 - o(1)$, and the clock sampler avoids all predictive failures on those prehistories. There is therefore a realization with both odd and even column bounds. The even column sums

are $O(2^n/N) = o(1)$, hence at most 1. Its rows are probability laws, so Hall's theorem gives distinct even labels. Together with the odd injection they form a monochromatic cube, proving the lemma. $\square$

Corollary 7.2 (Initial discrepancy). *After passing to a subsequence of a bad sequence and discarding a total of $o(N)$ labels, there is a fixed $\eta_0 > 0$ such that for every pair of probability laws μ, ν on the retained sides and either color G ,*

$$\text{width}(\mu), \text{width}(\nu) \leq n^{\eta_0} \implies |d_G(\mu, \nu) - 1/2| \leq n^{-\eta_0} \quad (7.2)$$

Proof. Take $\beta = \gamma = d/4$ in Lemma 4.1, with its corresponding $h > 0$. Stabilize its bias property at widths n^γ and its purity property at doubled width budgets. If purity were available at widths $2n^\gamma$ and defect $\leq e^{-n^h}$, first retain an available color on a subsequence. Restrict its second law to columns of defect at most $e^{-n^{h/2}}$; Markov's inequality loses only $o(1)$ mass. The restricted family is still available and has both widths at most $2n^{d/4} + O(1) \leq n^{d/2}$. It would satisfy Lemma 7.1 with $p = h/2$, a contradiction. Stabilization therefore makes this purity property absent.

With purity absent, Lemma 4.1 rules out availability of absolute bias $\geq n^{-h}$ at widths n^γ . Stabilization makes that property absent as well. Choosing a fixed $0 < \eta_0 \leq \min(\gamma, h)$ therefore gives (7.2) for all laws on the retained sets and either color, after a subsequence and a total of $o(N)$ discards. $\square$

8 Asymmetric purity and discrepancy

The initial discrepancy estimate (7.2) will now exclude pure patches whose two widths can be very different. We work along a sequence satisfying that estimate, and all patch laws use the retained labels. Set

$$\eta = \min(\eta_0/2, .04), \quad \tau = \eta/4.$$

Lemma 8.1 (Asymmetric purity exclusion). *Suppose there is a finite balanced tag mixture Λ in one color with laws μ_i on X , ν_i on Y , of respective widths $\leq n^\gamma, n^\beta$, where $\gamma < 1$, $0 < \beta < \tau/4$ are fixed. Every label in the second support has degree into μ_i at least $1 - \epsilon$ with $\epsilon = \exp(-n^p)$, $p > 0$ fixed. These hypotheses imply a monochromatic cube. The same works with sides reversed. Balance here means pointwise tag averages $\leq K/N$ as before.*

Proof. We construct probability rows for the odd vertices and posterior rows for the even vertices. The required edges will follow from their supports. To obtain injective assignments, we must also keep both sets of column sums bounded.

Step 1: the key grid and trimming. Take $s = \lceil n^\tau \rceil$ chunks of length $\lfloor n^2 \rfloor$, using m bits altogether. Bin each chunk count into consecutive intervals of mass at most $2n^{-.04}$, as in the preceding grid construction. Write g for the vector of bins and $E(g)$ for its distinct one-bin-step neighbors, so $|E(g)| \leq 2s$. Keep the full residual word $a \in Q_{n-m}$. We use both an even and an odd task at every cell (g, a) . The padded even neighbors of an odd task are

$$(g, b) \quad (d_H(a, b) \leq 1), \quad (u, a) \quad (u \in E(g)).$$

Call the first set ordinary neighbors and the second set cross neighbors. These lists contain every actual cube edge. For locality statements, $B_{\text{grid}}(g, k)$ denotes the radius- k ball in the grid graph, and $B_Q(a, R)$ a residual Hamming ball.

Fix $\delta = 10^{-4}$. We will choose a sufficiently large integer h , independent of n ; the list counts below have exponents independent of h . We first trim the mixture so that the survival probability of a first-side label under one independent h -tuple is nearly constant:

$$\alpha_x := \mathbb{E}_\Lambda d_G(x, \nu_i)^h = 2^{-h}(1 + o(1/s)) \quad \text{for every retained first-side label } x.$$

To see this, let

$$b(x) = \Lambda\{i : |d_G(x, \nu_i) - 1/2| > 2n^{-\eta}\}.$$

For a fixed i , (7.2), applied to uniform laws on either exceptional set, bounds their total size by $2Ne^{-n^\eta}$. Hence

$$\sum_x b(x) \leq 2Ne^{-n^\eta}, \quad D := \{x : b(x) > e^{-n^\eta/2}\}, \quad |D| \leq 2Ne^{-n^\eta/2}.$$

Balance gives $\mathbb{E}_\Lambda \mu_i(D) \leq 2Ke^{-n^\eta/2}$. Discard tags with $\mu_i(D) > 1/2$, whose total mass is at most $4Ke^{-n^\eta/2}$, normalize the remaining tag law, and replace each μ_i by its restriction to D^c . Continue to use the same symbols for these laws. Balance changes only by constant factors, first-side widths increase by $O(1)$, and each own-color defect is at most 2ϵ . The remaining tag law differs from the original one by $e^{-\Omega(n^\eta)}$ in total variation. Thus, uniformly over retained x ,

$$\alpha_x = 2^{-h}(1 + O(hn^{-\eta}) + O(2^h e^{-\Omega(n^\eta)})) = 2^{-h}(1 + o(1/s)),$$

because h is fixed and $\tau < \eta$.

Step 2: hidden tuples, center laws, and base gates. At each grid key sample an independent hidden tuple

$$\Theta_g \sim R' := \mathbb{E}_\Lambda \nu_i^{\otimes h}.$$

Let η_g be the posterior law of its mixture tag. For the following quantities the key g is fixed. Define two unnormalized masses under μ_i :

$$\begin{aligned} d_i^- &= \mu_i\{x : x \text{ hits every coordinate of } \Theta_u, u \in E(g)\}, \\ d_i^+ &= \mu_i\{x : x \text{ hits every coordinate of } \Theta_u, u \in E(g) \cup \{g\}\}, \\ A_g &= 2^{-h|E(g)|}, \quad \Delta = e^{-n^{p/2}}. \end{aligned}$$

Here and below a hit means adjacency in the chosen color. Tilt the tag law by the mass of cross hits, retaining tags for which adding the own tuple costs at most a Δ -fraction:

$$S_g(i) = \frac{\eta_g(i)d_i^-}{Z_g} 1[d_i^- \geq e^{-n^{2\tau}}, d_i^+ \geq (1 - \Delta)d_i^-].$$

The normalizer is Z_g . These laws are used where their denominators are positive; local fallbacks will be specified below. For a retained tag, the corresponding single-anchor law is

$$U_{g,i}(x) = \frac{\mu_i(x) 1[x \text{ hits } \Theta_u, u \in E(g) \cup \{g\}]}{d_i^+}.$$

We next establish the bounds on hidden tuples that will be used throughout this construction. At a fixed key their combined exception probability is at most $e^{-n^{c'}}$, for some fixed $c' > 0$:

$$\eta_g \leq e^{hn^\beta + n^{\tau/2}} \Lambda, \quad .8A_g \leq Z_g \leq 1.2A_g.$$

We also require each of $\int d_i^- d\eta_g(i)$ and $\int d_i^- d\Lambda(i)$ to be within a factor 1.1 of A_g . With any one cross key omitted, the analogous requirement is within a factor 1.1 of $2^h A_g$. We call these the base gates.

For the posterior bound, the event $R'(\Theta_g) < N^{-h} e^{-n^{\tau/2}}$ has probability at most $e^{-n^{\tau/2}}$. Outside it,

$$\frac{\eta_g(i)}{\Lambda(i)} = \frac{\nu_i^{\otimes h}(\Theta_g)}{R'(\Theta_g)} \leq e^{hn^\beta + n^{\tau/2}}.$$

The aggregate first laws $\bar{\mu}_\eta = \int \mu_i d\eta_g(i)$ and $\bar{\mu}_\Lambda = \int \mu_i d\Lambda(i)$ consequently satisfy

$$N \max \bar{\mu}_\eta \leq K e^{hn^\beta + n^{\tau/2}}, \quad N \max \bar{\mu}_\Lambda \leq K.$$

Their widths are therefore less than $n^\eta/2$ for large n , even though individual μ_i 's may have much greater width.

Filter either aggregate successively by the independent cross tuples. Conditional on a cross tuple's latent tag, its coordinates are independent with law ν_i . On preceding regular hits, the aggregate width has increased by only $O(hs)$, and remains less than n^η . A next hit fraction outside $1/2 \pm 2n^{-\eta}$ has probability at most $2e^{-n^\eta}$: restricting the next ν_i to a larger exceptional mass would produce, together with the current aggregate, a pair contradicting (7.2). Thus

$$\prod_{\text{cross coordinates}} \left(\frac{1}{2} \pm 2n^{-\eta} \right) = A_g(1 + o(1)).$$

The same calculation omitting one cross key gives $2^h A_g(1 + o(1))$. A union over both aggregates and all these deletions costs at most $O(hs^2)e^{-n^\eta}$.

It remains to control the cutoffs in S_g . Put $L_g = \int (d_i^- - d_i^+) d\eta_g(i)$. Given Θ_g , every tag in the support of η_g has own-miss mass at most $2h\epsilon$. A fixed retained x survives the independent cross tuples with probability $\alpha_x^{|E(g)|} = A_g(1 + o(1))$. Hence

$$\mathbb{E}[L_g \mid \Theta_g] \leq O(h\epsilon A_g).$$

The weight removed by $d_i^+ < (1 - \Delta)d_i^-$ is at most L_g/Δ , so

$$\Pr(L_g/\Delta > .1A_g \mid \Theta_g) \leq O(h\epsilon/\Delta) = \exp(-n^p + n^{p/2} + O_h(1)).$$

The other cutoff removes at most $e^{-n^{2\tau}} = o(A_g)$, since $\log A_g^{-1} = O(hs)$. Together with the cross-normalizer bounds these give $.8A_g \leq Z_g \leq 1.2A_g$. This calculation uses no ordering between p and τ .

Step 3: local centers and the future anchors. In each grid slice use the height device on Q_{n-m} , independently sampling prospective positions and activations across slices. Each prospective center in slice g receives an independent tag with law S_g , conditional on the hidden tuples. Take step length $D = 2$, radius $r = \lfloor \rho n \rfloor$, and $\lambda = n^{10}$. The choices

$$b = \tau/16, \quad b_0 = b/4, \quad \rho = .01$$

are suitable; the height parameters $\zeta, \sigma, 1 - \theta$ can be chosen sufficiently small in terms of b_0 to meet Lemma 3.8. At good Lipschitz heights, the ordinary neighbors of an odd cell occupy at most two levels. The crowd bound at one cell of each level then limits their distinct center IDs to $2n^b$, which is at most $T = \lceil n^{\tau/8} \rceil$ for large n .

We specify eligibility below before drawing activations or actual anchors. Once a center has been selected at every even cell, write $i(g, a)$ for its tag and draw $W_{g,a} \sim U_{g,i(g,a)}$, independently between cells at this stage. Cells choosing the same center share its tag, but receive independent anchor samples. This distinction permits the subsequent product likelihoods at odd cells.

8.1 Presentations at odd cells

Step 4: a fixed-presentation reference experiment. Consider an odd cell (g, a) with a prescribed list of at most T distinct own-slice IDs and one ID in each cross slice. The observations are the tags at the own-slice IDs and the pairs $(i, W_{u,a})$ at the cross IDs. Ordinary anchors are not observed. For now these are independent hypothetical observations: conditional on hidden tuples, their laws are S_g internally and $S_u(i)U_{u,i}$ externally.

Replace the target tuple Θ_g by a candidate ξ with prior R' , holding all other hidden tuples fixed. Retain the base gates at g and its grid neighbors; these gates depend on the candidate but not on the hypothetical tag and anchor outcomes. Use the following candidate-independent probability references for the data:

- For an internal tag, use $\Lambda(i)d_i^-$, normalized. Here the cross-hit mass d_i^- does not involve ξ . The density of S_g against this reference is at most $\exp(hn^\beta + n^{\tau/2} + O(1))$.
- For a cross pair at $u \in E(g)$, use $\eta_u(i)\mu_i(x)$ restricted to hits of the cross tuples of u other than g , and normalize. Its normalizer is at most $1.1 \cdot 2^h A_u$. On the base gate,

$$S_u(i)U_{u,i}(x) \leq \frac{\eta_u(i)\mu_i(x)}{(1-\Delta)Z_u} 1[x \text{ hits } \Theta_w, w \in E(u) \cup \{u\}],$$

because $d_i^-/d_i^+ \leq (1-\Delta)^{-1}$. Dividing by the indicated reference therefore costs at most 2^{h+2} .

The second calculation is the reason for tilting center tags by d_i^- . At undefined reference normalizers use fixed fallbacks independent of the omitted tuple; validity will require the legitimate positive cases.

Let Q be the product of these references, and let F_ξ be the product observation density against Q , multiplied by the candidate's base-gate indicator. Define

$$M = \int F_\xi dR'(\xi), \quad \varepsilon_0 = e^{-\delta h s \log n}.$$

The true-gated data have subdensity M against Q , so the joint probability of $M < \varepsilon_0$, including $M = 0$, is at most ε_0 . Require $M \geq \varepsilon_0$ on a valid presentation. The posterior $F_\xi dR'(\xi)/M$ then has log density against the uniform law on Y^h at most

$$hn^\beta + T(hn^\beta + n^{\tau/2} + O(1)) + O(hs) + \delta h s \log n \leq 1.5\delta h s \log n.$$

Here $T = n^{\tau/8+o(1)}$, $\beta < \tau/4$, and h is fixed.

Every candidate in this posterior's support hits all observed cross anchors, coordinate by coordinate. Each observed ordinary tag i has a positive $S_g(i)$ factor in F_ξ . Its defining cutoff $d_i^+ \geq (1-\Delta)d_i^-$ therefore gives

$$\mu_i\{\text{hits } \xi \mid \text{hits } \Theta_{E(g)}\} \geq 1 - \Delta.$$

The conditioning set on the left does not change as ξ varies. These two support properties will provide the required edges.

Step 5: hidden-history conditioning and local selection. We now arrange that the fixed-list denominator test remains likely after selecting centers. There are two successive conditional probabilities. For fixed hidden tuples, let $q_{g,k}$ be the true-gated probability of $M < \varepsilon_0$ over the

hypothetical tags and cross anchors, with $k \leq T$ distinct internal IDs. The preceding estimate and Markov's inequality give

$$\Pr(q_{g,k} > \varepsilon_0^{1/2}) \leq \varepsilon_0^{1/2}.$$

Only k matters here, not positions or residual words. Use a local lemma on the independent hidden tuples to impose the base gates and $q_{g,k} \leq \varepsilon_0^{1/2}$ for every $k \leq T$. One combined event per grid key suffices. It uses hidden keys in $B_{\text{grid}}(g, 2)$, has probability

$$q_H \leq C e^{-n^{c'}} + (T+1)\varepsilon_0^{1/2} \leq e^{-n^{c_H}}$$

for some $c_H > 0$, and dependency degree $O((1+2s)^4)$. Thus charges $x_H = 2q_H$ suffice for large n . All posterior formulas continue to use the raw references of Step 4, rather than references conditioned on this avoidance event.

Fix such a hidden history and generate prospective positions and tags. For a candidate ID list at an odd cell, let q_L be the denominator-failure probability over hypothetical cross anchors after fixing its tags. The hidden-history bound gives

$$\Pr_{\text{tags}}(q_L > \varepsilon_0^{1/4}) \leq \varepsilon_0^{1/4}.$$

Call precisely these lists bad. At each odd cell consider all lists using present IDs in the balls of its incident even cells, at all levels, with at most T internal IDs and one per cross key. No assignment of internal IDs to particular ordinary neighbors is needed. On prospective ball counts between $\lambda/2$ and 2λ , their number is at most

$$L_n = \exp(C(s+T) \log n),$$

where C is independent of h . Lists with disjoint ID sets have independent tag outcomes, conditional on the hidden tuples and positions. Therefore

$$\Pr(n \text{ disjoint bad lists at a cell}) \leq L_n^n \varepsilon_0^{n/4} = \exp(n[C(s+T) - \delta h s/4] \log n).$$

Choose the fixed integer h so that $\delta h/4 > 2C$. Since $T = o(s)$, the exponent is negative of order $n s \log n$, which dominates the logarithm $O(n + s \log n)$ of the number of cells.

Choose a maximal disjoint family of bad lists at each odd cell, using a fixed ordering of its local lists, and mark their IDs forbidden at the incident even sites whose corresponding balls contain them. On the event just proved, each family contains fewer than n lists. An even site has $O(n+s)$ incident odd cells, so the total number of forbidden IDs per site-level is at most

$$O(n(n+s)(s+T)) = o(\lambda).$$

Prospective ball counts lie in the required range simultaneously with high probability by binomial tails. At least $\lambda/3$ choices therefore remain eligible at every site-level. Any used list whose IDs are all eligible cannot be bad: it would be disjoint from the marked maximal family.

Only now activate centers and select them by the long height rule. Eligibility is independent of every slice's activations. Lemma 3.8 gives good Lipschitz heights simultaneously in all slices with probability $1 - o(1)$: its failure exponent exceeds n by a power, whereas there are only $\exp(O(s \log n))$ grid keys. In particular every used list passes the conditional bound $q_L \leq \varepsilon_0^{1/4}$, and ordinary fans have at most T IDs.

We record the inputs of this selection rule. Eligibility at (t, b) examines only the bad-list families at its incident odd cells. Their lists read center positions and tags in

$$B_{\text{grid}}(t, 2) \times B_Q(b, r + 2)$$

and hidden tuples in $B_{\text{grid}}(t, 3)$. The conditional list test uses the fixed-presentation experiment, not any selected IDs or adjusted label kernels. The height rule stays within slice t and consults statuses at residual distance at most $4H$. Consequently selection at (t, b) uses center positions, tags, activations and ties within

$$B_{\text{grid}}(t, 2) \times B_Q(b, r + 4H + 2), \quad \text{and hidden tuples in } B_{\text{grid}}(t, 3).$$

All levels are included. This is a finite sequence of consultations: a list test never asks for another selection. Define the same local rules off successful histories, using local fallbacks when a law or test is undefined and independently randomized uniform choices among eligible active IDs when usable. Such fallbacks do not inspect success elsewhere.

Step 6: selection adjustment under candidate replacement. Fix positions and all hidden tuples except Θ_g . In a raw experiment draw $\Theta_g \sim R'$, generate the center tags and selections needed at the incident even cells under their resulting laws, and then draw their cross anchors. There is no further conditioning. A valid presentation requires the true base gate of Step 4, legitimate local selections with at most T internal IDs, incident-ball position counts at all levels at most 2λ , and $M \geq \varepsilon_0$. Ordinary anchor values are still absent.

Group presentations by their distinct internal IDs and their one cross ID at each key; call this list L . Its observed tags and cross pairs have product reference Q_L from Step 4. Given a candidate ξ , the subevent of validly presenting L has density

$$F_\xi a_\xi \quad \text{against } Q_L, \quad 0 \leq a_\xi \leq 1.$$

Indeed the listed centers must first have the observed independent tag outcomes. Selection and validity can only reduce their probability. After selection, the cross anchors have the indicated independent laws conditional on the selected tags. Thus the unselected product density F_ξ dominates the valid-presentation subdensity.

The factor a_ξ is obtained by disintegrating this raw experiment. In particular, when Θ_g changes, every affected center law S_u changes with it. The unobserved center tags and the local selections are integrated under those new laws, rather than held at their realized values. The cross-key reference deletes the entire dependence on Θ_g . The different lists are disjoint outcomes of this same candidate-dependent experiment, so

$$\sum_L \int F_\xi a_\xi dQ_L \leq 1 \quad \text{for each } \xi.$$

Step 7: posterior truncation and mean recovery. For the realized presentation put $M^a = \int F_\xi a_\xi dR'(\xi)$. Use the selected-presentation posterior $F_\xi a_\xi dR'/M^a$ when $M^a \geq \varepsilon_0 M$; otherwise use the base posterior $F_\xi dR'/M$. In either case its log density against uniform tuples is bounded by

$$hn^\beta + T(hn^\beta + n^{\tau/2} + O(1)) + O(hs) + 2\delta hs \log n < .003hs \log n.$$

The second denominator costs at most one further factor ε_0^{-1} .

Let q be the average coordinate marginal of this tuple posterior and $\mathcal{H} = \{y : Nq(y) > e^{.01s \log n}\}$. Then $|\mathcal{H}|/N \leq e^{-.01s \log n}$, and the joint density cap implies

$$\Pr(\text{at least } h/2 \text{ coordinates lie in } \mathcal{H}) \leq 2^h e^{.003hs \log n} (|\mathcal{H}|/N)^{h/2} = o(1).$$

It follows that $q(\mathcal{H}) \leq 1/2 + o(1)$. Delete $\mathcal{H}$ and normalize; the normalization factor is bounded by a constant. Denote this law by $p_{g,a}^0$, setting it to zero on invalid presentations.

For later loads, the important conclusion is its raw mean bound

$$\mathbb{E} p_{g,a}^0(y) \leq O(1) \mathbb{E}_\Lambda \nu_i(y) \leq O(K)/N.$$

Here the expectation is in the experiment of Step 6, at fixed positions and non-target hidden tuples. To verify it, valid data for list L have subdensity M_L^a against Q_L . On the selected-posterior cases, cancellation gives the following inequality of measures on the candidate tuple:

$$\sum_L \int_{\{M_L^a \geq \varepsilon_0 M_L\}} M_L^a \frac{F_\xi a_\xi dR'(\xi)}{M_L^a} dQ_L \leq dR'(\xi) \sum_L \int F_\xi a_\xi dQ_L \leq dR'(\xi).$$

On the other cases, $M_L^a < \varepsilon_0 M_L$ bounds their contribution by

$$\varepsilon_0 dR'(\xi) \sum_L \int F_\xi dQ_L \leq \varepsilon_0 L_n dR'(\xi) = o(dR'(\xi)).$$

The list count follows from the local position gate, and h was chosen large enough that $\varepsilon_0 L_n = o(1)$. Taking average coordinate marginals and paying the bounded truncation factor proves the claim.

The input scope of this whole computation will matter when its means are multiplied. The incident even selections are at grid and residual distance at most one from (g, a) . By Step 5, their raw center inputs lie in

$$B_{\text{grid}}(g, 3) \times B_Q(a, r + 4H + 3), \quad \text{and their hidden inputs lie in } B_{\text{grid}}(g, 4).$$

The only realized anchors read by $p_{g,a}^0$ are $W_{u,a}$, $u \in E(g)$. Its evaluation of a_ξ integrates unobserved tags and selection randomness inside the displayed region, at the fixed positions there. Candidate replacement changes these local laws and tests but does not enlarge their scope. Thus, after cross-anchor integration, the composite raw experiment has grid radius 4 and residual radius $r + 4H + 3 = r + o(n)$, and reads no realized anchors. Neither its validity gate nor a_ξ uses the later anchor-avoidance tests. We use this raw-computed posterior rule on the constrained histories as well.

Step 8: the ordinary-anchor hit test. The support of $p_{g,a}^0$ already hits every cross anchor. Now restrict it to labels hitting the realized anchors at all padded ordinary even neighbors, and require the retained mass to be at least .98. Call the resulting normalized law $p_{g,a}$.

To bound this test's failure probability, fix valid presentation data and cross anchors before conditioning the anchors. A support label of $p_{g,a}^0$ belongs to a coordinate of a supported candidate tuple. For any observed internal tag i , Step 4 bounds its miss probability under μ_i conditioned on cross hits by Δ . Conditioning additionally on the true own tuple increases this by at most $(1 - \Delta)^{-1}$. Thus its miss probability under $U_{g,i}$ is at most $\Delta/(1 - \Delta)$. Summing over the $O(n)$ ordinary anchors and applying Markov's inequality gives failure probability $O(n\Delta)$. On validity,

$$p_{g,a} \leq p_{g,a}^0 / .98, \quad N \max p_{g,a} \leq e^{.02s \log n},$$

and this law hits every incident actual even anchor. The reference $p_{g,a}^0$ still omits all ordinary anchor values.

8.2 Taking the assignment

Step 9: selected-anchor load bounds. Before sampling actual anchors we need, with high probability on successful hidden and selection histories,

$$\text{avg}_{v \in A} NU_{g(v), i(g(v), a(v))}(x) = O(1) \quad \text{for every } x.$$

There are two successive averages to control: over center selection at fixed hidden tuples, and over the hidden tuples themselves.

For the first, retain local validity, including the eligibility-size bounds on the height consultation domain. Given hidden tuples, force a specified center present and fix its tag. The height lemma bounds its gated selection probability, averaged over positions, by $O(1/\lambda)$ at level zero and e^{-n^c} at a positive level. The sum over possible centers has weight

$$\lambda O(1/\lambda) + H\lambda e^{-n^c} = O(1).$$

On the base gate the tag and anchor laws satisfy

$$S_g(i)U_{g,i}(x) \leq \frac{\eta_g(i)\mu_i(x) 1[x \text{ hits } \Theta_{E(g)}]}{(1 - \Delta)Z_g}.$$

Choose a fixed constant C_1 large enough for the preceding selection bound, and define, on the base gate,

$$B_g(x) := C_1 A_g^{-1} 1[x \text{ hits } \Theta_{E(g)}] \int N\mu_i(x) d\eta_g(i).$$

Set $B_g(x) = 0$ off that gate. The conditional mean of the gated normalized selected law is at most $B_g(x)$, whose cap is

$$L_B = \exp(O(hs + hn^\beta + n^{\tau/2})).$$

Under the raw hidden law, averaging Θ_g returns the posterior tag law to its prior, $\mathbb{E}\eta_g = \Lambda$. Independently averaging the cross tuples gives $\alpha_x^{|E(g)|} = O(A_g)$. Therefore

$$\mathbb{E}_{\text{raw}} B_g(x) \leq O(1) A_g^{-1} \alpha_x^{|E(g)|} \mathbb{E}_\Lambda N\mu_i(x) = O(K).$$

Apply Lemma 3.6 first to these comparison means on the hidden-history avoidance law. Declare rows near when their grid keys are within distance eight. Their fraction is at most $f_{\text{grid}} = n^{O(1)}(2n^{-.04})^s$, and $nf_{\text{grid}}L_B = o(1)$. For separated rows the relevant hidden neighborhoods are disjoint. Removing hidden constraints touching one such neighborhood costs $\exp(O((1+2s)^3 x_H)) = 1 + o(1)$; the remaining raw expectations are bounded by $O(K)$. The scattered-moment estimate and a union over x show that the row average of $B_g(x)$ is bounded simultaneously for all x , with high probability.

Fix a hidden history with these bounds and now average the selected laws over center randomness. Their cap is

$$L_U = \exp(n^\gamma + n^{2\tau} + O(1)),$$

by the lower cutoff on d_i^- and $d_i^+ \geq (1 - \Delta)d_i^-$. Two selections have disjoint center inputs when their residual words are more than $2r + 8H + 4$ apart, by Step 5. Up to a constant parity factor, the fraction of nearer words is at most

$$f_{\text{res}} = 2^{-(n-m)} \sum_{j \leq 2r+8H+4} \binom{n-m}{j} = e^{-\Omega(n)}.$$

Here $r = .01n + O(1)$, $H = o(n)$, and the binomial-sum exponent is strictly less than $n \log 2$. Since $\log L_U = o(n)$, we have $nf_{\text{res}}L_U = o(1)$. Conditional independence at separated words and the comparison means $B_g(x)$ now give the asserted selected-anchor load bound by a second application of the scattered-moment lemma. Both applications keep successful histories as indicators until the raw comparisons have been made.

Step 10: anchor avoidance and predictive alarms. Fix a successful hidden and selection history with the selected-law load bound. Failures of these earlier stages may be treated as aborts; their laws will not be reweighted by later avoidance probabilities. Start with independent anchors $W_{g,a} \sim U_{g,i(g,a)}$. At each odd cell the denominator test fails with probability at most $\varepsilon_0^{1/4}$, by its selected list's bound, and the ordinary-hit test has the $O(n\Delta)$ bound of Step 8. We will avoid these events together with the following predictive alarms.

At an actual even position v , write U_v for its selected anchor law and let its anchor z vary, holding all other anchors fixed. For the vector $\mathbf{y}$ of labels at its actual odd neighbors set

$$F_z(\mathbf{y}) = 1_{\mathcal{V}} \prod_{u \sim v} p_{g(u),a(u)}(y_u), \quad M_v(\mathbf{y}) = \int F_z(\mathbf{y}) dU_v(z).$$

Here $\mathcal{V}$ requires those neighboring presentations and ordinary-hit tests to be valid; it contains no predictive alarms. All these gates and kernels, including the p^0 lookups at observed cross anchors, are reevaluated when z varies. Center histories remain fixed.

Choose a product reference Q_v independent of z . At a neighbor of the same grid key use p^0 , or a fixed independent fallback if necessary: z is an ordinary anchor there and hence was omitted in the definition of p^0 . At all other neighbors use the uniform law. There are at most m such cross neighbors, so

$$F_z \leq .98^{-n} (e^{.02s \log n})^m Q_v \leq e^{.03n} Q_v, \quad ms \log n = o(n).$$

Define predictive failure by $M_v = 0$ or $M_v < e^{-.04n} Q_v$. The joint probability of this failure on $\mathcal{V}$, under the raw target anchor and product neighbor sampling, is at most $e^{-.04n}$, since the data subdensity is M_v . Markov's inequality therefore bounds by $e^{-.02n}$ the probability over anchors that its conditional product-sampling probability exceeds $e^{-.02n}$.

At a fixed cell only $\exp(O(s \log n))$ actual multiplicity profiles occur: the numbers of special-bit neighbors in each cross bin or the own bin determine the formula up to a permutation of the sampled coordinates. Combine their alarm failures into one event per cell. Its probability is at most $\exp(O(s \log n) - .02n)$. In the cell graph with grid or residual one-step moves, denominator and hit events have anchor radius one and predictive events have anchor radius two. The maximum degree of this graph is $O(n)$, so the grouped events have dependency degree $O(n^4)$. A uniform event bound is

$$q_A \leq \varepsilon_0^{1/4} + O(n\Delta) + \exp(O(s \log n) - .02n) \leq e^{-n^{c_A}}$$

for some $c_A > 0$. The conditional local lemma applies with charges $x_A = 2q_A$, at each fixed entering history. Sample the anchors from its avoidance law.

Step 11: odd loads by successive removal of constraints. We show that the odd kernel column sums are at most θ_0 , as required by the clock lemma, with high probability on success. Let $\mathcal{S}$ denote successful earlier histories and selections, and consider $l \leq n$ odd rows $b_j = (g_j, a_j)$ with pairwise grid distance greater than eight.

First use $p_{b_j} \leq p_{b_j}^0 / .98$. One anchor variable is touched by $O(n^2)$ of the grouped events from Step 10. Removing all events touching the at most $2s$ cross anchors of a row costs $\exp(O(sn^2 x_A)) = 1 + o(1)$. The cross anchors can then be integrated independently under their selected raw laws. No ordinary anchor remains in the integrand.

Next discard global selection success while retaining each local valid-presentation indicator in p^0 . Finally remove the hidden-history constraints touching the target tuples Θ_{g_j} . At most $O((1+2s)^2)$ hidden events touch one target, so this costs $\exp(O((1+2s)^2 x_H)) = 1 + o(1)$ per row. The radius-four scopes of Step 7 are disjoint. Conditional on positions and the other hidden tuples, the target tuples have their independent priors and the local center experiments use disjoint primitive inputs. Each experiment regenerates its center laws and selections when its candidate tuple changes, exactly as in Step 6. The raw mean bound of Step 7 therefore gives

$$\mathbb{E} \left[1_S \prod_{j=1}^l N p_{b_j}(y) \right] \leq \left(\frac{O(K)}{.98} \exp\{O(sn^2 x_A + (1+2s)^2 x_H)\} \right)^l \leq C^l.$$

Both charge sums are $o(1)$, even after multiplication by n .

For the remaining, grid-near rows, the cap is $L = e^{.02s \log n}$ and

$$n f_{\text{grid}} L \leq n^{O(1)} (2n^{-.04})^s e^{.02s \log n} = \exp(-.02s \log n + O(s + \log n)) = o(1).$$

Lemma 3.6, followed by Markov and a union over labels, now bounds every normalized average odd load by a constant with high probability. The actual column sums are $O(2^n/N) = o(1)$, because $C_n \rightarrow \infty$, and are therefore at most θ_0 . This is a joint estimate with earlier successes, rather than a claim that conditioning on those successes preserves independence.

Step 12: injection, even loads, and Hall. On a successful prehistory with the odd column bound, apply Lemma 3.10 with $B = 3$ to the actual odd rows. The predictive-failure predicate at an even position involves its at most n odd neighbors, each odd row occurs in at most n such predicates, and their product-law failure probabilities are at most $e^{-.02n}$. The atom cap from Step 8 is more than sufficient for the lemma's polynomial requirement. We obtain an odd injection avoiding all predictive failures and retaining the joint upper comparison on at most n^2 queries.

At an even position use the posterior row

$$w_v(x; \mathbf{y}) = \frac{U_v(x) F_x(\mathbf{y})}{M_v(\mathbf{y})}.$$

Its support consists of common neighbors of the assigned odd labels. In auxiliary experiments set it to zero unless local validity and predictive success hold. Since

$$N \max U_v \leq e^{n^\gamma + n^{2\tau} + O(1)} = e^{o(n)},$$

the likelihood comparison $F_x \leq e^{.03n} Q_v$ and predictive lower bound $M_v \geq e^{-.04n} Q_v$ give $N \max w_v \leq e^{.1n}$ for large n .

For its load estimate fix the entire center history, including the selected-law bound of Step 9; only anchors and odd outputs are varied. Select at most n even rows whose residual words have pairwise distance greater than four. Their odd-neighbor sets are disjoint, and their anchor calculations have residual radius at most two. Transfer their neighbor outputs jointly to product odd sampling, keeping the successful-prehistory indicator for this comparison. Then discard nonlocal requirements on the anchors and odd outputs and remove the anchor constraints touching each target anchor. Each removal costs $\exp(O(n^2 x_A)) = 1 + o(1)$, and another retained row's calculation does not consult that target.

Fix all other anchors. After integrating the target anchor, the product neighbor data with the local validity indicator have subdensity M_v . Thus posterior cancellation gives, for each label x ,

$$\sum_{\mathbf{y}} M_v(\mathbf{y}) w_v(x; \mathbf{y}) 1[\text{predictive success}] \leq U_v(x) \sum_{\mathbf{y}} F_x(\mathbf{y}) \leq U_v(x).$$

The same identity multiplies over the separated rows. Their comparison means are thus the selected laws whose average was bounded in Step 9. Residual-near rows have fraction at most $n^{O(1)}2^{-(n-m)}$, and

$$n \cdot n^{O(1)}2^{-(n-m)}e^{1n} = o(1).$$

The scattered-moment estimate shows that the joint probability of prior success and any normalized even load exceeding a fixed constant is $o(1)$. Intersecting its complement with the earlier high-probability events gives a successful history and odd injection for which all even loads are bounded. Their actual column sums are $O(2^n/N) = o(1)$, hence at most one for large n . Hall's theorem supplies distinct even representatives. Together with the odd injection and the common-neighbor support of every even row, this gives the monochromatic cube. $\square$

Corollary 8.2 (Asymmetric power discrepancy). *Consequently in a bad sequence with (7.2), after further stabilization for countably many budgets, we can assume: for every rational $0 < \beta < \tau/4$, $0 < \gamma < 1$ there is a constant $h_{\beta,\gamma} > 0$ such that on retained sets all pairs at widths $\leq n^\beta, n^\gamma$ respectively have both-color densities $1/2 \pm n^{-h_{\beta,\gamma}}$ eventually, and similarly in reversed orientation. The exponent may depend on the budget pair; no uniform exponent over all rational pairs is asserted.*

Proof. We may replace γ by $\max(\beta, \gamma)$, so assume $\beta \leq \gamma$. Apply Lemma 4.1 at the budgets n^β, n^γ , and write $h_* > 0$ for its bias and purity exponent. Choose $\beta < \beta^+ < \tau/4$ and $\gamma < \gamma^+ < 1$. For large n , the doubled budgets $2n^\beta, 2n^\gamma$, including an $O(1)$ trimming cost, fit within $n^{\beta^+}, n^{\gamma^+}$. Stabilize the bias property with threshold n^{-h_*} and the purity property with defect $e^{-n^{h_*}}$ at the doubled budgets.

If that purity is available, pass to one available color and trim the small-budget side for minimum degree with defect $e^{-n^{h_*/2}}$. A balanced mixture then meets Lemma 8.1 with the enlarged exponents β^+, γ^+ , placing the small-budget side on its odd side. That lemma gives a cube, so the purity alternative is absent. An available bias of at least n^{-h_*} at the original budgets would now give a cube by Lemma 4.1. Stabilization excludes that alternative too. Applying this to each rational budget pair, and to the reversed orientation, gives the stated discrepancy; the exponent is allowed to depend on the pair. $\square$

9 Intermediate powers of bias

The preceding reductions give discrepancy at small and unequal power widths. We now compare available bias across those budgets and budgets that are linear in n . Two limiting exponents record the transition. The goal is to exclude intermediate values, leaving the possible jump at a linear budget for Section 11, after the cluster exclusion.

Definition 9.1 (Limiting bias exponents). Stabilize the absolute-bias properties at all rational budgets and exponents used below. In a specified orientation X, Y , let $H(x, y)$ be the infimum of rational $h > 0$ for which bias of magnitude at least n^{-h} is available at widths n^x, n^y , where $0 < x, y < 1$. Set the infimum to $+\infty$ if there is no such exponent. If $h < H(x, y)$, stabilization makes this witness property eventually absent: every pair of laws within these budgets then has bias less than n^{-h} . Let $H_L(x, \alpha)$ denote the corresponding infimum with second budget αn , for $\alpha > 0$.

Both functions are nonincreasing in their width parameters, since enlarging a budget preserves availability. Taking limits through rational arguments, put

$$H^\dagger = \lim_{x \downarrow 0} \lim_{y \uparrow 1} H(x, y), \quad H_L^\dagger = \lim_{x, \alpha \downarrow 0} H_L(x, \alpha).$$

Proposition 9.2 (Exclusion of intermediate bias). *In the stabilized contradiction sequence, neither of the following situations can occur:*

- some $H^\dagger < 1$;
- both $H^\dagger \geq 1$, and some $0 < H_L^\dagger < 1$.

Proof.

Parameter selection. Here are parameter choices and what is then available, using the orientation in question. We arrange $0 < x_s < x_d < .1$, $u = x_s/2$, exponents $0 < h_- < h_+ < 1$, $x_d < (1 - h_+)/10$, and small $\sigma, \chi > 0$ with

$$\sigma < x_s/10, \quad \chi < \min(x_s, h_-, 1 - h_+)/100, \quad h_+ - h_- < \chi/10.$$

Write $a_* = \frac{1}{2}n^{-h_+}$, $b_* = n^{-h_-}$. At shallow budgets n^{x_s} , S_s absolute bias $\geq 2a_*$ is available; at deep budgets n^{x_d} , S_d all densities are $1/2 \pm b_*$ (meaning within the indicated error). Further choices:

Sublinear case. $S_s = n^{y_s}$, $S_d = n^{y_d}$, with $y_s < 1 - \sigma < y_d$, $\chi < \sigma/10$; use $m = \lfloor n^{y_m} \rfloor$ special bits with $y_s < y_m < 1 - \sigma$;

Linear case. $S_s = \alpha_s n$, $S_d = \alpha_d n$, with $100\alpha_s < \alpha_d < .01$, $\sigma < \chi/10$; use $m = \lfloor \alpha_d n/10 \rfloor$. Also densities against any law of width $O(1)$ on Y , for all first laws up to width n^{x_d} , are $1/2 \pm o(a_*)$.

The two cases use different orders of parameter selection: the inequalities specific to either case are imposed only in that case.

For the sublinear case, the thresholds we need are

$$h_- < H(x_d, y_d) \leq H(x_s, y_s) < h_+.$$

Choose a small rational interval $[x_1, x_2]$ within the first-budget range of Corollary 8.2, also small relative to the margin between $H^\dagger$ and 1. Choose a rational y -interval near, but bounded away from, 1. We may arrange that H is bounded above by a fixed number below 1 throughout the resulting rectangle, that x_2 is less than a small constant times this margin, and that $1 - y \ll x_1$. The value at the deep corner is positive by the asymmetric discrepancy corollary; monotonicity gives a positive lower bound throughout the rectangle.

Fix χ using these positive margins, including $\chi < (1 - y)/10$ throughout the rectangle. Subdivide its ascending diagonal into sufficiently many equal intervals. The total decrease of H is bounded, so one adjacent pair has decrease less than $\chi/20$. Use its endpoints as the shallow and deep budgets, and place h_- , h_+ strictly below and above their respective H -values, with $h_+ - h_- < \chi/10$. We can then choose $1 - \sigma$ strictly between y_s, y_d , and choose y_m between y_s and $1 - \sigma$. These strict choices provide the required width slack.

In the linear case, first choose fixed positive margins around $H_L^\dagger \in (0, 1)$. The nonincreasing function $x \mapsto \lim_{\alpha \downarrow 0} H_L(x, \alpha)$ tends to $H_L^\dagger$ as $x \downarrow 0$. Choose a small x -interval on which these values stay within the fixed margins. Since $H^\dagger \geq 1$, the interval can also be chosen so that a sublinear discrepancy bound with exponent $h_{\text{broad}} > h_+$ holds there. In particular, tests against laws of width $O(1)$ have error $n^{-h_{\text{broad}}} = o(a_*)$.

Fix χ using this interval and the margins. As above, monotonicity gives $x_s < x_d$ for which the decrease of the limiting linear exponent is less than $\chi/30$. Choose α_d sufficiently small, and then $\alpha_s < \alpha_d/100$ sufficiently small, to obtain

$$h_- < H_L(x_d, \alpha_d) \leq H_L(x_s, \alpha_s) < h_+.$$

Here σ may be chosen arbitrarily small after χ , so the linear-case conditions on it are compatible.

In either case, availability at the shallow threshold permits one color to be fixed along a subsequence. Restrict each first law to labels of degree at least $1/2 + a_*$ into its second law. Since the original surplus is at least $2a_*$, the retained mass is at least a constant times a_* ; this costs $O(\log n)$ width. Lemma 3.3 now gives a balanced finite mixture of patches (μ_i, ν_i) . Each first support has the stated minimum degree, the shallow widths have only this logarithmic extra cost, and the deep discrepancy bounds hold on all retained labels.

9.1 Core and read bound for center IDs

Write a cube word as a special word $z \in Q_m$ and a residual word in Q_{n-m} . A center ID records a slice z , a residual location and a level. We first construct and then fix a coloring-independent map from every cube site, of either parity, to an ID in its own slice at residual distance at most $r = \lfloor n^\sigma \rfloor$. Randomness is used here only to prove existence of this map. The anchor variables introduced later are fresh, independent draws.

The map will have the following properties:

- At an odd row, the adjacent even sites use at most $T + m$ distinct IDs, where $T = O(n^{1-\sigma+\varepsilon})$ and $\varepsilon > 0$ can be arbitrarily small.
- At an even row $v = (z, v')$, the IDs observed in all neighboring odd stars have a common core of size at most n^χ , including the ID of v . Every other ID occurs in at most $r + 3$ of these stars.

The adapted height construction. Let V_j be the volume of a radius- j ball in the residual cube, and put $V = V_r$. Independently at every slice/location/level, put a prospective center with probability n^{10}/V , and activate it with probability n^{b_0-10} . Here $b_0 > 0$ will be small. All prospective centers in the same-slice radius- r ball are eligible. Besides the event that this ball contains no active center, use the following crowd tests:

active centers counted	mean	upper threshold
same slice, radius r	n^{b_0}	$n^{\chi/2}$
adjacent slices, radius $r - 1$	$O(mn^{b_0}r/n)$	$n^{\chi/2}$
same slice, radius $r + 1$	$O(n^{b_0}n/r)$	$n^{1-\sigma+\varepsilon}$

Require these bounds at the levels within 2 of the level being tested, clipping that window at its endpoints. The mean estimates use $V_{r-1}/V = O(r/n)$ and $V_{r+1}/V = O(n/r)$. Choose $0 < \varepsilon < \sigma$ within the filter-budget slack below, and then $b_0 < \varepsilon$ sufficiently small in terms of χ . In the linear case also require $b_0 + \sigma < \chi/2$; in the sublinear case $y_m + \sigma < 1$ provides the corresponding slack. Thus each crowd threshold exceeds its mean by a fixed power of n .

We adapt Steps 2–4 of the proof of Lemma 3.8, using paths with step bound 2 in the full cube. The position counts in all eligibility balls are at least $n^{10}/2$ with high probability. The hole and crowd probabilities, including the bounded level windows, have the same base estimates as in that proof. The following overlap check permits the paths to cross slices.

A child domain of metric radius R' consults slices within $O(R')$ special-coordinate distance and residual locations within $r + O(R')$. For starts separated by KR' , whose level ranges overlap, either the consulted slice ranges are disjoint or the residual separation is $\Omega(KR')$. If the residual balls also intersect, their centers are at distance at most $2r + O(R')$, so $R' = O(r)$ for large fixed K . The shell and hypergeometric estimates from the height proof then bound their residual overlap

fraction by $\exp(-\Omega(KR' \log n))$. Enlarging the residual balls and enumerating the consulted slices costs only $\exp(O(R' \log n))$. Taking K large therefore leaves pair-level overlap volume at most $V e^{-\Omega(R')}$.

Use degraded thresholds at successive scales for every crowd count in the table and for the eligible-size count. Delete overlaps to obtain private child regions. Deletion preserves holes; its loss from each size or crowd count fits the corresponding threshold gap unless an overlap contains too many prospective or active centers. With q separated child domains, the binomial bound for each such exception is at most

$$\exp(-\Omega(n^{\chi/2} R'/q)).$$

The minimum crowd exponent is $\chi/2$. Denote the height proof's scale-spacing exponent by σ_h , to distinguish it from the present radius exponent. Choose its $a, \zeta, 1 - \theta, \sigma_h$ small enough that

$$0 < \sigma_h < \zeta, \quad b_0 > a > \zeta + \sigma_h + (1 - \theta), \quad \chi/2 > a + 4\sigma_h.$$

The base bound, the overlap bound and independence on private regions now give the same scale induction as in Lemma 3.8. This argument uses the three displayed crowd tests; it does not require the original lemma's full-cube radius- $(r + 2)$ crowd test. It gives good heights differing by at most one at full-cube distance at most 2. Choose a covered active center at every such height and fix the resulting map.

The core and its read multiplicity. At an odd star, the residual-flip neighbors use at most $O(n^{1-\sigma+\varepsilon})$ IDs by the radius- $(r + 1)$ crowd bound and the bounded level window. Its special-flip neighbors use at most m more. At v , put in the core every observed ID in the same-slice radius- r ball or in an adjacent-slice radius- $(r - 1)$ ball, at the relevant heights within 1. The two smaller crowd bounds make the core size at most n^χ .

To verify the read bound, let $\Delta(c, v')$ be the set of residual coordinates where a center c differs from v' . Suppose a same-slice ID outside the core is seen at the residual-flip neighbor b_i through an even neighbor $v'^{i,j}$. The case $j = i$ returns to v' and would put the ID in the core. Otherwise

$$d_H(c, v'^{i,j}) = d_H(c, v') + 2 - 2(1_{i \in \Delta(c, v')} + 1_{j \in \Delta(c, v')}).$$

The left side is at most r , while $d_H(c, v') > r$. Hence $i \in \Delta(c, v')$, and $d_H(c, v') \leq r + 2$. Such an ID is therefore seen at no more than $r + 2$ residual-flip neighbors. For an adjacent-slice ID outside the core, the one-flip identity gives $i \in \Delta(c, v')$ and $d_H(c, v') \leq r + 1$. At special-flip neighbors, same-slice IDs are already in the core, an adjacent slice contributes only at its own special flip, and a distance-two slice contributes at most twice. These bounds give $r + 3$.

Finally choose ε so that

$$S_s + O(n^u) + \log(1/.49)(T + m) < S_d.$$

In the sublinear case both T and m have exponent below y_d ; in the linear case $T = o(n)$ and $m = \lfloor \alpha_d n / 10 \rfloor$, while $100\alpha_s < \alpha_d$. Thus this choice is possible.

Tags and masks. Assign independent tags i_z from the balanced mixture, one per special word. Their averaged first and second laws have pointwise normalized loads $O(1)$ with high probability. In the n -th moment, a repeated word costs a fraction at most $n2^{-m}$, multiplied by its normalized atom cap. This product tends to zero: $m \gg S_s$ in the sublinear case, and $m \log 2 > S_s$ with linear slack in the linear case. The first-law cap is smaller still.

In the linear case we also choose the tags so that, for every z , except on μ_{i_z} -mass $\exp(-\Omega(n^u))$,

$$\sum_{j \leq m} (d_G(w, \nu_{i_{zj}}) - 1/2) \geq -.05a_*n.$$

Here z^j is obtained by flipping bit j . To justify this choice, first use the raw product law of the tags and draw $w \sim \mu_{i_z}$. The broad-test bound gives degree $1/2 \pm o(a_*)$ into the tag-average second law outside $\exp(-\Omega(n^u))$ first-law mass. Indeed, conditioning on a larger exceptional set would stay inside the first deep budget. Deep discrepancy also gives individual degrees $1/2 \pm 2b_*$ except with the same exponential scale of joint (w, i_{zj}) -probability. By Markov, outside an exponentially small set of w 's, the conditional probability of such an irregular neighboring tag is exponentially small.

Clip every summand to $[-2b_*, 2b_*]$. Conditional on w , these clipped summands depend on independent adjacent tags. Their downward deviation of order na_* has probability $\exp(-\Omega(na_*^2/b_*^2))$. Adding the clipping exceptions proves a joint tail $\exp(-\Omega(n^u))$. If $E_z(w)$ denotes failure of the displayed sum, choose a small fixed $c > 0$ and put

$$B_z = \{\mu_{i_z}(E_z) > e^{-cn^u}\}.$$

Markov gives $\Pr(B_z) = \exp(-\Omega(n^u))$. The event B_z reads only the tag at z and its neighbors, so the dependency degree is polynomial. Lemma 3.5 allows all B_z 's to be avoided with exponentially small charges. The preceding tag-load moment still applies: removing constraints touching at most n separated tag neighborhoods has charge sum $\text{poly}(n)e^{-c'n^u} = o(1)$. Fix tags satisfying these conditions and the load bounds.

Each ID c in slice z now carries an independent anchor $W_c \sim \mu_{i_z}$. At an odd row b , let $\nu = \nu_{i_{z(b)}}$. Choose a mask of ν -mass at least $\frac{1}{2}e^{-n^u}$, normalize its restriction, and require hits of the anchors at every distinct neighboring ID. On a nonempty filter let p_b be the normalized result; otherwise use the initial masked law as an auxiliary fallback.

Choose the mask distribution separately at each row by convex separation, before drawing the anchors. For any nonnegative price, the labels costing at most $1 + e^{-n^u}$ times its ν -mean have mass at least $\frac{1}{2}e^{-n^u}$, and hence form an allowed mask. Every output from that mask has the same price bound, including the fallback. Separation therefore supplies a distribution of masks for which

$$R_b := \mathbb{E}p_b \leq (1 + e^{-n^u})\nu.$$

The expectation here averages all anchors used at b and its mask, before any further conditioning. Masks are drawn independently across rows. Since R_b and ν are probability laws, their total-variation distance is exponentially small. This is an unconditional mean identity; it is not asserted after a core has been fixed.

9.2 Gains after conditioning the core

All probabilities in this subsection use independent anchors and masks, with the tags and ID map fixed. For an even role v , write $W_* = W_{c(v)}$. At each $b \sim v$, let q_b be the fraction retained by imposing the hit of W_* after all other hits in that filter. We will prove, with failure probability $\exp(-\Omega(n^u))$ per even star, that all these filters have width less than S_d and

$$\sum_{b \sim v} \log q_b \geq -n \log 2 + c_4 na_*, \quad (9.1)$$

where $c_4 > 0$ is a small absolute constant.

The required regularity tests. Fix an ordering of all ID names. For each $b \sim v$, order its IDs outside the core first, then its core IDs other than $c(v)$, and finally $c(v)$, keeping the fixed order inside the first two blocks. We use this order with the masked starting law and with the unmasked starting law. We also use the order of the core IDs other than $c(v)$, starting from the unmasked law. Include all prefixes of these three orders. The masked prefixes give the outer filter, the deletion filter and the target-last fraction; the unmasked core prefixes are the laws used in the covariance telescoping argument below. There are $O(n(T + m))$ such prefix tests per star. The degree and signed-density tests below are applied at these same prefixes, so their number is polynomial as well. No assertion over all permutations or all subsets of IDs is needed.

At a fixed prefix whose preceding hits each retain at least .49, the current second-law width is at most

$$S_s + O(n^u) + \log(1/.49)(T + m) < S_d,$$

with room for the further $O(n^u)$ restrictions below. If the next hit fraction differs from $1/2$ by more than $2b_*$ on anchor mass at least $e^{-O(n^u)}$, condition its first law on one signed exceptional set. Its width is at most

$$n^{x_s} + O(n^u) + O(\log n) < n^{x_d},$$

and its density against the current second law contradicts deep discrepancy. Thus each next hit fraction is $1/2 \pm 2b_*$, except with probability $\exp(-\Omega(n^u))$. Conditional application at each prefix and a union over the stated family prove all required sequential tests. We enlarge the constant implicit in T if necessary.

We will also use the reverse form of this reasoning: for any first law within the deep budget, an exceptional set of second labels of mass at least e^{-n^u} on which its degree differs by more than $2b_*$ could be normalized and used as a second test law. The available width room excludes this. In particular, each actual $q_b = 1/2 \pm 2b_*$ except with the stated tail. The remaining issue is a positive sum of their surpluses.

For the fixed even role v , let C_v be the core specified above, common to all its neighboring stars. If I_b is the ID set observed at $b \sim v$, put

$$D_b = I_b \cap C_v, \quad O_b = I_b \setminus C_v.$$

Condition on the entire anchor vector $W_{C_v} = (W_c)_{c \in C_v}$, keeping the tags and the deterministic ID map fixed. At a given b , write $D = D_b$, $O = O_b$, and $k = |D| \leq n^x$; the target ID $c(v)$ belongs to D . Let J_-, J_D be the sets of second labels hitting D except the target, or all of D , respectively. Let P_O be the masked law filtered just by outer hits, with a width-preserving fallback (initial masked law) if their retained mass is below $.49^{|O|}$. Write $Q = \mathbb{E}P_O$, averaging the outer anchors and this row's independent mask; thus Q is independent of W_{C_v} . Core coordinates outside I_b do not enter this filter. Clip q_b to $[1/2 - 2b_*, 1/2 + 2b_*]$, with a fixed local handling of invalidity there (using only this filter's inputs), calling it $\hat{q}_b$.

Conditional Markov applied to the outer-then-core tests shows that, outside a set of core histories of mass $\exp(-\Omega(n^u))$, their conditional failure probability is also $\exp(-\Omega(n^u))$. On these histories, outside that conditional exception,

$$P_O(J_-) = 2^{1-k}(1 + O(kb_*)), \quad P_O(J_D) = P_O(J_-)\hat{q}_b.$$

Consequently

$$d_G(W_*; Q|_{J_-}) = \frac{\mathbb{E}[P_O(J_-)\hat{q}_b \mid W_{C_v}]}{\mathbb{E}[P_O(J_-) \mid W_{C_v}]} + O(2^k e^{-\Omega(n^u)}).$$

The fraction is a weighted mean of $\hat{q}_b$. Its weight differs from the constant 2^{1-k} by relative $O(kb_*)$, while $\hat{q}_b$ has oscillation $O(b_*)$. Removing this weight therefore costs $O(kb_*^2)$, rather than $O(kb_*)$. Since $k \leq n^\chi = o(n^u)$, the exceptional error remains exponentially small. We obtain

$$\mathbb{E}[\hat{q}_b \mid W_{C_v}] = d_G(W_*; Q|_{J_-}) + O(kb_*^2) + \exp(-\Omega(n^u))O(1).$$

Here $d_G(w; \lambda)$ is the degree of one label into a probability law, and every restriction bar denotes normalization.

Erasing the core from the mean filter. The law Q averages only the outer anchors and the mask. To compare it with ν , temporarily average the core as well, returning to the unconditional law R_b above. Independence of distinct core anchors gives the exact identity

$$2^k \mathbb{E}[P_O(y)1_{J_D}(y)] = a_D(y)Q(y), \quad a_D(y) = 2^k \prod_{c \in D} d_G(\mu_{i_{z(c)}}, y).$$

On regular histories the full filter is $P_O|_{J_D}$, and its normalizing multiplier is $2^k(1 + O(kb_*))$. Averaging this bounded multiplier pointwise yields a function θ with $|\theta| \leq Ckb_*$ such that

$$R_b = (1 + \theta)a_D Q + e, \quad \|e\|_1 = \exp(-\Omega(n^u))O(1).$$

To see the error estimate, extend the good-history multiplier divided by 2^k by 1 on bad histories. Replacing the actual kernel there costs at most their probability times $1 + 2^k$, which preserves the exponential scale.

The fallback in P_O gives it, and therefore Q , width at most $S_s + O(n^u) + \log(1/.49)|O|$. The reverse degree tests just proved apply to Q . Outside a Q -set of mass $\exp(-\Omega(n^u))$, every doubled degree factor $2d_G(\mu_{i_{z(c)}}, y)$ is $1 + O(b_*)$, so $a_D = 1 + O(kb_*)$. On the exceptional set use $a_D \leq 2^k$. Since $kb_* = o(1)$, we can invert the resulting multiplier and use the exponentially small distance between R_b and ν . This gives

$$Q = (1 + \psi)\nu + \text{err}, \quad |\psi| \leq O(kb_*), \quad \|\text{err}\|_1 = \exp(-\Omega(n^u))O(1).$$

All measures and multipliers in this identity are independent of the core data.

Now fix the core except for W_* . The unmasked regularity tests give $\nu(J_-) \geq .49^k$ outside an exponentially small exception. Restricting the last identity to J_- magnifies its total-mass error by at most $O(.49^{-k})$, still negligible. Put $\lambda = \nu|_{J_-}$. The normalized multiplier changes λ by a signed density

$$s = \frac{\psi - \mathbb{E}_\lambda \psi}{1 + \mathbb{E}_\lambda \psi}, \quad \mathbb{E}_\lambda s = 0, \quad |s| \leq Ckb_*.$$

Thus $\tau = (1 + s/(Ckb_*))\lambda$ is a probability law whose width exceeds that of λ by at most $\log 2$. The target anchor W_* is independent of these two laws. The individual-degree test gives $d_G(W_*; \tau) - d_G(W_*; \lambda) = O(b_*)$, except with the stated exponential tail. Rescaling back to the actual modulation gives

$$d_G(W_*; (1 + s)\lambda) - d_G(W_*; \lambda) = Ckb_*(d_G(W_*; \tau) - d_G(W_*; \lambda)) = O(kb_*^2).$$

Therefore replacing $Q|_{J_-}$ by $\nu|_{J_-}$ in the conditional mean costs $O(kb_*^2)$.

The effect of each core hit. It remains to compare $d_G(W_*; \nu|_{J_-})$ with $d_G(W_*; \nu)$. At a prefix of the unmasked core order, let λ be the current law, let w be the next anchor, and write $x = W_*$. These two anchors are independent of λ and each other. With $g_w(y) = 1[w \sim_G y]$, the degree change on imposing the next hit is

$$d_G(x; \lambda|_{N_G(w)}) - d_G(x; \lambda) = \frac{\text{cov}_\lambda(g_w, g_x)}{d_G(w; \lambda)}.$$

The denominator is at least .49 on regularity. We claim that the covariance has magnitude at most $a_* n^{-2\chi}$, except with probability $\exp(-\Omega(n^u))$.

Suppose instead that its bad-pair probability is at least e^{-n^u} . Choose one sign and an x -set X_0 of mass at least $\rho = e^{-O(n^u)}$ such that, for every $x \in X_0$, the w -set B_x with that signed violation has mass at least ρ . Draw $t = \lceil n^{8\chi} \rceil$ independent w_j 's from the anchor law conditioned on B_x . This conditioned law still fits the first deep budget. Its mean adjacency indicator is $1/2 \pm 2b_*$ outside a λ -set of exponentially small mass, by the reverse degree test. Hence, for

$$S = \sum_{j \leq t} (g_{w_j} - \mathbb{E}_\lambda g_{w_j}),$$

independence gives

$$\mathbb{E} \|S\|_{L^2(\lambda)}^2 = O(t + t^2 b_*^2) = O(t),$$

since $tb_*^2 = o(1)$. For an absolute K , the tuple event $\mathcal{G} = \{\|S\|_{L^1(\lambda)} \leq K\sqrt{t}\}$ thus has conditional probability at least a fixed $c > 0$.

The event $\mathcal{G}$ depends on the tuple and the fixed law λ , not on x . Returning to the raw independent tuple law gives

$$\int_{X_0} \Pr(\mathcal{G}, w_j \in B_x \text{ for all } j) d\mu(x) \geq c\rho^{t+1}.$$

Here μ denotes the target anchor law. Fubini therefore supplies one fixed norm-good tuple whose simultaneous signed-violation set of x 's has mass at least $c\rho^{t+1} = e^{-O(tn^u)}$. Restricting μ to that set costs $O(n^{u+8\chi})$ width, which fits the first deep budget because $u + 8\chi < x_s$.

For this fixed tuple, $\mathbb{E}_\lambda S = 0$, so the laws

$$\lambda_\pm = \frac{\sqrt{t} + S_\pm}{Z} \lambda, \quad Z = \sqrt{t} + \mathbb{E}_\lambda S_+ = \sqrt{t} + \mathbb{E}_\lambda S_- = O(\sqrt{t})$$

have equal normalizers. Here $S_\pm = \max(\pm S, 0)$. Their multipliers are $O(\sqrt{t})$, costing only $O(\log n)$ second width. Deep discrepancy, with the target law restricted to the simultaneous violation set, bounds the average over that law of $\mathbb{E}_\lambda(g_x S)$ in magnitude by $O(b_* \sqrt{t})$. The simultaneous violations force it to exceed $ta_* n^{-2\chi}$ in the chosen sign. This is impossible, since

$$\frac{\sqrt{t} a_* n^{-2\chi}}{b_*} \asymp n^{2\chi - (h_+ - h_-)} \longrightarrow \infty.$$

This proves the covariance tail.

There are at most $k \leq n^\chi$ core hits. Their total degree shift is $O(ka_* n^{-2\chi}) = o(a_*)$. Also $kb_*^2 = o(a_*)$. Combining this with the erasure comparison, and taking the union over $b \sim v$, gives outside exponentially rare core data

$$\mathbb{E}[\hat{q}_b \mid W_{C_v}] = d_G(W_*; \nu_{i_{z(b)}}) + o(a_*).$$

Concentration with the core fixed. The comparisons above hold for the fixed tags. Every ordinary neighbor uses the same slice as v , so its conditional mean surplus is at least $a_* + o(a_*)$. Their sum is at least $(1 - o(1))(n - m)a_*$. In the linear case the additional tag condition bounds the sum over special neighbors below by $-.06na_*$. In the sublinear case the bound $mb_* = o(na_*)$ gives the required estimate without that condition.

After fixing W_{C_v} , the remaining inputs are independent outside-core anchors and independent row masks. Each clipped fraction reads one row mask, and each remaining anchor is read at most $r + 3$ times. Exponential Markov and the product-space Hölder inequality [11] therefore bound a downward deviation of $.1na_*$ by

$$\exp\left(-\Omega\left(\frac{na_*^2}{(r+3)b_*^2}\right)\right).$$

Here the product-space inequality bounds the expectation of a product of nonnegative functions by the product of their d -norms when each independent input is read at most d times. We apply it with $d = r + 3$ in this raw conditional experiment, before avoidance is imposed. The parameter choices give

$$\frac{na_*^2}{(r+3)b_*^2} = n^{1-\sigma-2(h_+-h_-)+o(1)} \gg n^u.$$

Thus the tail has the required strength. On individual regularity, the clipped fractions equal the actual q_b 's. Finally

$$\log(\tfrac{1}{2} + t) = -\log 2 + 2t + O(t^2), \quad |t| \leq 2b_*.$$

The quadratic loss is $O(nb_*^2) = o(na_*)$, so the positive surplus proves (9.1).

9.3 Taking the assignment

Anchor avoidance and the predictive tests. At every even star require the neighboring filter tests just enumerated and the log-gain bound (9.1). For a target v , fix all anchors except W_* and all masks. For candidate $W_* = x$, define the data sublikelihood

$$F_x(\mathbf{y}) = 1_{\mathcal{V}} \prod_{b \sim v} p_b(y_b),$$

where $\mathcal{V}$ is this star's filter-validity and log-gain event, recomputed with x . Let P^- be the product of the deletion kernels, each obtained by normalizing the same filter without the target ID. A zero deletion denominator uses a fixed fallback independent of x . On $\mathcal{V}$, (9.1) gives

$$F_x \leq 2^n e^{-c_4 na_*} P^-.$$

Put $M_v = \int F_x d\mu_{i_z}(x)$, where $z = z(v)$. Define predictive failure by $M_v = 0$ or $M_v < e^{-c_4 na_*/4} P^-$. Lemma 3.7 bounds its true-gated probability in the raw anchor-and-product-data experiment by $e^{-c_4 na_*/4}$. Markov then shows that the anchor event

$$1_{\mathcal{V}} \Pr \left(\text{predictive failure} \right) > e^{-c_4 na_*/8}$$

has probability at most $e^{-c_4 na_*/8}$.

Each star event reads IDs within residual distance $r + O(1)$ and bounded special-coordinate distance. Its dependency degree, and the number of events touching any one local list, are at most

$\text{poly}(n)e^{O(r \log n)}$. Since $\sigma < u$, charges e^{-cn^u} , with small fixed $c > 0$, meet Lemma 3.5 for the star and predictive events. Moreover,

$$\text{poly}(n)e^{O(r \log n)}e^{-cn^u} = o(1),$$

so removing the touching constraints for any polynomial number of queried local lists costs $1 + o(1)$.

Odd column loads and injection. Fix a second label y and use normalized weights $Np_b(y)$ in Lemma 3.6. A row has cap e^{S_d} on success. Declare rows near when their special words are at bounded distance and their residual words at distance $O(r)$, with constants large enough to contain the declared input lists. The near fraction is at most $2^{-n} \exp(O(r \log n))$, and

$$\exp(O(r \log n) + S_d - n \log 2) = o(1).$$

For separated rows remove the avoidance constraints touching their independent local anchor and mask inputs. Their raw means are $R_b \leq (1 + e^{-n^u})\nu_{i_z(b)}$, whose normalized average is bounded by the fixed tag balance. The charge sum just estimated supplies the separated-product comparison. Scattered moments therefore give column sums $O(2^n/N) = o(1)$, simultaneously, with high probability.

At such a prehistory apply Lemma 3.10 to the odd rows, with the predictive failures as predicates. Each predicate reads at most n rows and each row occurs in at most n predicates; the atoms and failure probabilities are smaller than any required inverse power. This gives an odd injection avoiding every predictive failure and satisfying the required joint product comparisons.

Even posterior rows and Hall. At v , use the posterior $F_x d\mu_{i_z}(x)/M_v$ as a row on its common neighbors. Set this row to zero off its local gate or predictive success when estimating moments. The likelihood and denominator bounds, together with $n^{x_s} = o(na_*)$, give normalized cap

$$L_X \leq 2^n e^{-c_4 na_*/2}.$$

The same near relation has repeat cost $\exp(O(r \log n) - c_4 na_*/2) = o(1)$, since $r \log n = o(na_*)$.

For separated even rows, retain the successful entering-history indicator while using the clock comparison on their disjoint odd neighbor sets. Only after replacing those outputs by their product laws do we remove nonlocal success indicators. Keep every true local star gate. Remove the anchor-avoidance constraints touching the target IDs, hold the other anchors and masks fixed, and integrate a target and its data. The cancellation is

$$\sum_{\mathbf{y}} M_v(\mathbf{y}) \frac{F_x(\mathbf{y}) \mu_{i_z}(x)}{M_v(\mathbf{y})} = \mu_{i_z}(x) \sum_{\mathbf{y}} F_x(\mathbf{y}) \leq \mu_{i_z}(x),$$

with zero terms where $M_v = 0$. The fixed-map locality ensures that no other retained calculation consults this target ID. These integrations consequently multiply. The avoidance cost is $1 + o(1)$, and the normalized comparison means have bounded average by tag balance. Scattered moments now give even column sums $O(2^n/N) = o(1)$, hence at most 1, with high probability on success. All even rows have mass 1 there and are supported on common neighbors of the odd injection. Hall's theorem finishes the cube and excludes both stated intermediate situations. $\square$

10 Full-dimensional cluster patches

We now exclude patches in which diffuse clusters of second-side labels have uniformly large common-neighbor mass on the first side. Under the initial discrepancy bound, the next proposition turns an available family of such patches into a cube. Their eventual absence will be used twice: to exclude the linear-budget jump in Section 11 and to control the residual cluster scale during patch extraction in Section 13.

Proposition 10.1 (Full-dimensional cluster exclusion). *Assume (7.2) on the retained sets in both orientations. Fix constants $\zeta > 0$ and $0 < \delta < \min(\eta_0, \zeta, 1)/2000$. Suppose there are available patches of **one** color G as follows:*

- *a law μ_i on the first side and a finite probability mixture λ_i of cluster laws D on the second side, with $\nu_i = \mathbb{E}_{\lambda_i} D$; both μ_i, ν_i have width $\leq n^\delta$, and each D has maximum atom $\leq \exp(-n^\zeta)$;*
- *within the support of each D all pairs y, y' (repetitions allowed) have G -codegree (common-neighbor mass) under μ_i at least $1/4 + a$, $a = n^{-\delta}$.*

Then there is a cube.

Proof.

The finite patch menu. In proving this we can limit patch tags i to a finite menu still available with fixed discard fraction (retain, for each allowed discard set, a witness off it). Use X, Y for first and second sides. Take

$$m = \lfloor n^{200\delta} \rfloor, \quad k = \lceil n^{300\delta} \rceil, \quad T = \lceil n^{141\delta} \rceil.$$

We split off m bits, writing z for their exact word. Tags i_z will be independently sampled from product profiles to be fixed later.

10.1 Projections and lists

Write $d = n - m$ as a sum of distinct powers of two and split the residual coordinates into chunks of those sizes H' . Index the bits in one chunk by the elementary binary group of order H' . For a word s , its syndrome is $u(s) = \sum_l l s_l$. Define its projection to be $s \oplus e_{u(s)}$. The projected syndrome is $u(s) + u(s) = 0$. Conversely, a zero-syndrome word has exactly the H' preimages obtained by flipping one of its bits. This uses the binary syndrome correction rule of Hamming's parity-check construction [21, Sections 3–4]. The index 0 is also a bit position, so exactly one bit is always flipped.

Use the product projection ψ on residual words of both parities. It toggles parity by the number of chunks modulo two. In a fixed slice z , call the actual odd roles with the same projected state a group. The projected even states are the sites where centers will be selected. Across a residual edge only one projected chunk changes, in at most three coordinates; a special-coordinate flip leaves the projected residual word fixed. Thus the even projected sites incident to one odd group have pairwise residual distance at most 6 in its own slice, and there is one incident site in each adjacent slice.

A changed chunk admits at most $O(H'^3)$ neighboring projected states. Hence each even projected site belongs to at most $O(\sum H'^3) = O(n^3)$ group neighborhoods. The size of a projection fiber is $\prod H' \leq n^{O(\log n)}$, since there are $O(\log n)$ chunks. In particular the group sizes have this bound.

At a single actual even role v , partition its ordinary neighbors by odd group. Different chunks of the flip give different groups (the changed projected chunk has opposite parity to that at v). Within one chunk the neighbor projections are $s \oplus e_l \oplus e_{u(s)+l}$. If the syndrome $u(s) \neq 0$, multiplicities are 2; if zero, the single multiplicity is H' . Thus these ordinary incidences have multiplicities $j \geq 2$, with at most one singleton (possible unit chunk). Special incidences are all singleton groups.

Use Lemma 3.8 independently in each slice, with step bound 6 among even projected sites and parameters

$$r = \lfloor n^{1-10\delta} \rfloor, \quad \lambda = n^{10}, \quad b_0 = 50\delta, \quad b = 140\delta.$$

For the height lemma's parameters $\zeta, \sigma, 1 - \theta, a$, unrelated to the cluster parameters here, take $8\delta, \sigma_g, \delta, 20\delta$, with $0 < \sigma_g \ll \delta$. The sublinear-radius condition is satisfied, and the height horizon H obeys $r + H = O(n^{1-7\delta})$. An ID c consists of a slice, a residual location and a level. Independently pre-generate $W_c \sim \mu_{i_z}^{\otimes k}$ at every possible ID in slice z , including IDs that will be absent from the position field.

At each odd group in slice z , choose a mask of the cluster prior. The finite menu of permitted masks is defined as follows. Choose a subset of second labels, retain clusters assigning it mass at least $1/2$, provided their total prior probability is at least $1/2$, and condition each retained cluster on that subset. Normalize the mixture weights as well. The resulting aggregate is at most $4\nu_{i_z}$; codegrees on the retained supports are unchanged, and cluster atoms increase by at most a factor 2. The trivial mask is permitted. The distribution over masks will be specified below. It depends only on the tags at z and its adjacent slices; conditional on tags, masks are independent across groups and independent of the positions and tuple arrays.

The randomization order is now fixed. After the tags, generate prospective positions and the independent tuple arrays, then the group masks. The eligibility rule below uses these data but no activations. Afterwards generate independent activations and site-level selection ties. Only after the centers have been selected do we draw clusters and then labels. Uniform ties are pre-randomized separately at every site-level.

On successful heights, own-slice IDs in a group neighborhood number at most T . Indeed the incident projected sites have heights differing by at most one, and a crowd ball about one representative at each occupied height contains all choices at that height. There is one further ID per special neighboring slice. Every group therefore sees at most $T + m$ distinct IDs.

Tests for hypothetical lists. Consider at a group a hypothetical list $\mathcal{D}$ of distinct involved IDs, at most T of its own slice and exactly one from each adjacent slice. Write F for the set of labels hitting (in G) all tuple entries there; use F_{-c} on deleting one of the tuples. For the masked cluster mixture put

$$A(F) = \mathbb{E}_D D(F)^2.$$

Test the following bounds:

$$A(F) \geq e^{-2k|\mathcal{D}|}, \quad A(F)/A(F_{-c}) \geq \begin{cases} e^{(-\log 4 + .4a)k} & c \text{ own,} \\ e^{-2k} & c \text{ external.} \end{cases} \quad (10.1)$$

We claim that, conditional on positions, masks and tags, a fixed list fails these tests with probability at most $\exp(-c_5 a^2 k)$, for an absolute $c_5 > 0$. Only its independent tuple entries remain random. Fix one order of its tuples, and for each tuple also use the order obtained by putting it last. Thus at most $|\mathcal{D}| + 1$ orders suffice for the absolute bound and all deletion bounds.

After a set J of second labels has survived the preceding hits, exposing the next first label x multiplies $A(J)$ by

$$q(x) = \mathbb{E}_D \text{ tilted by } D(J)^2 d_G(x; D|J)^2.$$

The aggregate law of $D|_J$ under this tilt is

$$\overline{D}_J = \frac{\mathbb{E}_D[D(J)1_J D]}{A(J)} \leq \frac{4\nu_{i_z}}{A(J)}.$$

Jensen's inequality gives $q(x) \geq d_G(x; \overline{D}_J)^2$. As long as preceding ratios are at least .24, the width of $\overline{D}_J$ is at most

$$n^\delta + O(1) + 2k(T + m), \quad k(T + m) = O(n^{500\delta}) = o(n^{\eta_0}).$$

The initial discrepancy bound (7.2) therefore implies $q(x) \geq 1/4 - O(n^{-\eta_0})$, except with probability at most $e^{-n^{\eta_0}/2}$. Indeed, conditioning an incoming first law on a larger exceptional set would still fit that discrepancy bound and contradict the corresponding degree estimate.

For an own-slice entry, the codegree hypothesis additionally gives, at every such prefix,

$$\mathbb{E}_{x \sim \mu_{i_z}} q(x) = \mathbb{E}_D \text{ tilted by } D(J)^2 \mathbb{E}_{y, y' \sim D|_J \mu_{i_z}} (N_G(y) \cap N_G(y')) \geq 1/4 + a.$$

Clip the logarithmic increment below at $\log(.24)$. Its typical lower bound is $-\log 4 - O(n^{-\eta_0})$. Its expected surplus is at least $.7a$ for large n : use the last mean bound and $\log v - \log u \geq v - u$ for $0 < u \leq v \leq 1$, with the rare exceptional values charged separately. Thus the conditional mean increment for an own tuple is at least $-\log 4 + .7a$.

Expose the k independent entries of the tuple placed last. The clipped log increments are bounded, so their deviation below $(-\log 4 + .4a)k$ has probability $\exp(-\Omega(a^2 k))$. Stop this calculation if a preceding regularity bound fails and continue with fixed good increments after the stop; the stopped-prefix failures are bounded by the discrepancy exceptions already estimated. For an external last tuple the typical lower bound gives the weaker threshold $-2k$; processing a whole list gives the absolute threshold $-2k|\mathcal{D}|$. A union over the stated orders proves the fixed-list claim.

Eligibility and local validity. For each group, enumerate the hypothetical lists of the prescribed form using present IDs in its incident even-site balls, at any levels. A failed list is one that violates (10.1). Choose a maximal pairwise disjoint family of failed lists and mark every ID in that family as forbidden at each incident even site where it could be selected. The enumeration and maximal-family rule can be fixed deterministically. Every failed list meets the marked family by maximality; thus a list consisting entirely of eligible IDs passes the tests.

With probability $1 - o(1)$, all radius- r position counts at all site-levels are within relative error .002 of λ . On these position bounds, the number of lists per group is at most $L_n = \exp(O((m + T) \log n))$. After fixing masks and tags, failures on disjoint lists use disjoint tuple entries and are independent. If $p_n = e^{-c_5 a^2 k}$, the chance of n disjoint failures at one group is at most

$$L_n^n p_n^n = \exp(-\Omega(n a^2 k)), \quad (m + T) \log n = o(a^2 k).$$

This bound also permits the union over all groups. A site lies in $O(n^3)$ group neighborhoods. Fewer than n failed lists, each of size at most $T + m$, at every such group remove at most $O(n^4(T + m)) = o(\lambda)$ IDs from its ball. Hence at least $.99\lambda$ choices remain at every site-level, with high probability.

Now activate prospective centers independently and apply the height rule. Its largest-scale estimate gives simultaneous success in all slices on these eligibility sizes. Eligibility may depend on other slices; the height bound is uniform over all fixed eligible sets of the required sizes. The actual lists then consist of eligible IDs and pass (10.1).

Let $\mathcal{V}_g$ be the following local event for a group g : its incident position counts obey the stated bounds at every level; all eligibility sets consulted by the needed height rules have size at least $.99\lambda$; the long-truncated rules select legitimate active IDs at levels less than H , at sites and heights that are not bad; the own-slice fan is at most T ; and its realized list passes (10.1). We have proved that all $\mathcal{V}_g$'s hold simultaneously with probability $1 - o(1)$. On a local failure the later kernels use zero subprobabilities or the specified local fallback.

We record an input domain for these events and their later recomputations. Relative to the projected state of g , its queried even sites are within residual distance 3 and special-coordinate distance 1. A height calculation remains in its queried slice and consults sites within residual distance $12H$. Eligibility at one of those sites reads only the failed-list tests at incident groups, whose projected states are at distance at most 3 and whose slices differ by at most one special coordinate. Such a test reads tuples and positions in the radius- r balls of that group's incident sites, and its mask. A mask uses tags one additional special step away. Thus a group calculation reads only tags, tuple and position fields, masks, activations and ties within special-coordinate distance 3 and residual distance $r + 12H + 9$.

An even star combines groups at projected distance at most 3 and special-coordinate distance at most 1 from it. For all group and star calculations below, including every candidate-tuple replacement, we may therefore use the common domain

$$d_H(z, z_0) \leq 4, \quad d_H(t, t_0) \leq R_{\text{loc}}, \quad R_{\text{loc}} = 4r + 40H + 40\lceil \log_2 n \rceil.$$

All levels are included. The final term covers passage between actual and projected words. Replacing a tuple only reevaluates the same list, marking, height and kernel functions on this fixed primitive domain; it does not initiate a recursive search through other eligibility rules. In particular heights remain slice-local. This domain will justify the independence of sufficiently separated calculations.

10.2 Squared tilt, masks and presentations

On validity, sample at each group independently (given all preceding data) a masked cluster D tilted by $D(F)^2$, further restricted to

$$D(F) \geq e^{-1.5k|\mathcal{D}|}, \quad D(F)/D(F_{-c}) \geq \begin{cases} e^{(-\log 2 + .08a)k} & c \text{ own,} \\ e^{-1.2k} & c \text{ external.} \end{cases}$$

Let h_F be the probability retained by these restrictions under the squared tilt. For any ratio threshold t_c ,

$$\Pr_{D(F)^2 \text{ tilt}} \left\{ \frac{D(F)}{D(F_{-c})} < t_c \right\} \leq t_c^2 \frac{A(F_{-c})}{A(F)}.$$

Substitution from (10.1) bounds this by $e^{-.24ak}$ for an own tuple and $e^{-.4k}$ for an external tuple. The absolute-mass restriction fails with probability at most $e^{-k|\mathcal{D}|}$. Their union is $o(1)$, so $h_F \geq 1/2$.

Conditional on the chosen cluster, every odd role in the group has row law $D|_F$. The reference labels are independent given the clusters. For a prescribed list $\mathcal{D}$, let $p_{\mathcal{D}}$ be the marginal law obtained by averaging this row over the restricted cluster draw, and set it to zero when (10.1)

fails. The restriction costs at most 2; cancelling one factor $D(F)$ in the squared tilt and using its aggregate bound gives

$$p_{\mathcal{D}} \leq 8e^{2k(T+m)}\nu_{i_z}.$$

Call all data through the actual center selections the pre-cluster history. It includes tags, positions, tuple arrays, masks, activations and ties. For an odd role b , write p_b^W for the cluster-averaged row at its realized list, multiplied by $1_{\mathcal{V}_g}$ for its group. The superscript does not condition only on W ; it denotes this full pre-cluster history. The same cap holds for p_b^W .

We also define a reference sampling law at every pre-cluster history, without requiring global success. Draw clusters independently at the locally valid groups by the stated rule, then draw role labels independently conditional on those clusters. Invalid groups use independent local defaults. Later sublikelihoods retain $\mathcal{V}_g$ for every group they query.

We now specify masks and bound tag-conditional means. For fixed neighboring tags and a deterministic configuration (the $\mathcal{D}$ inputs with iid respective tuples), the mean of $p_{\mathcal{D}}$ depends only on the number $t \leq T$ of own IDs in addition to the tag/mask laws. Choose a distribution of masks so these hypothetical means, for **all** t , are each bounded by $O(T+1)\nu_{i_z}$ pointwise. Indeed for nonnegative prices $c_t(y)$ put $S = \sum_{t,y} c_t(y)\nu_{i_z}(y)$ and discard labels with $\sum_t c_t(y) > 10S$. One can mask as specified to retain only cheap labels (they have ν_{i_z} -mass at least .9, also when $S = 0$, so at most .2 of the prior on clusters has cheap mass below half); summed cost is $\leq 10(T+1)S$. Separation in the finite menu proves feasibility. Fix such strategies as lookups, independent of the choices of tag profiles.

From a prescribed list to the actual selection. Let $\hat{p}_b = \mathbb{E}[p_b^W \mid (i_z)_z]$, averaging the pre-cluster randomness at fixed tags. We claim $N \max \hat{p}_b \leq e^{.1m}$. Condition first on the own-slice positions of the group of b . On its position gate there are at most $n^{O(T)}$ possible own lists: enumerate all subsets of at most T names in its incident balls. This enumeration depends only on positions. In each adjacent slice enumerate one external ID, requiring it to be present.

Fix the tuple and mask arrays. For an external queried slice z' , a specified present ID c at level ℓ , and its position field $P_{z'}$, let $w_{\ell,c}(P_{z'})$ be the supremum, over fixed legal eligibility sets, of the conditional activation-and-tie probability that this ID is selected on the queried slice's size and height gates. In this upper bound discard all other components of the group event $\mathcal{V}_g$. This supremum removes dependence on the particular tuple and mask values used to form eligibility. At level 0 uniform eligible-active selection gives the pointwise bound $w_{0,c} \leq 1/ (.99\lambda)$. At a higher level selection requires positive height, so the forced-center estimate in Lemma 3.8 gives

$$\mathbb{E}[w_{\ell,c}(P_{z'}) \mid c \text{ present}] \leq e^{-n^{c_0}}$$

for some fixed $c_0 > 0$.

Conditional on all positions, the activation and tie fields for the external queries are independent because they lie in distinct slices. Their bounds $w_{\ell,c}$ depend only on the respective position fields. Those fields are independent, also with one center forced present in each slice. Hence the expectations of the bounds multiply, although the original eligibility rule uses cross-slice data. Summing named choices in one external slice, with their presence probabilities, gives

$$\sum_{\ell,c} \Pr(c \text{ present}) \mathbb{E}[w_{\ell,c} \mid c \text{ present}] \leq 1/.99 + H\lambda e^{-n^{c_0}} = 1/.99 + o(1).$$

We have ignored any requirement to present the enumerated own list, which only enlarges the probability.

The resulting weights are independent of tuple and mask values. For each deterministic list, we may now average $p_{\mathcal{D}}$ using its hypothetical mean bound. Summing the own lists and external vectors gives

$$\hat{p}_b \leq n^{O(T)}(1/.99 + o(1))^m O(T+1)\nu_{iz}.$$

Its normalized logarithm is

$$O(T \log n) + m \log(1/.99 + o(1)) + n^\delta + O(\log(T+1)) < .1m.$$

This proves the claim and explains why eligibility had to be fixed before activation.

10.3 Internal weights and load transfer

Fix an actual even role v in slice z and a specified present candidate ID c in its center ball. Hold fixed the tags, positions, masks, activations and choice randomizers, and all tuples other than W_c . The candidate tuple $w = W_c$ has prior $\mu_{iz}^{\otimes k}$.

Define L_w to be the sublikelihood of the ordered odd-neighbor labels: require the position-count gate at v , the selection of c there, and every neighboring group event $\mathcal{V}_g$, all recomputed with w . On this gate use the reference law that draws one newly tilted cluster in each group and then independent queried labels within that cluster. Thus L_w integrates the clusters; it does not fix any realized cluster or impose later injectivity constraints. We claim that

$$L_w \leq \exp((\log 2 - .06a)kn) Q \tag{10.2}$$

for a reference probability Q independent of w .

To prove this comparison, first prescribe the ID list at one neighboring group. Delete c from that list and keep the other tuples and the mask fixed. Reference its j queried labels by the masked cluster prior tilted by $D(F_{-c})^2$, without the extra restrictions, followed by independent samples from $D|_{F_{-c}}$.

Before integrating the cluster, the likelihood ratio on actual support is

$$h_F^{-1} \frac{A(F_{-c})}{A(F)} \left(\frac{D(F)}{D(F_{-c})} \right)^{2-j} \leq 2 \frac{A(F_{-c})}{A(F)} \left(\frac{D(F)}{D(F_{-c})} \right)^{2-j}.$$

The three factors come from the retained tilt mass, the two tilt normalizers, and the j conditional label densities. For an ordinary group with $j \geq 2$, the deletion bound in (10.1) and the lower ratio gate give

$$2 \exp((j \log 2 - .08aj - .24a)k) \leq 2e^{(\log 2 - .08a)kj}.$$

For a singleton use $D(F)/D(F_{-c}) \leq 1$, giving at most $2e^{2k}$. There are at most $m+1$ singleton groups.

The realized lists may change with w . At each group average the deletion references uniformly over all possible name lists containing c . This probability law is independent of w , and dominates any one such reference at a cost at most $\exp(O((m+T) \log n))$ on the position gate. Nonexistent or undefined references use fixed independent fallbacks. Taking the product over groups yields a reference Q for which

$$\log(L_w/Q) \leq (\log 2 - .08a)kn + O(km) + O(n(m+T) \log n) + O(n) \leq (\log 2 - .06a)kn.$$

The last inequality uses $m = o(an)$ and $(m+T) \log n = o(ak)$, proving (10.2).

On the true local gates with c actually selected, test

$$Z = \int L_w(\mathbf{y}) d\mu_{i_z}^{\otimes k}(w) \geq e^{-.01akn} Q(\mathbf{y}), \quad Z > 0$$

at the observed star data $\mathbf{y}$. Under reference odd sampling the unconditional probability of such a true-gated failure is $\leq e^{-.01akn}$ per candidate test with the conditioning just described, by predictive mass. On passing, use the resulting posterior proportional to $L_w d\mu_{i_z}^{\otimes k}$. It is supported on tuples of common neighbors, and has normalized joint atom cap (relative to uniform k -fold sampling on original X)

$$\exp((\log 2 - .04a)kn)$$

using $n^\delta = o(an)$. Let $\bar{p}$ be the average coordinate marginal of this posterior and put

$$B = \{x : N\bar{p}(x) > e^{(\log 2 - .02a)n}\}.$$

Then $|B|/N \leq e^{-(\log 2 - .02a)n}$. If $q = \lceil (1 - .01a)k \rceil$, the joint tuple cap and a union over coordinate subsets give

$$\Pr\{\#\{j : W_j \in B\} \geq q\} \leq 2^k e^{(\log 2 - .04a)kn} (|B|/N)^q \leq e^{-\Omega(akn)}.$$

Indeed the coefficient gained in the exponent is at least $.02 - .01 \log 2 + .0002a > 0$, while $k \log 2 = o(akn)$. Thus the expected light-coordinate fraction is at least $.01a(1 - o(1))$. For an absolute $c_6 > 0$, the light part of $\bar{p}$ has mass at least c_6a .

Normalize this light part to obtain the even row p_v^X . In estimates set it to zero off the true local gates or predictive success. It is supported on common neighbors, and

$$N \max p_v^X \leq e^{(\log 2 - .01a)n},$$

since normalizing costs $1/(c_6a)$ and $\log(1/a) = o(an)$.

Conditional means and tag balance. Let $\hat{p}_v^X$ be the mean of this row over the raw local randomizations and reference neighbor draws, with tags fixed. Its normalized cap is at most $e^{.1m}$. For a specified candidate c , the subdensity L_w includes exactly the event that c is selected and the neighboring groups are locally valid. Its data marginal is Z . Consequently, as measures in the candidate tuple,

$$\int Z(\mathbf{y}) \frac{\mu_{i_z}^{\otimes k}(dw) L_w(\mathbf{y})}{Z(\mathbf{y})} d\mathbf{y} = \mu_{i_z}^{\otimes k}(dw) \int L_w(\mathbf{y}) d\mathbf{y} \leq \mu_{i_z}^{\otimes k}(dw),$$

with zero terms at $Z = 0$. The last integral is a gate probability. Coordinate averaging and light-part normalization cost at most $1/(c_6a)$. On the position gate there are only polynomially many possible candidates at v . Summing their bounds and using $N \max \mu_i \leq e^{n^\delta}$ proves the claimed cap.

Both $\hat{p}_v^X$ and $\hat{p}_b$ depend only on the tags in the declared bounded special-coordinate neighborhood. Each row is supported on labels in its own patch.

Choose independent tag profiles, one per slice, so that the expected odd and even mean rows, averaged over their respective roles within that slice, are bounded coordinatewise by $O(1)/N$. For fixed profiles elsewhere, price separation makes these bounds feasible for one slice: availability supplies a tag whose two supports avoid expensive labels, regardless of the remaining draws. The mask lookups and all other rules conditional on realized tags have already been fixed, independently

of profile probabilities. The response sets are therefore nonempty, convex and have closed graph on the finite profile simplices. Lemma 3.3 gives simultaneous profiles.

Under these product profiles, apply Lemma 3.6 to the normalized tag-conditional means. Their cap is e^{1m} . Two roles are near for this calculation when their special words lie within a sufficiently large fixed distance, so the near fraction is at most $n^{O(1)}2^{-m}$. At separated words their mean functions use disjoint tags. The product comparison is thus independence, and $n n^{O(1)}2^{-m}e^{1m} = o(1)$. With high probability their averaged normalized means are bounded simultaneously for all labels. Fix such typical tags.

Next expose the pre-cluster history at these tags. Declare actual parity roles residual-near when their projected words are within $2R_{\text{loc}}$, enlarging the fixed constant if necessary. The primitive domains for separated roles are disjoint. Since $R_{\text{loc}} = O(n^{1-7\delta})$, its ball volume is $\exp(o(an))$; the m special bits and the $O(\log n)$ projection changes also cost only this amount. Thus the near fraction is

$$f_{\text{res}} \leq 2^{-n} \exp(o(an)).$$

For a fixed second label, the normalized row $Np_b^W(y)$ has cap $8e^{2k(T+m)+n^\delta}$, which is $e^{o(n)}$. Its mean is $N\hat{p}_b(y)$. At separated roles the raw local experiments are independent given tags, even when events elsewhere fail, because all rules were extended locally. Scattered moments and the bounded average of these comparison means give odd column sums $o(1)$ with high probability. Intersect this event with the already proved simultaneous pre-cluster validity.

On these good prehistories, draw the independent group clusters. The conditional individual laws $D|_F$ then have column sums $\leq \theta_0$ with high probability, where θ_0 is as in the clock lemma. Indeed their sums have the preceding small expectations; the maximum contribution from a group is

$$n^{O(\log n)} 2 \exp(-n^\zeta + 1.5k(T+m)),$$

even including its entire group size. This is $\exp(-\Omega(n^\zeta))$ by $500\delta < \zeta$. Bounded-summand exponential concentration for independent groups beats the union over columns: if L_* is the displayed summand bound, a column sum with conditional mean $\bar{s} = o(1)$ exceeds θ_0 with probability at most $\exp(((e-1)\bar{s} - \theta_0)/L_*)$, by exponential Markov (use $e^x \leq 1 + (e-1)x$ on $[0, 1]$). On this success use a clock-lemma injection of all odd labels with joint upper domination through n^2 queries (no additional predicates needed).

Transfer to even rows. At a successful cluster history the clock law compares each queried star, and any collection of at most n separated stars, to independent odd-row sampling conditional on those realized clusters. Retain the prehistory and cluster-column success indicators while applying this comparison; it queries at most n^2 odd rows. Once the comparison is made, remove the nonlocal cluster-column indicator and integrate the independent group clusters. This gives exactly the product-by-groups reference law used to define L_w , with the local gates still present.

For one star, this order of integration permits the true-gated predictive bound above to be used. Summing it over all 2^{n-1} even roles and their polynomially many possible candidates gives $o(1)$, because its exponent is $.01akn \gg n$. Thus predictive failures occur with total probability $o(1)$ on the entering successes.

For the even column estimate, use normalized weights $Np_v^X(x)$ with these success indicators. Their cap is $e^{(\log 2 - .01a)n}$, and

$$nf_{\text{res}} e^{(\log 2 - .01a)n} = o(1).$$

At retained separated rows, first make the same clock comparison and integrate clusters in the same order. Then remove nonlocal prehistory restrictions. The true local gates remain in each

likelihood, and the declared primitive domains are disjoint. Integrating those local fields therefore gives the product of the fixed-tag comparison means $N\hat{p}_v^X(x)$. Their average is bounded by the chosen tags. Lemma 3.6 yields even column sums $O(2^n/N) = o(1)$, hence at most 1, with high probability on the successes.

On their joint success every even row has mass 1 and consists of common neighbors of the odd injection. Hall's theorem supplies distinct even representatives, completing the cube and the proposition. $\square$

Corollary 10.2 (Eventual absence of cluster witnesses). *After stabilization, cluster witnesses from Proposition 10.1 are eventually absent for every fixed rational admissible parameter choice, in both orientations and both colors.*

Proof. We may stabilize (for countably many fixed choices of these parameters) and hence assume no such cluster witnesses at all on the retained sets, either orientation or color, since availability gives the just-proved contradiction. More precisely we take in particular all fixed rational choices meeting the parameter conditions, and for each such choice we have eventual absence. Further $o(N)$ discards do not affect previous availability conclusions. $\square$

11 Excluding the linear-budget jump

Proposition 11.1 (Exclusion of the linear-budget jump). *In the stabilized contradiction sequence, it is impossible that both orientations have $H^\dagger \geq 1$ while one has $H_L^\dagger = 0$.*

Proof.

The slice experiment. Suppose both orientations have $H^\dagger \geq 1$, but one has $H_L^\dagger = 0$; denote that orientation by X, Y . We will use a biased patch inside a small cube slice to leave large first-side common-neighbor sets. Independent labels from the other slices will then pay for the remaining cube edges. The compatibility argument below ensures that those outer labels do not consume almost all of an internal set.

Fix cluster-exclusion parameters $\zeta = 1/100$ and rational δ as in Proposition 10.1, and put $s_c = \lceil e^{n^\zeta} \rceil$, $v = \delta/4$. In the reversed orientation, $H^\dagger \geq 1$ permits a rational $x_0 > 0$ with $H_{\text{rev}}(x_0, 1 - v/4) > .95$. Indeed the near-1 second-budget limit exceeds .95 for a sufficiently small first argument, and monotonicity gives the same lower bound at $1 - v/4$. Stabilized absence at exponent .95, followed by reversing the sides, gives

$$|d_G(\sigma, \pi) - 1/2| \leq b_* := n^{-1+.05} \quad \text{for} \quad \text{width}(\sigma) \leq n^{1-v/4}, \quad \text{width}(\pi) \leq n^{x_0}, \quad (11.1)$$

on X, Y , in either color.

Set $g = n^{-.01}$. Since $H_L^\dagger = 0$, monotonicity and nonnegativity give $H_L(.01, .01) = 0$. Choose an available exponent $h_0 < .01$. Its surplus n^{-h_0} is much larger than g . After fixing a color along a subsequence, the columns of degree at least $1/2 + 2g$ have second-law mass at least cn^{-h_0} . Conditioning on them costs only $h_0 \log n + O(1)$ width. Lemma 3.2 therefore supplies an available finite menu of tags i , all of one color G , such that

$$\text{width}(\mu_i) \leq n^{.01}, \quad \text{width}(\nu_i) \leq .011n, \quad d_G(\mu_i, y) \geq 1/2 + 2g \quad (y \in \text{supp } \nu_i).$$

All supports lie in the retained sets.

One slice and its odd rows. Split the cube into inner slices Q_h , where $h = \lfloor n^{.1} \rfloor$, indexed by outer words $s \in Q_{n-h}$. Put $k = \lceil n^{.2} \rceil$, and first consider one slice with fixed tag i . Define the hit-conditioned first law

$$\rho_y(w) = \frac{\mu_i(w)1[w \sim_G y]}{d_G(\mu_i, y)}.$$

Provisionally draw $Y_0 \sim \nu_i$. Given Y_0 , draw the tuples $W_v \sim \rho_{Y_0}^{\otimes k}$ independently at its even roles. At an odd role b , the observed data are the tuples at its h internal even neighbors. Their likelihood density, relative to independent μ_i -entries, is

$$L_b(y) = \prod_{\text{observed entries } w} \frac{1[w \sim_G y]}{d_G(\mu_i, y)}.$$

Let $Z_b = \int L_b d\nu_i$, and let $Z_{b,-v}$ be the corresponding predictive density with one neighboring tuple omitted. Require positivity and

$$Z_b \geq e^{-kh}, \quad Z_b/Z_{b,-v} \geq e^{-.2gk} \quad \text{for each internal } v \sim b.$$

On passing, set $p_b^W = \nu_i L_b/Z_b$; otherwise set $p_b^W = \nu_i$.

Lemma 3.7 applies first to all kh entries, using reference $\mu_i^{\otimes kh}$. For a deletion test, fix the other entries and use their posterior on Y_0 as the prior; the next tuple has reference $\mu_i^{\otimes k}$, and predictive density $Z_b/Z_{b,-v}$. Hence

$$\Pr\{b \text{ fails a test}\} \leq e^{-kh} + he^{-.2gk} = \exp(-\Omega(gk)).$$

The row laws always have width at most $.02n$. On a passing row,

$$\log(N \max p_b^W) \leq .011n + kh(1 - \log(1/2 + 2g)) \leq .02n,$$

and the fallback has the smaller width $.011n$.

For each fixed y_0 , the conditional test-failure probability is the same at every odd role by translation symmetry of the slice. Averaging over Y_0 therefore selects one deterministic y_0 for this tag for which the per-row bound holds. Fix it from now on and sample independent $W_v \sim \rho_{y_0}^{\otimes k}$. The likelihood rule L_b is unchanged. This selection gives the same bound for each row, not simultaneous success of all rows at this stage. Conditional on W , the slice reference law draws the odd labels independently from the rows p_b^W .

The first-side common-neighbor set. At an even v , let σ_v be uniform on the labels of $\text{supp } \mu_i$ hitting all its actual internal odd neighbors, if this set has size at least $N \exp(-(\log 2 - .5g)h)$; otherwise let $\sigma_v = 0$. We claim that σ_v has mass 1 except with probability $\exp(-\Omega(gk))$ in the slice reference law.

Fix every tuple except W_v , and vary $w = W_v$ under prior $\rho_{y_0}^{\otimes k}$. For a vector t of internal odd outputs, let $F_w(t)$ be the product of their row masses, multiplied by the indicator that all their tuple tests pass. Recompute both rows and tests at (W_{-v}, w) . Let $Q(t)$ be the product of the deletion-posterior rows, with fixed w -independent fallbacks if a deletion denominator is zero. Each neighboring row costs at most $(1/2 + 2g)^{-k} e^{.2gk}$, so on the gate

$$F_w(t) \leq e^{(\log 2 - 2g)kh} Q(t).$$

Put $m(t) = \int F_w(t) d\rho_{y_0}^{\otimes k}(w)$. The true-gated event $m = 0$ or $m < e^{-.5gkh}Q$ has probability at most $e^{-.5gkh}$. Outside it, the posterior on w has atom cap

$$N^{-k} \exp((\log 2 - 1.5g)kh + kn^{.01} + O(k)) \leq N^{-k} e^{(\log 2 - g)kh},$$

where $n^{.01} = o(gh)$.

If the common-neighbor set has size T_0 , this posterior is supported on at most T_0^k tuples. Its total mass 1 forces

$$T_0 \geq N e^{-(\log 2 - g)h},$$

which exceeds the cutoff defining σ_v . Adding the h possible neighboring row-test failures proves the claim.

11.1 Compatibility

Define the tag's two mean rows by

$$\pi_i = \mathbb{E}_W p_b^W, \quad \alpha_i = \mathbb{E}_{W, Y_{N_{\text{in}}(v)} | W} \sigma_v.$$

Here $N_{\text{in}}(v)$ is the set of internal odd neighbors of v . Their outputs in the second expectation are independent with rows p_b^W . Thus π_i is a probability law, whereas α_i can be a subprobability. Translation symmetry makes these means independent of the chosen role of the indicated parity, also when the outer word reverses the slice's internal parity. They are supported on the two sides of the tag, and π_i has width at most $.02n$.

The next two auxiliary lemmas use the slice experiment, patch menu, and parameter choices fixed in this proof.

Lemma 11.2 (A compatible balanced profile). *There is one profile p for independent tag choices with*

$$\pi = \sum p_i \pi_i \leq K/N, \quad \sum p_i \alpha_i \leq K/N, \quad (11.2)$$

where K is constant, and the following compatibilities. Write $f_x(y) = 2\mathbf{1}[x \sim_G y] - 1$, $m_x = \mathbb{E}_\pi f_x$, $K_\pi(x, x') = \mathbb{E}_\pi f_x f_{x'}$; fix $\eta = 10^{-8}$. For every tag used by p :

- on $\text{supp } \mu_i$, $|m_x| \leq 4b_*$ and $p\{i' : d_G(x; \pi_{i'}) > .8\} \leq \eta$;
- on that support there is no clique of size s_c of pairs of distinct labels with $K_\pi(x, x') > 8n^{-\delta}$.

Proof. Fix K sufficiently large and consider the compact convex set of profiles satisfying (11.2). It is nonempty by availability and price separation. For an input profile in this set, let $\pi = \sum p_i \pi_i$. We will discard $o(N)$ first labels so that every patch supported off the discards meets all compatibility requirements for this input.

First discard labels with $|m_x| > 4b_*$. Their number is $o(N)$ by (11.1): the uniform law on a larger signed exception, tested against the broad law π , would contradict that estimate. On the remaining labels greedily remove disjoint cliques of size s_c for the graph $K_\pi(x, x') > 8n^{-\delta}$. Let V be the union removed. If $|V| \geq N e^{-n^\delta}$, use π as a first law in the reversed orientation and the uniform laws on these disjoint cliques as clusters, weighted in proportion to their sizes. Their aggregate is uniform on V . Its width is at most n^δ , the width of π is at most $\log K$, and each cluster atom is at most $1/s_c \leq e^{-n^\delta}$.

For distinct members of a clique,

$$d_G(\pi; x, x') = \frac{1 + m_x + m_{x'} + K_\pi(x, x')}{4} \geq \frac{1}{4} + n^{-\delta}$$

for large n , since $b_* = o(n^{-\delta})$. A diagonal pair has codegree $(1 + m_x)/2$ and also meets this bound. This is a witness forbidden by Corollary 10.2. Therefore $|V| < Ne^{-n^\delta}$.

Excluding too many high-degree tags. Before clique removal, let U be the good-degree labels violating the η -condition. Suppose $|U| \geq Ne^{-n^\delta}$, and let μ_U be uniform on U . Write $D_i(x) = d_G(x; \pi_i)$. For every $x \in U$, the mean of $D_i(x)$ over tags is $1/2 + O(b_*)$, while tag mass greater than η has $D_i(x) > .8$. Hence

$$\mathbb{E}_i D_i(x)^2 = (\mathbb{E}_i D_i(x))^2 + \text{Var}_i D_i(x) \geq \frac{1}{4} + c\eta$$

for an absolute $c > 0$. Integrate this inequality over μ_U .

In $L^2(\mu_U)$, put $v_y(x) = 1[x \sim_G y]$ and $v_i = \mathbb{E}_{y \sim \pi_i} v_y$. Then $\|v_i\|_2^2 = \int D_i(x)^2 d\mu_U(x)$. A fixed positive fraction of the tags therefore have $\|v_i\|_2 > 1/2 + c'$, for a fixed $c' > 0$. For each such tag, a fixed positive π_i -mass of labels has projection on $v_i/\|v_i\|_2$ at least $1/2 + c''$, with fixed $c'' > 0$. Let $\rho_i \leq C\pi_i$ be π_i conditioned on these labels.

Join two of these labels when their codegree under μ_U is at least $1/4 + c'''$, with fixed $c''' > 0$ sufficiently small. This graph has bounded independence number. Indeed, for an independent set of size t_0 , the average of its adjacency vectors has projection at least $1/2 + c''$, but squared norm at most

$$1/t_0 + 1/4 + c'''.$$

These are incompatible for large fixed t_0 . The diagonal codegrees also exceed the threshold because the squared projection of each vector is at least $(1/2 + c'')^2$.

Draw $r_0 = \lfloor e^{.03n} \rfloor$ labels from ρ_i , conditioned to be distinct. Since $\max \rho_i \leq Ce^{.02n}/N$, the collision probability before conditioning is

$$O(r_0^2 e^{.02n}/N) = o(1).$$

The elementary Ramsey binomial bound with the fixed independence threshold permits greedy packing of all but $s_c^{O(1)}$ sampled labels into cliques of size s_c . A positive fraction of the sample, in fact $1 - o(1)$, is retained. Make each clique a uniform cluster and weight these clusters in proportion to their sizes. If R_i is the retained sample, its aggregate obeys

$$U(R_i) \leq \frac{2}{r_0} \sum_{j \leq r_0} \delta_{Y_j}.$$

Average over the distinct sample and over the selected positive-mass set of tags. The conditional sampling and tag restriction each cost only a constant in this inequality, so the final aggregate is bounded by $O(1)\pi$.

We have obtained a cluster witness with first law μ_U of width at most n^δ , aggregate second law of width $O(1)$, atom cap $1/s_c$, and codegree at least $1/4 + c''' \geq 1/4 + n^{-\delta}$. Corollary 10.2 again gives a contradiction. Thus the labels violating the high-degree-tag condition also number less than Ne^{-n^δ} .

It follows that outside $o(N)$ discards all the input-dependent requirements for output tags can be met along with arbitrary price exclusions by availability. Separation gives an output profile satisfying the same balance constraints (11.2), increasing their fixed K at the outset as needed for discard slack. Require it supported only on tags compatible with the **input** as listed. These responses are convex and have closed graph (cliques used strict inequalities), so Kakutani proves

the claim. To make the domain choice explicit, fix the balance constant large enough for both separation arguments before forming the compact profile domain. The single-tag laws and support sets are fixed throughout. For a fixed label, the exceptional-tag set in the degree condition is therefore fixed, and its profile mass is linear in the input profile. For each fixed candidate clique, exclusion of the strict simultaneous inequalities is a closed condition. There are finitely many labels, tags and candidate cliques at each stage, so compatibility of any fixed output tag is closed in the input profile. Restricting output mass to these compatible tags gives precisely the closed response graph used above. $\square$

11.2 Outer moment

Lemma 11.3 (Outer mass lower tail). *Fix a compatible tag and any probability σ on its first support with*

$$\max \sigma \leq N^{-1} \exp((\log 2 - .5g)h).$$

Take $d = n - h$ independent labels $Y_l \sim \pi$. Write $D_x = d_G(x; \pi)$ and

$$a_x(y) = \frac{\mathbf{1}[x \sim_G y]}{D_x} - 1 = \frac{f_x(y) - m_x}{1 + m_x}.$$

Then for any fixed P ,

$$\Pr \left\{ \int \prod_{l \leq d} (1 + a_x(Y_l)) d\sigma(x) < 1/2 \right\} \leq n^{-P} \quad (11.3)$$

eventually, uniformly.

Proof. Let

$$Z(Y_1, \dots, Y_d) = \int \prod_{l \leq d} (1 + a_x(Y_l)) d\sigma(x).$$

We will bound a high even moment of $Z - 1$. For independent $x_1, \dots, x_u \sim \sigma$, where u is a fixed length to be chosen, define

$$M_J = \int \prod_{j \in J} a_{x_j} d\pi.$$

Singleton interactions vanish because $\int a_x d\pi = 0$. A free label below means one involved tuple coordinate integrated with its original σ law, the other specified coordinates being fixed. All constants may depend on u .

One free label. With one label free, $|M_J| \leq O_u(b_*)$ except with probability at most $2e^{-n^{1-v/4}/2}$. To see this, write the interaction as

$$M_J = \frac{\langle f_x, H_0 - \mathbb{E}_\pi H_0 \rangle_\pi}{1 + m_x},$$

where H_0 is the product of the fixed factors and is bounded by a constant depending on u . If one signed exceptional set had σ -mass at least $e^{-n^{1-v/4}/2}$, conditioning on it would give a first law of width at most $O(h) + n^{1-v/4}/2 < n^{1-v/4}$. Reweight that law by $(1 + m_x)^{-1}$ and normalize, at width cost $O(1)$. The centered second function is a difference of two probability tests obtained by adding a fixed positive offset to its positive and negative parts. Their widths are $O(1) < n^{x_0}$. Applying (11.1) to them contradicts an interaction larger than a sufficiently large constant times b_* .

Two free labels. With two involved labels x, z free, we claim

$$\Pr\{|M_J| > n^{-1-.03}\} \leq e^{-n^4}.$$

Write $M_J = \langle a_x, H_0 a_z \rangle_\pi$, with fixed bounded H_0 . If the claim fails, choose one sign and a set A of x 's of mass at least $\rho = e^{-O(n^4)}$ such that the signed violation set E_x of z 's also has mass at least ρ for every $x \in A$. For a fixed x , draw $t = \lceil n^{.25} \rceil$ independent z_j 's from $\rho_x = \sigma|_{E_x}$.

The width of ρ_x is $O(h + n^4)$. After the bounded reweighting $(1 + m_z)^{-1}$, (11.1) says that its mean adjacency indicator is within $O(b_*)$ of $1/2$ outside a π -set of mass $e^{-\Omega(n^{x_0})}$. Otherwise that exceptional second set, conditioned with the corresponding sign, would still fit the second budget. Boundedness and the degree estimates on the first support therefore give

$$\|\mathbb{E}_{\rho_x} a_z\|_{L^2(\pi)} = O(b_*).$$

For $S = H_0 \sum_{j \leq t} a_{z_j}$, independence yields

$$\mathbb{E}\|S\|_2^2 = O_u(t + t^2 b_*^2) = O_u(t).$$

A fixed positive fraction of these conditioned tuples thus has $\|S\|_2 \leq C_u \sqrt{t}$.

Let $\mathcal{G}$ denote that norm event. It depends only on the tuple and H_0 , not on x . Under the raw product law $\sigma^{\otimes t}$,

$$\int_A \Pr(\mathcal{G}, z_j \in E_x \text{ for every } j) d\sigma(x) \geq c_u \rho^{t+1}.$$

Fubini supplies a fixed norm-good tuple whose simultaneous signed-violation set has x -mass at least $e^{-O(tn^4)}$. Let q be σ restricted to that set. Its width is $O(h + tn^4) = O(n^{.65}) < n^{1-v/4}$.

For this fixed tuple put

$$r_x = (1 + m_x)^{-1}, \quad c_q = \mathbb{E}_q r_x, \quad \tilde{q} = r_x q / c_q, \quad T = S - \mathbb{E}_\pi S.$$

The exact projection identity is

$$\mathbb{E}_q \langle a_x, S \rangle_\pi = c_q \mathbb{E}_{\tilde{q}} \langle f_x, T \rangle_\pi.$$

The bounded summands give $\|T\|_\infty = O_u(t)$, while centering preserves $\|T\|_2 = O_u(\sqrt{t})$. Thus the positive test laws

$$\pi_\pm = \frac{\sqrt{t} + T_\pm}{Z_T} \pi, \quad Z_T = \sqrt{t} + \mathbb{E}_\pi T_+ = \sqrt{t} + \mathbb{E}_\pi T_- = O_u(\sqrt{t})$$

have equal normalizers and widths $O_u(\log n)$. Applying (11.1) to $\tilde{q}, \pi_\pm$ bounds the projection by $O_u(b_* \sqrt{t}) = O_u(n^{-.825})$. But the simultaneous violations give magnitude at least $tn^{-1-.03} = n^{-.78+o(1)}$ in one sign. This contradiction proves the two-free estimate.

Mean interaction size. $\mathbb{E}|M_J| \leq n^{-.4|J|}$ for large n .

For $r_J = |J|$, independence and Fubini give

$$\mathbb{E} M_J^2 = \mathbb{E}_{y, y' \sim \pi} \left(\int a_x(y) a_x(y') d\sigma(x) \right)^{r_J}.$$

Fix y . The inner kernel is a bounded signed reweighting of σ tested against $f_x(y')$, plus an $O(b_*)$ term arising from m_x . Adding positive offsets to this signed reweighting costs only $O(1)$ first width. By (11.1), the kernel is $O(b_*)$ except on a y' -set of π -mass $e^{-\Omega(n^{x_0})}$, uniformly in y . Hence

$$\mathbb{E} M_J^2 \leq (C_u b_*)^{r_J} + C_u e^{-\Omega(n^{x_0})}.$$

Cauchy-Schwarz, with $.95/2 > .4$, gives the stated first absolute moment.

Counting large extensions. Fix the other labels of an interaction. There are at most $e^{n^{.03}}$ labels in the support extending it to $|M_J| > n^{-v}$. Such an extension has projection of f_x on a fixed bounded direction of magnitude at least $c_u n^{-v}$. Split by sign. For t_0 candidates of one sign with mutual $K_\pi \leq 8n^{-\delta}$, the squared norm of their average is at most

$$1/t_0 + 8n^{-\delta},$$

whereas its squared projection is at least $c_u^2 n^{-2v}$. As $\delta = 4v$, this forces $t_0 = O_u(n^{2v})$. The compatibility hypothesis forbids an s_c -clique of larger inner products. The elementary Ramsey bound obtained from the one-vertex recursion therefore bounds the number of candidates by a binomial coefficient, with logarithm at most

$$\log \binom{s_c + t_0}{t_0} = O_u(n^{\zeta+2v}) < n^{.03}.$$

The sign split and the fixed number of interaction choices fit the slack in this bound.

Exponential weights for moderate interactions. For a tuple let $w = \max_{|J| \geq 2} |M_J|$, taking an empty maximum to be zero. On $w \leq n^{-v}$, the integral of $e^{2^u n w}$ against the original product tuple law is bounded. Its contribution from $w > n^{-1-.03}$ is smaller than any fixed negative power of n . Indeed the two ranges are bounded respectively by

$$\begin{aligned} \int_{\{n^{-1-.03} < w \leq n^{-1+.06}\}} e^{2^u n w} d\sigma^{\otimes u} &\leq C_u e^{2^u n^{.06} - n^{.4}}, \\ \int_{\{n^{-1+.06} < w \leq n^{-v}\}} e^{2^u n w} d\sigma^{\otimes u} &\leq C_u e^{2^u n^{1-v} - n^{1-v/4}/2}. \end{aligned}$$

The first line uses the two-free estimate, and the second the one-free estimate; the finite union over interactions contributes C_u . Both exponents tend to minus infinity faster than $\log n$. On the remaining range the weight is bounded by $e^{2^u n^{-.03}}$. The same statements hold for every subtuple of this fixed length.

The centered expansion. Choose $L = L(P)$ sufficiently large and then a sufficiently large even u . The moment $\mathbb{E}(Z - 1)^u$ is the integral of

$$\Phi_u(x_1, \dots, x_u) = \sum_{I \subseteq [u]} (-1)^{u-|I|} \left(\int \prod_{j \in I} (1 + a_{x_j}) d\pi \right)^d$$

against $\sigma^{\otimes u}$. On $w \leq n^{-1-.03}$, expand each of the d factors into its nonempty interactions and the constant 1. A term selects interactions from distinct factors. If their union is U_0 , its coefficient in the alternating sum is $\sum_{I \supseteq U_0} (-1)^{u-|I|}$, which vanishes unless $U_0 = [u]$. Thus every surviving list covers all tuple coordinates.

Lists with more than L interactions have total absolute contribution bounded by the geometric tail in $2^u n^{-.03}$. Choose L so that this is smaller than the negative power needed for the final Markov bound. For a list of at most L interactions, some selected interaction contains at least u/L coordinates. Bound its integral by the mean-interaction estimate and the other factors by their small-set bounds. The total is at most

$$O_u(n^{L-4u/L}).$$

Choosing the even u with $.4u/L > L + P + O(1)$ makes this small enough as well. On $n^{-1-.03} < w \leq n^{-v}$, bound each positive product in Φ_u by $e^{2^u n w}$; the preceding exponential-weight estimate gives superpolynomial decay without cancellation.

Removing labels involved in large interactions. It remains to integrate tuples with $w > n^{-v}$. From such a tuple choose a maximal retained set of coordinates R whose interactions all have magnitude at most n^{-v} . Fix one possible R first, and later sum over the finitely many choices. Once the retained labels are fixed, maximality implies that every omitted label lies in a set B_R of extensions causing a large interaction with them. The count just proved gives

$$|B_R| \leq 2^u e^{n^{.03}}.$$

This is a necessary condition on each omitted coordinate, not a claim of independence after conditioning on maximality. Under the original product law the separate necessary tests can be integrated independently, because B_R depends only on the retained coordinates.

For each positive product term in Φ_u , an omitted label involved in that term costs at most D_x^{-d} pointwise. We may use the same upper bound when it is not involved. Combining this factor with its necessary extension condition gives a charge at most

$$\begin{aligned} |B_R| \max_x \sigma(x) D_x^{-d} &\leq \frac{2^n}{N} \exp(-.5gh + O(n^{.05}) + n^{.03} + O_u(1)) \\ &= n^{-\omega(1)}. \end{aligned}$$

Here $D_x^{-d} \leq 2^d e^{O(nb_*)}$, $gh = n^{.09+o(1)}$, and $2^n/N \leq 1$ eventually. The retained positive product has bounded integral by the exponential-weight estimate. At least one coordinate was omitted, so summing over retained sets and positive terms still gives a superpolynomially small contribution.

All parts of $\mathbb{E}(Z - 1)^u$ are therefore smaller than any prescribed negative power, after the stated choices of L, u . Since $Z < 1/2$ implies $|Z - 1|^u > 2^{-u}$, Markov proves (11.3). $\square$

11.3 Taking the assignment

Reference failure probability. Choose the slice tags $i(s)$ independently from the compatible profile p , draw the independent tuples at each tag's fixed y_0 , and use independent odd-row sampling conditional on those tuples. Denote the odd outputs by Y_b . At an even v , define the unnormalized row

$$M_v(x) = \sigma_v(x) \prod_{j \text{ outer}} \frac{1[x \sim_G Y_{vj}]}{D_x},$$

where v^j is obtained from v by flipping outer bit j ; the row is zero off the indicated first support. Its mass is at least $1/2$ except with arbitrarily strong polynomially small probability in this raw reference law. The internal failure probability for σ_v is already exponentially small. Given internal data with σ_v of mass 1, the outer neighbors lie in distinct slices. Their unconditional laws are independent copies of π , so Lemma 11.3 applies.

Avoidance at the tag stage. Fix a large constant P , exceeding all needed polynomial dependency exponents and the requirements of Lemma 3.10 with $B = 4$. At each outer word s , impose two conditions on the product tag law:

- the conditional raw probability that $\|M_v\|_1 < 1/2$, for a given even role in this slice, is at most n^{-2P} ;
- for every x on the first support of its tag, at least half the outer neighboring tags satisfy $d_G(x; \pi_{i(sj)}) \leq .8$.

The first conditional probability is identical for all even roles of the slice by inner translation. It reads the tags in the radius-one outer neighborhood. Markov applied to a stronger raw bound from the preceding paragraph makes its failure probability at most n^{-2P} . For the second condition, compatibility gives exceptional tag probability at most η for every such x . Independence of adjacent tags and a binomial union bound give total failure probability at most

$$N(4\eta)^{(n-h)/2} = e^{-\Omega(n)}.$$

The tag events have dependency degree $O(n^2)$. Lemma 3.5, with charges of order n^{-2P} , therefore supplies a law avoiding both kinds of failure.

We next choose a typical outcome of this tag-avoidance law. Write $d = n - h$, and for a fixed host label define the normalized comparison means

$$A_s(y) = N\pi_{i(s)}(y), \quad B_s(x) = N\alpha_{i(s)}(x) \prod_{j \leq d} \frac{d_G(x; \pi_{i(sj)})}{D_x}.$$

The second expression is zero where its first-side row vanishes. These weights have cap at most $2^n e^{-cn}$ on the tag gates for a fixed $c > 0$. For A_s use the width $.02n$. For B_s , use $N\alpha_i \leq e^{h \log 2}$, $D_x^{-d} \leq 2^d e^{O(n^{.05})}$, and the second gate, which contributes at most $.8^{d/2}$ in the numerator.

Apply scattered moments on the outer words, with words at bounded distance declared near. Their near fraction is $n^{O(1)}2^{-d}$, and the product of this fraction, n , and the cap tends to zero. At up to n separated neighborhoods remove the touching tag constraints; there are $O(n^3)$ of them, with total charge $o(1)$. In the restored product law,

$$\mathbb{E}A_s \leq K, \quad \mathbb{E}B_s \leq K.$$

For the second inequality, condition on $i(s)$; each independent neighboring tag integrates its ratio to 1 on the compatible support, and then (11.2) bounds $\mathbb{E}_i N\alpha_i$. Thus, with high probability, both comparison means have bounded averages simultaneously for all labels. Fix tags having these bounds and all tag gates.

Avoidance at the tuple stage. At these fixed tags, the raw W -law is a product of the laws $\rho_{y_0}^{\otimes k}$ at the even roles. Require at every even star that its conditional mass-failure probability under independent odd-row sampling be at most n^{-P} . The tag gate and Markov make each tuple-stage bad event have probability at most n^{-P} . It reads W -variables within full-cube distance 2, so the dependency degree is $O(n^4)$. The conditional avoidance lemma applies again, now with charges of order n^{-P} .

For a fixed second label y , apply scattered moments to $Z_b(y) = Np_b^W(y)$ under this tuple-avoidance law. The cap is $e^{.02n}$, and a sufficiently large fixed full-cube neighborhood has fraction $n^{O(1)}2^{-n}$. Their repeat cost is $o(1)$. At separated rows remove the touching tuple constraints; at most $O(n^5)$ constraints are removed for n queried neighborhoods, with charge sum $o(1)$. Their raw means are $A_{s(b)}(y)$, whose average is bounded by the selected tags. The resulting actual column sums are $O(2^n/N) = o(1)$, simultaneously with high probability.

The odd injection. At a tuple history with these column bounds, apply the clock lemma, retaining this entering success indicator in subsequent comparisons. Its label count is N , with $\log N = O(n)$; column sums are eventually below θ_0 ; and row atoms are at most $e^{.02n}/N < n^{-A^*}$. A mass-failure predicate reads the n odd neighbors of one even role, and each odd row occurs in n such predicates. Their product-law probabilities are at most n^{-P} , as required by the chosen P . The clock law therefore gives an odd injection avoiding every mass failure and joint upper comparison with the independent odd-row law.

Even loads and normalization. The normalized weights $NM_v(x)$ have cap

$$NM_v(x) \leq 2^n \exp(-.5gh + O(n^{.05})) \leq 2^n e^{-.4gh}.$$

A fixed-radius full-cube near relation again has fraction $n^{O(1)}2^{-n}$, so its repeat cost is $o(1)$, because $gh = n^{.09+o(1)}$.

For up to n separated even stars, retain the successful tuple-history indicator and apply the clock comparison to their at most n^2 distinct odd neighbors. This replaces the injection outputs by independent odd-row draws at fixed W . We may then discard the nonlocal column-load indicator for an upper bound. Each remaining local row and weight reads only the radius-two W -scope of its star. Removing the avoidance constraints touching those scopes costs $1 + o(1)$ and restores independent raw tuple fields in the separated domains.

The resulting product means are exactly $B_{s(v)}(x)$: the internal weight integrates to $\alpha_{i(s(v))}$, and each outer label contributes its degree into the corresponding $\pi_{i(s^j)}$. Their bounded averages give, by scattered moments, simultaneous column sums

$$\sum_v M_v(x) = O(2^n/N) = o(1),$$

hence at most $1/2$ with high probability on the entering successes. This uses only $N/2^n \rightarrow \infty$, with no rate assumption. Clock avoidance has ensured that every row has mass at least $1/2$. Normalizing each row thus increases column sums by at most 2. The resulting probability rows lie on common neighbors of the odd injection and have column sums at most 1, so Hall's theorem completes the cube. This excludes the linear-budget jump. $\square$

Corollary 11.4 (The remaining deep regime). *The possibility remaining in this analysis has both $H_L^\dagger \geq 1$, i.e., for each fixed error exponent slack $\epsilon > 0$ one has deep discrepancy $\leq n^{-1+\epsilon}$ on retained labels at some budgets $n^x, \alpha n$ with constants $x, \alpha > 0$, in both orientations.*

Proof. Proposition 9.2 leaves both $H^\dagger \geq 1$ and excludes $0 < H_L^\dagger < 1$. Proposition 11.1 also excludes $H_L^\dagger = 0$; since these exponents are nonnegative, both linear limits are at least 1. Given $0 < \epsilon < 1$, choose a rational h_0 with $1 - \epsilon < h_0 < 1$, and then positive rational budgets with $H_L(x, \alpha) > h_0$ in each orientation. Stabilized absence at h_0 gives discrepancy less than $n^{-h_0} \leq n^{-1+\epsilon}$. Larger error slacks follow by weakening such a bound. $\square$

12 Deep discrepancy and interaction estimates

Corollary 12.1 (Deep discrepancy in both orientations). *In the remaining case, on the simultaneous retained sets with the full cluster exclusions, there are constants $0 < x_* < .01$, $0 < \alpha < .01$ for which*

$$|d_G(\sigma, \pi) - 1/2| \leq b_* := n^{-1+.04} \quad \text{at widths } \alpha n, n^{x_*}, \text{ in either order.} \quad (12.1)$$

This is for either color, including arbitrary probability laws at the indicated widths. The analogous assertion for any fixed smaller positive exponent slack instead of .04 uses possibly smaller positive budget constants.

Proof. Apply Corollary 11.4 with the chosen positive error slack, and shrink the two budget constants to work simultaneously in both orientations. $\square$

We next bound the lower tail of common-neighbor mass after independent second-side draws. Small interactions are controlled by centered moments; in the homogeneous case, clique exclusion also bounds the number of unusually correlated labels.

Below, either orientation of the retained sides can be denoted X, Y . Let $f_x(y) = 2\mathbf{1}_G(x, y) - 1$. Write $m_\pi(x) = \int f_x d\pi$, $D_\pi(x) = (1 + m_\pi(x))/2$, $K_\pi(x, z) = \int f_x f_z d\pi$.

Here and later, constants for small failure probabilities can be fixed in the following order. Take a cell-size exponent $A_c = 200$, a sufficiently large constant integer P (arbitrarily large in terms of A_c , and of the absolute constants and polynomial exponents for the clock lemma with $B = 5$), then $R = P^2$. We describe estimates with this much slack even where a smaller power suffices. Call laws of widths $\leq n^{x_*/4}$ small-width in this section. We use this terminology only where later fixed multiples or moderate conditionings still fit (12.1).

Lemma 12.2 (Heterogeneous centered-moment identity). *Use $d \leq n$ independent inputs with possibly different small-width laws π_l on Y , and a small-width probability τ on X with $|D_{\pi_l} - .5| \leq O(b_*)$ on its support for each l . Write $a_{l,x} = (f_x - m_{\pi_l}(x))/(1 + m_{\pi_l}(x))$, so $1 + a_{l,x} = \mathbf{1}_G(x, \cdot)/D_{\pi_l}(x)$. Put*

$$Z(\mathbf{y}) = \int \prod_{l \leq d} (1 + a_{l,x}(\mathbf{y}_l)) d\tau(x).$$

For even u , the expectation $\mathbb{E}_{\mathbf{y}}(Z - 1)^u$, over the independent $y_l \sim \pi_l$, is the integral for $x_1, \dots, x_u$ iid from τ of

$$\sum_{I \subseteq [u]} (-1)^{u-|I|} \prod_{l \leq d} \int \prod_{i \in I} (1 + a_{l,x_i}) d\pi_l. \quad (12.2)$$

Interactions are $M_{l,J} = \int \prod_{i \in J} a_{l,x_i} d\pi_l$, $|J| \geq 2$ (an empty maximum of interaction magnitudes is 0).

Proof. Expand the centered power, introduce independent copies of the first label for its factors, and integrate the independent second labels one coordinate at a time. The alternating sum is exactly (12.2); the individual centered factors have zero mean, so singleton interactions vanish. $\square$

Lemma 12.3 (Interaction tails). *In the setting of Lemma 12.2, the following estimates hold for each fixed u . A free label is an unfixed tuple coordinate, sampled with its original law τ ; the other tuple coordinates are held fixed.*

- With one free label and the other labels fixed, the magnitude of any interaction involving it exceeds $O_u(b_*)$ with probability at most $e^{-\alpha n/3}$.
- With two involved labels free and the others fixed, it exceeds $n^{-1-.03}$ with probability at most e^{-n^4} . The same two assertions hold for K_{π_l} on the labels at a pair of distinct tuple coordinates, even without the degree assumption in that raw test.
- $\mathbb{E}|M_{l,J}| \leq n^{-.4|J|}$.

- *The one-free bound also holds with the degree requirements retained as a gate: if the other coordinates satisfy them, the unnormalized probability that the free coordinate satisfies them and has interaction magnitude above $O_u(b_*)$ is at most $e^{-\alpha n/3}$. Here the incoming support need not satisfy the degree requirements everywhere; the interaction is used only on the gate.*

Proof.

One free coordinate. Fix the column l and the other coordinates. The interaction is the projection of $a_{l,x}$ onto a fixed bounded function of the second label. Equation (12.1) applies to bounded signed tests: add a fixed positive offset, normalize to a probability law, apply discrepancy, and subtract the offset term. If a one-sign exceptional set of first labels had τ -mass at least $\frac{1}{2}e^{-\alpha n/3}$, conditioning on it would have width

$$n^{x_*/4} + \alpha n/3 + O(1) < \alpha n.$$

The signed-test estimate would then bound the alleged exceptional projection by $O_u(b_*)$, a contradiction. This also proves the gated assertion: condition on the signed exception intersected with the degree gate, and never divide by the probability of the gate alone.

Two free coordinates. Write the interaction as $\langle a_{l,x}, H a_{l,z} \rangle_{\pi_l}$, with H a fixed bounded function supplied by the other coordinates. Suppose its magnitude exceeds $n^{-1-.03}$ on a set of τ^2 -mass greater than e^{-n^4} . Splitting signs and then the conditional masses gives a set A of first coordinates, of mass at least $e^{-O(n^4)}$, such that for every $x \in A$ the same signed violation holds on a set E_x of z 's with $\tau(E_x) \geq e^{-O(n^4)}$.

Let $\rho_x = \tau(\cdot \mid E_x)$. Its width is $O(n^4) < \alpha n$. Applying (12.1) to this first law shows that $\mathbb{E}_{\rho_x} f_z(y) = O(b_*)$ outside a π_l -set of mass $e^{-\Omega(n^{x_*})}$. The degree bounds change f_z to $a_{l,z}$ by $O(b_*)$, and boundedness on the exceptional set therefore gives $\|\mathbb{E}_{\rho_x} a_{l,z}\|_{L^2(\pi_l)} = O(b_*)$. Take $t = \lceil n^{.25} \rceil$ independent $z_j \sim \rho_x$. For $S = H \sum_{j \leq t} a_{l,z_j}$, independence gives

$$\mathbb{E}\|S\|_2^2 = O_u(t + t^2 b_*^2) = O_u(t).$$

A fixed positive fraction of these tuples consequently satisfy $\|S\|_2 = O_u(\sqrt{t})$.

Return now to the original product law $\tau^{\otimes t}$. For each $x \in A$, the tuples just obtained have raw mass at least $e^{-O(tn^4)}$, and all their coordinates lie in E_x . Fubini therefore supplies one fixed norm-good tuple for which the set of x 's satisfying all the signed violations has mass at least $e^{-O(tn^4)}$. Conditioning τ on that set costs width $O(n^{.65}) < \alpha n$. Its mean $a_{l,x}$ has $L^2(\pi_l)$ -norm $O(b_*)$ by the preceding argument. Its projection onto this fixed S is thus at most

$$O_u(b_* \sqrt{t}) = O_u(n^{-.835}),$$

whereas the simultaneous violations make that projection at least $tn^{-1-.03} = n^{-.78+o(1)}$ with one sign. This contradiction proves the two-free estimate. For the raw correlation $K_{\pi_l}(x, z) = \langle f_x, f_z \rangle_{\pi_l}$, the same proof uses f in place of a and requires no degree gate.

Mean interaction size. Independence of the tuple coordinates yields

$$\mathbb{E} M_{l,J}^2 = \mathbb{E}_{y,y' \sim \pi_l} \left(\int a_{l,x}(y) a_{l,x}(y') d\tau(x) \right)^{|J|}.$$

For fixed y , the inner expression is a bounded signed test on the first side applied to $f_x(y')$, with an $O(b_*)$ error from the centering and denominators. Deep discrepancy bounds it by $O(b_*)$ outside a π_l -set of y' 's of mass $e^{-\Omega(n^{x*})}$. Boundedness elsewhere gives

$$\mathbb{E}M_{l,J}^2 \leq (C_u b_*)^{|J|} + C_u e^{-\Omega(n^{x*})}.$$

Cauchy-Schwarz now gives $\mathbb{E}|M_{l,J}| \leq n^{-.4|J|}$, since $.96/2 > .4$. $\square$

Put $L = \lceil 500R \rceil$, and choose an even constant $u > 10L^2$. Take successively tiny constants $\xi, \theta > 0$, with $\xi < \alpha 2^{-10u-100}$ and θ sufficiently small in terms of ξ, u . All can be fixed before patch and geometry choices below.

Lemma 12.4 (The moderate-interaction contribution). *Let Φ_u be the signed integrand in (12.2), and put $w = \max_{l,J} |M_{l,J}|$. Then*

$$\left| \int_{\{w \leq 2\xi\}} \Phi_u d\tau^{\otimes u} \right| \leq n^{-3R}.$$

Each positive product term in that integrand has bounded integral on the same set. This latter assertion also holds for subtuples of the fixed length bound.

Proof. Every positive product term is at most $\exp(2^u n w)$. The union over the $O_u(n)$ interactions and Lemma 12.3 bound its integral on the two ranges above $n^{-1-.03}$ by

$$\begin{aligned} O_u(n) e^{2^u n^{.06} - n^{.4}}, & \quad n^{-1-.03} < w \leq n^{-1+.06}, \\ O_u(n) e^{2^{u+1} \xi n - \alpha n/3}, & \quad n^{-1+.06} < w \leq 2\xi. \end{aligned}$$

In the second range the lower endpoint exceeds the one-free threshold $O_u(b_*)$. Both bounds decay faster than any fixed inverse power of n ; the second uses the stipulated smallness of ξ .

On $w \leq n^{-1-.03}$, expand each column factor into interactions and then sum over I . A term with chosen interactions whose union is $S \subseteq [u]$ has coefficient $\sum_{I \supseteq S} (-1)^{u-|I|}$. It vanishes unless $S = [u]$. Thus every surviving term covers all u coordinates, and its interactions come from distinct columns. Terms choosing more than L interactions have a geometric tail bounded by powers of $2^u n^{-.03}$, which is smaller than the required n^{-3R} . Among at most L interactions covering $[u]$, one has at least u/L coordinates. Its mean absolute size is at most $n^{-.4u/L}$; the remaining factors are bounded on this set. The column and interaction choices cost $O_u(n^L)$, giving $O_u(n^{L-.4u/L})$, again smaller than n^{-3R} because $u > 10L^2$.

Without the alternating sum, the small range has weight $\exp(O_u(n^{-.03}))$, and the two displayed tail bounds still apply. This proves the uncentered assertion, including for subtuples. $\square$

Lemma 12.5 (Homogeneous extension counts and peeling). *Use the homogeneous version $\pi_l = \pi$. Suppose the support in use has no clique on $\lceil e^Q \rceil$ labels with $K_\pi > 3\theta$ between all distinct members. Then given any other labels of a tuple, the number of extensions by one label from the support to any interaction of magnitude $> \xi$ involving that label is at most $\exp(C_* Q)$, for a fixed C_* (we use Q large). Also for any fixed x , the number of labels z there with $|K_\pi(x, z)| > \xi$ satisfies this bound. If in addition*

$$\max_x \tau(x) D_\pi(x)^{-d} \exp(C_* Q) \leq \Gamma < 1$$

on the support, then the absolute contribution of the large-interaction tuples to the positive product terms of (12.2), summed over those terms, is $O_u(\Gamma)$.

Proof. After the other coordinates are fixed, a large extension gives a projection of f_z onto a bounded fixed direction whose magnitude is bounded below by a positive constant depending only on ξ, u . Split the candidates by projection sign. If t vectors in one sign class have all mutual inner products at most 3θ , their average has squared norm at most $1/t + 3\theta$, while its projection has that fixed positive lower bound. For sufficiently small θ , this forces $t < t_0$ for a fixed $t_0 = t_0(\xi, u)$. The graph joining pairs with $K_\pi > 3\theta$ has neither an independent set of size t_0 nor a clique of size $\lceil e^Q \rceil$. The elementary Ramsey binomial bound gives

$$\log \binom{\lceil e^Q \rceil + t_0}{t_0} \leq C_* Q.$$

Increase the fixed C_* , chosen after u, ξ, θ , to include the two signs and the finitely many interactions. The same projection argument gives the stated count for $|K_\pi(x, z)| > \xi$.

For a tuple with a large interaction, choose a maximal retained index set whose subtuple has every interaction at most ξ . Sum over the finitely many possible retained index sets. Once its labels are fixed, each excluded coordinate must create a large interaction when added to them; otherwise maximality fails. Its possible labels therefore lie in a set of size at most $\exp(C_* Q)$, determined by the retained labels alone.

In a positive product term of (12.2), deleting an excluded coordinate costs at most $D_\pi(x)^{-d}$. Under the original product measure, each excluded coordinate consequently contributes at most

$$\exp(C_* Q) \max_x \tau(x) D_\pi(x)^{-d} \leq \Gamma.$$

These necessary extension conditions can be integrated independently; no independence after conditioning on maximality is asserted. The remaining positive product has bounded integral by Lemma 12.4. Every tuple under consideration excludes at least one coordinate, so summing the finitely many retained sets and positive terms gives $C_u \Gamma$.

In particular, for the homogeneous mass Z defined in Lemma 12.2, the moderate bound and even centered-power Markov give the explicit consequence

$$\Pr\{Z < 1/2\} \leq 2^u (n^{-3R} + C_u \Gamma).$$

The peeling bound on the set with an interaction above ξ covers the complement of the moderate set. In heterogeneous applications the large-interaction part will be controlled separately. $\square$

The small- and moderate-interaction estimates of Lemmas 12.3 and 12.4 allow a different law at each second-label coordinate. The clique-count argument of Lemma 12.5 is used only with a single common law. In particular, conditioning on internal data later does not convert a heterogeneous collection into a homogeneous one.

Lemma 12.6 (Simultaneous row trimming). *For a uniform small-width law on a set of first labels, and any collection of at most $\exp(o(n^2))$ small-width second laws, we can, by removing an $o(1)$ fraction, arrange (for each law) for every kept x , with z uniform on the original first set:*

$$\Pr_z(|K_\pi(x, z)| > n^{-1-.03}) \leq e^{-n^2}, \quad \mathbb{E}_z[\mathbf{1}_{\{n^{-1-.03} < |K_\pi(x, z)| \leq 4\xi\}} e^{150n|K_\pi(x, z)|}] \leq e^{-n^2}. \quad (12.3)$$

For parameters subsequently interpolated at extremely fine mesh we can use $n^{-1-.02}$, 2ξ , $100n$, $e^{-n^{.19}}$ instead at the realized laws, by choosing the mesh fine enough.

Proof. Let x, z be independent uniform labels in the original first set. The raw two-free estimate gives $\Pr(|K_\pi(x, z)| > n^{-1-.03}) \leq e^{-n^{.4}}$. For the weighted expression, split at $n^{-1+.06}$. The two-free and one-free bounds respectively give

$$\mathbb{E}_{x,z} \left[\mathbf{1}_{\{n^{-1-.03} < |K_\pi(x,z)| \leq 4\xi\}} e^{150n|K_\pi(x,z)|} \right] \leq e^{150n^{.06}-n^{.4}} + e^{600\xi n - \alpha n/3} \leq e^{-cn^{.4}}$$

for a fixed $c > 0$ and large n . Markov, applied to each conditional expectation over z at threshold $e^{-n^{.2}}$, removes at most $e^{-cn^{.4}+n^{.2}}$ of the first labels per test. The union over $\exp(o(n^2))$ laws still removes an $o(1)$ fraction.

For interpolation, choose simplex diameters at most n^{-3} in ℓ^1 . Then $|K_\pi - K_{\pi'}| \leq \|\pi - \pi'\|_1 \leq n^{-3}$. The realized event with thresholds $n^{-1-.02}, 2\xi$ is contained in the corner event with thresholds $n^{-1-.03}, 4\xi$. Its exponential weight is at most the old weight times $e^{100n^{-2}}$. The relaxation from $e^{-n^{.2}}$ to $e^{-n^{.19}}$ pays for this factor. A finer mesh can simultaneously preserve any other finitely many strict margins needed later. $\square$

Parameters and normalizations in the final regime. The constants and scales in Sections 12–18 have the following separate roles.

- The failure-probability constants are fixed as $A_c = 200$, then sufficiently large P , then $R = P^2$. The moment estimates use a sufficiently large even u depending on R , followed by small ξ and then θ , as specified above. Their homogeneous extension bound supplies the fixed constant C_* .
- Section 13.1 chooses width exponents within the available discrepancy budgets. Section 13.2 fixes the extraction and geometry constants in the order required by the construction, ending with a sufficiently large fixed scale threshold. Estimates above that threshold are uniform; low cluster scales need not tend to infinity.
- Extraction fixes residual bias and cluster scales, patch sizes, internal dimensions, and gains. Mode, orientation, and color selection precedes prefix allocation and the later choice of internal coordinate sets. These quantities may depend on n . The low regime uses order $\log n$ late classes and $\lceil \log^2 n \rceil$ resampling rounds. Its late-error exponent is fixed small after the low-mode constants, as specified in Section 18.
- Widths remain normalized by the original N , including after labels are discarded. The later local normalizations are introduced where used: patch loads refer to the extracted side size, endpoint kernels to that size divided by the number of palettes, and late kernels to the size of the current late pool. Temporary notation has the scope declared at its use.

13 Patch extraction and cube allocation

We enter this construction with the initial discrepancy (7.2), the deep discrepancy (12.1) in both orientations, and the eventual cluster-witness absence of Corollary 10.2, all on the same retained sides. The persistent discards in Corollary 7.2 and the subsequent stabilizations of Lemma 3.2 total $o(N)$, so each retained side has size $N - o(N)$. We will extract disjoint host patches, select one common construction for them, allocate cube roles by prefixes, and then clean their first supports for prescribed comparison laws.

13.1 Measured bias and cluster scales

Use a tiny $\iota > 0$, say less than $\min(x_*, \eta_0, .01)/1000$, and $0 < a_C < \min(\eta_0, 1)/10^6$. Choose $a_B > 0$ much smaller than a_C , also small enough that (12.1) with exponent slack $\iota/2$ applies at widths n^{a_B} on both sides. Reserve a disjoint part of each retained side of size, say, $\lfloor N/3 \rfloor$ for possible later use.

Definition 13.1 (Residual scales). On the labels left after this reserve and any earlier patch removals, measure two dyadic scales (values that are powers of two with integer exponent). Both maxima are recalculated on each current residual pair:

- g is the largest dyadic value $b \geq 2$ for which two uniform sets each of size $\geq Ne^{-b^{a_B}}$ have absolute bias at least b/n .
- q is the largest dyadic $b \geq 2$ for which, in some orientation, there is a uniform first set of size $\geq Ne^{-b^{a_C}}$ and a family of disjoint bins on the second side, each of size at least e^b , with union of size $\geq Ne^{-b^{a_C}}$, such that all distinct pairs in each bin have inner product of their f -vectors $> \theta$ under that first law. This inner product does not depend on color choice.

Use 1 for an empty maximum.

These scales measure actual uniform-set witnesses at the current dimension; the exponents of Definition 9.1 measure availability along the stabilized sequence.

Lemma 13.2 (Bounds for the residual scales). *Always $g \leq n^{\iota/2}$ eventually by its elementary cap n and the budget just chosen. Also $q < n^\gamma$ eventually for each fixed $\gamma > 0$, uniformly on the residual sets.*

Proof. A nontrivial bias witness has $g \leq n$, so its two uniform laws have widths at most n^{a_B} . The discrepancy estimate chosen with slack $\iota/2$ gives

$$g/n \leq n^{-1+\iota/2},$$

which proves the first bound. For the cluster-scale bound, if not, pass to maxima with $\log_n q \rightarrow \gamma' \in [\gamma, 1]$ on a subsequence ($q \leq \log N$ when nontrivial). Remove column degree outliers into the stated first set using (7.2); this costs negligible mass even relative to the union size (exception count at most $2Ne^{-n^{\eta_0}}$, testing their uniform laws). Here we can take deviation $2n^{-\eta_0} = o(1)$, since the first width exponent is at most $\gamma' a_C + o(1)$. Keep the bins retaining at least half their mass, restricted to the survivors.

For distinct retained labels, their codegree in the red color is

$$\frac{1 + m(y) + m(y') + K(y, y')}{4},$$

where the means and inner product are under the first law. Here $|m(y)|, |m(y')| \leq 4n^{-\eta_0}$ and $K(y, y') > \theta$; blue changes the signs of the two means and keeps K . Thus both codegrees exceed $1/4 + \theta/8$ eventually. A diagonal codegree is the degree itself, close to $1/2$. All pairs therefore have codegree $\geq 1/4 + n^{-\delta}$, where one can fix rational numbers ζ, δ with $0 < \zeta < \gamma'$ and $\gamma' a_C < \delta < \min(\eta_0, \zeta, 1)/2000$. Uniform bin clusters with weights proportional to retained sizes now give a full-dimensional cluster witness with atom cap $\exp(-n^\zeta)$ and both widths $\leq n^\delta$, contradicting Corollary 10.2. $\square$

13.2 Extraction and cube allocation

Choose the parameters in the following order. There is room in the inequalities, and very large *fixed* thresholds suffice.

1. For extraction, use a large constant M_1 compared to the conflict-count constant C_* above, so $2C_*/\sqrt{M_1} \ll 10^{-3}$. Let $C_b > 100a_C/a_B + 100$, $M_{lo} > C_b + 100$, $0 < c_q < 1/(20M_{lo})$, and $M_{hi} \gg M_{lo} + 10/c_q$. Fix tiny $\omega > 0$ with $5\omega M_{hi} < a_C/100$. Here M_1 ensures that the homogeneous extension count in direct mode fits its gain; C_b separates the larger bias-test width from the input-law width. The constants M_{lo}, M_{hi} set internal dimensions, and ω keeps tuple-filter costs below the available width and bin-size budgets.
2. For cluster patches, use the fixed gain parameter $a = \theta/100$. For the geometric constructions take a tiny $\rho > 0$ so that h -bit balls of radii up to $1000\rho h$ have log volume $< (a/10^9)h$ for large h .
3. Use large constants K_B, A_0 compared to $10^6 R$; A_0 serves for order-log n late classes. Fix all the small-surplus geometric and probability constants before taking a large scale threshold Q_0 . Increase Q_0 as needed so estimates in positive powers of q, g, h below already have the required slack for arguments bounded below by Q_0 . They need not all grow along the sequence.

For a patch (X_i, Y_i) , write $M_i = |X_i| = |Y_i|$. Its internal dimension h_i is the number of cube directions to be handled inside an internal slice; s_i is the gain budget for support and prefix losses. The following three patch types specify the output of one extraction. Their scales g, q are those of the residual pair at that pass, and the constants in the size bounds are independent of n .

- In the *bounded case*, $\max(g, q) < M_1 Q_0$. We use one patch of size $M_i \asymp N$, with uniform laws on its two sides, either orientation and color, and $h_i = s_i = 0$. It belongs to low direct mode in the later assignment.
- A *direct bias patch* is used when the bounded case does not apply and $g > M_1 q$. It has

$$M_i \geq cNe^{-g^{a_B}}, \quad d_G(x, U(Y_i)) \geq \frac{1}{2} + \frac{g}{4n} \quad (x \in X_i),$$

$$h_i = 0, \quad s_i = 10^{-3}g,$$

where $U(Y_i)$ is uniform on Y_i . It is high when $g > K_B \log n$, and low otherwise.

- A *cluster patch* is used in the remaining case $g \leq M_1 q$, where $q \geq Q_0$. It has $M_i \geq cNe^{-q^{a_C}}$, and Y_i is partitioned into bins of common size d_i . Every pair in a bin, including repeated labels, has G -codegree at least $1/4 + 3a$ under the uniform law on X_i . It is low when $q \leq (\log n)^{c_q}$, and high otherwise. High patches use high-small-bin mode when $q \leq (\log n)^2$, and high-large-bin mode otherwise. Set

$$d_i = \lfloor e^{q/2} \rfloor,$$

except in high-small-bin mode, where this is replaced by the smaller of $\lfloor e^{q/2} \rfloor$ and $\lfloor \exp(\sqrt{\log n}) \rfloor$. Choose a power of two $q^M \leq h_i < 2q^M$, taking $M = M_{lo}$ in low mode and $M = M_{hi}$ in high mode, and put

$$k_i = \lceil h_i^{3\omega} \rceil, \quad T_i = \lceil h_i^\omega \rceil, \quad s_i = (a/10^6)h_i.$$

The internal solver of Section 14 will turn the within-bin codegree surplus into a gain on the h_i internal directions.

In the proposition, S denotes the total size of the selected mode/orientation/color family before a subcollection is retained; the dyadic sum and prefix claims concern that allocated subcollection.

Proposition 13.3 (Extraction and allocation). *There are pairwise disjoint patches (X_i, Y_i) with $|X_i| = |Y_i| = M_i$ and $\sum_i M_i = S \geq cN$ for a fixed $c > 0$, all of one mode, orientation and color. They have one of the three patch types just specified. A subcollection admits dyadic prefix lengths ℓ_i with*

$$\sum_i 2^{-\ell_i} = 1, \quad M_i/S \leq 2^{-\ell_i} < 2M_i/S.$$

In its prefix allocation, patch i has $\Theta(2^{\ell_i} M_i/N)$ roles of each parity. Each star in that patch has exactly ℓ_i crossing flips, visiting distinct other leaves, and a nested tail coordinate set of size h_i supplies its internal directions. Eventually $\max_i(h_i, \ell_i) < n^t$. Except in the bounded case,

$$\ell_i, \log(N/M_i) \leq s_i/(1000u).$$

The bounded case uses one patch, with $\ell_i = h_i = s_i = 0$.

Proof.

At every pass, work on the labels left after removing both supports of the earlier patches from the working sides, and recompute g, q on the current residual pair. A new pass is started only while the extracted equal sizes sum to less than $.1N$. Thus each residual side then has size at least

$$N - o(N) - \lfloor N/3 \rfloor - .1N \geq cN$$

for a fixed $c > 0$, as needed by the bounded branch.

Extracting one patch. If $\max(g, q) < M_1 Q_0$ during extraction, take the bounded case right away on those residual sets, still of sizes $\geq cN$; this uses just one equal-side-size patch, by truncating to arbitrary sets of the smaller size, with uniform laws, either orientation and color, and low direct mode below. Otherwise proceed as follows (later g, q for an extracted patch refer to the values here at its own extraction):

- If $g > M_1 q$, use direct bias mode. Extract first and second sets X_i, Y_i of equal size $M_i \geq cNe^{-g^{aB}}$, with the second law uniform and every first label having degree at least $1/2 + g/(4n)$ into it in some fixed color of that patch. To see this, let U, V witness g and choose their denser color. The top eighth U_1 of rows by degree has size at least $Ne^{-(2g)^{aB}}$, since the fixed threshold makes $(2g)^{aB} - g^{aB} > \log 8 + O(1)$. The absent $2g$ budget therefore gives $d_G(U_1, V) < 1/2 + 2g/n$. If fewer than an eighth of all rows had degree at least $1/2 + g/(2n)$, the remaining seven eighths would be below that threshold, and

$$d_G(U, V) < \frac{1}{8}(1/2 + 2g/n) + \frac{7}{8}(1/2 + g/(2n)) = 1/2 + 11g/(16n),$$

contrary to the witness. Let U^+ be the rows meeting this degree threshold and take $M_i = \min(|U^+|, |V|)$. Keep any M_i of the good first labels, and choose an M_i -subset of V . A uniform such choice preserves all their degrees to $o(1/n)$ simultaneously: $M_i = e^{\Omega(n)}$, so sampling concentration permits a union over at most N rows. One realization gives the asserted $g/(4n)$ surplus.

- Otherwise use the cluster mode ($q \geq Q_0$), with orientation from its witness and, say, red color. Extract X_i, Y_i of equal size $M_i \geq cNe^{-q^{aC}}$, where Y_i is partitioned into bins of the prescribed size d_i . Every within-bin pair (including diagonals) has codegree $\geq 1/4 + 3a$

under uniform X_i . First remove the degree outliers as in Lemma 13.2. Retain original bins losing at most half their mass, restrict them to the survivors, and partition them into bins of the stated size, discarding the final remainders. This loses only a negligible or fixed small fraction. Choose a common size that is a multiple of d_i , no larger than either the surviving first set or the whole-bin union, and maximal with these properties. It remains at least $cNe^{-q^{a_C}}$. Keep that many whole-bin labels on the second side and sample the same number of first labels. Since this number is $e^{\Omega(n)}$, concentration followed by a union over the at most N^2 within-bin pairs preserves every codegree with the displayed slack.

The chosen cluster parameters satisfy

$$k_i T_i = O(q^{4\omega M}) \ll q^{a_C/2}, \log d_i.$$

For high-small-bin mode, $\log d_i = \min(q/2, \sqrt{\log n}) + O(1)$; the bound $q \leq (\log n)^2$ and the choice of ω give the second comparison also at the cutoff. Likewise h_i is smaller than any fixed positive power of d_i needed below, uniformly after increasing Q_0 . The inequalities $c_q M_{lo} < 1/20$ and $c_q M_{hi} > 5$ give, respectively, $h_i \leq (\log n)^{-1}$ in low mode and $h_i > (\log n)^5$ in high mode eventually. After either nonbounded extraction, remove its two supports before the next pass and retain its measured g, q values with that patch.

Repeating extraction and selecting one type. Continue until the sum of the equal sizes first reaches $.1N$, or until the immediate bounded branch fires. In the former case, select one of the finite mode/orientation/color possibilities, distinguishing also high-small-bin and high-large-bin. Let I be its patch indices and put $\sum_{i \in I} M_i = S \geq c'N$. In the immediate bounded case, ignore earlier extractions and take I to contain only the bounded patch, with $S = M_i$ and $s_i = h_i = 0$. From now on all patches used are of this one type, orienting notation accordingly. The color is selected together with the mode and orientation, before any cube allocation; thus every edge required by the subsequent construction is tested in this single selected color.

Dyadic prefix allocation. For $i \in I$, put $p_i = M_i/S$ and round it **up** to a dyadic weight $r_i = 2^{-\ell_i}$. Then

$$p_i \leq r_i < 2p_i, \quad \sum_{i \in I} r_i \geq 1.$$

Order the r_i decreasingly, and let J be the first segment whose sum reaches 1. Both 1 and all preceding terms are integer multiples of the last term in this segment. The preceding partial sum is below 1, so adding that last term reaches 1 exactly. Hence $\sum_{i \in J} r_i = 1$. We retain only these patches for the allocation, but S continues to mean the total over I before this restriction. In particular $\sum_{i \in J} M_i > S/2$.

The allocation uses subcubes obtained by fixing initial coordinates, as in the cube tilings of Conlon, Fox, Lee and Sudakov [6, Section 2.2] and their adaptation by Fiz Pontiveros, Griffiths, Morris, Saxton and Skokan [13, Section 2]. Here the sizes are chosen from the patch masses, and the later embedding uses the patch laws. Use the weights r_i , $i \in J$, as disjoint complete prefix leaves in the first cube coordinates. Positions in leaf i belong to patch i for both parities. The leaf has exactly $2^{n-\ell_i-1} = \Theta(2^n M_i/N)$ roles of each parity eventually. Leading flips, called crossings, number ℓ_i per star of patch i , visiting distinct other leaves: the first prefix mismatch identifies the flipped coordinate.

Internal and bulk coordinates. In the tail, beyond all leading coordinates, dedicate nested coordinate sets of sizes h_i as internal for patch i ; by a slice in that patch we mean exactly fixing their complementary coordinates. Other nonleading, noninternal flips there are called bulk. Here nonleading is relative to the current leaf: a coordinate beyond its prefix may still belong to a longer leaf's prefix. The size bounds and dyadic rounding give

$$\log(N/M_i) \leq \begin{cases} g^{a_B} + O(1) & \text{direct,} \\ q^{a_C} + O(1) & \text{cluster,} \end{cases} \quad \ell_i \leq \log_2(S/M_i) + 1.$$

Together with $g \leq n^{\iota/2}$, and Lemma 13.2 used with a fixed $\gamma < \iota/(2M_{\text{hi}})$, these show that the maximum of the h_i over I and the ℓ_i over J is $< n^{\iota}$ eventually. In particular all asserted dimension allocations fit. Except in the bounded case we can ensure

$$\ell_i, \log(N/M_i) \leq s_i/(1000u)$$

because $g^{a_B} = o(g)$ and $q^{a_C} = o(q^M)$ as their arguments grow; choose the fixed threshold so these inequalities hold with the displayed constants for every argument above it. In the bounded case these quantities are $O(1)$ (actually $\ell_i = 0$). $\square$

13.3 Cleaning at a prescribed input profile

Section 14 will choose the cluster comparison laws through a fixed point. We first clean the first supports at any prescribed input, with enough stability for nearby inputs. Here X_i continues to denote the extracted set of size M_i ; μ_i is uniform on the support left after cleaning.

Lemma 13.4 (Stable cleaning). *Here are the cleaned first laws. In direct and bounded modes the comparison laws π_i are uniform on Y_i . In cluster mode for now consider any vector of input comparison laws π_i on Y_i of normalized cap*

$$M_i \max \pi_i \leq \exp(10k_i T_i).$$

At such input, take μ_i uniform on X_i after the following cleanings, which need waste at most a tiny fraction (less than a). Its support has

$$\begin{aligned} 1/2 + g/(4n) &\leq D_{\pi_i} \leq 1/2 + 4g/n && \text{direct bias,} \\ |D_{\pi_i} - .5| &\leq 10q^{C_b}/n && \text{cluster mode,} \\ |D_{\pi_i} - .5| &\leq K/n && \text{bounded case, fixed } K. \end{aligned} \tag{13.1}$$

In cluster mode the right deviation budget $10q^{C_b}$ can be made smaller than $s_i/(100u)$. Furthermore, for a parameter Q_i , each cleaned support has no set of $\lceil e^{Q_i} \rceil$ distinct labels with $K_{\pi_i}(x, z) > 3\theta$ on every pair, as required by Lemma 12.5. This parameter satisfies $\Delta_i := \exp(C_ Q_i) \leq e^{s_i}$, or just $\Delta_i \leq K$ fixed in the bounded case. All first labels there have $|D_{\pi_j} - .5| \leq 3b_*$ for $j \neq i$, and (12.3) with its relaxed thresholds $n^{-1-.02}, 2\xi, 100n, e^{-n^{.19}}$, with z uniform on the original extracted X_i , for every patch law π_j . Cluster mode retains all within-bin codegrees $\geq 1/4 + a$ into μ_i . In cluster mode the same support has these properties for all inputs in a sufficiently small ℓ^1 -neighborhood of the input used for cleaning. The no-clique assertion is about each such support separately, not their union over inputs.*

Proof.

Approximating a capped law by a uniform set. The measured budgets concern uniform sets, so first fix a law π on the residual labels of width $W = \log(N \max \pi)$, and a larger powered width budget B . Set

$$t = Ne^{-(B+W)/2}, \quad \Pr(y \in V) = t\pi(y),$$

independently over labels. Then $t \max \pi = e^{-(B-W)/2} < 1$ and $\mathbb{E}|V| = t \gg Ne^{-B}$. In the applications $B, W = o(n)$, so $t = e^{\Omega(n)}$. Concentration of the cardinality and all bounded degree and pair sums, followed by a union over $O(N^2)$ tests, gives one realization whose uniform law approximates every such test against π to $o(1/n)$, and whose size is at least Ne^{-B} .

Own-patch degrees. At the residual pair used to extract a cluster patch, choose a dyadic bias budget in $[q^{C_b}, 4q^{C_b}]$, larger than the measured $g \leq M_1 q$. Its powered width exceeds the input width with ample slack:

$$q^{C_b a_B} \gg q^{a_C} + q^{4\omega M}, \quad \log(N \max \pi_i) \leq \log(N/M_i) + 10k_i T_i.$$

Use the uniform approximant just constructed for the second law. If the rows of X_i exceeding $8q^{C_b}/n$ in either bias direction numbered at least the higher budget's size threshold, they and this approximant would form a rectangle witnessing that absent budget. Each exceptional row set is therefore smaller than its threshold, a tiny fraction of M_i . In direct mode π_i is already uniform and fits the absent $2g$ budget; the same rectangle test gives the upper degree trimming while preserving the lower bound. In bounded mode choose fixed dyadic budgets above $M_1 Q_0$, large enough that their size thresholds are less than $aM_i/10$. They give a fixed degree constant K .

Own-patch clique exclusion. For cluster mode choose a dyadic $Q_i \in [q^2, 2q^2]$; for direct mode choose one in $(g/(2\sqrt{M_1}), 2g/\sqrt{M_1}]$. Both exceed the measured cluster maximum. Their powered widths exceed the input widths because

$$q^{2a_C} \gg q^{a_C} + k_i T_i, \quad (g/\sqrt{M_1})^{a_C} \gg g^{a_B}$$

above the fixed threshold. Approximate π_i by a uniform law at this budget. Greedily remove disjoint cliques of size $\lceil e^{Q_i} \rceil$ in X_i , with pair inner products under π_i greater than 2θ . If their union had size at least $Ne^{-Q_i^{a_C}}$, it would, together with the approximating first law in the reverse orientation, witness the absent cluster budget Q_i : the approximation retains pair inner products greater than θ . The removed union is thus smaller than that threshold. Its fraction is tiny, and $\Delta_i = e^{C_* Q_i} \leq e^{s_i}$ by the powers of q and the choice of M_1, Q_0 . In bounded mode a sufficiently large fixed Q_i gives the same small-loss conclusion and a fixed Δ_i .

Other-patch degrees and row tails. In cluster mode the input widths satisfy

$$\log(N/M_i) + 10k_i T_i \leq n^{x^*/4}, \quad \#I \leq e^{n^{2\epsilon}} = e^{o(n^2)}.$$

For direct and bounded modes omit the term $10k_i T_i$; their input laws are uniform. For each $j \neq i$, conditioning uniform X_i on a putative outlier fraction e^{-cn} , with fixed $c < \alpha/2$, still fits the linear width in (12.1). Thus the fraction with deviation greater than $2b_*$ against π_j is $e^{-\Omega(n)}$. Lemma 12.6 supplies the row-tail removal for the whole collection. Their unions have vanishing relative mass. Choose the fixed thresholds so own-degree and clique removals together cost less

than, say, $a/2$; the other two removals together cost less than $a/2$ eventually. If a total fraction $r < a$ is removed, each original within-bin codegree of at least $1/4 + 3a$ becomes at least

$$\frac{1/4 + 3a - r}{1 - r} \geq 1/4 + a.$$

Finally hold this cleaned support fixed while varying the inputs. Degrees and pair inner products are Lipschitz in the ℓ^1 distance of the comparison laws. A sufficiently fine mesh, of diameter at most n^{-3} , preserves the degree margins 8 to 10, $2b_*$ to $3b_*$, the clique threshold 2θ to 3θ , and the relaxed row-tail bounds proved in Lemma 12.6. Cleaning is performed separately at each corner; it requires no union over mesh corners. The later union of corner supports inherits only the bounds that hold label by label. $\square$

The threshold Q_0 is fixed before the asymptotic sequence is taken. In particular, estimates in powers of a cluster scale are required uniformly above this threshold; low-mode scales are not assumed to diverge. For profile interpolation, clique exclusion is a property of each cleaned corner support separately. The later union of those supports is used only for the degree and row-tail bounds that hold label by label.

14 Internal slice solver and balanced profiles

Section 13 fixed the internal cube slices and then supplied cleaned first supports for prescribed comparison laws. This section first constructs the internal odd and even laws for a cluster patch, then chooses the comparison laws consistently through a fixed point.

A parameter point consists of one input law π_i and one nonnegative price vector z_i for each cluster patch, with

$$M_i \max \pi_i \leq e^{10k_i T_i}, \quad z_i \in \Delta(Y_i) := \{z \geq 0 : \sum_{y \in Y_i} z(y) = 1\}.$$

This is a compact polytope. Triangulate it finely enough for Lemma 13.4, and so that coordinate price changes within a simplex are $o(1/M_i)$. At every vertex of this mesh, store the cleaned first laws and the price masks defined below. A fixed parameter point supplies its barycentric probabilities on the vertices of its simplex. Each center and each odd group will draw a vertex independently with these same probabilities; they do not share one vertex draw. The output laws and the eventual fixed point are constructed after the slice experiment.

Proposition 14.1 (The internal slice solver). *Fix a cluster patch from Proposition 13.3, cleaned by Lemma 13.4 at the corners of a sufficiently fine parameter mesh. For any common choice of barycentric corner probabilities, there is a slice experiment with primitive data W , odd bin probabilities q_g^W , conditional in-bin laws U_g^W , odd marginal laws p_g^W , and even subprobability rows σ_v . The odd bin probabilities, individual label masses, and in-bin support sizes obey (14.1). The even rows have the bounds*

$$N \max_x \sigma_v(x) \leq 2^{h_i} e^{-500s_i}, \quad \mathbb{E} \sigma_v(x) \leq K/M_i,$$

where the expectation uses unrestricted primitive data and internal reference sampling. Whenever an even row is nonzero, it is a probability law supported in one cleaned corner support and on common neighbors of every internal odd label. The local tests in (14.3) assert local validity and

bound the internal reference probability of $\sigma_v = 0$, conditional on W , by $\epsilon_i = e^{-.001ak_i h_i}$. They hold simultaneously in a slice except with probability $\exp(-h_i^{1+\Omega(1)})$. All estimates are uniform in the corner probabilities, with the deterministic local consultation domains specified below.

Proof.

14.1 Primitive data and selection geometry

Fix an h_i -slice and abbreviate $h = h_i, k = k_i, T = T_i$. Conditional on the common parameter point, different slices have independent copies of the following experiment. At every potential center c , indexed by an internal location and a level, draw a mesh vertex, use its cleaned law μ_c , and draw $W_c \sim \mu_c^{\otimes k}$. These variables are generated even when the center is absent. Each odd group independently draws its mask vertex. Prospective positions, the orders used in searches, activations, and site-level choice ties are further independent primitive variables. Eligibility is computed before using activations or choice ties. We write W for this whole pre-bin history, including the selections derived from the primitives. Its derived fields need not be independent. Before success conditioning, the records at distinct centers and groups and the remaining randomizers are independent; each tuple is sampled conditional on its own corner.

Here is the projection defining groups and sites. Index the h internal coordinates by the binary group of order h , and for a word z set

$$s(z) = \sum_l l z_l, \quad \psi(z) = z \oplus e_{s(z)}.$$

The map has syndrome-zero image and fibers of size h , and always flips one bit. An odd group is a fiber on the odd roles of the slice; a selection site is $\psi(v)$ for an even role. The parity convention includes the shift from the slice's fixed complementary word. A group's site neighborhood consists of the projected even roles adjacent to one of its odd roles. As in Section 10.1, these sites have mutual distances at most 6, and each site lies in $O(h^3)$ such neighborhoods.

For a fixed even role,

$$\psi(v \oplus e_l) = v \oplus e_l \oplus e_{s(v)+l}.$$

If $s(v) \neq 0$, the flips l and $l + s(v)$ form pairs with the same image; if $s(v) = 0$, all h flips have the same image. Thus each group queried by this star has multiplicity $j \geq 2$, and the multiplicities sum to h . This is the incidence property used in the posterior calculation.

For each group and mesh vertex, form a mask from that vertex's price vector on Y_i . Take the uniform prior on bins with uniform in-bin laws; retain bins having at least half their labels of price $\leq 10/M_i$, then restrict to those cheap labels in each bin, normalizing both stages. At most a tenth of labels overall exceed the cut, so most bins have enough cheap mass; aggregate is bounded by 4 times uniform on Y_i . A physical bin is a member of the fixed partition of Y_i . Write D for its masked uniform probability law when it occurs inside a formula such as $D(F)$, and $r_g(D)$ for the masked prior probability of choosing that bin at group g . Thus a bin identity determines a group-dependent masked law. All these prior lookups and draws up to the actual assignment of clusters are local. In particular different centers can have different cleaned first laws, all still giving codegrees $\geq 1/4 + a$.

We give details of the template, including its gates used later. Use the geometric height device on Q_h with step bound 6, radius $\lfloor \rho h \rfloor$, $\lambda = h^{10}$, activation and crowd exponents $b_0 = \omega/8$, $b = \omega/2$, height-proof exponents $a_g = \omega/12$, $\zeta_g = 1 - \theta_g = \omega/30$, $0 < \sigma_g \ll \omega$ (subscripts refer here to geometric parameters only). Positions and activations are independent of tuples, masks and tie

randomizers. The consultation distance for paths is $o(h)$ in this geometric estimate, in particular tiny compared to ρh by the large threshold. Throughout, asymptotic slack in powers of h itself need only hold for $q \geq Q_0$, choosing that fixed threshold sufficiently large.

14.2 List tests and odd bin laws

Fix corners, the group's mask, prospective positions, and a hypothetical set $\mathcal{D}$ of 1 through T distinct candidate IDs in its neighborhood. Let F be the set hitting all entries of their tuples and let F_{-c} omit tuple c . Set $A(J) = \mathbb{E}_{D \sim r_g} D(J)^2$. The list must satisfy the absolute and deletion bounds from (10.1), with every ID treated as an own-slice ID:

$$A(F) \geq e^{-2k|\mathcal{D}|}, \quad \frac{A(F)}{A(F_{-c})} \geq e^{(-\log 4 + .4a)k} \quad (c \in \mathcal{D}).$$

To bound failure, expose tuple coordinates successively; for each deletion test choose an order in which the indicated tuple is last. If the entries already exposed impose the hit set J , the next ratio is

$$\frac{A(J \cap N_G(x))}{A(J)} = \mathbb{E}_{D \text{ tilted by } D(J)^2 d_G(x; D|J)^2}.$$

The next x has its center's cleaned first law. All-pairs codegrees give conditional mean ratio at least $1/4 + a$, even though different centers can use different corner laws. While preceding ratios are at least .24, the second aggregate law has width at most $\log(N/M_i) + O(1) + 2kT < n^{\eta_0}/2$. Jensen and (7.2) then bound the probability of a ratio below $1/4 - O(n^{-\eta_0})$ by $e^{-\Omega(n^{\eta_0})}$.

As in the proof of (10.1), clip log ratios below at $\log(.24)$. Their conditional means are at least $-\log 4 + .7a$. The deletion inequality asks the last k increments to sum to at least $(-\log 4 + .4a)k$, leaving a deviation margin $.3ak$. Bounded-increment concentration gives failure $e^{-\Omega(a^2k)}$. For the absolute inequality, sum all $k|\mathcal{D}|$ increments and use its still larger margin above $-2k|\mathcal{D}|$. Stop on an earlier invalid ratio, continue the clipped process with good values, and separately add the negligible probability of an unclipped catastrophe. The finite collection of orders gives the stated failure bound for one list.

In each group neighborhood search failed lists using present IDs in range of incident even sites, at any levels, and mark a maximal disjoint family's union forbidden for choices by those incident sites. On polynomial position bounds there are at most $\exp(O(T \log h))$ lists per search and disjoint tests in one search are independent conditional on positions, masks and corner realizations. Here disjoint lists have disjoint ID sets. A union bound for h disjoint failures, using $T \log h = o(a^2k)$, shows that each search has fewer than h of them except with probability $\exp(-h^{1+\Omega(1)})$ per slice. Such a search marks fewer than hT IDs. Since a site participates in $O(h^3)$ searches, it loses at most $O(h^4T) = o(\lambda)$ choices. Binomial position concentration puts the original counts close to λ ; hence the remaining eligible count is at least $\lambda/2$, while the original count is at most 2λ , with the same slice failure bound. Now apply the activation/path mechanism with these eligibilities. Choose one ID per site at the obtained height by a uniform active eligible tie (that same choice used everywhere it is queried). The largest scale proof gives heights and non-bad-layer choices everywhere there except with probability $\exp(-h^{1+\Omega(1)})$ on size success. Realized lists at each group then have size $\leq T$ by the crowd bound at the (at most two) levels used, and pass (10.1): a failed realized list uses no marked ID and would enlarge the maximal disjoint family.

At each group on validity, given W , sample independently a masked D tilted by $D(F)^2$, conditional further on

$$D(F) \geq e^{-1.5k|\mathcal{D}|}, \quad D(F)/D(F_{-c}) \geq e^{(-\log 2 + .08a)k} \quad \text{for all } c.$$

The extra restrictions retain probability at least $1/2$ by (10.1). Thus the first stage selects a physical bin by a restricted squared-mass tilt of r_g . Conditional on that bin, the internal reference law draws all queried role labels independently from its uniform law $D|_F$. Write $q_g^W(D)$ for the induced bin probabilities at group g , $U_g^W(D)$ for the resulting individual conditional uniform-subset law given bin, $p_g^W = \sum_D q_g^W(D)U_g^W(D)$. On needed fallbacks just draw from the group's masked prior and laws. The label-marginal cap uses cancellation of one hit-mass factor:

$$p_g^W(y) \leq \frac{2}{A(F)} \sum_D r_g(D) D(F) D(y) \mathbf{1}_F(y) \leq \frac{8e^{2kT}}{M_i}.$$

The masked bin prior is at most $2d_i/M_i$, so its squared tilt and extra restrictions give the first cap below. The absolute mass restriction and the masked support size give the third. These bounds also hold for the prescribed fallbacks:

$$q_g^W(D) \leq 4e^{2kT} d_i/M_i, \quad p_g^W(y) \leq 8e^{2kT}/M_i, \quad |\text{supp } U_g^W(D)| \geq \frac{1}{2} d_i e^{-1.5kT} \quad (14.1)$$

for used bins. In this notation g as subscript is a group, not bias budget.

Local validity and fallback rules. The local rules are defined even when some tests fail. At a site, use the height rule restricted to its long consultation domain. Choose no ID if that height reaches the horizon or its site-level is bad; otherwise choose a uniform active eligible ID, with the pre-randomized tie for that level.

A group is valid when all its incident sites make choices, every candidate ball at those sites and levels has at most 2λ positions, and the resulting list has size between 1 and T and passes (10.1). An invalid group uses the masked-prior fallback above.

For an even star at v , let $\mathcal{V}_v$ require validity of its neighboring groups and the position upper bound 2λ and eligibility lower bound $\lambda/2$ at every site-level in its projected site's long consultation domain. This event depends on W , before bins or labels are drawn. It ensures a genuine selection at the projected site. A positive selected level then requires a path reaching positive height, and selection uses the actual independent uniform active tie. All these events hold simultaneously on the slice success already proved.

Fallback/mark/choice tie rules can use separate independent order permutations per queried group or site-level, as needed (on ID lists or on IDs, respectively) to be translation-invariant in law under syndrome-zero shifts, including shifts interchanging the two local parity conventions. Choice ties themselves are independent of eligibility and activation data. All validity and comparison rules may and will use this symmetry. The height lemma with ambient parameter h gives $2DH + O(\rho h + 1) < 10\rho h$ after increasing the threshold. Thus the rules consult only this slice within $10\rho h$ of the inner queried state, including in the counterfactual calculations below (eligibility calculations themselves just use incident group tests in range, not their selections).

14.3 Even posteriors and predictive tests

Fix a present candidate ID c for v 's projected site. Hold all other primitive data fixed, including the corner selecting its prior μ_c , and vary $w = W_c$. For ordered internal neighbor labels $\mathbf{y}$, define

$$L_w(\mathbf{y}) = \mathbf{1}_{\{c \text{ selected, } \mathcal{V}_v\}}(W_{-c}, w) \Pr_{\text{int}}\{\mathbf{Y} = \mathbf{y} \mid W_{-c}, w\}.$$

The bin at each group is integrated out in this probability. Selections, validity, and all bin laws are recomputed at w .

We construct a reference independent of w . For each queried group g , let $\mathcal{L}_g$ be the family of all lists of at most T present IDs in its candidate ranges that contain c . This family uses only the frozen positions. For $\mathcal{D} \in \mathcal{L}_g$, delete c , tilt the masked bin prior by $D(F_{-c})^2$, and draw its j_g queried labels independently from $D|_{F_{-c}}$. Denote this probability by $Q_{g,\mathcal{D}}$, using a fixed symmetric fallback if its normalization is zero. Average uniformly over the family, with a fallback also for an empty family, and set

$$\mathcal{Q} = \bigotimes_g \left(\frac{1}{|\mathcal{L}_g|} \sum_{\mathcal{D} \in \mathcal{L}_g} Q_{g,\mathcal{D}} \right).$$

This notation uses the fallback convention when the family is empty. In particular the actual list may depend on w , but the reference mixture does not.

For one fixed list, the likelihood ratio before integrating its bin is at most

$$2 \frac{A(F_{-c})}{A(F)} \left(\frac{D(F)}{D(F_{-c})} \right)^{2-j_g} \leq 2e^{(\log 2 - .08a)kj_g}.$$

The factor 2 pays for the extra bin restrictions; the other factors are the squared-tilt normalizer and the j_g conditional label densities. The inequality uses $j_g \geq 2$ and the tested ratio bounds. Passing to the fixed mixture costs $|\mathcal{L}_g|$, with logarithm $O(T \log h) = o(ak)$ on the position gate. Since $\sum_g j_g = h$ and there are no singleton incidences,

$$L_w \leq e^{(\log 2 - .06a)kh} \mathcal{Q}.$$

If a required position bound fails, the sublikelihood is zero.

With c actually selected on $\mathcal{V}_v$, require

$$Z(\mathbf{y}) = \int L_w(\mathbf{y}) d\mu_c^{\otimes k}(w) > 0, \quad Z(\mathbf{y}) \geq e^{-.01akh} \mathcal{Q}(\mathbf{y}).$$

The posterior $P_{c,\mathbf{y}}(dw) = L_w(\mathbf{y})d\mu_c^{\otimes k}(w)/Z(\mathbf{y})$ has density at most $e^{(\log 2 - .05a)kh}$ relative to its prior. Since $\max \mu_c \leq 2/M_i$ and the allocation gives $\log(2N/M_i) \leq .01ah$, its joint atom cap relative to N^{-k} is $e^{(\log 2 - .04a)kh}$.

Let $\bar{P}$ be its average coordinate marginal, and let $B = \{x : N\bar{P}(x) > 2^h e^{-.02ah}\}$. Then $|B|/N \leq 2^{-h} e^{.02ah}$. For any specified $r = \lceil (1 - .01a)k \rceil$ coordinates, the joint cap bounds the probability that all lie in B by

$$e^{(\log 2 - .04a)kh} (|B|/N)^r = e^{-\Omega(akh)}.$$

The union over at most 2^k coordinate subsets is negligible. Hence the expected fraction of coordinates outside B , namely $\bar{P}(B^c)$, is at least ca for a fixed $c > 0$. Define σ_v by restricting $\bar{P}$ to B^c and normalizing; this costs $O(1/a)$, which the fixed threshold absorbs in the cap below.

On failures of the gates let $\sigma_v = 0$; otherwise it is supported in a single cleaned corner law's support and on common neighbors of **all** internal labels. We have

$$N \max_x \sigma_v(x) \leq 2^h e^{-500s_i}, \quad \mathbb{E} \sigma_v(x) \leq K/M_i \quad (14.2)$$

where expectation is for unrestricted W and internal reference sampling and K a fixed constant. To prove the mean bound, first fix the data other than W_c . If $\bar{P}_{c,\mathbf{y}}$ is the posterior coordinate marginal, cancellation of its denominator gives

$$\sum_{\mathbf{y}} Z(\mathbf{y}) \bar{P}_{c,\mathbf{y}}(x) = \int \frac{1}{k} \sum_{r \leq k} \mathbf{1}_{\{w_r=x\}} \sum_{\mathbf{y}} L_w(\mathbf{y}) d\mu_c^{\otimes k}(w).$$

The inner sum is the probability of selecting c and passing $\mathcal{V}_v$ in this candidate experiment. Thus, after the $O(1/a)$ restriction cost, the contribution is bounded by the actual tuple-coordinate incidence with that selection gate.

An entry equals x with prior probability at most $2/M_i$; conditioning on this equality leaves positions and activations with their original laws. Given presence of c , the gated selection probability at level zero is $O(1/\lambda)$ by the uniform active tie bound. At higher levels the position-averaged, eligibility-uniform positive-height bound is $e^{-h^{\Omega(1)}}$, with the one specified center forced present. These are precisely the bounds of Lemma 3.8; they do not condition on global success. The sum of candidate presence probabilities is λ per level. Summing over the polynomial number of levels proves $\mathbb{E}\sigma_v(x) \leq K/M_i$.

Let E_v^{in} denote failure ($\sigma_v = 0$). For each candidate, Lemma 3.7 bounds failure of the displayed predictive inequality, on its selection and validity gate, by $e^{-.01akh}$. The position-count gate permits only polynomially many candidates. Consequently the nonnegative function of primitive data

$$B_v(W) := \mathbf{1}_{\mathcal{V}_v} \Pr_{\text{int}}(E_v^{\text{in}} \mid W)$$

has raw mean at most $h^{O(1)}e^{-.01akh}$. For $\epsilon_i = e^{-.001ak_ih_i}$,

$$\mathcal{H}_v : \quad \mathcal{V}_v, \quad \Pr_{\text{int}}(E_v^{\text{in}} \mid W) \leq \epsilon_i \quad (14.3)$$

holds simultaneously per slice except with probability $\exp(-h_i^{1+\Omega(1)})$: apply Markov to B_v at threshold ϵ_i , union over the 2^{h-1} even roles, and add the separate slice-wide validity failure. The exponent akh , with $k = h^{3\omega} + O(1)$, pays for this union. The individual tests (14.3) are local as well. All these estimates are uniform in the corner probabilities. $\square$

14.4 The simultaneous profile choice

Use the parameter polytope and mesh defined at the start of this section. For each cluster patch, define its unrestricted mean odd law by

$$\pi_i^0 = \mathbb{E}_W p_g^W.$$

In high mode set $\pi_i^{\text{out}} = \pi_i^0$. In low mode first condition the primitive slice history W on (14.3) at every even role. For such a history, let $\mathcal{B}_g(W)$ be the positive bins satisfying

$$\Pr_{\text{int}}(E_v^{\text{in}} \mid W, D_g = D) \leq \sqrt{\epsilon_i} \quad \text{for every internal star incident to } g.$$

There are at most h^2 incident stars. Markov and (14.3) give $q_g^W(\mathcal{B}_g(W)) \geq 1 - h^2\sqrt{\epsilon_i}$, so the pretrimmed bin law

$$q_g^{\text{in},W}(D) = \frac{q_g^W(D)\mathbf{1}_{\mathcal{B}_g(W)}(D)}{q_g^W(\mathcal{B}_g(W))}$$

is well defined. Leave the in-bin laws $U_g^W(D)$ unchanged, and define the low-mode output π_i^{out} as the mean of $\sum_D q_g^{\text{in},W}(D)U_g^W(D)$ under the conditioned slice law. The tests here evaluate the original raw failure rule; no posterior is recomputed with a trimmed prior. Syndrome-zero translation symmetry makes these means independent of the group g , including across the two parity conventions.

Proposition 14.2 (Balanced patch profiles). *In the capped product domain of cluster-patch input laws and price simplices, there is a parameter point (π, z) such that $\pi_i = \pi_i^{\text{out}}$ and z_i maximizes $z'_i \cdot \pi_i$ over $\Delta(Y_i)$, for every patch i . At this point,*

$$\max \pi_i \leq 11/M_i. \quad (14.4)$$

Every cleaned corner support used with positive probability satisfies Lemma 13.4 at the realized comparison laws $(\pi_i)_i$. Their union in a patch, called its envelope, inherits the degree and row-tail bounds; clique exclusion is asserted separately on each corner support. In low cluster mode the unrestricted mean satisfies

$$\|\pi_i^0 - \pi_i\|_{\text{TV}} \leq \exp(-h_i^{1+\Omega(1)}) + O(h_i^2 \sqrt{\epsilon_i}).$$

Proof. The output means are continuous in the parameters. Barycentric probabilities are continuous and piecewise affine; varying them changes the weights of the finite primitive experiments but leaves every corner lookup, recomputed validity gate and posterior rule for a fixed realization unchanged. In particular, each posterior retains its realized corner law as prior. The slice-conditioning denominators are bounded below, as are the pretrim normalizers just estimated. Equation (14.1) puts the outputs in the capped input domain: the bounded conditioning and trim factors fit the gap from $2k_i T_i$ to $10k_i T_i$.

Apply Kakutani to the response

$$(\pi, z) \mapsto \left(\pi^{\text{out}}(\pi, z), \prod_i \arg \max_{z'_i \in \Delta(Y_i)} z'_i \cdot \pi_i \right).$$

This response is nonempty, convex and has closed graph, so it has a fixed point. There $z_i \cdot \pi_i = \max_y \pi_i(y)$. Every label in the output support is cheap at some mesh vertex used by the point, whose price differs coordinatewise from z_i by $o(1/M_i)$. Hence $z_i(y) \leq 11/M_i$ on that support. Integrating against $\pi_i = \pi_i^{\text{out}}$ proves (14.4).

The mesh was chosen for the stability conclusion of Lemma 13.4. Each used corner support therefore has the cleaned properties at the fixed-point laws. Degree and row-tail bounds hold label by label and pass to the envelope; the no-clique condition remains a property of each corner support separately.

In low mode, conditioning on all slice gates changes the raw history law in total variation by at most their failure probability, $\exp(-h_i^{1+\Omega(1)})$. For each conditioned history, the bin pretrim changes the label marginal by at most $h_i^2 \sqrt{\epsilon_i}$. Averaging and the triangle inequality give the stated comparison with π_i^0 . These bounds are uniform above the fixed threshold Q_0 ; they do not require a low-mode scale to tend to infinity. $\square$

Lemma 14.3 (Finite low-mode data). *At the fixed point, the low-mode primitive slice data can be represented with log cardinality at most $n^{1+o(1)}$.*

Proof. Encode W by its finite categorical outcomes, not by continuous random seeds. A parameter point uses at most dimension plus one active vertices, hence $O(N)$ possibilities for each corner draw. In low mode $h \leq (\log n)^1$, so the number $h^{O(1)} 2^h$ of position, group, and level records is $n^{o(1)}$. Each tuple and vertex record costs at most $O(k \log N) = n^{1+o(1)}$ in log cardinality, using $N \leq n 2^n$.

The order universes for IDs and lists have sizes $U = \exp(O(T(h + \log h))) = n^{o(1)}$. An order contributes at most $\log(U!) \leq U \log U = n^{o(1)}$. Multiplying by the number of records, and including the binary presence and activation variables, gives total log cardinality $n^{1+o(1)}$. $\square$

In direct modes none of the internal mechanism is needed: σ_v always denotes the uniform cleaned first law in the patch and π_i the fixed uniform second law.

The calibrated near-product bound (3.2), from Lemma 3.9, will be used for assignments within bins.

15 High modes of the tiling

In the high modes we assign distinct odd labels first, then construct even rows supported on their common neighbors. We must retain positive mass in every even row and keep each column sum small enough that normalization permits Hall's theorem.

For an even v of patch i call its bulk and crossing neighbor sets L_v and H'_v respectively. Write $i(b)$ for an odd neighbor's own patch, $D_j = D_{\pi_j}$, $I_x(y) = \mathbf{1}_G(x, y)$ (using the common color). Ratio weights are set to zero off their specified support/gates, requiring positive denominators on support.

Lemma 15.1 (High direct assignment). *In high direct mode, for all sufficiently large members of the counterexample sequence, there is an odd injection whose hit-supported even rows have mass at least one half and column sums at most one half. Consequently these rows admit distinct representatives.*

Proof.

Mass estimate. In high direct mode define the unnormalized even row

$$Z_v(x) = \sigma_v(x) \prod_{b \sim v} I_x(y_b) / D_{i(b)}(x).$$

First sample the odd labels independently from their $\pi_{i(b)}$. We claim that $\sum_x Z_v(x) < 1/2$ has probability at most n^{-R} .

Expose crossings first. After successful earlier filters, their normalized first law is at most $4^{\ell_i} \sigma_v$, hence still has small width. Deep discrepancy (12.1), applied after conditioning the next second law on a putative exceptional set, puts its hit mass at $1/2 \pm O(b_*)$ except with probability exponentially small in a power of n . The cleaned crossing denominators have the same bounds. Each normalized ratio filter therefore changes total mass by $1 + O(b_*)$, and the product is $1 + o(1)$, since $\ell_i b_* = o(1)$.

Let τ be the normalized first law after these filters. Its remaining bulk draws all have law π_i . Its support retains the own-patch degree and clique conditions from Lemma 13.4. The degree error is $O(b_*)$, because $g \leq n^{\ell/2}$, and the positive surplus gives

$$D_i(x)^{-|L_v|} \leq 2^{|L_v|} e^{-.4g}.$$

Here $|L_v| = n - o(n)$. The support and crossing losses from allocation, together with $N \geq 2^n$ and $s_i = 10^{-3}g$, yield

$$\Delta_i \max_x \tau(x) D_i(x)^{-|L_v|} \leq e^{-200s_i}.$$

Apply Lemmas 12.4 and 12.5 to this homogeneous bulk mass. Their even centered-moment bound also controls any fixed lower-tail threshold near $1/2$. Since $s_i > 10^{-3}K_B \log n$, the peeling term and the crossing failures give the asserted n^{-R} bound.

Injection and column loads. Apply the clock lemma with $B = 5$, using these mass failures as predicates. Disjoint patches and the allocation counts give odd column sums $o(1)$; atoms are exponentially small, each predicate reads n rows, and each row belongs to at most n predicates. The sampler therefore enforces every even-row mass bound.

Column loads require a separate estimate under this sampler. For fixed $x \in X_i$, expand the n -th moment of the average of $M_i Z_v(x)$ over even roles in patch i . Its cap is $2^n e^{-200s_i}$. Stars can intersect only at positions within distance 2, a fraction $O(n^2)2^{-n+\ell_i}$; their cap contribution tends to zero because the high gain exceeds the polynomial and prefix losses. For the remaining positions the stars are disjoint. The clock comparison on their at most n^2 queried rows restores independent labels, under which each row has mean $M_i \sigma_v(x) = O(1)$. Lemma 3.6 gives a moment bound K^n .

If this normalized average is A_x , the actual column sum is $O(2^n M_i / N) A_x / M_i = O(A_x / C_n)$. Markov and the union over at most $N \leq n 2^n$ first labels bound the probability of any column sum above $1/2$ by $N(K'/C_n)^n = o(1)$. Thus one injection has both the enforced mass bounds and these column bounds. Normalize the hit-supported rows, increasing columns by at most a factor two, and apply Hall. $\square$

15.1 Cluster history alarms

Now use either high cluster mode, with the internal experiments and high profiles above. The stages are: raw primitive histories W ; avoidance of the local history failures defined below; a global test of the odd column loads; bin sampling conditional on each passing history; and label injection conditional on each successful bin history. These last samplers are normalized separately at their entering histories and do not reweight earlier stages. Let $D_b^W(x) = \int I_x dp_{g(b)}^W$, $g(b)$ the group of b . Define $J_v^0 \subseteq X_i$ by

$$|D_b^W(x) - .5| \leq 2b_* \quad (b \in L_v), \quad 1/2 \leq \prod_{b \in L_v} D_b^W(x) / D_i(x) \leq 2$$

(requiring defined ratios); define J_v by further requiring $|D_b^W(x) - .5| \leq 2b_*$ for crossings. Use

$$Z_v(x) = \sigma_v(x) \mathbf{1}_{J_v}(x) \prod_{b \in L_v \cup H'_v} I_x(y_b) / D_b^W(x). \quad (15.1)$$

The row σ_v depends on W and the observed internal labels, while J_v^0 depends only on the bulk histories. Those bulk inputs are in distinct own-patch slices different from v 's internal slice. Crossings are in other, mutually distinct patches. Neither of the next two history tests reads a crossing history; the further crossing removal will have a deterministic uniform-mass bound.

The product-of-degrees test. Along with (14.3), require the uniform X_i -mass of $(J_v^0)^c$ to be at most e^{-n^2} . Its raw failure probability is at most e^{-n^2} , as follows. Each law $p_{g(b)}^W$ has small width for every history, so (12.1) bounds its bulk degree-outlier fraction in X_i by $e^{-\Omega(n)}$. Averaging in W and applying Markov shows that, outside an $e^{-\Omega(n)}$ fraction of x 's, the probability of any such outlier is $e^{-\Omega(n)}$.

For a remaining x , the variables $D_b^W(x)$, $b \in L_v$, are independent by slices, with means $D_i(x)$ by the high-profile identity. Clipping them to $.5 \pm 2b_*$ changes those means negligibly. Their ranges are $O(b_*)$, so concentration bounds a constant deviation of their sum by e^{-n^5} . Expanding the logarithms costs only $O(nb_*^2) = O(n^{-.92})$. Thus their normalized product lies in $[1/2, 2]$ except with this probability and the clipping exceptions. Integration over x , then Markov at e^{-n^2} , proves the test bound. For each fixed crossing history, deep discrepancy also makes its extra removed X_i -mass $e^{-\Omega(n)}$.

The large-interaction test. For $\mathbf{x} = (x_1, \dots, x_u)$, let $M_{b,J}^W(\mathbf{x})$ be the interactions of (12.2) for the conditional bulk law $p_{g(b)}^W$, with its own degree denominator. They are used only where all coordinates satisfy its degree gates. Define the positive function

$$\Phi_v^W(\mathbf{x}) = \sum_{I \subseteq [u]} \prod_{b \in L_v} \int \prod_{j \in I} \frac{I_{x_j}(y)}{D_i(x_j)} dp_{g(b)}^W(y)$$

and the following nonnegative function of W :

$$T_v(W) = \mathbb{E}_{\text{int}|W} \left[\int \mathbf{1}_{(J_v^0)^u}(\mathbf{x}) \mathbf{1}_{\{\max_{b,J} |M_{b,J}^W(\mathbf{x})| > 2\xi\}} \Phi_v^W(\mathbf{x}) d\sigma_v^{\otimes u}(\mathbf{x}) \right].$$

The integral is zero when $\sigma_v = 0$. Our second test is $T_v(W) \leq e^{-100s_i}$. It bounds an average over the internal reference experiment, not each realized internal posterior.

We prove the raw mean bound $\mathbb{E}T_v \leq e^{-300s_i}$ by comparison with the nominal interactions $M_J^{\pi_i}$, which use the fixed law π_i and denominator D_i . If some nominal interaction exceeds ξ , discard the conditional anomaly and degree indicators in this nonnegative integral. Averaging the independent bulk histories then replaces each $p_{g(b)}^W$ by π_i . Conditional on the internal data, the nonzero σ_v is supported on one cleaned corner support. Its widths and degrees fit the homogeneous estimates, and

$$\Delta_i \max_x \sigma_v(x) D_i(x)^{-|L_v|} \leq e^{-450s_i}$$

by (13.1), (14.2), the allocation slacks, and $h_i + |L_v| \leq n$. Lemma 12.5 bounds this part.

On tuples with all nominal interactions at most ξ , choose a bulk column witnessing a conditional interaction above 2ξ , and keep that column's degree gates. Freeze its history and all but one involved first coordinate. The raw internal experiment, and hence σ_v conditional on it, is independent of this bulk history. The gated one-free bound in Lemma 12.3 therefore charges at most $e^{-\alpha n/4}$ for the remaining coordinate. Meanwhile the other bulk histories integrate to their nominal laws; expanding each nominal factor bounds their product by $e^{2^u \xi n}$. The singled-out column's factor is bounded by a constant depending on u . Summing over columns and interactions, the choice of ξ makes this contribution exponentially small in n , hence smaller than the required gain-scale bound. Together the two parts give the asserted raw mean.

The two test failures and the failure of (14.3), at each v , thus have probabilities $\leq e^{-100s_i}$ by Markov and the high margins. Their W -scopes use only patch i , at bounded complementary-word distance (outside internal axes) and inner distance $< 20\rho h_i$. Dependence degrees are at most $n^{O(1)}$ times the volume of an inner ball of radius $50\rho h_i$. Charges e^{-s_i} work for product-local-lemma conditioning, by the volume bound and $s_i/\log n \rightarrow \infty$ uniformly here. Use this conditioned W law; dropping constraints touching polynomially many local consultations of the same sizes (possibly at different patches) costs $1 + o(1)$.

Lemma 15.2 (Conditional high-cluster mass). *Fix a history satisfying the two alarms above and (14.3). Given such W , in product-by-groups reference odd sampling,*

$$\Pr(\sum_x Z_v(x) < 1/2 \mid W) \leq n^{-R}. \quad (15.2)$$

Proof. By (14.3), the internal row σ_v is a probability except with probability ϵ_i . When it is nonzero, its mass removed by J_v is at most

$$\frac{M_i}{N} 2^{h_i} e^{-500s_i} (e^{-n \cdot 2} + e^{-\Omega(n)}) = o(1),$$

using the first test, the deterministic crossing-removal bound, and $h_i < n^\ell$. Now expose crossings. At each step the current normalized first law has small width, the new conditional law $p_{g(b)}^W$ has small width, and its denominator is $.5 \pm 2b_*$ on J_v . The deep-discrepancy calculation from the direct case therefore makes each ratio filter change mass by $1 + O(b_*)$, outside exponentially small exceptions. At the end the retained mass is $1 + o(1)$, and its normalized law satisfies $\tau \leq 5^{\ell_i+1}\sigma_v$ on J_v .

The remaining bulk inputs are independent, each in a distinct group and slice. Apply (12.2) with their laws $p_{g(b)}^W$ and denominators D_b^W . Lemma 12.4 bounds the small and moderate part of the centered moment by n^{-3R} . For the large-interaction part, the domination of τ and the bulk product-ratio gates give, after averaging the internal and crossing data with their preceding success indicators,

$$\text{large-interaction contribution} \leq 5^{u(\ell_i+1)} 2^u T_v(W).$$

The factor 2^u changes the conditional denominators to the nominal D_i 's in Φ_v^W . Since $\ell_i \leq s_i/(1000u)$, the alarm bounds this expression by n^{-3R} . Even centered-power Markov, adding the preceding exceptions, gives (15.2). $\square$

15.2 Bin and injection stages

We first prove that, with high probability under the history-avoidance law, every odd-label column sum is less than a fixed $\theta_* > 0$, chosen sufficiently small compared with the clock constant θ_0 . For a label in patch i , take the n -th moment of the average of $M_i p_{g(b)}^W(y)$ over its odd roles. Equation (14.1) bounds each term by $8e^{2k_i T_i}$. Treat two roles in the same slice as near; their fraction is at most $2^{-n+o(n)}$, so the repeated-slice contribution in Lemma 3.6 tends to zero. For retained roles in distinct slices, dropping avoidance constraints touching their local consultations costs $1 + o(1)$ and restores independent raw histories. Their means are $M_i \pi_i(y) \leq 11$ by (14.4). The resulting moment is K^n , and the patch-count conversion used in the direct case makes every actual column sum $o(1)$ with high probability. Denote this global load success by $\mathcal{H}$. It is not an additional local conditioning of W . The following samplers are used only on $\mathcal{H}$; off it extend the experiment arbitrarily and set all tested weights to zero. Thus later moment estimates retain $\mathbf{1}_{\mathcal{H}}$ until the conditional sampling comparisons have been applied.

Proposition 15.3 (High-cluster sampling stages). *Fix an admissible W -history with the preceding column-load success. There is a conditional bin sampler satisfying the star mass tests, the small-bin distinctness tests when relevant, and the prescribed label-column capacity. On the stated local scopes it has upper comparison with independent group choices. For every successful bin history there is then an odd injection satisfying all even-row mass gates, with the label-stage comparisons described below. These stages are conditional samplers; completing a stage does not reweight its incoming history.*

Proof.

Bin predicates. Given such prehistory, start the bin stage with independent group choices by q_g^W (bin identities below refer to physical bins, with their particular further masked/filtered laws as in each group). Require per v that conditional on bins its independent-role label probability of failure in (15.2) is $\leq n^{-R/2}$; exception costs $\leq n^{-R/2}$ by Markov. In small-bin mode also require no bin chosen twice by distinct groups in its star, with exponentially small exception by (14.1). These are star constraints on bin choices. Finally require given-bin label column sums $\leq \theta_0$.

Capacity certificates. Fix a physical bin D of size $d = d_i$, a label $y \in D$, and the entering history W . If group g chooses this bin, its h_i roles contribute

$$a_g(y) := h_i U_g^W(D)(y) \leq d^{-1/2}$$

to the column, by (14.1) and the scale choices. Put groups with $a_g(y) \in (\lambda_j/2, \lambda_j]$ in bucket j , where $\lambda_j = d^{-1/2} 2^{-j}$. Its expected load under independent bin choices is $m_j = \sum_{g \text{ in bucket } j} q_g^W(D) a_g(y)$. For

$$l_j = \left\lfloor (K' m_j + c' 2^{-j/2}) / \lambda_j \right\rfloor + 1,$$

forbid every event that a specified l_j -subset of this bucket all choose D . Avoidance leaves at most $l_j - 1$ chosen groups in the bucket, so its load is at most $K' m_j + c' 2^{-j/2}$. First choose K' large, then c', θ_* sufficiently small. Since $\sum_j m_j < \theta_*$, the sum of these allowed loads is at most θ_0 .

Give such a certificate charge 2^{l_j} times its product probability. To sum charges touching a fixed group, first pin its bin choice to D . For each $y \in D$, that group belongs to only one bucket. In that bucket put $z = \sum_g q_g^W(D) \leq 2m_j / \lambda_j$. The chosen constants and the additive allowance ensure

$$l_j - 1 \geq 20z, \quad l_j - 1 \gtrsim d^{1/2} 2^{j/2}.$$

Summing over subsets containing the pinned group costs at most

$$2^{l_j} \frac{z^{l_j-1}}{(l_j-1)!} \leq e^{-\Omega(d^4)}.$$

The union over at most d columns still has this negligible bound; averaging the pinned target gives the unconditional touching bound. Here $q > (\log n)^{c_q}$, so these errors are smaller than every inverse polynomial in n .

Use charge n^{-5P} for each star constraint. At most $O(nh_i)$ of them touch one group, and their raw probabilities have much stronger decay. Thus the total charge touching any group is $\delta_n = o(n^{-4P})$. For a certificate involving l groups, the reciprocal neighbor product is at most $\exp(O(l\delta_n)) \leq 2^l$, which is paid by its chosen charge inflation. The polynomial scopes of star constraints likewise have reciprocal neighbor product $1 + o(1)$. Lemma 3.5 now supplies all bin requirements. Dropping constraints touching $O(n^3)$ queried groups costs $1 + o(1)$, giving the asserted comparison with independent bin choices.

Conditional label assignment. In high-large-bin mode individual laws conditional on successful bins now meet the clock lemma (atoms $\leq 2 \exp(1.5k_i T_i) / d_i$ uniformly superpolynomially small). Draw an odd injection avoiding every mass failure. In high-small-bin mode instead start independently per assigned bin with (3.2) for its individual laws (maximum atom $\leq d_i^{-.95}$ by (14.1)). Here $d_i \leq n^{o(1)}$ and $h_i = o(d_i^{0.25})$ uniformly. In each star a queried bin belongs to just one participating group by bin-stage success; multi-observations in it are possible only for internal incidences. Using exact singleton comparisons and (3.2) their overall joint upper-error factor is $\leq \exp(d_i^{-.04} h_i)$, so failure still costs $\leq 2n^{-R/2}$. Each bin serves at most d_i roles by the column-sum requirement, hence bin-variable dependence degrees are $n^{2+o(1)}$ or less. Condition to avoid all these failures by the local lemma (charges e.g. n^{-5P}), retaining upper comparison to the preconditioning bin-wise sampling at $1 + o(1)$ cost on $O(n^2)$ queried bins. Thus either procedure gives an odd injection with the mass gates, without further reweighting earlier histories. $\square$

15.3 High cluster load transfer

Proposition 15.4 (High-cluster column loads). *Let $\mathcal{E}_i$ be the even positions of patch i , and let $\mathcal{H}$ denote the prerequisite W -load success. Under the conditioned history law and the conditional samplers of Proposition 15.3, uniformly for x in the cleaned envelope,*

$$\mathbb{E} \left[\mathbf{1}_{\mathcal{H}} \left(\frac{1}{|\mathcal{E}_i|} \sum_{v \in \mathcal{E}_i} M_i Z_v(x) \right)^n \right] \leq K^n.$$

Together with the mass gates, the patch counts and divergence of the ambient ratio, this gives an embedding in each high cluster mode.

Proof.

Caps and geometric separation. Fix x in patch i 's cleaned envelope; only its even patch positions can use it. We bound n -th moments of the patch average of $M_i Z_v(x)$, with indicator of the prerequisite W -load success. On successes use both

$$M_i Z_v(x) \leq 2^n e^{-200s_i}, \quad Z_v(x) \leq 4\sigma_v(x) \prod_{b \in L_v \cup H'_v} I_x(y_b)/D_{i(b)}(x). \quad (15.3)$$

Indeed off J_v the weight is zero; on it the bulk ratio gate and crossing degree bounds for both kernels ($\ell_i b_* = o(1)$) give the product comparison. Its individual factors are ≤ 3 on this envelope. Also use (13.1), (14.2) with $10q^{C_b} \leq s_i/(100u)$ for the cap.

There are three possible sources of repeated information in this moment: intersecting geometric core scopes, crossing scopes shared by different stars, and, for small bins, distinct groups choosing the same physical bin. We handle them in that order.

Expand by independent uniform even patch indices (positions, ordered by draw number). Call a position's core scopes those for σ_v and its internal and bulk samples/kernels: the needed own-slice and nearby own-patch W consultations, groups and odd roles. Its core bin-choice variables are just the groups actually sampled for its internal and bulk labels; even the counterfactual calculations for σ_v use the observed internal labels and W , not bin draws elsewhere. Remove by the cap each index core-near an earlier index (earlier removals included), using proximity of complementary-word distance ≤ 4 **and** inner distance $\leq 100\rho h_i$. This covers core intersections, and partner fraction per fixed index is at most $2^{-n} e^{.01s_i}$ by the inner volume, prefix and high slacks. On retained indices form a graph by full distance $\leq n^{2\ell}$, partner fraction $\leq 2^{-n+o(n)}$. It covers any crossing-scope intersections between them (all internal dimensions $< n^\ell$); crossing consultations themselves are never in patch i , and within a star are in different patches. When using the second bound in (15.3) below, drop all crossing factors at nonisolated indices of this graph. If this graph has rank j , meaning its number of vertices minus its number of connected components, at most $2j$ vertices are nonisolated. Deleting their crossing factors therefore costs at most $3^{2j\ell_i}$.

Label and bin transfers. For large bins transfer the retained indices' needed data in that product bound to independent labels given bins by the clock bound (at most n^2 distinct queries), then use independent-bin upper comparison on the involved groups.

For small bins we need the following extra splice before these transfers. Given the bin choices, remove by the cap each geometrically retained index whose core groups' bin choices intersect the core choices of an earlier geometrically retained index (even one that these bin repeats remove).

In using (15.3) thus take the cap on all indicated removals, using the product bound and crossing-graph drops only on the remaining indices. For a core-bin removal keep its necessary repeat condition; against fixed earlier choices under independent bin sampling given W this costs per index

$$n^{O(1)} 4e^{2k_i T_i} d_i / M_i \leq 2^{-n} \exp(o(s_i)),$$

by distinctness of group variables across the geometrically retained cores and the high scale bounds. Among the remaining indices' crossing queries not excised by the graph, use a fixed order and drop by factor 3 each hit factor whose bin repeats an earlier bin of these queries. Their group variables are all distinct and outside patch i ; the necessary repeat condition costs $\leq 2^{-n+o(n)}$ per drop against earlier choices under independent bins.

Small-bin removal patterns. Fix the position tuple and then one outcome of the ordered removal procedure just described: core-near rows are capped first; among the remaining rows, later core-bin repeats are capped; among crossing queries still allowed by the crossing graph, later bin repeats lose only their hit factor. Fix the resulting removal masks before comparing probability laws.

Let $\mathcal{R}$ be the rows whose factors remain, and $\mathcal{B}_v$ their remaining bulk and crossing queries. After paying the row caps and the factors 3, the nonnegative integrand has the form

$$\mathbf{1}_{\text{rep}} \prod_{v \in \mathcal{R}} \left[M_i \sigma_v(x) \prod_{b \in \mathcal{B}_v} \frac{I_x(y_b)}{D_{i(b)}(x)} \right].$$

Here $\mathbf{1}_{\text{rep}}$ retains the necessary earlier-bin repeat for every removal. Initially one also retains the mask-consistency and successful-history indicators. The rule for σ_v uses its observed internal labels and local W ; its candidate calculations integrate hypothetical bins and never consult other actual bin choices. Consequently the displayed product reads actual bins only through its remaining label queries.

First, conditional on a successful bin history, remove touching label-stage constraints and apply (3.2). Different remaining groups use distinct bins; several role queries can share a bin only within one internal group. Exact singleton marginals handle the other queries. The total comparison cost is

$$(1 + o(1)) \exp(d_i^{-.04} h_i n) \leq O(1)^n.$$

Integrating the resulting independent role labels leaves a nonnegative function only of the kept bin choices and local W , multiplied by the repeat indicators.

Next apply the bin-stage upper comparison to every group read by this function or a repeat condition. Its independent-role formula extends nonnegatively off the bin gates. Thus, after this comparison, discard the other bin gates and mask-consistency restrictions, retaining just the necessary earlier repeats.

At this point the bin variables have their independent raw laws given W . Integrate removed variables in reverse order. Earlier removed choices can be held fixed while charging the necessary repeat of a later one; the surviving label integral does not read that later variable. A core-row repeat costs at most $2^{-n} e^{o(s_i)}$, paying its cap $2^n e^{-200s_i}$. A crossing repeat costs at most $2^{-n+o(n)}$, paying its factor 3. The two types concern disjoint patches. There are at most 2^n core masks, while the crossing-repeat costs sum to $1 + o(1)$ over masks on at most n^2 queries. All these bounds are uniform in the kept choices and W .

Restoring the reference product. Reverse integration has therefore used independence only after both the label and bin comparisons, not under the avoidance-conditioned bin law. The prerequisite W -load indicator has remained in place to license those comparisons.

After these transfers, integrate kept independent bin choices and reference label draws. Keep only the internal reference weights and used external factors in the product test; in particular nonlocal W -load restrictions can now be dropped. Use the W upper comparison on the kept local scopes. In unrestricted sampling they give independent core experiments across positions, internal data independent of the used external inputs, and independence across those individual bulk and remaining crossing inputs. This follows by core separation and removal of crossing overlaps as above and the slice/patch structure within each star. Thus each $I_x(y_b)/D_{i(b)}(x)$ left integrates to 1 by profile definition, while $M_i\sigma_v(x)$ on internal reference data has mean $\leq K$ by (14.2). The product-mean costs here are $O(1)^n$.

Moment summation and Hall. Finally enumerate geometric removal masks (at most 2^n); in independent uniform-position sampling conditional on the values at a specified mask's retained indices, enforcing just necessary earlier proximities for its removed indices and summing those variables backwards costs per removal at most $n2^{-n}e^{.01s_i}$. The graph cost on retained positions (before bin splices) for rank j needs a forest of j edges paying fraction $(2^{-n+o(n)})^j$ under unrestricted uniform-position sampling, with at most n^{2j} choices. This absorbs $3^{2j\ell_i}$. Thus all normalized moments in question are $O(1)^n$ uniformly. For the normalized column average $A_x = |\mathcal{E}_i|^{-1} \sum_v M_i Z_v(x)$, the actual column sum is

$$\sum_{v \in \mathcal{E}_i} Z_v(x) = \frac{|\mathcal{E}_i|}{M_i} A_x = O(C_n^{-1}) A_x.$$

The moment bound with $\mathbf{1}_{\mathcal{H}}$, Markov, and the union over at most $N \leq n2^n$ labels give failure probability at most $N(K'/C_n)^n = o(1)$. Meanwhile $\Pr(\mathcal{H}) = 1 - o(1)$, and both conditional samplers enforce their mass gates at every passing entering history. Their intersection with column sums at most $1/2$ is therefore nonempty. Normalizing row masses, which are at least $1/2$, and applying Hall gives the cube. This excludes the high modes of the tiling. $\square$

16 Low modes: geometry and cell states

It remains to treat the bounded, low direct-bias and low cluster modes, one mode at a time. We first assign all odd roles within small cells, reserving some of these roles for replacement later. A cell keeps a fixed pool of host bins. One *fresh cell state* consists of its primitive slice data, its chosen group bins, its injective odd labels, and the internal even priors computed from those data. We will construct a sampler whose bin and label stages preserve their specified conditional singleton marginals and satisfy upper comparisons for joint observations. Section 17 uses independent repetitions of this sampler to prepare the early lists.

Recall $C_n = N2^{-n}$. In direct modes take singleton bins $d_i = 1$ and singleton groups on odd roles. Let $\mathcal{X}_i$ be the cleaned envelope of patch i , which is the single cleaned support in direct modes. The scale bounds give, uniformly in the present cases,

$$\begin{aligned} h_i &\leq (\log n)^1, & \ell_i &= o(\log n), \\ M_i &= Nn^{-o(1)}, & d_i &= n^{o(1)}, \\ n|D_i - .5| &= O(\log n) \quad \text{on } \mathcal{X}_i, & s_i &= O(\log n). \end{aligned}$$

The constants may depend on the fixed large thresholds. In low cluster mode, neither d_i nor h_i need tend to infinity; the choice of Q_0 ensures the required bin and internal-parameter inequalities. Call x, z conflicting for patch i when $|K_{\pi_i}(x, z)| > \xi$, including the diagonal. By Lemma 12.5 and the cleaning, the number of labels conflicting with a fixed x in any one used corner support is at most $\Delta_i \leq Ke^{s_i}$. This assertion concerns each corner support separately. Fixed factors may be incorporated into K below.

Lemma 16.1 (Late classes and cell inputs). *The low-mode cube admits the late-class and cell allocations described below, with $A_0 \log n \leq r < 2A_0 \log n$ late classes. For $0 \leq j \leq r$, every even star has between $\lfloor j/2 \rfloor$ and j neighbors in the last j classes, at most one in each class, and at most one late internal neighbor. Distinct slices of a cell have the separation stated below. Pool inputs are uniform permutation slots; conditional on the pools, fresh cell states can be drawn using independent local tapes.*

Proof.

Syndrome classes. Let $\mathcal{H}$ be a binary group of power-of-two size $n \leq H_* < 2n$. Give the n coordinates distinct IDs $a_1, \dots, a_n \in \mathcal{H}$, including every element of a linear hyperplane $\mathcal{H}_0$. Choose a subspace $\mathcal{L}$ with $A_0 \log n \leq r := |\mathcal{L}| < 2A_0 \log n$, not contained in $\mathcal{H}_0$. The IDs of all axes in the nested internal allocation can be put in distinct cosets modulo $\mathcal{L}$. There are enough such cosets even using IDs in $\mathcal{H}_0$, since the total number of internal axes is at most $(\log n)^1$, whereas $|\mathcal{H}/\mathcal{L}| \asymp n/\log n$.

For a cube word z , set $s(z) = \sum_k a_k z_k$. An odd role is late when $s(z) \in \mathcal{L}$; the r possible syndromes give its late classes. Each class has size $2^{n-1}/H_*$: the joint map recording syndrome and parity has full rank, since the IDs include zero and span $\mathcal{H}$. Order the classes so that every suffix divides as evenly as possible between syndromes in and outside $\mathcal{H}_0$.

At an even v , flipping coordinate k reaches class t exactly when $a_k = s(v) + t$. If $s(v) \in \mathcal{H}_0$, every $t \in \mathcal{L} \cap \mathcal{H}_0$ has its required ID in $\mathcal{H}_0$; if $s(v) \notin \mathcal{H}_0$, the same is true of every $t \in \mathcal{L} \setminus \mathcal{H}_0$. Thus any last j classes supply at least $\lfloor j/2 \rfloor$ neighbors. Distinct IDs allow at most one per class, giving the upper bound j . Write $p(v)$ for the total number of late neighbors, so $r/2 \leq p(v) \leq r$. All eligible flip IDs lie in the single coset $s(v) + \mathcal{L}$, and hence at most one is internal.

Split the reserved Y -labels into disjoint late pools of size $\asymp N/r$. The initial assignment nevertheless labels every odd role, using only the Y_i 's outside this reserve. The initial labels of late roles are dummies, to be replaced later. The remaining odd roles are called early. The initial lists will impose only the early external hits; an internal prior may still retain the dummy hit at the one late internal neighbor. Deferring the other late hits saves a factor $2^{-p(v)}$, up to a fixed constant, in the initial atom estimate of Lemma 17.1. This supplies polynomially small initial failure bounds even when the patch gain s_i is bounded.

Separated cells and persistent pools. Partition each patch into cells of at most n^{A_c} positions, each a union of entire slices, so that distinct slices of one cell have cube distance greater than $(\log n)^3$. To do this, greedily color the slices with $\exp(O(\log^4 n))$ colors so that near slices have different colors, then batch each color. This gives $O(2^{n-\ell_i}/n^{A_c})$ cells in patch i .

Give each cell

$$L_i^{\text{slot}} = \lceil K_{\text{cell}} n^{A_c} / d_i \rceil = n^{A_c + o(1)}$$

slots, where K_{cell} is a sufficiently large fixed constant compared with $1/\theta_*$. The slot count is distinct from all neighbor counts. There are $B_i = M_i/d_i$ physical bins in the patch. The patch counts and

$C_n \rightarrow \infty$ ensure that all cells' slots fit disjointly among these bins. A uniform permutation of the bins, independent for each patch, supplies the slot images and thus the cell pools.

On any $\exp(\text{polylog}(n))$ specified slots, with fixed powers of $\log n$, this law has upper comparison $1 + o(1)$ with iid uniform bins. The same comparison holds after fixing any one global slot to a prescribed bin, retaining that pin in both laws. Indeed the without-replacement factors differ from one by a total $\exp(\text{polylog}(n))/B_i = o(1)$, since $B_i = 2^{n+o(n)}$.

The pool persists when a cell is resampled. Its tape is a sequence of independent randomizers for the conditional fresh-state sampler constructed below. A drawing reads only this cell's pool and the selected tape entry; it uses local fallback rules on invalid inputs. Later schedules may depend on other data when deciding which entry to read. $\square$

A fixed table of permitted bins. We need a label in an externally prescribed slot to be compatible with a fresh internal prior. For an external early incidence (b, v) , meaning that the edge is noninternal, remove from the entire group of b every bin containing a label y for which

$$\Pr_v^{\text{base}} \{ \sigma_v \neq 0, |d_G(\sigma_v, y) - .5| > 2b_* \} > e^{-cn}. \quad (16.1)$$

Here $\Pr_v^{\text{base}}$ is the unrestricted internal W experiment and reference label sampling at v , or the deterministic prior in direct modes. It is an independent comparison experiment, not the eventual cell state at v . The constant $c > 0$ is sufficiently small.

Each nonzero σ_v has small width by (14.2) or the direct bounds. Deep discrepancy therefore bounds the probability in (16.1), averaged over uniform $Y_{i(b)}$, by e^{-3cn} . Markov, followed by multiplication by the bin size and the number of incidences of a group, shows that an $e^{-\Omega(n)}$ fraction of all bins is removed. The remaining bins form a deterministic permission table, fixed before the cells are sampled. Write $\mathcal{B}_g^{\text{perm}}$ for the bins permitted at group g .

In cluster mode begin with the internally pretrimmed bin law $q_g^{\text{in}, W}$ used to define the low output π_i in Proposition 14.2. Its further restriction to permitted bins is

$$\bar{q}_g(D) = \frac{q_g^{\text{in}, W}(D) \mathbf{1}_{\{D \in \mathcal{B}_g^{\text{perm}}\}}}{q_g^{\text{in}, W}(\mathcal{B}_g^{\text{perm}})}.$$

This additional loss is $e^{-\Omega(n)}$, by (14.1). The in-bin law $U_g^W(D)$ is unchanged. The same notation applies in direct modes, starting from uniform singleton bins and using point in-bin laws. Although the permission table is fixed, $\bar{q}_g$ still depends on W in cluster mode. Throughout these changes of sampling law, σ_v and E_v^{in} remain the raw posterior and failure rules of Section 14.3.

16.1 Pool and history conditions

Fix one cell. In cluster mode first draw its slice histories independently, each conditioned on (14.3) throughout that slice. Given these histories and the cell pool $\mathcal{P}$, define, whenever the denominator is positive,

$$\tilde{q}_g(D) = \frac{\bar{q}_g(D) \mathbf{1}_{\{D \in \mathcal{P}\}}}{\sum_{D' \in \mathcal{P}} \bar{q}_g(D')}.$$

A zero denominator uses the local fallback. The same restriction defines the direct-mode laws without W . A pool is individually typical when the following hold:

- Its slots have distinct images, and every denominator above is $(L_i^{\text{slot}}/B_i)(1 + O(n^{-4}))$, with relative error at most n^{-4} , uniformly over every group and every admissible history of its slice.
- In cluster mode, for every such history and internal star v , independent group choices from $\tilde{q}_g$, followed by independent reference role labels given those bins, have $\Pr(E_v^{\text{in}}) \leq \epsilon_i^{1/4}$. This also holds if one participating group's bin is fixed to any target of positive restricted mass.

The latter is a *group-bin pin*. A *slot pin* instead fixes one pool slot to a physical bin. Both are allowed simultaneously in the estimates that follow. Later a *role-label pin* will fix an output of the within-bin injection.

Lemma 16.2 (Typical pools and the history-load gate). *The preceding pool requirements fail at a cell with probability at most $e^{-n^{c_0}}$, for a fixed $c_0 > 0$, also conditional on any one prescribed slot pin. On a typical pool the additional history-load gate below has negligible relative conditioning cost. Direct modes already have the required load slack.*

Proof.

Empirical normalizers. Use the iid-slot comparison, keeping a prescribed slot pin if present. For fixed admissible slice data and slots $D_1, \dots, D_{L_i^{\text{slot}}}$, write

$$\hat{q}_g = \frac{B_i}{L_i^{\text{slot}}} \sum_l \bar{q}_g(D_l) \delta_{D_l}.$$

Its total mass is the pool restriction denominator in units of L_i^{slot}/B_i . Changing one slot changes this mass by at most $O(\exp(O(k_i T_i))/L_i^{\text{slot}})$; in direct modes the exponential factor is constant. Its expectation is one, up to the negligible change caused by a slot pin. Bounded differences gives the required relative n^{-4} accuracy.

Empirical failure probabilities. Let $g_1, \dots, g_m$ be the participating groups at an internal star, and set

$$f_v^W(D'_1, \dots, D'_m) = \Pr_{\text{independent roles}}(E_v^{\text{in}} \mid W, D'_1, \dots, D'_m).$$

Before pool restriction, its integral against the $\bar{q}_g$'s is at most $2\sqrt{\epsilon_i}$, including after a positive group-bin pin. This follows from the internal pretrim and (14.3); the normalization losses over the other groups fit because $h_i^3 \sqrt{\epsilon_i} \ll 1$, and the permission trim is exponentially small.

Consider instead the unnormalized integral

$$\int f_v^W d\hat{q}_{g_1} \cdots d\hat{q}_{g_m},$$

replacing one factor by a point mass if a group-bin pin is imposed. Its change under one slot replacement is $n^{-A_c+o(1)}$, since $\exp(O(h_i k_i T_i)) = n^{o(1)}$. To compute its expectation, expand the empirical measures by their slot indices. Distinct unpinned indices give back the original independent bin laws. Coincident indices, or occurrences of a pinned slot, contribute at most $n^{-A_c+o(1)}$. Thus its expectation is at most $2\sqrt{\epsilon_i} + n^{-A_c+o(1)}$.

Bounded differences puts this integral below $\epsilon_i^{1/4}/2$ except with exponentially small probability; normalizing the empirical masses then gives the claimed bound. Here $\epsilon_i = n^{-o(1)}$, and $2\sqrt{\epsilon_i} \ll \epsilon_i^{1/4}$, uniformly for the fixed large threshold. Both concentration exponents can exceed n^{100} . By

Lemma 14.3, the logarithm of the number of slice histories is at most $n^{1+o(1)}$; adding group-bin targets costs only $O(n)$, and there are polynomially many slices in a cell. The union therefore preserves an $e^{-n^{c_0}}$ bound. Repeated slot images have probability $e^{-\Omega(n)}$, including after one slot pin.

The load gate on slice histories. For a fixed typical pool in cluster mode, let

$$\Lambda_y(W) = \sum_{b \text{ in the cell}} \tilde{q}_{g(b)}(D(y)) U_{g(b)}^W(D(y))(y)$$

be the conditional expected load on label y . Condition the slice histories further on $\Lambda_y(W) \leq \theta_*$ for every label. To bound the cost, normalizer control and the permission-trim bound compare each role's law with $(1 + O(n^{-3}))(B_i/L_i^{\text{slot}})$ times its internally pretrimmed law. Averaging the latter gives π_i , so (14.4) bounds the role's mean at a pool label by $12/(L_i^{\text{slot}} d_i)$. Hence $\mathbb{E}\Lambda_y = O(1/K_{\text{cell}})$.

Different slices are independent under the slice-conditioned law. Each slice contributes at most $n^{-A_c+o(1)}$ to a column, by (14.1) and the restriction normalizers. Concentration and a union over labels show that the load gate fails with probability $\delta(\mathcal{P}) \leq e^{-n^{c_0}}$, decreasing c_0 if necessary. For every nonnegative test F , consequently,

$$\mathbb{E}[F \mid \mathcal{P}, \text{ load gate}] \leq \frac{1}{1 - \delta(\mathcal{P})} \mathbb{E}[F \mid \mathcal{P}].$$

The expectations here already use slice-conditioned histories. This conditioning is performed separately at each fixed pool; it does not reweight the pool law. In direct modes typicality itself gives the same load slack, with no history conditioning. $\square$

16.2 Cell calibrations

We now complete the sampler at a typical pool and a passing history. The two calibration steps keep exact single-coordinate marginals while imposing the internal tests and distinct labels.

Proposition 16.3 (Calibrated group bins). *Given those successful data in a cluster cell, there is a bin sampler across groups with exact individual marginals $\tilde{q}_g$ and the following properties:*

- *At every internal star, different participating groups choose distinct bins, and the independent-role conditional probability of E_v^{in} given bins is at most $\epsilon_i^{1/8}$.*
- *Given-bin label column sums are at most .2.*
- *For targets on l distinct groups, the joint probability is at most $\exp(d_i^{-.05}l)$ times their product $\tilde{q}_g$ -mass.*

Proof.

Fix the data throughout. Let $\mathcal{C}$ be the set of laws on bin assignments that satisfy the first two requirements and every joint upper inequality in the third, always relative to the original $\tilde{q}_g$'s. This is a compact convex set. We construct members of $\mathcal{C}$ for arbitrary small price-directed perturbations, then separate its marginal image to obtain exact calibration.

Avoidance for perturbed inputs. Perturb each input by a relative factor $1 + O(d_i^{-1})$ on its support, and begin with independent group choices. The failure probability of a star requirement, also after any positive group-bin pin, is at most $2\epsilon_i^{1/8} + n^{-A_c+o(1)}$. Indeed pool typicality and

Markov control the conditional label failure, the perturbation is bounded even across h_i groups, and $\max_D \tilde{q}_g(D) \leq n^{-A_c + o(1)}$ bounds repeated bins. Each group belongs to only $O(h_i^2)$ such stars.

Enforce the column bound by the capacity subset certificates of Section 15.2, using the present nominal loads and the target .2. The incoming bound θ_* leaves room for the fixed bucket multiplier. A group's contribution to a column when it chooses that column's bin is at most $d_i^{-1/2}$. The same factorial calculation, inflating each certificate probability by 2 per participating group, bounds the sum touching any specified group by $d_i \exp(-\Omega(d_i^4))$, even after fixing that group's bin. Star charges $\epsilon_i^{1/16}$ together with these certificate charges have total at most d_i^{-10} per group. This uses the fixed choice of Q_0 : h_i is a large power of $\log d_i$ and is small compared with $d_i^{.01}$. The products of neighboring avoidance factors therefore fit the certificate inflations and the star charges. Lemma 3.5 supplies a law avoiding all the constraints. For l group targets, its upper comparison, including the input perturbation, is within $\exp(d_i^{-.05}l)$. Thus every constructed law lies in the same $\mathcal{C}$.

Relative singleton bounds and separation. Write $\hat{q}_g$ temporarily for a perturbed probability input. Fix a group g , and let A be avoidance of all constraints not reading it. Under the product law D_g is independent of A . For every supported target D , the pinned charge estimates and the reciprocal avoidance factors give

$$\Pr(\text{some constraint touching } g \text{ fails} \mid A, D_g = D) \leq d_i^{-2}.$$

Indeed a touching event's nonneighbors remain independent of it after this pin; avoiding its neighbors costs the reciprocal factors from Lemma 3.5. Summing the pinned inflated probabilities gives the displayed bound with slack. For the upper bound, removing the touching constraints costs at most $\exp(O(d_i^{-10}))$, by their total charge. Thus each marginal after full avoidance is $(1 \pm d_i^{-2})\hat{q}_g(D)$.

For real prices $c_g(D)$, put $\bar{c}_g = \sum_D c_g(D)\tilde{q}_g(D)$. Multiply the original input by $1 + d_i^{-1} \text{sign}(c_g(D) - \bar{c}_g)$ and normalize. As in Lemma 3.9, its centered-price gain is at least half of $d_i^{-1} \sum_D |c_g(D) - \bar{c}_g| \tilde{q}_g(D)$. The two-sided singleton error is smaller than this gain. Hence no linear functional separates the original marginal vector from the marginal image of $\mathcal{C}$. A law in $\mathcal{C}$ with the exact original marginals exists. $\square$

Proposition 16.4 (Calibrated role labels). *In low cluster mode, given the successful bins, there is an injective label assignment with exact singleton marginals $U_g^W(D)$, avoiding every E_v^{in} . Its joint probability on l distinct queried roles is at most $\exp(d_i^{-.01}l)$ times the product of their laws, provided at most h_i roles are queried per bin.*

In bounded and low direct-bias modes, fix a typical cell pool $\mathcal{P}$ and put $L = L_i^{\text{slot}}$. Identifying singleton bins with labels, there is an injective assignment with exact role marginals $\tilde{q}_{g(b)}$. For any $l \leq L^{.025}$ distinct roles and specified labels, it satisfies

$$\Pr(y_b \text{ is assigned at every queried } b \mid \mathcal{P}) \leq \exp(L^{-.04}l) \prod_{b \text{ queried}} \tilde{q}_{g(b)}(D(y_b)).$$

In particular, this comparison holds for every $l \leq n^3$.

Proof. First work in cluster mode. Perturb each role law relatively by $1 + O(d_i^{-.02})$. In each physical bin independently, apply Lemma 3.9 to its roles and perturbed laws. The atoms are at most $d_i^{-.95}$ by (14.1), and the column sums retain the lemma's slack from the .2 bound. These

within-bin assignments have exact perturbed singleton marginals. Here the independent product variables are whole injective assignments within bins, rather than individual role labels.

A star failure has probability $O(\epsilon_i^{1/8})$ under this product of bin assignments. Its probability conditional on any supported role-label pin in a touched bin is at most $O(d_i \epsilon_i^{1/8})$. To see the two cases, a pin outside the star can be included in the joint upper comparison for at most $h_i + 1$ roles and then canceled by division by its exact singleton mass. If the pinned role is in the star, division costs $O(d_i)$, since every supported atom of its perturbed uniform-subset law is $\gtrsim 1/d_i$. The comparison factors in both cases are bounded.

Each bin serves at most d_i roles and touches at most $d_i h_i$ star tests. Local-lemma conditioning therefore avoids all failures with total charge at most d_i^{-10} per bin. The same pinned argument as for group bins gives two-sided relative singleton control, now with “touching” meaning sharing the pinned role’s bin variable. Let $\mathcal{C}'$ be the compact convex set of laws on injective, failure-free assignments satisfying all the stated joint upper bounds relative to the unperturbed role laws. The perturbation and avoidance costs fit $\exp(d_i^{-0.01} l)$, so all price-directed constructions belong to $\mathcal{C}'$. Centered-price separation, with the $d_i^{-0.02}$ gain dominating the singleton error, yields the exact unperturbed marginals.

In direct modes the physical bins are singletons, but the label universe for Lemma 3.9 is the whole cell pool of $L_i^{\text{slot}} = n^{A_c + o(1)}$ labels. The role-specific permitted restrictions have atoms $O(1/L_i^{\text{slot}})$, and typicality gives column sums below .4. Thus that lemma applies with $d = L_i^{\text{slot}}$, not with $d_i = 1$. It gives the exact $\tilde{q}_{g(b)}$ marginals and the displayed joint upper bound, which applies through at least n^3 queries since $(L_i^{\text{slot}})^{.025} = n^{5+o(1)}$. There are no internal tests. Fix the resulting cell strategies as lookup kernels, so fresh drawings read only the declared local data. $\square$

16.3 Fresh comparisons

Remark 16.5 (Exact marginals and fixed-scale errors). The low cluster parameters need not diverge along the counterexample sequence. The joint comparison errors from the two calibration propositions are controlled by the fixed large threshold; they need not vanish as $n \rightarrow \infty$. The singleton estimate below instead uses exact calibration at both stages, together with the pool normalizers and the negligible history-conditioning and permission-trim losses.

Lemma 16.6 (Fresh singleton comparison). *The cell procedures give probability priors σ_v valid internally throughout each typical cell. If $D(y)$ is the physical bin containing y , then each odd role b of patch $i = i(b)$ has fresh law*

$$p_b(y \mid \mathcal{P}) \leq (1 + O(n^{-3}))(B_i/L_i^{\text{slot}})\pi_i(y)\mathbf{1}_{\{D(y) \in \mathcal{P}, \text{ permitted}\}}. \quad (16.2)$$

Proof. At a fixed typical pool, exact calibration at both stages gives the identity

$$p_b(y \mid \mathcal{P}) = \mathbb{E}_{W \mid \mathcal{P}, \text{ slice and cell gates}}[\tilde{q}_{g(b)}(D(y))U_{g(b)}^W(D(y))(y)].$$

Omit the expectation in direct modes. The load-gate denominator costs $1 + e^{-n^{c_0}}$. Pool restriction then costs $(B_i/L_i^{\text{slot}})(1 + O(n^{-4}))$, retaining the presence and permission indicators. Undoing the permission trim costs $1 + e^{-\Omega(n)}$. Finally the internally pretrimmed marginal, averaged over slice-conditioned data, is exactly π_i by the low profile definition in Proposition 14.2. These factors prove (16.2).

For an iid uniform pool, a fixed bin belongs to the pool with probability at most L_i^{slot}/B_i . Thus averaging (16.2) conditional on individual typicality gives at most

$$\frac{1 + O(n^{-3})}{\Pr(\mathcal{P} \text{ is typical})} \pi_i(y) \leq (1 + O(n^{-3}))\pi_i(y).$$

Both measures are probabilities, so this also gives total-variation closeness. This is an iid-pool comparison; it does not assert independence of the actual permutation pools. $\square$

Lemma 16.7 (Fresh internal-prior comparison). *On unforced iid slot pools, with individual typicality retained, the law of a fresh internal prior has bounded upper comparison with its unrestricted base experiment. More precisely, for every nonnegative test Φ of the prior, with $\Phi(0) = 0$,*

$$\mathbb{E}_{\mathcal{P} \text{ iid}} \left[\mathbf{1}_{\{\mathcal{P} \text{ typical}\}} \mathbb{E}_{\text{fresh}(\mathcal{P})} \Phi(\sigma_v) \right] \leq K \mathbb{E}_{\text{base}} \Phi(\sigma_v).$$

Consequently its mean and its permission-exception estimates hold up to fixed factors. True internal validity may be retained in the comparison.

Proof. Remove the cell-load conditioning at its negligible upper cost, keeping its gate until the conditional samplers have been compared. Apply first the label upper comparison and then the group-bin upper comparison on the internal star. The queried groups choose distinct physical bins; retain that restriction while averaging the pool. For each target bin replace its $\tilde{q}_g$ factor by the upper bound consisting of B_i/L_i^{slot} , its pool-presence indicator, and the raw q_g^W factor, with the trim and normalizer costs.

If there are m distinct target bins, simultaneous containment in unforced iid slots has probability at most $(L_i^{\text{slot}}/B_i)^m$: sum over distinct slot indices realizing those targets. This cancels all m restriction factors. The remaining nonnegative integral can now discard distinctness and use the raw bin and reference label kernels. Removing slice conditioning on (14.3) restores the unrestricted base experiment. The raw posterior rule itself has stayed unchanged throughout.

All costs together are bounded: besides the normalizers, the relevant quantities are $h_i^{O(1)} \sqrt{\epsilon_i} \ll 1$ for the pretrims and $h_i d_i^{-0.1} \ll 1$ for the joint comparisons. In direct modes the internal prior is deterministic. The same proof can retain the true internal-validity indicator. Taking $\Phi(\sigma) = \sigma(x)$ gives the mean estimate from (14.2); taking the permission exception indicator gives (16.1) up to the same fixed factor. Conditioning the iid pool on typicality costs its probability's reciprocal. An own-pool slot pin is excluded here because it would change the containment calculation. $\square$

17 Initial list gates and resampling estimates

The fresh cell sampler, used on typical pools with its required history and bin conditions, makes every internal prior valid. We now require that the early labels in other cells leave a substantial common-neighbor list at each even role. After proving a failure estimate conditional on typical pools, we will resample cells for finitely many rounds. The final states must also retain a multiplicative comparison with fresh states; this comparison will later be applied to products of list weights.

Let B_v^e be the external early neighbors of an even role v . Here external means noninternal, including both bulk and crossing edges. Using the initial cell states, define the unnormalized row and its mass by

$$U_v^0(x) = \sigma_v(x) \prod_{b \in B_v^e} \frac{I_x(y_b)}{D_{i(b)}(x)}, \quad M_v^0 = \sum_x U_v^0(x).$$

The internal prior includes hits on the initial dummy labels at any late internal neighbors. For $J \subseteq B_v^e$, let

$$L_v(J) = \{x \in \mathcal{X}_{i(v)} : I_x(y_b) = 1 \text{ for every } b \in B_v^e \setminus J\}.$$

This unweighted set imposes no internal hits. The event to be avoided is

$$S_v = \{M_v^0 < 1/2\} \cup \left\{ \max_{\substack{J \subseteq B_v^e \\ |J| \leq (\log n)^4}} |L_v(J)| > \exp((\log n)^8) \right\}.$$

The second clause keeps a small set of possible even labels after selected early data are withheld. This is the support bound used in the late-exposure comparison of Section 18.2.

Lemma 17.1 (Independent list estimate with pins). *Fix $s \leq s_0 := \lceil 20R \rceil$ distinct members of B_v^e and arbitrary labels for them. Also fix a valid probability prior σ_v whose mass on the intersection of these pinned neighborhoods is at least $.9 \cdot 2^{-s}$. Draw the remaining external early labels independently from their $\pi_{i(b)}$'s. Then $\Pr(S_v) \leq n^{-2R}$.*

Proof.

Retained mass. Write $i = i(v)$. Partition the external early incidences into the s pins, the c' unpinned crossings, and the d unpinned bulk edges. There is at most one late internal neighbor, so

$$s + c' + d = n - h_i - \#\{\text{external late neighbors}\} \leq n - h_i - p(v) + 1, \quad d = n - O(\log n).$$

Applying the pinned hit factors and their denominators first leaves mass at least $.9 - o(1)$, since each denominator is $.5 + o(1)$ and s is fixed. Normalize this remaining row, then expose the unpinned crossing labels successively. At each step deep discrepancy (12.1) gives hit mass $.5 \pm O(b_*)$ except with probability exponentially small in a power of n . The intermediate normalized laws remain small-width, and the cleaned crossing denominators have the same estimate. Since $c'b_* = o(1)$, the mass before the bulk steps exceeds $.8$ outside this exception. Its normalized law τ satisfies

$$\tau \leq K 2^s 4^{c'} \sigma_v.$$

The remaining bulk labels all have law π_i . For the homogeneous peeling estimate their parameter obeys

$$\Gamma := \Delta_i \max_x (\tau(x) D_i(x)^{-d}) \leq K e^{-200s_i} 2^{-p(v)}.$$

In cluster mode this follows by combining $N \|\sigma_v\|_\infty \leq 2^{h_i} e^{-500s_i}$, $\Delta_i \leq e^{s_i}$, the degree-drift bound (13.1), and $c' \leq \ell_i \leq s_i/(1000u)$. The preceding edge count and $N \geq 2^n$ give the factor $2^{-p(v)}$, with the possible extra factor 2 incorporated into K . In direct bias mode the positive bulk degree gain absorbs the patch-width, conflict and crossing factors. In the bounded case those factors are fixed, so the same inequality holds with a fixed K and $s_i = 0$. Finally $p(v) \geq r/2 \geq A_0 \log n/2$; the chosen $A_0 \gg R$ therefore makes $\Gamma \leq n^{-3R}$ for large n .

Let Z be the bulk mass starting from τ . The centered-moment identity, the moderate-interaction bound and homogeneous peeling give

$$\mathbb{E}|Z - 1|^u \leq n^{-3R} + O_u(\Gamma), \quad \Pr(Z < 3/4) \leq 4^u \mathbb{E}|Z - 1|^u.$$

Together with the crossing exception this is below the required n^{-2R} bound with room to spare. On its complement, $M_v^0 \geq .8(3/4) > 1/2$.

Support size after omissions. For a fixed J , ignore all internal hits, crossings and pins. The remaining unpinned bulk hits are independent. The bound $n|D_i - .5| = O(\log n)$, the edge count above, and $N \leq n2^n$ give

$$\mathbb{E}|L_v(J)| \leq \exp(O((\log n)^5)).$$

There are at most $\exp(O((\log n)^5))$ deletion sets with $|J| \leq (\log n)^4$. Markov at $\exp((\log n)^8)$ and a union over these sets give a probability much smaller than n^{-2R} . Adding it to the mass estimate proves the lemma. $\square$

17.1 Pool estimate for S_v

At one star, fresh sampling uses the own cell of v for its internal prior and at most one external early label from each other involved cell. These cells are distinct: bulk neighbors lie in different nearby slices, which cannot share a cell by the slice separation, and crossings visit distinct other patches.

Lemma 17.2 (Uniform pool estimate). *Put $T_s = \lceil \log^2 n \rceil$. Except on a pool event of probability at most $n^{-RT_s/2}$ per star, all involved individual pools are typical and*

$$p_S(v \mid \text{pools}) := \Pr_{\text{fresh cells}}(S_v \mid \text{pools}) \leq n^{-P}. \quad (17.1)$$

The same exception bound holds conditional on any one prescribed global slot-to-bin pin.

Proof.

Compatibility with a few prescribed slot labels. In addition to individual typicality, define the pool event $\mathcal{A}_v$ as follows. For every choice of at most s_0 distinct members of B_v^e , one slot in each of their cells, and one label in each selected bin permitted at that member, require

$$\Pr_{\text{fresh own-cell tape}} \left(\sigma_v \left(\bigcap_{b \text{ chosen}} N_G(y_b) \right) < .9 2^{-s} \mid \text{own pool} \right) \leq n^{-2R},$$

where s is the number of chosen members and $N_G(y) = \{x : I_x(y) = 1\}$. This is a predicate of the pools: it bounds a conditional probability over a later own-cell draw.

Its failure, on individual typicality, has probability $\exp(-n^{\Omega(1)})$. To prove this, fix members, slot indices and within-bin label indices, and compare the involved permutation slots to iid slots, retaining a possible global pin. There are three cases. If the pinned slot is one of the selected external slots, expose its label first. The permission test (16.1) and Lemma 16.7, applied to the unforced own pool, give an exponentially small averaged probability of a bad first degree. The other selected slots are unforced. If the pin lies in the own cell, all selected external slots are unforced, and only validity and the width bound of the own prior are needed. If the pin lies in neither place, the same argument uses entirely unforced selected data.

A prescribed within-bin index in a uniform bin has small width. Hence at each unforced slot, (12.1) puts its degree into the current normalized first law in $.5 \pm 2b_*$, except with probability exponentially small in a power of n . Successful restrictions preserve small width through these at most s_0 steps, and their intersection mass is at least $(.5 - 2b_*)^s \geq .9 2^{-s}$. Thus the incompatibility probability, averaged over the iid pools and the own tape with the needed typicality indicators, is exponentially small. Markov at n^{-2R} , followed by a union over the $n^{O(s_0(A_c+1))}$ choices of members, slots and within-bin indices, proves the asserted bound for $\mathcal{A}_v$.

Independent trials sharing their pools. Let $\mathcal{T}_v$ require individual typicality and $\mathcal{A}_v$. Take T_s independent fresh-cell trials at these same pools. Then

$$\mathbb{E}_{\mathcal{P}}[\mathbf{1}_{\mathcal{T}_v} p_S(v \mid \mathcal{P})^{T_s}] = \Pr(\mathcal{T}_v \text{ and every fresh trial has } S_v).$$

Use the permutation-to-iid upper comparison, maintaining a global pin if present. In each trial, (16.2) bounds each external singleton by a uniform choice of a slot index, with subsequent bin-label weight

$$B_{i(b)} \pi_{i(b)}(y) \mathbf{1}_{\{D(y)=D_{\text{slot}}, \text{ permitted}\}}.$$

Indeed averaging this weight over the $L_{i(b)}^{\text{slot}}$ indices recovers the right side of (16.2) without its error factor. The accumulated factor over at most nT_s singletons is $1 + o(1)$.

There are now three remaining layers of randomness: selected slot indices, the bin images of those slots, and the within-bin labels under the displayed weights. Let j be the number of selected slot occurrences minus the number of distinct unpinned slots visited. The fraction of index choices with overlap rank j is at most

$$(2nT_s)^{3j} (\min_i L_i^{\text{slot}})^{-j}.$$

One obtains this bound by specifying which choices repeat earlier indices or hit the pinned index. A repeated unpinned slot of multiplicity q contributes $q - 1$ to j , and its q occurrences number at most $2(q - 1)$. A pinned slot contributes its entire multiplicity to j . Thus at most $2j$ occurrences use repeated or pinned slots. Call their labels the trial pins.

Fix the images of those slots, their trial-pin labels, and the own pool. Each such occurrence has total bin-label weight at most 11, because $\pi_i \leq 11/M_i$ and a bin contains d_i labels; thus all trial pins cost at most 11^{2j} . The unique unforced slot images are still independent. Integrating each restores $\pi_{i(b)}$, after discarding its permission indicator for an upper bound. For a trial with at most s_0 pins, retain from $\mathcal{A}_v$ only its own-tape compatibility assertion. This assertion does not read any of the unique unforced slots. Its incompatible own-tape probability is at most n^{-2R} ; on compatibility, Lemma 17.1 gives another n^{-2R} . Hence the trial's bad probability is at most n^{-R} . The fresh own tapes and the remaining unforced slots are independent across trials, so these bounds multiply.

At most $2j/s_0$ trials have too many pins and require the trivial bound 1. Summing overlap ranks therefore gives

$$\begin{aligned} \mathbb{E}[\mathbf{1}_{\mathcal{T}_v} p_S^{T_s}] &\leq O(1) n^{-RT_s} \sum_{j \geq 0} \left[\frac{(2nT_s)^3 11^2 n^{2R/s_0}}{\min_i L_i^{\text{slot}}} \right]^j \\ &\leq O(1) n^{-RT_s}. \end{aligned}$$

Here $2R/s_0 \leq .1$ and $L_i^{\text{slot}} = n^{A_c + o(1)}$, with $A_c = 200$, so the bracket tends to zero. This proves the moment bound with its typicality indicator. Finally

$$\Pr(p_S > n^{-P}, \mathcal{T}_v) \leq n^{PT_s} \mathbb{E}[\mathbf{1}_{\mathcal{T}_v} p_S^{T_s}] \leq O(1) n^{-(R-P)T_s}.$$

Since $R = P^2$, this and the exponentially smaller individual and auxiliary pool exceptions are at most $n^{-RT_s/2}$ for large n . $\square$

17.2 Palettes

Lemma 17.3 (Palette retention and pair estimates). *There is a fixed palette assignment with the mass retention and separation properties below, uniform over every valid local history. It satisfies (17.2) and the accompanying second-moment bound.*

Proof.

Separating internal words. Choose a binary linear map on the internal words of patch i whose kernel contains no nonzero word of weight at most $500\rho h_i$. Random linear checks and the ball-volume bound give such a map with image size $\chi_i \leq e^{s_i}$. Set $\chi_i = 1$ when there are no internal bits. Its colors separate distinct words at the indicated distance. Among even roles the colors have equal sizes, since a free outer bit supplies either parity for each internal word.

Retaining list mass. Independently color the labels of X_i uniformly by these χ_i colors. We can fix a realization with each palette of size at most $2M_i/\chi_i$, such that for every possible valid local history with $M_v^0 \geq 1/2$, its assigned palette satisfies

$$\sum_{x \text{ in the palette of } v} U_v^0(x) \geq 1/(4\chi_i).$$

At the same time, every palette has σ_v -mass at most $2/\chi_i$ for every possible valid internal prior.

For this simultaneous assertion the deterministic atom estimate is

$$\max_x U_v^0(x) \leq KC_n^{-1} e^{-200s_i} 2^{-p(v)}.$$

It follows from the cap and degree calculation in the pinned-list proof. The logarithm of the number of histories is at most $n^{2+o(1)}$: Lemma 14.3 counts the own slice data, and the star labels add at most $n \log N = O(n^2)$. Weighted-sum concentration for the random label colors has exponent at least a fixed multiple of

$$C_n e^{199s_i} 2^{p(v)},$$

using $\chi_i \leq e^{s_i}$ and $M_v^0 \geq 1/2$. The bound $p(v) \geq A_0 \log n/2$ makes this dominate the logarithmic history count. The much smaller atoms of σ_v give the simultaneous upper palette-mass bound, and ordinary size concentration gives the palette sizes. Thus one coloring has all the asserted properties. After restriction to the assigned palette and normalization, the starting row has cap at most $4KC_n^{-1} e^{-199s_i} 2^{-p(v)}$.

Pair estimates. For all these valid internal priors and all $x \in \mathcal{X}_i$,

$$\sum_{\substack{z \text{ in assigned palette} \\ |K_{\pi_i}(x,z)| \leq \xi}} \sigma_v(z) \prod_{b \in B_v^e} \frac{\int I_x I_z d\pi_{i(b)}}{D_{i(b)}(x) D_{i(b)}(z)} \leq K/\chi_i. \quad (17.2)$$

To verify this, write the ratio for a law π_j as

$$R_j(x, z) = 1 + \frac{K_{\pi_j}(x, z) - m_{\pi_j}(x)m_{\pi_j}(z)}{(1 + m_{\pi_j}(x))(1 + m_{\pi_j}(z))}.$$

First consider pairs typical in every relevant relaxed row-tail test (12.3). Each bulk ratio is $1 + O(n^{-1-.02} + \log^2 n/n^2)$; each crossing ratio is $1 + o(1/\log n)$. There are at most n bulk factors

and $\ell_i = o(\log n)$ crossings, so their product is bounded. The palette's σ_v -mass is at most $2/\chi_i$, proving the claim on this part of the sum.

For bulk exceptions within the nonconflict cutoff, the weighted row tail in relaxed (12.3) controls the product of bulk ratios. Its $e^{-n^{19}}$ bound also absorbs the crude $\exp(O(\ell_i))$ crossing factor. For crossing exceptions with typical bulk, the unweighted row tail suffices. In each case replacing uniform $z \in X_i$ by σ_v costs only $\exp(O(\log n))$, by the low-mode cap. The resulting exception contribution is negligible compared with $1/\chi_i$, proving (17.2).

We also record the precise uniform integral used later. Let $\mathcal{A}(x, z)$ mean that both labels belong to $\mathcal{X}_i$ and are nonconflicting. Then, for $0 \leq d \leq n$,

$$\mathbb{E}_{x, z \text{ iid uniform on } X_i} \left[\mathbf{1}_{\mathcal{A}(x, z)} \left(4 \int I_x I_z d\pi_i \right)^{2d} \right] \leq \exp(O(\log n)).$$

Indeed the degree terms contribute $\exp(O(\log n))$; the typical pair term is bounded, and the same weighted tail controls the exceptions inside the cutoff. This is an unconditioned uniform integral with an indicator, rather than a law conditioned on the envelope or a palette. $\square$

17.3 Parallel initial resampling

We use the independent sample tapes and backward execution witnesses of the Moser–Tardos resampling method [28]. Haeupler, Saha and Srinivasan [19, Theorem 2.2] established multiplicative comparisons for fixed events determined by the current assignment during a resampling execution; Harris and Srinivasan [22] developed this distributional analysis further. The parallel-round theorem of Moser and Tardos uses a maximal independent set in each round. The fixed-priority rule below need not choose a maximal independent set, so we prove its finite-round and final-state estimates directly.

Fix the pools, and give each cell its independent tape of fresh states. The scope of S_v is the set of cells it reads. Form the scope graph with one vertex per event S_v , joining events whose scopes share a cell, and fix an ordering of its vertices. At each of T_s rounds, execute every currently true event having no earlier-numbered true neighbor. An execution advances the tape index of every cell in its scope. The executing scopes are disjoint, although some true events may have no executing neighbor. Scope sizes, cell incidences and graph degrees are at most n^{A_c+4} .

A restricted simulation uses a deterministic subset of events, its induced scope graph, and the same priorities and tapes. Truth outside that subset is not consulted. During one round, update and priority influences travel at most two graph steps. Hence the final state of a target cell, or the final truth of a target S_v , agrees with the global simulation whenever the restricted set contains the graph ball of radius $2T_s + 3$ about that cell's incidence events or about v , respectively. The same holds for any larger deterministic set.

Proposition 17.4 (Finite resampling and multiplicative comparison). *Fix, before drawing tapes, a deterministic restricted event set, a defining event S_v in that set, and a deterministic set of at most $n^{10(A_c+10)}$ target cells. Suppose the pools are individually typical and satisfy (17.1) for every event tested in the restricted simulation. All conclusions below concern that restricted process. The defining event's final failure probability is at most $n^{-PT_s/2}$. The probability of more than T_s executions in the backward closure of the target cells is also at most $n^{-PT_s/2}$. For every nonnegative test of those target states, the final-state expectation is at most $1 + o(1)$ times its independent fresh-cell expectation at the same fixed pools.*

Proof.

All probabilities in this proof condition on the fixed pools. Every fresh truth test has probability at most n^{-P} .

A witness for final failure. If the defining event S_v is true at the finish, consider its component in the restricted scope graph induced by all sites ever true, at round starts or at the finish. This component has a true site at every round start. Indeed, if it had none at some round, no outside execution could change any of its truth values: an executing outside site would be true and adjacent to it, and hence belong to the same component. It would then stay entirely false, contradicting final truth. The least-numbered true site in the component therefore executes each round.

List all execution occurrences in this component, say $M \geq T_s$. Add a maximal collection of J further ever-true sites nonadjacent to any execution site and pairwise nonadjacent. The added sites have untouched inputs. The distance-one neighborhoods of the listed sites cover the component; consequently the listed sites are connected by graph steps of length at most three and anchored near v . Rooted tree traversals, with occurrence multiplicities, types and execution rounds specified, bound the number of such lists by

$$(n^{O(A_c+4)}T_s^{O(1)})^{M+J}.$$

For each possible list, determine the tape entries used by its truth tests from the listed prior executions. Every execution touching one of these sites belongs to the same component and has been listed. The event that this list is the actual witness is contained in the event that all these prescribed truth tests pass. These necessary tests use independent entries: an execution consumes each entry it tests, while the extra sites read untouched, mutually disjoint scopes. Their joint probability is at most $n^{-P(M+J)}$, before any consistency conditions on the list are imposed. The enumeration and this probability form a geometric sum with per-occurrence factor $n^{-P+C(A_c+4)+o(1)}$. The sufficiently large choice of P , and $M \geq T_s$, bound the sum by $n^{-PT_s/2}$.

Backward executions and terminal entries. For the fixed target cells, list every execution touching them, then recursively include every earlier execution whose scope meets a listed later one. This is their backward closure. Its occurrence lists are counted in the same way, joining all target incidences to one auxiliary root. The polynomial number of targets is included in the constant in $n^{O(A_c+4)}$. A list of size m has at most $(n^{O(A_c+4)}T_s^{O(1)})^m$ possibilities, and its necessary truth tests have probability at most n^{-Pm} . Summing $m > T_s$ gives the second assertion.

For a fixed occurrence list, the final entry of each target cell is the entry immediately after its last listed touch, or its initial entry if there was no touch. These indices are prescribed by the list. None is used in a necessary truth test: every execution touching a target has been included, and all its consumed entries precede the terminal one. The terminal entries are thus independent fresh states, independent also of the necessary truth tests. For any nonnegative target test Ψ , consistency of the list may be discarded, leaving

$$\begin{aligned} \mathbb{E}[\Psi(\text{final targets})] &\leq \mathbb{E}[\Psi(\text{fresh targets})] \left(1 + \sum_{m \geq 1} [n^{-P+C(A_c+4)+o(1)}]^m \right) \\ &= (1 + o(1))\mathbb{E}[\Psi(\text{fresh targets})]. \end{aligned}$$

The first term accounts for an empty closure. This factorization uses only the pool requirements in the specified deterministic horizons and proves the multiplicative comparison. $\square$

Remark 17.5 (Scope of the comparison). The target cells and simulation horizons in Proposition 17.4 are fixed before their tapes are drawn. No buffer is needed for the restricted-process estimates; the preceding radius condition is used only when identifying these states with those of the global run. The comparison applies directly to nonnegative list products, regardless of their size. An additive rare-history bound would not give this conclusion: the proof instead separates the consumed truth-test entries from the fresh terminal entries before discarding occurrence-pattern consistency.

18 Late exposure and the final matching

Write S_{fin} for the cell-state configuration after the T_s resampling rounds of Section 17. These post-round states have not yet been conditioned on the terminal avoidance requirements introduced below. The remaining task is to assign the late odd roles injectively while preserving the common-neighbor lists at even roles, and then to choose distinct labels from those lists.

Which law is being used. The argument uses the following processes and comparison laws.

1. *Initial states.* The initial process draws permutation pools and runs the finite resampling procedure to produce S_{fin} . Terminal avoidance later conditions this same process. For nonnegative tests on the specified local regions and deterministic horizons, expectations are bounded using independent fresh cells at the same fixed pools while the required pool conditions are retained, and then using fresh cells on iid uniform slots.
2. *Reference late transitions.* Given one entering history, each late class uses independent row outputs consisting of masks, sketches, and labels. The mask profiles are fixed independently of sampled histories. The local gates are evaluated on actual data and the specified reference kernels, and are retained as indicators in comparisons.
3. *Actual late assignments.* Given a history satisfying the incoming bounds and column-load condition, a conditional injective sampler assigns the current class while avoiding its bad events and future alarms. For the specified row tests, its upper comparison is to that class's product reference law, and its normalizer applies only to the current draw at the fixed entering history.

For choosing the mask profiles, the annealed baseline averages over iid slots, the full initial process including resampling and fallbacks, and unconstrained product reference late transitions. Thus the iid input there refers to the slots; the cell states still come from the full initial process. The profiles are fixed under this baseline before terminal conditioning.

18.1 Reference transitions and local requirements

Initial and current lists. Initial validity at an even w requires a valid internal prior σ_w as above (a probability on initial internal hits, with its indicated cap and single-corner cleaned support), S_w false including the masked-list bound, and the indicated starting palette mass $\geq 1/(4\chi_{i(w)})$. These hold everywhere after the desired initial successes. They are gate predicates evaluated on the data. The reference computations below are defined from direct data, with local fallbacks only for invalid or zero computations.

Let $\mathcal{P}_w^X$ be the host palette assigned to w . On initial validity define

$$\tau_w^r(x) = \frac{U_w^0(x) \mathbf{1}_{\{x \in \mathcal{P}_w^X\}}}{\sum_z U_w^0(z) \mathbf{1}_{\{z \in \mathcal{P}_w^X\}}}, \quad \mathcal{D}_w^0 = \text{supp } \tau_w^r.$$

Number the late classes $r, \dots, 1$, so that j classes remain at stage j . The current probability τ_w^j is τ_w^r conditioned to hit every previously assigned late neighbor of w . It reads only the initial data in w 's own slice, the external early labels at w , and its previous late neighbor labels. In particular, it does not read the masks or sketches of previous late draws.

All reference kernels are defined on invalid histories as well. Fix a label on each relevant host side in advance; an invalid initial list or a zero normalizer is replaced by the point mass at that label. Use the same lookup on every subsequent occurrence of the same direct data. This choice affects no gated calculation below. More generally any fixed probability fallback with these same inputs gives the comparisons used here. Let $p_w(j)$ count the neighbors of w in the remaining classes. For a sufficiently small fixed $\beta > 0$, put $i = i(w)$ and define

$$e_{w,j} = C_n^{-\beta} \exp(-\beta s_i) 2^{-\beta j}.$$

The inequalities $C_n \leq n$, $s_i = O(\log n)$ and $r = O(\log n)$ give $\min e_{w,j} \geq n^{-.02}$ when β is chosen after the low-mode constants. Also

$$\sum_{j=1}^r e_{w,j} \leq K_\beta C_n^{-\beta} e^{-\beta s_i} = o(1).$$

This remains $o(1)$ after multiplication by $\max(1, h_i)$: in cluster mode $s_i = (a/10^6)h_i$, and $h_i e^{-\beta s_i}$ is bounded even if the scale does not grow.

The product reference transition. Fix the entering history for class j . The reference transition is a product of independent row draws under that conditional law. At a late odd role b of this class, draw the row as follows.

1. A mask profile is a probability distribution, fixed in advance, over subsets of the current late pool of size at least half that pool. Draw a subset from this profile and let μ_{mask} be its uniform law. For every even $w \sim b$, draw an iid sketch of τ_w^j of length $\lceil n^{.25} \rceil$. These side draws are independent given history.
2. Test a candidate y by requiring empirical sketch hit fraction $\geq .5 - e_{w,j}$ for each neighbor w . Sample y_b by the masked law conditioned on passing all tests if their retained mass is at least $\exp(-g_0 n)$, $g_0 = \alpha/100$; otherwise just use the masked law.

Write K_b^j for this full row kernel and $K_b^j(\cdot \mid \text{side data})$ for its conditional label law. The mask and sketches belong to the side data in this output. A deletion kernel, given these same side values, uses the same threshold rule after omitting the indicated tests; it does not redraw the side data. No success checks are inputs to these draws themselves. Masks with prescribed independent profiles (specified for balance later) do not adapt to sampled histories.

Desired local conclusions. We want true hit mass for y_b under each τ_w^j , $w \sim b$, at least $.5 - 3e_{w,j}$. We also want, given actual mask/sketch values, pointwise label-kernel upper comparison to each specified deletion kernel at factor $\exp(1000le_{w,j})$, where one deletes any one external neighbor test (w that neighbor, $l = 1$), or all internal neighbor tests at b (w any such internal neighbor, $l = h_i(b)$). Internal/external refers to the edge; an internal neighbor lies in the same patch.

The prerequisite gate. Let G_b be the following predicate of the entering history: initial validity holds at every even site at Hamming distance at most $6r$ from b , and the true-hit and deletion

conclusions hold at every earlier late role in that radius. We retain $\mathbf{1}_{G_b}$ in each bad event below. These history gates check actual data: they evaluate the specified **reference** conditional kernels at those data, without recursively verifying prior gate statuses. Thus raw late-event simulation from S_{fin} needs only direct data within $O(r)$ spatial radius, and similarly for forward influences (initial direct data may of course require cell consultations there). On the prerequisite gate, summable losses and at most one late hit per class at $w \sim b$ give a valid probability with

$$\max \tau_w^j \leq 8KC_n^{-1} \exp(-199s_i) 2^{-p_w(j)} \quad (18.1)$$

for K as in the initial cap. Indeed a previously imposed hit costs at most $(.5 - 3e_{w,s})^{-1}$. Their product is at most $2^{p_w(r)-p_w(j)} \exp(O(\sum_s e_{w,s}))$; the last factor is at most two eventually.

The three requirements are as follows.

Requirement 1 (sketch conflicts). For a sketch $(X_1, \dots, X_m)$ of τ_w^j , where $m = \lceil n^{.25} \rceil$, require

$$m^{-2} \sum_{a,c \leq m} \mathbf{1}_{\{X_a, X_c \text{ conflicting}\}} \leq 2e_{w,j}^4.$$

The ordered sum includes its diagonal.

Requirement 2 (prefix moments). Fix an order with the external tests singly, followed by the internal tests as one batch; omit an empty internal batch. Include also the orders obtained by moving any specified external test to the end. For a batch in one of these orders let A_q be the intersection of the preceding sketch tests, excluding the current batch. Whenever $\mu_{\text{mask}}(A_q) \geq e^{-g_0 n}$, set $U = \mu_{\text{mask}}(\cdot \mid A_q)$. Thus U is a law on late host labels. For each even role v in the current batch require

$$\int I_x dU = .5 \pm 10b_*, \quad \int I_x I_z dU = .25 \pm 10b_* \quad (18.2)$$

for all $x \in \mathcal{D}_v^0$ and all nonconflicting pairs there.

Requirement 3 (true hits). Require $d_G(\tau_w^j, y_b) \geq .5 - 3e_{w,j}$ for every $w \sim b$.

The sketch-conflict and prefix-moment bad events are their respective failures intersected with G_b . The true-hit bad event is $G_b \cap \{\text{Requirements 1 and 2 hold}\}$ intersected with a failure of Requirement 3. These are events in the reference process, not new conditionings of its kernels.

The next lemma also records the single-label and pair-hit kernel bound used later in the endpoint comparison.

Lemma 18.1 (Consequences of the local transition requirements). *Fix an entering history satisfying the prerequisite gate, and use the product reference transition at b . There is a fixed $c > 0$ such that, uniformly over the fixed mask profiles,*

$$\begin{aligned} \Pr(\text{Requirement 1 fails}) &\leq e^{-n^c}, \\ \Pr(\text{Requirements 1 and 2 hold but a true-hit conclusion fails}) &\leq e^{-n^c}. \end{aligned}$$

If the first two requirements hold, every tested prefix is broad, the specified deletion comparisons hold, and the full conditional label kernel satisfies (18.3). The true-hit conclusion is enforced by the third requirement.

Proof.

Sketch concentration. The conditional expected conflict fraction is at most

$$m^{-1} + \Delta_i \|\tau_w^j\|_\infty \leq m^{-1} + 8K^2 C_n^{-1} e^{-198s_i} 2^{-p_w(j)} \leq e_{w,j}^4$$

eventually. The last inequality uses $p_w(j) \geq \lfloor j/2 \rfloor$ and small β ; the conflict degree is used only on the active single-corner support. Write $m = \lceil n^{.25} \rceil$ for the sketch length and $e = e_{w,j}$ for a fixed target. Changing one sample changes the empirical conflicting-pair fraction by at most $2/m$, so its excess above $2e^4$ costs $\exp(-\Omega(me^8))$ by bounded differences. Since $e \geq n^{-.02}$, this is $\exp(-\Omega(n^{.09}))$. A union over at most n targets gives the first asserted bound for some fixed $c > 0$.

Broad prefixes and deletion kernels. Suppose Requirements 1 and 2 hold. For a target sketch, its empirical hit fraction $H(y) = m^{-1} \sum_a I_{X_a}(y)$, with $y \sim U$, has mean $.5 + O(b_*)$. In its second moment, nonconflicting pairs contribute $.25 + O(b_*)$ by (18.2), and conflicting pairs have total fraction at most $2e_{w,j}^4$. Hence

$$\text{Var}_U H = O(b_* + e_{w,j}^4), \quad U\{H < .5 - e_{w,j}\} \leq O((b_* + e_{w,j}^4)/e_{w,j}^2) = O(e_{w,j}^2).$$

Here $b_* \ll e_{w,j}^4$. Multiply the retention factors for external tests, and use a union bound for the internal batch. The total retained mass is $\exp(-o(n))$, and the internal batch loses $o(1)$. Starting with the masked law, this proves inductively that every tested prefix exceeds $e^{-g_0 n}$. Thus neither the full label kernel nor the relevant deletion kernel uses its fallback.

Place the deleted external test last, or use the base order with the internal batch last. The density of the full kernel relative to the leave-out law is the reciprocal retained mass of that last test or batch, at most $\exp(1000le_{w,j})$. Applying (18.2) under this leave-out law now bounds a single hit or a nonconflicting pair hit from $\mathcal{D}_w^0$ by

$$2^{-u'} \exp(O(le_{w,j})), \quad u' = 1, 2 \tag{18.3}$$

respectively, where l is the size of the deleted external test or internal batch.

True hits. On the first two requirements, labels meet the sketch tests, so compare to the corresponding deletion laws while retaining those requirements. The comparison label for w no longer depends on that neighbor's sketch. Its pointwise multiplier $\exp(1000le_{w,j})$ is bounded by the uniform error-sum estimate, including when $l = h_i$. We may now drop the success restrictions in the nonnegative upper integral. For a comparison label whose true hit mass is below $.5 - 3e_{w,j}$, the independent target sketch passes the threshold $.5 - e_{w,j}$ with probability at most $\exp(-\Omega(me_{w,j}^2)) \leq \exp(-\Omega(n^{.21}))$, by the bounded-summand estimate. The union over targets proves the second asserted bound, decreasing c if necessary. This bounds the intersection with Requirements 1 and 2; it does not condition the reference law on their success. $\square$

Avoiding all these gated bads after globally good initial states suffices by induction (our stage assignments with joint upper comparison will indeed use only values in reference support given history). The failure still needing a transfer through fresh initial data is Requirement 2, namely (18.2): a broad U is history-adaptive and the supports have used early data.

18.2 Transfer for a late prefix

We call the bad events just defined the late events F . For Requirement 2 use one event for each fixed role, target and prefix, bundling all its single-label and pair tests. Fix such an event at $v \sim b$, with $i = i(v)$. Let $\mathcal{I}_i$ be the internal coordinate set of patch i , and t the word of v on its complement. Throughout this proof outer words are relative to $\mathcal{I}_i$, even at sites in other patches. Write A_k for the coset modulo $\mathcal{L}$ of coordinate k 's ID, C_w for the syndrome coset of a cube vertex w , and w^k for a coordinate flip.

The critical cells are those containing v^k for bulk coordinates k with $A_k \neq C_v$. These are distinct cells of patch i , and the indicated roles are external early neighbors of v , in slices with outer words t^k . There are $d = n - O(\log n)$ of them. One such cell may be omitted; if so, put it among the fixed noncritical inputs and impose no witness hit from it in the estimate. The experiment for the lemma is:

- fix arbitrary initial states in all other needed cells;
- draw each critical whole-cell state independently, using an unpinned iid slot pool conditioned on individual typicality and then a fresh cell sample on that pool;
- run the product reference late transitions from these states.

On the gated prefix failure there is a label $x \in \mathcal{D}_v^0$, or a nonconflicting pair $x, z \in \mathcal{D}_v^0$, violating (18.2). Every retained critical label hits every label in this witness, because $\mathcal{D}_v^0$ satisfies all early hits at v . This necessary condition is what permits us to sum over possible witnesses below.

Lemma 18.2 (Late-prefix transfer). *In the preceding experiment, the fixed prefix-moment bad event F has probability at most $e^{-n^{c_1}}$, for a fixed $c_1 > 0$. The estimate is uniform over the fixed states of all noncritical cells and the independent mask profiles, and still holds when one critical cell is omitted as allowed above.*

Proof.

The proof first compares reference path factors with deletion kernels on the required path event, and then integrates the erased sketches. Only afterward does it fix independent late-draw seeds and construct a deterministic block simulation. A survival tilt and a stopped likelihood estimate then bound the rare prefix deviation.

Critical labels. Let $y_k = y_{v^k}$, and denote its marginal law by q_k . Integrating the fresh singleton comparison over an unpinned iid pool conditioned on individual typicality gives $q_k \leq (1 + O(n^{-3}))\pi_i$, and therefore total-variation distance $O(n^{-3})$. Direct spatial consultations for this event lie within $O(r)$ of b . Cell spacing implies that they reach each critical cell only in its indicated slice. This does not assert independence of that slice from the other slices of its cell.

18.2.1 Erasing part of a history

The predecessor set. Start with the mask and prefix sketches at b . Whenever a needed sketch at an even site w uses a previous late neighbor's label, include that neighbor's full reference transition and all sketches needed by it. Continue backwards through the class order, generating each late role once. Call these the direct roles, and call the even sites queried by their sketches the direct even sites. This finite set is closed under label predecessors, so its marginal law is exactly the product of the corresponding conditional reference kernels. Gate-only consultations outside it may be discarded when taking an upper bound.

There are at most r backward class steps, so every direct role is within distance $2r$ of b . Let $P(w)$ record, over the binary field, the parity of the number of active coordinates in each ID coset. A two-edge passage between late roles flips two IDs in the same coset, because both late syndromes belong to $\mathcal{L}$. Consequently $P(b') = P(b)$ for every direct late role b' , and

$$P(w) = P(b) + e_{C_w}, \quad P(w) + P(v) = e_{C_w} + e_{C_v}$$

for every direct even site. The prerequisite gate supplies initial validity, positive current lists and earlier deletion comparisons on this whole set, which lies within its stated radius.

Erase the exposed outer word. Let $H_i = \{A_a : a \in \mathcal{I}_i\}$; these cosets are distinct. If $D = \{q \notin \mathcal{I}_i : w_q \neq v_q\}$, the parity identity is

$$\sum_{\substack{a \in \mathcal{I}_i \\ w_a \neq v_a}} e_{A_a} = e_{C_w} + e_{C_v} + \sum_{q \in D} e_{A_q}. \quad (18.4)$$

Its right side must be supported on H_i , and then uniquely determines the internal bits of w .

Let V_0 consist of direct even sites with outer word t . If $C_v \notin H_i$, setting $D = \emptyset$ in (18.4) forces $C_w = C_v$ and $w = v$. If $C_v \in H_i$, let k_0 be its unique internal coordinate. The same identity gives $w = v^{k_0 a}$ for some internal a , so V_0 is contained in the internal neighbors of $b_0 = v^{k_0}$. A direct late role with outer word different from t can meet V_0 only through the unique external flip returning its word to t . A direct late role with outer word t must be b_0 , by $P(b') = P(b)$, and all its internal tests lie in V_0 .

At each earlier direct role meeting V_0 , replace its label kernel by the deletion kernel omitting that external test or internal batch. Keep its side-draw factors unchanged. On the original gated failure, the earlier deletion comparisons bound the product of original path factors by the product of modified factors times

$$\exp \left(1000 \max(1, h_i) \sum_s e_{v,s} \right) = O(1).$$

Indeed at most $\max(1, h_i)$ tests are deleted in each class. The comparison is made at path values, before fixing any randomization seeds. At the current role b , the tested prefix already excludes the single target v in the external case, or the entire internal batch in the other case. It therefore contains no sketch from V_0 .

The event retained after deletion. For each remaining direct sketch at w , form the set L_w of envelope labels at w hitting all its unaffected external early labels. Ignore the own prior, all critical external early inputs, and all late hits in this definition. Thus L_w uses only fixed noncritical data. We retain the following weaker event $\mathcal{E}_{\text{keep}}$: these sets have size at most $\exp((\log n)^8)$ at every affected sketch site; each remaining sketch there lies in L_w ; the tested prefix is broad; and there is an envelope witness of the required kind which survives every critical hit and violates (18.2) for that prefix. The one-block calculation below shows that the sets L_w omit at most $(\log n)^4$ external early inputs. Thus the initial masked-list requirement and the supported current lists on the original gate imply these retained conditions.

The original failure is contained in $\mathcal{E}_{\text{keep}}$, after the pointwise comparison has been applied. This weaker event reads only initial data, generated labels and remaining sketches. The erased sketches no longer enter any label kernel, and earlier side outputs are never inputs to later current lists.

They enter neither the retained event nor a remaining transition, so their conditional probability factors integrate to one. In particular no deletion test on an erased sketch is being retained implicitly.

One-block locality. Partition the critical coordinates by their ID cosets, merging all cosets in H_i into one block. Each block has at most $m_* := \max(1, h_i)r$ coordinates. We show that a remaining direct sketch consults at most one such block.

If its own slice is critical, its outer word is t^k . An external flip produces either t , whose sketches have been erased, or a word t^{kj} differing in two external coordinates; it cannot reach another critical slice. Internal flips stay inside the same patch and belong to the own slice computation. Hence this case reads only the block containing k .

Otherwise a critical external observation at w^j , in slice t^k , requires outer word t^{kj} at w , with $j \notin \mathcal{I}_i$ and $j \neq k$. An internal flip would stay in that same patch, whereas $j = k$ would put w in the erased word t . There can be only one further critical observation: flipping k reaches t^j . If both observations are critical, then $A_k, A_j \neq C_w$; their being early gives $A_k, A_j \neq C_w$. If $A_k \neq A_j$, neither coordinate can cancel on the right of (18.4). Both therefore belong to H_i . They lie in the merged block; if their cosets are equal, they already lie in one block.

This calculation also bounds the number of affected sites. For fixed critical k , an external observation through j has $A_j \neq C_w$. Equation (18.4) then forces $A_j \in H_i \cup \{A_k, C_w\}$, giving at most $(h_i + 2)r$ choices of j . For each resulting outer word, at most $h_i + 3$ values of C_w are possible, each determining all internal bits. A block of size m therefore affects at most

$$m [1 + (h_i + 2)r](h_i + 3)$$

sites, with at most r sketch calls at each. These counts are polylogarithmic in n . They include cross-patch observations: $\mathcal{I}_i$ lies beyond every allocation prefix, so the outer word fixes the patch, and the parity identity remains global.

18.2.2 A bounded-response simulation

A total protocol after fixing seeds. Now fix independent seeds for all late draws in the modified process, independently of the initial states. Order the finite predecessor computation by class and then by a fixed role order. A block input is the collection of whole critical cell states in that block, including their pools. At an affected sketch call, send that block the preceding labels and fixed data needed for the current list. The block computes the sketch using the fixed seed. It replies with the sketch's $\lceil n^{.25} \rceil$ indices in the fixed list L_w , if $|L_w| \leq \exp((\log n)^8)$ and every sampled label belongs to it. Otherwise it replies with a single abort symbol. This is a total rule: the current-list fallbacks were fixed when defining the reference law.

On receiving an abort, halt and output the current role b 's masked law. Its mask is determined by its independent fixed seed. In the absence of an abort, use the replies to evaluate the remaining modified label transitions. At the end output the prefix-conditioned law if its mass is at least $e^{-g_0 n}$, and the masked law otherwise. Unaffected computations use only fixed noncritical inputs and preceding replies. Every request is therefore determined without reading any other block. On $\mathcal{E}_{\text{keep}}$ there is no abort, and induction in the chosen order shows that every reply and generated label agrees with the modified reference process. Its final output is the tested prefix law.

The list-size assertion used here follows from the preceding geometry: $m_* = O((\log n)^{1.1}) \leq (\log n)^4$ eventually, so omitting a block omits no more early inputs than the initial masked-list

bound allows. The output U of the protocol is always broad, with

$$\text{width}(U) \leq \log(2N/M_{\text{late}}) + g_0 n = O(\log r) + g_0 n < \alpha n/2.$$

Information returned by each block. Only $n \text{polylog}(n)$ response steps are needed. This counts calls involving critical blocks, not every node of the possibly larger predecessor computation. Each call returns a sketch with at most $n^{.25+o(1)}$ indices, and each index uses at most $(\log n)^8$ logarithmic range. The number of calls to any one block is polylogarithmic by the site count above. Including the abort symbol, the entire reply sequence of one block has log-range at most n^4 . No size bound on requests is needed.

Fix this protocol, including its fallbacks and response alphabets, before choosing witness labels. The event $\mathcal{E}_{\text{keep}}$ is now contained in the event that some envelope label, or nonconflicting envelope pair, survives all critical hits and violates (18.2) for the protocol's broad output. We bound this larger event. Its protocol is independent of the witness being tested.

Raw states and survival conditioning. Integrate a witness $\mathbf{x}$ against the uniform law on $X_i^{a'}$, where $a' = 1$ or 2 . Let $\mathcal{A}$ require envelope membership and, for a pair, nonconflict. Write $\mathbb{P}_0$ for the raw independent block law. For a block H , let $A_H(\mathbf{x})$ be the event that all its critical labels hit every witness label, and put

$$c_k(\mathbf{x}) = \Pr(y_k \text{ hits } \mathbf{x}), \quad c_H(\mathbf{x}) = \mathbb{P}_0(A_H) = \prod_{k \in H} c_k(\mathbf{x}).$$

On $\mathcal{A}$, each $c_k \geq .15$. For pairs this follows from

$$\int I_x I_z d\pi_i = \frac{1}{4}(1 + m_{\pi_i}(x) + m_{\pi_i}(z) + K_{\pi_i}(x, z)),$$

the cleaned degree bounds, $|K_{\pi_i}(x, z)| \leq \xi$, and the $O(n^{-3})$ perturbation from π_i to q_k . The single-label case follows directly from the degree bounds.

For fixed allowed $\mathbf{x}$, let $\mathbb{P}_{\mathbf{x}}$ condition each whole block independently on $A_H(\mathbf{x})$. Equivalently it conditions each critical whole-cell state, including its pool, on the required hits of its critical label. We first prove that under this law the probability of witness deviation into the protocol's output, averaged over $\mathbf{x}$ with $\mathbf{1}_{\mathcal{A}}$, is $\exp(-n^{\Omega(1)})$.

Transcript cylinders. For a transcript prefix t , let $C_H(t)$ be the set of block inputs giving its prescribed replies to the prescribed requests. Since requests depend only on preceding replies and fixed data,

$$\mathbf{1}_{\{T=t\}} = \prod_H \mathbf{1}_{C_H(t)}.$$

Conditional on this prefix, raw blocks are therefore independent, each restricted to its own cylinder. Fixing all other blocks, one block's reply sequence determines the full transcript. There are at most e^{n^4} possible such sequences. At any specified update, the raw probability that the selected block's cylinder has mass below $e^{-\alpha n/8}$ is consequently at most $e^{n^4 - \alpha n/8}$, with a further polynomial factor harmless if the block is selected adaptively. This is an integrated raw bound, not a bound conditional on every transcript.

Off this exception, if the block has m critical labels their joint conditional law satisfies

$$Q \leq e^{\alpha n/8 + o(n)} \pi_i^{\otimes m}.$$

The unconditioned tuple has a product upper bound because its labels belong to distinct independent cells. Independence of those labels is not asserted after restriction to a cylinder.

Survival under a cylinder. For any such Q , let $q_j = dQ_{\leq j}/d\pi_i^{\otimes j}$. The Q -mass of value prefixes with $q_j < e^{-\alpha n/8}$ is at most $e^{-\alpha n/8}$. On the other prefixes the next-coordinate density relative to π_i is at most $e^{\alpha n/4 + o(n)}$, leaving its width below $\alpha n/2$. For any second-side law with that width, deep discrepancy implies that a uniform $x \in X_i$ has degree $.5 \pm 2b_*$, except on mass $\exp(-\Omega(n^{x_*}))$. Indeed a larger exception of one sign could be normalized to a first-side law within the other budget of (12.1). For two witness labels, first impose the typical hit of x , then apply the same estimate to the normalized restriction for z . This costs only bounded width.

Thus each next coordinate survives with probability $2^{-a'} + O(b_*)$, apart from a tiny set of witness tuples for each fixed value prefix. Fubini bounds the mean, over witness tuples, of the Q -mass of exceptional value prefixes. Markov and a union over the m coordinates make this mass $\exp(-n^{\Omega(1)})$ outside witness mass of that same form. The small-density prefixes have already been bounded in absolute Q -mass. For a witness outside the resulting exceptional set, let $A_{\leq j}$ mean that the first j coordinates hit every witness label, with $A_{\leq 0}$ the whole space. At each step the bad value prefixes have total Q -mass at most $\varepsilon_n = \exp(-n^{\Omega(1)})$. Integrate the conditional hit bound with the indicator of $A_{\leq j-1}$, bounding its contribution on those bad prefixes by their total mass. This gives

$$Q(A_{\leq j}) = (2^{-a'} + O(b_*))Q(A_{\leq j-1}) + O(\varepsilon_n).$$

Since $m \leq m_*$ is polylogarithmic, iteration makes the additive errors negligible relative to 2^{-2m} , and gives

$$Q(A_H(\mathbf{x})) = 2^{-a'm}(1 + O(mb_*))$$

outside an exceptional uniform witness mass $\exp(-n^{\Omega(1)})$. Apply this estimate both to a transcript-conditioned block and to its raw law before any transcript. The latter application controls the denominator c_H ; the cutoff alone supplied only its positive lower bound $.15^m$. The transcript-conditioned estimates are averaged over the raw transcript distribution. There is no union over all transcripts.

The likelihood process. Include the fixed witness in the transcript filtration $\mathcal{F}_t$, and define

$$r_H^t = \frac{\mathbb{P}_0(A_H(\mathbf{x}) \mid C_H(T_t))}{c_H(\mathbf{x})}, \quad R_t = \prod_H r_H^t = \frac{d\mathbb{P}_{\mathbf{x}}(T_t)}{d\mathbb{P}_0(T_t)}.$$

For fixed allowed witness, R_t is a nonnegative martingale under $\mathbb{P}_0$, starting at one. At an update only the responding block changes. The preceding estimates give $r_H^t = 1 + O(mb_*)$, except on an event whose raw probability averaged with $\mathbf{1}_{\mathcal{A}}$ is at most $e^{-n^{c_2}}$, for a fixed $c_2 > 0$. This exceptional event includes a small cylinder, a failed conditional survival estimate, or a witness with a failed no-transcript denominator estimate. The last set has small integrated mass even though every ratio starts at one on it.

Stop at the first exception to these factor bounds or the first $R_t > B_* := e^{n^{c_3}}$. Choose $c_3 > 0$ smaller than c_2 and than the positive exponent in the broad-law deviation estimate below. Before stopping, the next multiplier $Z_t = r_H^t/r_H^{t-1}$ has conditional mean one and is $1 + O(m_*b_*)$ off the next exception. On that exception, $c_H \geq .15^m$ and the previous good factor give the crude bound $Z_t \leq e^{O(m_*)}$. Hence

$$Z_t^2 \leq 1 + 2(Z_t - 1) + O(m_*^2 b_*^2) + e^{O(m_*)} \mathbf{1}_{\text{next exception}}.$$

Writing τ for the stopping time and averaging over both raw states and uniform witnesses gives

$$\begin{aligned} \mathbb{E}[\mathbf{1}_{\mathcal{A}} R_{t \wedge \tau}^2] &\leq (1 + O(m_*^2 b_*^2)) \mathbb{E}[\mathbf{1}_{\mathcal{A}} R_{(t-1) \wedge \tau}^2] \\ &\quad + B_*^2 e^{O(m_*) - n^{c_2}}. \end{aligned}$$

The linear term vanishes by the martingale property. The barrier is what permits the merely integrated exception estimate in the last term. There are at most $n \text{polylog}(n)$ updates, and $n \text{polylog}(n) b_*^2 = o(1)$; thus the stopped second moment is $O(1)$.

Under the tilted law, a barrier stop has integrated probability at most $B_*^{-1} \mathbb{E}[\mathbf{1}_{\mathcal{A}} R_\tau^2] = O(B_*^{-1})$. Exception stops cost $\exp(-n^{\Omega(1)})$, using the pre-stop barrier, the maximum next multiplier and the raw exception bound. Off stops, tilted deviation probability is at most B_* times raw deviation probability. For every fixed transcript the output U is broad and independent of the witness; deep discrepancy, applied sequentially for a pair, bounds its exceptional uniform witness mass by $\exp(-n^{\Omega(1)})$, with the slack in (18.2). The choice of c_3 leaves the same type of bound after multiplication by B_* . This proves the required integrated tilted estimate.

Returning to surviving witnesses. Let $p_{\text{tilt}}(\mathbf{x})$ be the tilted deviation probability. Undoing the survival conditioning supplies the factor $\prod_k c_k(\mathbf{x})$. The integrated singleton comparison, the low-mode degree bounds and the second-moment estimate after (17.2) give

$$\mathbb{E}_{\mathbf{x}} \left[\mathbf{1}_{\mathcal{A}} (2^{a'd} \prod_k c_k(\mathbf{x}))^2 \right] \leq e^{O(\log n)}.$$

For a singleton this uses the degree bounds; for pairs it uses the stated pair second moment. The relative $1 + O(n^{-3})$ perturbation of each critical marginal contributes a bounded factor. Cauchy-Schwarz, using $p_{\text{tilt}}^2 \leq p_{\text{tilt}}$, bounds the expected number of surviving deviating witnesses by

$$M_i^{a'} 2^{-a'd} \left(\mathbb{E} \mathbf{1}_{\mathcal{A}} (2^{a'd} \prod_k c_k)^2 \right)^{1/2} (\mathbb{E} \mathbf{1}_{\mathcal{A}} p_{\text{tilt}})^{1/2} = \exp(-n^{\Omega(1)}).$$

Indeed $M_i \leq n^{2^n}$ and $d = n - O(\log n)$, so the first factor is polynomial. The bounds are uniform in the fixed seeds and noncritical states. Averaging them and restoring the bounded path comparison proves the lemma. $\square$

18.3 Terminal conditioning of initial sampling

Fix any family of independent mask profiles of the kind specified above. Given S_{fin} , let p_F be the probability of each late bad F under the reference late process for these profiles, with no late constraints imposed. Include as terminal requirements:

- individual pool typicality and (17.1) everywhere;
- final S_v avoidance;
- $p_F \leq e^{-n^\delta}$ for every F , with sufficiently small fixed $\delta > 0$.

For each of the latter two kinds, gate the failure on the pool requirements throughout a deterministic region sufficient for its restricted resampling simulation. That is, start with all involved direct cell consultations and their incidence events (also the defining site as needed), then take simulation balls of radius $2T_s + O(1)$ large enough by deterministic locality, including all required event scopes and pool inputs. On success of the first kind globally these gates are passed. Slot and cell-tape scope sizes per requirement, and requirement incidences per slot or cell tape, are at most $n^{O(A_c+4)T_s+O(r)}$. Indeed graph neighborhoods per resampling round enlarge by polynomial factors as above. Before expanding those simulations the raw F consultations/influences only use spatial radius $O(r)$ (including paths to compute direct gate conditions), with polynomial

cell/position size ratios and polynomially many tests per late role. Here the absolute constants fit by taking P sufficiently large in terms of A_c as above, using $r = o(T_s)$.

Write $\mathcal{G}_F$ for this local pool gate when considering the terminal requirement for p_F .

Proposition 18.3 (Terminal avoidance and local comparison). *The terminal requirements admit simultaneous conditioning on avoidance. Each bad requirement has probability at most $n^{-PT_s/3}$ before conditioning, also after any one global slot pin. Under the resulting law, nonnegative tests on the stated unions of local regions have upper comparison with unconditioned initial sampling at a multiplicative cost tending to one.*

Proof.

Bounds with one prescribed slot. Pool failures and final S_v failures already have the required $n^{-PT_s/3}$ bound from Lemma 17.2 and Proposition 17.4, also after one prescribed slot-to-bin mapping. For a late event define

$$p_F(S_{\text{fin}}) = \Pr_{\text{ref}}(F \mid S_{\text{fin}}).$$

The sketch-conflict and true-hit events have the required tiny expectation uniformly in the entering history by Lemma 18.1. The prefix-moment event requires the following replay.

Use its critical cells, omitting the cell of the globally prescribed slot if it is among them. Mark every resampling event touching a critical cell and every neighbor of such an event in the cell-scope graph. Let the target cells be the union of scopes of events touching critical cells. Their number is polynomial, below $n^{10(A_c+10)}$. Every marked execution touches this target set, so belongs to its backward execution closure. Let B_F be the event that this closure has more than T_s executions. On the required pool gate, its probability is at most $n^{-PT_s/2}$. Put $\eta = e^{-n^\delta}$ and split before applying Markov:

$$\Pr(\mathcal{G}_F, p_F > \eta) \leq \Pr(\mathcal{G}_F \cap B_F) + \eta^{-1} \mathbb{E}[\mathbf{1}_{\mathcal{G}_F} \mathbf{1}_{B_F^c} p_F].$$

Thus the factor η^{-1} will apply only to the replay estimate on B_F^c .

Replay with a full neighbor buffer. On B_F^c , record the complete list ω of marked site/round executions. At most T_s entries gives at most $\exp(O((A_c + 4)T_s \log n))$ possible lists. For one fixed list, force precisely those marked executions in their recorded rounds, regardless of truth. At unmarked sites use the original priority rule, including its readings of neighboring truths. If off-path forced and unforced executions overlap, advance each involved cell only once in that round. This defines a replay on every input and agrees with the original process whenever its marked occurrence list is ω .

An event whose truth reads a critical cell is marked. Every decision that could read that truth is marked as well, because the entire neighbor buffer was included. Hence no unmarked decision reads a critical truth. The final critical tape indices are prescribed by ω ; noncritical final outputs depend only on noncritical inputs. For this fixed pattern, discard its consistency condition in the nonnegative upper integral, retaining individual typicality of critical pools. The consulted permutation slots compare to iid slots at $1 + o(1)$ cost, with the global slot pin maintained outside the critical cells. Conditional on the other inputs and on each critical pool's individual typicality, its prescribed terminal entry is a fresh cell state, independently across critical cells. Thus Lemma 18.2 gives

$$\mathbb{E}[\mathbf{1}_{\mathcal{G}_F} \mathbf{1}_{\{\text{pattern } \omega\}} p_F(S_{\text{fin}})] \leq (1 + o(1))e^{-n^{c_1}}.$$

The inequality uses the enlarged replay integral just described; normalizing the individual typicalities costs only negligible factors. Summing these pattern bounds in the truncated expectation above, with $\delta < c_1$, proves the terminal bad bound. Cell tapes may be represented by independent randomizers before pools are drawn, using fixed lookups for every possible local pool.

Leaves of the permutation experiment. Split each terminal bad event into leaves specifying the exact images of its consulted permutation slots and a measurable event of its consulted tapes. Declare two leaves neighbors if they share a domain slot, an image bin, or a cell tape. Image collisions matter even when their domain slots differ. For the negative-dependency precedent for partial mappings in a uniform injection see Lu and Székely [27, Theorem 1]. Here the following coupling also handles the independent tape variables.

Force a positive-probability leaf by successively swapping each prescribed image into its required slot, and condition its tape data on the prescribed tape event. This yields the conditional law given the leaf and preserves every occurring nonneighbor leaf: their domains, images and tapes avoid those used in the forcing. Consequently

$$\Pr(\text{avoid nonneighbors} \mid L) \leq \Pr(\text{avoid nonneighbors}), \quad \Pr(L \mid \text{avoid nonneighbors}) \leq \Pr(L).$$

Let $D_n = n^{C(A_c+4)T_s+O(r)}$ bound both scopes and incidences, for one fixed constant C . At a fixed domain slot or tape the sum of incident leaf probabilities is at most $D_n n^{-PT_s/3}$. For a fixed image bin D in patch i , its sum is at most

$$\sum_{E,s} \Pr(D_s = D, E) = B_i^{-1} \sum_{E,s} \Pr(E \mid D_s = D) \leq D_n n^{-PT_s/3},$$

where each event is charged through its consulted slots. There are at most B_i such slots, each with incidence at most D_n , and the one-slot bound applies to every summand. Charges twice the raw leaf probabilities satisfy the conditional local lemma when P is sufficiently large compared with $C(A_c + 4)$, since a leaf touches at most D_n slots, images and tapes.

The same argument compares tests supported on a polynomial union of these local regions, or on $n^{O(r)}$ direct consultations expanded by their deterministic simulation horizons. Split a test event by its consulted slot images. Each resulting leaf has the same nonneighbor property; the sum of touching charges is $o(1)$. Dropping those constraints therefore costs a reciprocal avoidance product $1 + o(1)$. Summing test leaves, and then integrating level sets, proves the stated comparison for every nonnegative test on these deterministic scopes. This includes the column moments and endpoint products below. $\square$

18.4 Staging and late injectivity

Proposition 18.4 (Completion of the late assignment). *There are independent mask profiles fixed in advance and conditional injective samplers for the successive late classes such that the full run completes with probability tending to one under terminal-conditioned initial sampling. On a full run all true-hit and deletion conclusions hold. Each executed class has the localized upper comparison with its reference transition used below.*

Proof.

Balance in the baseline. Use the full initial process, including finite resampling and fallbacks, on iid uniform slots before terminal conditioning, followed by unconstrained reference late transitions. This is the baseline for choosing profiles; its cell states are not fresh independent states. Fix profiles in processing order. For one row b , the preceding baseline history distribution is then fixed. Consider the convex hull of its baseline label-marginal vectors obtained from the allowed masks. Against any normalized nonnegative label prices, the cheapest $\lceil M_{\text{late}}/2 \rceil$ labels all have price at most $2/M_{\text{late}}$. Any conditional label kernel supported on this mask has expected price at most that value, for every entering history. Separation therefore gives a fixed mixture of masks with baseline marginal at most $2/M_{\text{late}}$ pointwise. Rows of the current class do not enter one another's kernels, so their profiles can be chosen independently. Use these same profiles for terminal avoidance.

Alarms and injective sampling at one class. After q classes have been completed, use the threshold

$$a_q = \exp(-n^\delta + qn^\delta/(2r))$$

for conditional reference probabilities of every still-future bad F . Terminal avoidance supplies a_0 . At the next class, condition on its entering history and require both avoidance of the current bads and that every future probability be at most a_{q+1} . Under its product reference law a future alarm costs at most $a_q/a_{q+1} = \exp(-n^\delta/(2r))$ by Markov; current bads satisfy the incoming bound. The last threshold is still $a_r = e^{-n^\delta/2}$.

Let $q_b(y | h)$ be the reference label marginal at history h , after integrating mask and sketch data. If $\sup_y \sum_b q_b(y | h) \leq \theta_0$, apply the clock lemma with parameter $n_{\text{clock}} = \lfloor e^{n^{\delta'}} \rfloor$, where $0 < \delta' \ll \min(\delta, 1)$, and $B = 5$. The required comparisons are

$$\begin{aligned} \log M_{\text{late}} &= O(n) \ll n_{\text{clock}}, & 2e^{g_0 n}/M_{\text{late}} &\ll n_{\text{clock}}^{-A_*}, \\ n^{O(r)} &\ll n_{\text{clock}}^5, & e^{-n^\delta/(2r)} &\ll n_{\text{clock}}^{-P_*}. \end{aligned}$$

The second bound is the deterministic reference atom cap, including fallbacks. The third covers both predicate scopes and row incidences, by direct spatial locality with history fixed. The sampler assigns distinct labels, avoids the current bads and future alarms, and has joint upper comparison at factor at most two for the localized outputs needed below. Side data are part of those outputs. If the column condition fails, stop the run instead.

Column loads on reached histories. Fix a class and label y . Its class size is $L_{\text{late}} = 2^{n-1}/H_*$. Put

$$Z_b = M_{\text{late}} q_b(y | h), \quad A_y = L_{\text{late}}^{-1} \sum_b Z_b.$$

Take the n -th moment with the indicator that the run has reached this prehistory. A row weight has cap $2e^{g_0 n}$. Declare positions near if their full baseline consultation scopes overlap, including reference ancestors, deterministic initial resampling horizons and slot variables. Scope and incidence counts give near fraction $f \leq 2^{-n+o(n)}$, so $nf 2e^{g_0 n} = o(1)$.

For at most n separated positions, integrate previous class outputs backwards. At each step the nonnegative integrand is the remaining product of queried conditional row masses. Retain the event that the run reached that earlier stage with its incoming probability and column bounds until its actual kernel has been compared with the reference product. This costs one factor two for

the entire queried set. After the comparison, remove that stage's load check and all later reach restrictions; retain only the reach condition needed to compare the next earlier kernel. Reference kernels consult direct local data, so the queried sets per stage remain of size $n^{O(r)}$.

After at most r such comparisons, apply terminal comparison on the required initial regions, then replace their permutation slots by iid slots, at $1 + o(1)$ cost. The resulting law is exactly the baseline in which profiles were balanced, including its finite resampling. Separated consultation scopes give independent computations in this baseline, each of normalized mean at most two. Thus scattered moments give

$$\mathbb{E}[\mathbf{1}_{\text{reach}} A_y^n] \leq 2^r K^n.$$

Since the actual column sum is $(L_{\text{late}}/M_{\text{late}})A_y$,

$$\Pr\left(\text{reach}, \sum_b q_b(y \mid h) > \theta_0\right) \leq 2^r \left(\frac{KL_{\text{late}}}{\theta_0 M_{\text{late}}}\right)^n.$$

Here $L_{\text{late}}/M_{\text{late}} \rightarrow 0$. Union over the $\exp(O(n))$ label/class choices gives probability $o(1)$ that any reached stage stops. On a complete run all gated bads have been avoided; induction from initial validity then supplies every true-hit and deletion conclusion. $\square$

18.5 Matching the low-mode lists

On a full run let q_v be the normalized final filtered law at even role v . Successive restrictions preserve its original cleaned corner support and assigned palette. Draw an ordered pair from $q_v \otimes q_v$ conditioned on nonconflict. Given the completed run, use independent pair draws at different rows. The conditional pair law at one row is

$$\frac{q_v(x)q_v(z)\mathbf{1}_{\{|K_{\pi_i}(x,z)| \leq \xi\}}}{Z_v}, \quad 1 - Z_v \leq \Delta_i \|q_v\|_\infty \leq 8K^2 C_n^{-1} e^{-198s_i} = o(1).$$

Thus $Z_v \geq 1/2$ eventually. Each pair has distinct endpoints in the row's assigned palette, and each endpoint hits every actual odd neighbor.

Fix one patch and palette, writing $M = M_i/\chi_i$. It has at most $2M$ labels. The allocated patch has $\Theta(2^n M_i/N)$ even roles, divided equally among its χ_i role colors by Lemma 17.3. Thus the even-role count in this palette satisfies $L \leq KM/C_n$, and the low-mode size bounds give $L = 2^{n-o(n)}$. Fix $t' \leq n$ distinct roles in it. Let $C(v)$ be the own cell of v together with the cells supplying its external early labels. Join two tested roles in the geometric overlap graph if their sets $C(v)$ intersect or they have a common late neighbor. Each role has at most $n^{A_c+O(1)}$ partners. Let j be this graph's rank, the number of vertices minus the number of components, and $m \leq 2j$ the number of nonisolates. Enlarged resampling horizons need not be disjoint: a joint comparison will remove that dependence before we use disjointness of the direct cell sets.

Proposition 18.5 (Joint endpoint kernels). *For every fixed tuple of positions just described there are deterministic nonnegative symmetric pair weights H_v , chosen before endpoint values, such that every assignment of two endpoints per row satisfies*

$$\Pr(\text{specified pairs, full run}) \leq e^{O(r)} K^{t'} \exp(.01nm + C't'j) \prod_v H_v(x_v, z_v), \quad (18.5)$$

where C' is fixed and the nonnegative symmetric kernels on palette labels (allowing different kernels according to the positions) satisfy

$$\sup_x \sum_z H_v(x, z) \leq K/M, \quad \max H_v \leq M^{-2} e^{.01n}.$$

In particular we can use $H_v = M^{-2}$ on nonisolates.

We specify the initial-state test in Figure 2. Let $J_v(S)$ assert initial validity at v : a valid internal prior with its cap and single-corner support, S_v false, and palette mass at least $1/(4\chi_i)$. Let

$$A_v(x, z) = \mathbf{1}_{\{x, z \text{ in the assigned palette and } \mathcal{X}_i\}} \mathbf{1}_{\{|K_{\pi_i}(x, z)| \leq \xi\}}.$$

Define $\Phi(S) = 0$ if any $J_v(S)$ fails; otherwise set

$$\Phi(S) = \chi_i^{2t'} \prod_v \left[A_v(x_v, z_v) \mathbf{1}_{\{x_v, z_v \in \mathcal{D}_v^0(S)\}} U_v^0(x_v; S) U_v^0(z_v; S) \right].$$

The support indicator is redundant when the list product is positive, but records precisely the support retained in the comparison. This test reads only $C_* := \bigcup_v C(v)$, with at most $n(n+1)$ cells. Let $\mathcal{G}$ require individual pool typicality and (17.1) throughout their deterministic resampling horizons. Let $\mathcal{T}$ retain only individual pool typicality. These scopes fit the terminal and resampling comparison propositions, even if their horizons overlap.

Proof.

Figure 2 records the comparison order for the initial-list reduction below.

Late factors and their side-data gates. Write $a_v \geq 1/(4\chi_i)$ for the initial palette mass and $m_{v,s} \geq .5 - 3e_{v,s}$ for each subsequent true hit mass. On a full run the final row is

$$q_v(x) = \frac{\mathbf{1}_{\text{palette}}(x) U_v^0(x)}{a_v} \prod_{s: v \sim b_s} \frac{I_x(y_{b_s})}{m_{v,s}}.$$

Pair normalization costs at most two, and the squared hit denominators cost $4 \exp(O(e_{v,s}))$ per late incidence. Summable errors therefore bound the conditional product of specified pair weights by $K^{t'} \Phi(S_{\text{fin}})$ times $\prod_{v \sim b_s} 4I_{x_v}(y_{b_s}) I_{z_v}(y_{b_s})$.

At a late role shared by several tested rows, retain one pair hit by a rule fixed from the position tuple and replace the other factors by four. Two distinct even roles have at most two common neighbors; inside geometric components this removes $O(t'j)$ incidences. For each retained pair keep the side-data predicate certifying (18.3). It reads the entering history, mask and sketches, not the current label.

Integrate backwards by class. Keep that class's reach and incoming bounds until applying its conditional sampler comparison at factor two. The side-data predicate remains while the corresponding reference label is integrated, so each retained factor $4I_x I_z$ has integral at most $\exp(O(le_{v,s}))$. Only then remove its predicate. The product of these costs is $K^{t'}$, since each row has at most one incidence in each class and $\max(1, h_i) \sum_s e_{v,s} = O(1)$. The stage comparisons contribute $2^r = e^{O(r)}$, and removed incidences contribute $e^{O(t'j)}$. What remains is exactly the nonnegative initial test Φ defined above, with none of the late side-data gates left in it.

Terminal and resampling comparisons. Retain $\mathcal{G}$ while comparing terminal-conditioned initial sampling to the unconditioned process. Conditional on these pools, Proposition 17.4 applies jointly to the nonnegative test Φ on C_* , replacing final states by independent fresh states at factor $1 + o(1)$. Only after this fixed-pool comparison weaken $\mathcal{G}$ to $\mathcal{T}$ and replace consulted permutation slots by iid slots.

Write $\mathbb{E}_{\text{term}}$ for terminal-conditioned initial sampling, $\mathbb{E}_{\text{perm}}$ for the original process with permutation pools and finite resampling, and $\mathbb{E}_{\text{fresh}}(\cdot \mid \mathcal{P})$ for independent fresh states on fixed

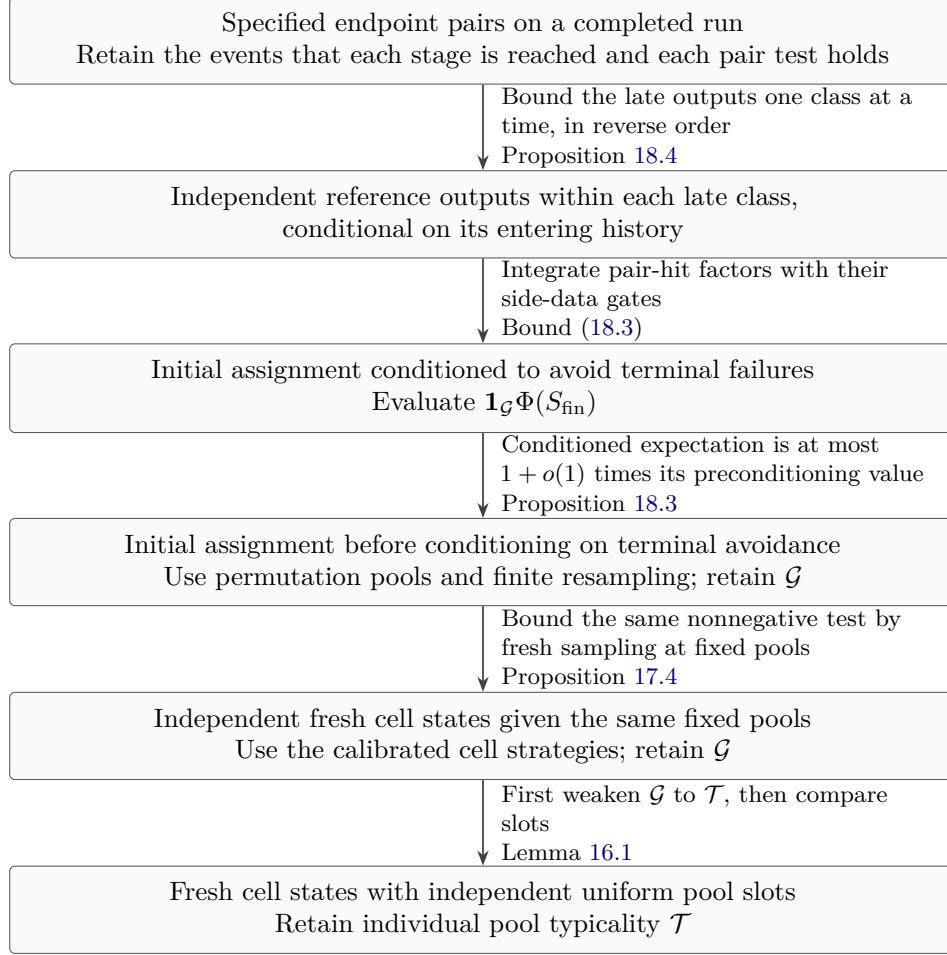


Figure 2: Upper comparisons for the endpoint product in Proposition 18.5. The first two operations alternate, one class at a time in reverse order. The late comparisons are conditional at each class, with the entering history indicator retained until that comparison; each pair side-data gate remains until its hit factor is integrated. The pool gate $\mathcal{G}$ includes individual typicality and (17.1) throughout the fixed deterministic simulation horizons. It is retained through the fixed-pool resampling comparison, then weakened to individual typicality $\mathcal{T}$ before the slot comparison. Every arrow bounds the indicated nonnegative test; none asserts equality of laws or independence after avoidance. The diagram stops before the later within-cell label, bin, pool and history comparisons for nonisolated rows.

pools. The zero convention in the definition of Φ makes it a nonnegative test on all configurations. These comparisons give

$$\begin{aligned}\mathbb{E}_{\text{term}}[\mathbf{1}_{\mathcal{G}}\Phi(S_{\text{fin}})] &\leq (1 + o(1)) \mathbb{E}_{\text{perm}}[\mathbf{1}_{\mathcal{G}}\Phi(S_{\text{fin}})] \\ &\leq (1 + o(1)) \mathbb{E}_{\text{perm}}[\mathbf{1}_{\mathcal{G}}\mathbb{E}_{\text{fresh}}(\Phi \mid \mathcal{P})] \\ &\leq (1 + o(1)) \mathbb{E}_{\text{iid}}[\mathbf{1}_{\mathcal{T}}\mathbb{E}_{\text{fresh}}(\Phi \mid \mathcal{P})],\end{aligned}$$

where the last expectation uses independent uniform slots, and each $1 + o(1)$ may absorb the preceding initial-comparison factors. The fixed-pool comparison is applied before weakening $\mathcal{G}$ to $\mathcal{T}$. The endpoint probability with full-run indicator is bounded by $\exp(O(r) + O(t'j))K^{t'}$ times the left side, including the cost of omitted shared late incidences in the backward integration above.

Isolated rows. After the three displayed comparisons, discard typicality conditions on pools outside C_* . The direct cells of an isolated row are disjoint from every other row's cells, so their iid-pool fresh expectation factors. An upper pair weight is

$$H_v(x, z) = \chi_i^2 A_v(x, z) \mathbb{E}_{\text{iid, fresh}} \left[\mathbf{1}_{\mathcal{T}_v} \sigma_v(x) \sigma_v(z) \prod_{b \in B_v^e} \frac{I_x(y_b) I_z(y_b)}{D_{i(b)}(x) D_{i(b)}(z)} \right],$$

where $\mathcal{T}_v$ requires typicality of the direct pools, and invalid internal-prior contributions are zero. We have removed the initial mass, masked-support and palette-retention gates and the redundant positive-support indicator from the row factor of Φ . Every removal increases the integral.

The external observation cells are distinct and separate from the own cell. Their fresh singleton comparisons give $(1 + O(n^{-3}))\pi_{i(b)}$ after integration. Then (17.2) and Lemma 16.7 give

$$\sum_z H_v(x, z) \leq K \chi_i \mathbb{E}[\mathbf{1}_{\text{typ}} \sigma_v(x)] \leq K \chi_i / M_i = K / M.$$

For the entrywise bound, each prior atom is at most $M_i^{-1} e^{O(\log n)}$; crossing factors contribute $e^{O(\log n)}$, and the nonconflict cutoff and degrees bound bulk pair-hit ratios by $1 + O(\xi + \log n/n)$. Hence $H_v \leq M^{-2} \exp(O(\log n) + O(\xi n)) \leq M^{-2} e^{.01n}$ by the prior choice of ξ .

Reducing nonisolates to pair-hit queries. For each of the other m rows first use, on J_v , the cap $\sigma_v \leq c_i := M_i^{-1} e^{O(\log n)}$. In particular

$$\mathbf{1}_{J_v} \mathbf{1}_{\{x, z \in \mathcal{D}_v^0\}} \sigma_v(x) \sigma_v(z) \leq c_i^2.$$

After this inequality remove all remaining state-dependent initial-validity, support, mass and palette-retention indicators for that row. Keep only the deterministic endpoint predicate A_v . Bound crossing denominators on the envelope and discard their hits. Bulk pair denominators cost $4(1 + O(\log n/n))$ each. The remaining random factors are pair-hit indicators of external early bulk labels. There are no residual tests of the removed own priors.

Index such a query by its even row, observed odd role, outer word, observed slice, cell and group. At each observed slice choose one even outer-word family to retain, by a rule depending only on the positions. Discard the other hit incidences there. Two different outer words have at most two common one-flip words, and a row has at most one observation in a fixed slice. Charging to pairs of rows within the geometric components removes $O(t'j)$ hits. Let $k \leq mn$ be the number of retained queries.

Queries in the same slice now arise from one even outer word. Their internal words have the same palette color and are distinct, so their distances exceed $500\rho h_i$. In cluster mode two members of one projected group differ in at most two internal bits, whereas the fixed threshold ensures $500\rho h_i > 2$. Thus these groups are distinct. Their raw primitive consultation radii are at most $10\rho h_i$, so those consultations are disjoint as well. When $h_i = 0$ there is no sharing. This independence will be used only after removing slice conditioning below.

Repeated bins and reverse summation. Apart from deterministic factors, the integral to bound is now

$$\mathbb{E}_{\text{iid, fresh}} \left[\mathbf{1}_{\mathcal{T}_{\text{qry}}} \prod_{a=1}^k f_a(Y_a) \right], \quad f_a(y) = I_{x_{v(a)}}(y) I_{z_{v(a)}}(y),$$

where $\mathcal{T}_{\text{qry}}$ requires typical pools for the queried odd cells. Conditional on these pools and their successful W , let D_a be the bin chosen for group g_a , and set

$$u_a^W(D) = \int f_a dU_{g_a}^W(D), \quad J(D) = \{a : D_a = D_b \text{ for some } b < a \text{ in the same cell}\}.$$

For fixed bin values discard $f_a \leq 1$ whenever $a \in J(D)$. The remaining queries use at most one label per bin in each cell, so the conditional label comparison costs $e^{d_i^{\cdot, 01} k}$. Next the group-bin comparison, applied to all k distinct group queries, costs $e^{d_i^{\cdot, 05} k}$. In direct mode bins are singleton labels; its calibrated cell sampler gives the corresponding upper comparison at $e^{o(1)k}$, since $k \leq n^2$.

For a fixed repeat mask J , the sum after these comparisons is bounded by the appropriate calibration factor times

$$\sum_{D_1, \dots, D_k} \mathbf{1}_{\{J(D)=J\}} \prod_{a=1}^k \tilde{q}_{g_a}(D_a) \prod_{a \notin J} u_a^W(D_a).$$

Enlarge the repeat indicator by retaining only distinctness of the kept targets within each cell and the necessary earlier-bin equality for each removed query. Sum removed bins in decreasing index order. At the summation for $a \in J$, all earlier candidate bins are still fixed, even if they will be removed later, so

$$\sum_{D_a \in \{D_b : b < a \text{ in the same cell}\}} \tilde{q}_{g_a}(D_a) \leq k \|\tilde{q}_{g_a}\|_\infty \leq n^{-A_c+3}.$$

No kept factor $u_a^W(D_a)$ reads a removed bin. This is why the initial indicators were discarded before the two sampler comparisons. The total repeat cost is $(n^{-A_c+3})^{|J|}$; kept-bin distinctness still remains for the pool integral.

Removing pool restrictions and counting the losses. On typical pools the kept restricted bin probabilities satisfy

$$\tilde{q}_g(D) \leq (1 + \varepsilon)(B_i/L_i^{\text{slot}})q_g^W(D)\mathbf{1}_{\{D \in \mathcal{P}\}}, \quad \varepsilon = 10^{-5}.$$

Here q_g^W is the raw bin law, uniform before trims in direct mode; the bound includes both trim losses. Undo the W -cell load conditioning, at negligible logarithmic cost per query, while its typical-pool hypothesis is still retained. Then discard the remaining pool tests. The slice-conditioned W 's

are now independent of the iid pools. For a distinct kept targets in one cell, their simultaneous presence costs at most

$$(L_i^{\text{slot}})_a / B_i^a \leq (L_i^{\text{slot}} / B_i)^a.$$

This cancels the pool-restriction factors. Discard kept-target distinctness and sum their raw bin kernels. Finally remove the (14.3) slice conditioning at cost at most another $(1 + \varepsilon)^k$. Raw primitive inputs are now independent between slices and between the separated consultations within them.

Their pair-hit means use π_i^0 in cluster mode, and are at most $(1 + 10^{-4})/4$, by the degree bounds, the pair cutoff and the total-variation estimate following (14.4). Let δ_{cal} include the two per-query calibration logarithms and the load-normalizer logarithm. With ξ already fixed small, increase the fixed threshold Q_0 so that

$$\delta_{\text{cal}} + 2 \log(1 + \varepsilon) + \log(1 + 10^{-4} + 4n^{-A_c+3}) \leq .005$$

eventually. This makes the fixed low-cluster errors small; it does not assume they tend to zero with n . Summing all repeat masks therefore bounds the k pair hits by

$$e^{(\delta_{\text{cal}} + 2 \log(1 + \varepsilon))k} ((1 + 10^{-4})/4 + n^{-A_c+3})^k \leq 4^{-k} e^{.005k}.$$

For $k = 0$ the bound is one. Each retained pair hit contributes its factor $1/4$, cancelling a bulk denominator factor four. Discarded hits leave $e^{O(t'j)}$; the prior caps, crossing factors and degree drift leave $e^{O(m \log n)}$. Since $k \leq mn$, the total is at most $M_i^{-2m} \exp(.01nm + O(t'j))$. The palette factor χ_i^{2m} turns this into M^{-2m} with the asserted error. Together with the isolate weights and the preceding late and initial comparisons, this proves (18.5). $\square$

Lemma 18.6 (Two-endpoint Hall estimate). *Assume that the full run has probability $1 - o(1)$, as supplied by Proposition 18.4. The joint endpoint bounds of Proposition 18.5, with the stated palette sizes and geometric partner counts, then imply simultaneous Hall matchings in every low-mode palette with positive probability of full success.*

Proof.

Geometric averaging. Choose a small constant $\eta > 0$ with $.02 + C'\eta < (\log 2)/2$, and put $t_0 = \lfloor \eta n \rfloor$. For a uniformly chosen distinct p -set of positions, $p \leq t_0$, a geometric overlap graph of rank j contains a forest of j necessary partner relations. Specifying its edges costs at most p^{2j} ; exposing along that forest uses the partner fraction $2^{-n+o(n)}$ once per edge. Since $m \leq 2j$, the correction in (18.5) is at most $e^{(.02+C'\eta)nj}$. Its average is bounded by a convergent geometric sum with ratio $e^{-(\log 2)n/2+o(n)}$.

The endpoint multigraph. This graph is different from the geometric overlap graph: its vertices are host labels, and each even row is an edge joining its two chosen endpoints. Hall fails precisely when some set of row edges has fewer incident endpoint vertices than edges. Such a set has a connected violating component. The actual distinct-endpoint condition is retained in the following sums, even for nonisolates whose convenient upper weight $H_v = M^{-2}$ is positive on the diagonal.

Connectivity of p distinct row edges is witnessed by a tree on those rows, with one endpoint identification along each tree link. There are at most $4^{p-1}p^{p-1}$ such diagrams. These identifications leave $p + 1$ abstract endpoint variables, and the row edges form a tree on them. Summing a product of the row weights by successively integrating leaves costs at most $2M(K/M)^p$, using the row-sum bound and at most $2M$ choices for the last variable.

A connected Hall-violating p -set, $p \geq 3$, uses fewer than p endpoint vertices. Thus at least two further mergers of distinct current endpoint classes occur, with $O(p^4)$ choices. Keep just these two mergers and discard any additional identifications for an upper bound. The resulting quotient has $p - 1$ vertices and p row edges. On the retained distinct-endpoint support it has no loops; parallel edges are allowed. A spanning tree uses $p - 2$ edges, leaving exactly two edges to which the maximum-weight bound applies. The corresponding sum is at most

$$\underbrace{2M}_{\text{root choices}} \underbrace{(K/M)^{p-2}}_{\text{tree edges}} \underbrace{(M^{-2}e^{.01n})^2}_{\text{two remaining edges}} \leq K^p M^{-p-1} e^{.02n}.$$

The mergers here are independent identifications of equivalence classes, not probabilistically independent events. Different rows may use different kernels; the same two uniform bounds suffice.

Summing obstructions. Equation (18.5), geometric averaging and $\binom{L}{p} \leq L^p/p!$ bound the expected number, with full-run indicator, of connected t_0 -row sets by $e^{O(r)} M(K/C_n)^{t_0}$. For smaller connected p -row sets with fewer than p endpoints the bound is

$$e^{O(r)} p^4 M^{-1} e^{.02n} (K/C_n)^p.$$

Sizes one and two cannot violate Hall because each row has distinct endpoints. For the large-set estimate,

$$\log(M(K/C_n)^{t_0}) = n \log 2 + o(n) + \eta n \log(K/C_n) + O(\log n),$$

which is eventually negative at a linear rate, however slowly $C_n \rightarrow \infty$. For the small-set sum, $M^{-1}e^{.02n} = e^{-(\log 2 - .02)n + o(n)}$, and $\sum_{p \geq 3} p^4 (K/C_n)^p$ is bounded. Both estimates still tend to zero after summing over the $e^{o(n)}$ patches and palettes.

Every connected component with at least t_0 row edges contains a connected t_0 -row set. Every Hall failure contains a connected violating component. Consequently the probability of a full run with no Hall obstruction is at least its probability $1 - o(1)$ minus the preceding expected obstruction counts, and is positive. Hall gives distinct representatives simultaneously in all palettes. Palettes and patches are disjoint, so these choices combine. The early odd labels are already injective, and the disjoint injective late pools replace their dummy labels. Every required edge is present, yielding the monochromatic cube and excluding the low modes. $\square$

Conclusion of Theorem 1.1. An unbounded ratio $R(Q_n)/2^n$ would supply the sequence of Lemma 2.1. The earlier exclusions reduce it to deep discrepancy in both orientations. Patch extraction then leaves high or low modes. Section 15 excludes the high modes, and the preceding matching excludes the low modes. No such sequence exists. The ratio therefore has finite supremum, giving an absolute C with $R(Q_n) \leq C2^n$ for all n . This existence argument does not provide a numerical value for C .

A Quantitative guide to the regimes

Table 1 records the cases of the proof in their logical order. All discrepancy and eventual-absence conclusions concern the retained sides of the stabilized counterexample subsequence, after the total $o(N)$ label removals specified in the proof, and hold eventually along that subsequence. Width always means $\log(N \max \mu)$, using the original host-side size N . Thus a normalized atom bound $N \max \mu \leq n^D$ gives width $D \log n$. Defect means density of the other color. Availability means that, for some fixed $\kappa > 0$, a witness can be chosen after every removal of at most κN labels from each side; the witness may depend on the removed sets. See Section 3. Every constant and exponent is fixed before $n \rightarrow \infty$, with the local meaning assigned in its cited section. In particular, parameters with the same name in different rows need not have the same value.

For the atom bounds below, p_b denotes an odd probability row and p_v an even probability row. A tuple posterior Π_v is a law on k first-side labels, so $N^k \max \Pi_v$ is its density bound relative to the uniform product law. This differs from the one-label bound $N \max p_v$. The limiting exponents $H^\dagger, H_L^\dagger$ are those of Definition 9.1; they delimit the available-bias ranges at power and linear width budgets.

The last six rows use initial discrepancy (7.2), deep discrepancy (12.1) in both orientations, and the eventual cluster-witness absence of Corollary 10.2. Here g, q are the residual bias and cluster scales of Definition 13.1; M_i is the size of either side of patch i , ℓ_i its prefix length, h_i its internal dimension, d_i its bin size, and s_i its gain parameter. The cleaned first-side degrees are measured by $D_{\pi_i}(x) = \sum_y \pi_i(y) \mathbf{1}_G(x, y)$, where π_i is uniform on Y_i in the bounded and direct branches and is the comparison law of Section 14 in the cluster branches. Every nonbounded row is reached only after $\max(g, q) \geq M_1 Q_0$. For such a patch the common allocation budget is

$$\ell_i, \log(N/M_i) \leq s_i/(1000u),$$

where u is the fixed even moment order from Section 12. The word *bounded* names the first extraction branch; it does not assert that the scales in every other branch tend to infinity.

Table 1: All regimes: hypotheses, quantitative costs, and the interface used by the next stage. In the first reductions an available witness produces a cube; its exclusion supplies the stated discrepancy or structural conclusion.

Regime and input	Width, density, and loss budget	Output and location
Counterexample sequence	$n \rightarrow \infty, C_n = N/2^n \rightarrow \infty, N \leq n2^n$. No rate is imposed on C_n .	The bipartite contradiction setting; Lemma 2.1.
Power-width bias versus purity	Widths (n^β, n^γ) , $0 < \beta \leq \gamma < 1$; an available patch with absolute bias $\geq n^{-h}$, and no pair of laws on the current sets with defect $\leq e^{-n^h}$ in either color at the respective doubled widths $(2n^\beta, 2n^\gamma)$. Here $\omega = \min(\beta, 1 - \gamma)/1000$, $h = \omega/10^6$. Filter widths fit the plateau margin; $N \max p_v \leq \exp((\log 2 - cn^{-h})n)$.	The two hypotheses together give a cube. The density gain pays for local repetitions and predictive tests; Lemma 4.1.

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Table 1 continued

Regime and input	Width, density, and loss budget	Output and location
Constant broad side	Balanced first laws of width n^γ , $0 < \gamma < 1$; at least χN columns of degree $\geq .95$ for each tag. Given the parents, the stream law relative to $R[u]$ is $\leq e^{a_0 u}$, $a_0 = 1/3$. The posterior satisfies $N^k \max \Pi_v \leq e^{\tau_0 k n}$, and $N \max p_v \leq O(e^{\tau_1 n})$, with $a_0 < \tau_0 < \tau_1 < \log 2$.	A cube in that color. The remaining gap below $\log 2$ pays for geometric overlap; Lemma 5.1.
Polynomial broad-side concentration and polynomial defect	Balanced laws of widths at most n^γ and $D_* \log n$, $0 < \gamma < 1$; every broad-side degree is $\geq 1 - n^{-p_0}$, for fixed $p_0 > 0$. After marginalizing the tag, $N^k \max \Pi_v \leq e^{4kn}$; $N \max p_v \leq O(e^{.55n})$. The constant $D_* > 0$ is universal.	A cube in one of the colors; Lemma 6.1. This excludes the sparse rectangles tested in (7.1).
Small-grid purity	Available patches have both widths $\leq n^{d/2}$ and every second-support degree $\geq 1 - e^{-n^p}$, $p > 0$. Every law pair of widths $(D_0 \log n, n^{1/4})$ has both color densities $> n^{-c}$, where $0 < D_0 < \min(.1, D_*/2)$. Use $d < D_0/1000$, $b = d/4$, $c = b/20$; $\log(N \max p_b) \leq n^{d/2} + 2cn^d \log n + O(\log n)$ and $\log(N \max p_v) \leq .04n^{4d} + o(n^{4d})$.	Purity gives a cube. Together with the power-width alternative this yields discrepancy $n^{-\eta_0}$ at both widths n^{η_0} , for a fixed $\eta_0 > 0$; Lemma 7.1 and Corollary 7.2.
Asymmetric purity	Under initial discrepancy (7.2), a finite balanced mixture has widths (n^γ, n^β) , $\gamma < 1$, $0 < \beta < \tau/4$, where $\tau = \min(\eta_0/2, .04)/4$; every second-support degree is $\geq 1 - e^{-n^p}$ for fixed $p > 0$. With $s = \lceil n^\tau \rceil$, $N \max p_b \leq e^{.02s \log n}$ and $N \max p_v \leq e^{.1n}$. No relation $p > \tau$ is required.	Purity gives a cube. Consequently, at every admissible fixed rational width pair, all weighted densities differ from $1/2$ by at most a fixed negative power of n , in both orientations; Lemma 8.1 and Corollary 8.2.
Intermediate sublinear bias: some $H^\dagger < 1$	Shallow widths (n^{x_s}, n^{y_s}) and deep widths (n^{x_d}, n^{y_d}) , $y_s < 1 - \sigma < y_d$. Available bias $2a_* = n^{-h_+}$; deep error $b_* = n^{-h_-}$, $0 < h_- < h_+ < 1$, $h_+ - h_- < \chi/10$. The gain cna_* pays for first-side width and $O(n^\sigma \log n)$ local entropy.	A cube, using a core of size $\leq n^\chi$ and an outside-core read bound $n^\sigma + 3$; Proposition 9.2. This excludes the whole range $H^\dagger < 1$, including zero.
Intermediate linear bias: both $H^\dagger \geq 1$, some $0 < H_L^\dagger < 1$	Shallow widths $(n^{x_s}, \alpha_s n)$ and deep widths $(n^{x_d}, \alpha_d n)$, $100\alpha_s < \alpha_d$. Use $a_* = \frac{1}{2}n^{-h_+}$, $b_* = n^{-h_-}$ with parameters chosen afresh in the linear case; $\sigma < \chi/10$ and broad-test error $o(a_*)$. The read-tail exponent is $na_*^2/((n^\sigma + 3)b_*^2)$.	A cube. The surplus survives the external-coordinate costs; Proposition 9.2. The two intermediate cases use different parameter orders.

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Table 1 continued

Regime and input	Width, density, and loss budget	Output and location
Full-dimensional clusters	Assume initial discrepancy in both orientations and available one-color patches. The first law and aggregate second law have widths $\leq n^\delta$; individual cluster atoms are $\leq e^{-n^\zeta}$. Codegrees, including repeated labels, are $\geq 1/4 + n^{-\delta}$. Here $\delta < \min(\eta_0, \zeta, 1)/2000$ and $k(T + m) = O(n^{500\delta})$ fits both discrepancy and atom budgets. The even bound is $N \max p_v \leq e^{(\log 2 - .01n^{-\delta})n}$.	A cube, hence eventual absence for every fixed admissible rational choice. Used in the linear jump and again in extraction; Proposition 10.1 and Corollary 10.2.
Linear-budget jump: both $H^\dagger \geq 1$, some $H_L^\dagger = 0$	Available widths $(n^{.01}, .011n)$ with column surplus $2n^{-.01}$. Reverse-orientation discrepancy $n^{-.95}$ at widths $(n^{1-\nu/4}, n^{x_0})$. The internal dimension n^1 gives gain $n^{.09}$, exceeding $O(n^{.05}) + n^{.03}$ losses.	A cube; Proposition 11.1. Thus both $H_L^\dagger \geq 1$ remain.
Deep discrepancy	Widths $(\alpha n, n^{x_*})$ in either order, for fixed $0 < \alpha, x_* < .01$; error $b_* = n^{-1+.04}$. Every fixed smaller positive error slack is available with possibly smaller budgets. The interaction tails use the setting of Lemma 12.2: the first law τ and second laws π_l have widths $\leq n^{x_*/4}$, with $ D_{\pi_l}(x) - 1/2 \leq O(b_*)$ on $\text{supp } \tau$. For each fixed u , the one-free threshold $O_u(b_*)$ has tail $e^{-\alpha n/3}$, and the two-free threshold $n^{-1-.03}$ has tail $e^{-n^{.4}}$.	Interaction moments, row trimming, and homogeneous clique peeling; Corollary 12.1 and Lemmas 12.3–12.6. Interaction bounds allow different laws at different coordinates; clique peeling uses a single common law.
Bounded extraction: $\max(g, q) < M_1 Q_0$	One patch $M_i \asymp N$, $\ell_i = h_i = s_i = 0$. On cleaned supports, $ D_{\pi_i}(x) - 1/2 \leq K/n$ and the number of labels with a large interaction is bounded by a fixed constant. The $r \asymp A_0 \log n$ late classes supply polynomial slack.	Two surviving candidates per even role, with joint bounds that give Hall; Proposition 13.3, Lemma 13.4, and Sections 16–18.
Low direct: $g > M_1 q$, $g \leq K_B \log n$	$M_i \geq cNe^{-g^{a_B}}$, $h_i = 0$, $s_i = 10^{-3}g$. On cleaned supports, $1/2 + g/(4n) \leq D_{\pi_i}(x) \leq 1/2 + 4g/n$. Prefix and width losses are $\leq s_i/(1000u)$; late classes pay for the remaining list losses.	Two surviving candidates per even role, with joint bounds that give Hall; Proposition 13.3, Lemma 13.4, and Lemma 18.6.

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Table 1 continued

Regime and input	Width, density, and loss budget	Output and location
High direct: $g > M_1 q$, $g > K_B \log n$	The same direct patch budgets and degree interval. The homogeneous peel parameter is $\leq e^{-200s_i}$, giving mass-failure probability $\leq n^{-R}$. The unnormalized row bound $M_i Z_v(x) \leq 2^n e^{-200s_i}$ pays for repeated local neighborhoods.	An odd injection and even weights Z_v supported on common neighbors, with row masses $\geq 1/2$ and column sums $\leq 1/2$. Normalizing gives Hall; Lemma 15.1.
Low cluster: $g \leq M_1 q$, $Q_0 \leq q \leq (\log n)^{c_q}$	$M_i \geq cN e^{-q^{aC}}$, $d_i = \lfloor e^{q/2} \rfloor$, $q^{M_{i_0}} \leq h_i < 2q^{M_{i_0}}$, $s_i = ah_i/10^6$. Within-bin codegree is $\geq 1/4 + a$; on cleaned supports, $ D_{\pi_i}(x) - 1/2 \leq 10q^{C_b}/n$, with $10q^{C_b} < s_i/(100u)$. Also $h_i \leq (\log n)^1$.	Exact group and label marginals preserve the mean estimates. The later assignment gives two candidates per even role and Hall; Proposition 14.1, Propositions 16.3 and 16.4, Lemma 18.6. The scales q, h_i, d_i may stay fixed.
High cluster with small bins: $g \leq M_1 q$, $(\log n)^{c_q} < q \leq (\log n)^2$	The same cluster width, gain, degree, and codegree bounds, with $q^{M_{hi}} \leq h_i < 2q^{M_{hi}}$; d_i is the smaller of $\lfloor e^{q/2} \rfloor$ and $\lfloor e^{\sqrt{\log n}} \rfloor$. Here $h_i > (\log n)^5$ and $h_i = o(d_i^{0.25})$. Exact singleton and calibrated joint estimates control repeated bins.	An odd injection and even weights Z_v with row masses $\geq 1/2$ and column sums $\leq 1/2$. Normalizing the rows gives Hall; Propositions 15.3 and 15.4.
High cluster with large bins: $g \leq M_1 q$, $q > (\log n)^2$	The same cluster budgets, now $d_i = \lfloor e^{q/2} \rfloor$ and $q^{M_{hi}} \leq h_i < 2q^{M_{hi}}$. With $k_i = \lceil h_i^{3\omega} \rceil$, $T_i = \lceil h_i^\omega \rceil$, all cluster modes have $k_i T_i \ll q^{aC/2}$, $\log d_i$. The conditional in-bin label law satisfies $\max_y U_g^W(D)(y) \leq 2e^{1.5k_i T_i}/d_i$, a superpolynomially small bound.	An odd injection and even weights Z_v with row masses $\geq 1/2$ and column sums $\leq 1/2$. Normalizing the rows gives Hall; Propositions 15.3 and 15.4.

Exhaustion and boundary conventions. The intermediate-bias proposition first excludes $H^\dagger < 1$ in either orientation. Once both $H^\dagger \geq 1$, each $H_L^\dagger$ belongs to $\{0\}$, $(0, 1)$, or $[1, +\infty]$. The first two possibilities are excluded separately, and the third gives deep discrepancy by taking strictly smaller error exponents. No conclusion at the threshold defining an infimum is needed.

During extraction the bounded case is tested first. Outside it, $g \leq M_1 q$ implies $q \geq Q_0$, so the cluster branch is defined. Equality at $g = M_1 q$ belongs to the cluster branch, equality at $g = K_B \log n$ to low direct, and equality at either cluster threshold to the lower range. The fixed choice of Q_0 makes estimates in its positive powers uniformly sufficient; it never licenses replacing a fixed-scale error by an error tending to zero with n .

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